SOA-C02

SYSOPS

LEARNER GUIDE SLIDES · WSQ

AWS Certified SysOps Administrator - Associate (SOA-C02)

WSQ Course Code: TGS-2024051413

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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It is mandatory for you to take both AM, PM and Assessment digital attendance for WSQ funded courses.

- The trainer or administrator will show you the digital attendance QR code generated from SSG portal.
- Please scan the QR code from your mobile phone camera and submit your attendance.
- Digital Attendance (Mandatory)

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ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with AWS or systems operations (if any).
- What you want to monitor, automate, or troubleshoot on AWS after this course.

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Ground Rules

- Set your mobile phone to silent mode
- Participate actively in the class. No question is stupid.
- Mutual respect. Agree to disagree.
- One conversation at one time.
- Be punctual. Back from breaks on time.
- Exit the class silently if you need to step out for phone call, toilet break etc.
- 75% attendance is required

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Lesson Plan

- Day 1
- Digital Attendance (AM)
- Trainer and Learners Introduction
- Learning Outcomes
- Course Outline
- Topic 1: ….
- Lunch Break
- Digital Attendance (PM)
- Topic 2: ….
- End of Day 1

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Lesson Plan

- Day 2
- Digital Attendance (AM)
- Topic 3 …
- Lunch Break
- Digital Attendance (PM)
- Revision
- Course Feedback and TRAQOM Survey
- Digital Attendance (Assessment)
- Final Assessment
- End of Class

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Skills Framework

1 TSC Title: Cloud Computing Application

2 TSC Code: LOG-TEM-3007-1.1

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TSC Abilities:

- A1 Implement data loading, security, performance monitoring and tuning, capacity planning, backup and recovery
- A2 Implement performance diagnostics and troubleshooting at scale in live environment
- A3 Maintain availability and performance of cloud computing operations and to ensure secure services with no unplanned downtime
- A4 Analyse processing issues and data anomalies and identify solutions
- A5 Implement and release cloud computing operations changes
- A6 Implement IT risk management measures for business continuity and disaster recovery
- A7 Implementation of service quality standards
- TSC Knowledge:
- K1 Cloud computing infrastructure components and technologies
- K2 Installation, configuration and troubleshooting of cloud computing infrastructures
- K3 Installation, configuration and troubleshooting of software packages supporting cloud computing infrastructures

Learning Outcomes

- LO1: Maintain business continuity of AWS cloud operations to ensure stable service delivery.
- LO2: Implement secure AWS cloud infrastructure setup such as data loading and performance monitoring
- LO3: Apply service quality standards to ensure security and compliance of AWS cloud operations.
- LO4: Analyze and resolve processing issues to ensure continuity and disaster recovery in AWS cloud applications.
- LO5: Troubleshoot networking performance issues at scale for optimal AWS infrastructure operation.

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LU1: Reliability and Business Continuity

- Topic 1 EC2 for SysOps
- Topic 2 Amazon Machine Image (AMI)
- Topic 3 Managing EC2 at Scale - Systems Manager (SSM)
- Topic 4 EC2 High Availability and Scalability
- Topic 5 Elastic Beanstalk for SysOps
- LU2: Deployment, Provisioning, and Automation
- Topic 6 CloudFormation for SysOps
- Topic 7 EC2 Storage and Data Management - EBS and EFS
- Topic 8 Amazon S3
- Topic 9 Advanced Amazon S3 and Athena
- Topic 10 Advanced Storage
- Topic 11 CloudFront
- Topic 12 Database for SysOps

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LU3: Security and Compliance

- Topic 13 Amazon S3 Security
- Topic 14 Security and Compliance for SysOps
- Topic 15 Identity
- LU4: Monitoring, Logging, and Remediation
- Topic 16 Monitoring, Auditing and Performance
- Topic 17 AWS Account Management
- Topic 18 Disaster Discovery
- LU5: Networking and Content Delivery
- Topic 19 Networking - Route 53
- Topic 20 Networking VPC
- Course Outline

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Final Assessment

- Written Assessment (SAQ) - 1 hr
- Practical Performance (PP) - 1 hr
- Assessment format: Open Book
- Open book assessment ONLY includes Slides, Learner Guide or any approved materials.
- Appeal process

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Briefing for Assessment

1. Place phones & other materials under the table or on the floor
2. No photos or recording of assessment scripts
3. No discussion during assessment
4. Use black/blue pen for assessment [hard copies]
5. No usage of liquid paper or correction tape
6. Assessment scripts will be collected when the time is up

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Criteria for Funding

1. Minimum attendance rate of 75% based on SSG Digital Attendance record.

2. Complete the assessment and be assessed as ‘Competent

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Access AWS Skills Builder

- https://skillbuilder.aws/
- Builder ID Login:
- training@tertiaryinfotech.com
- training2@tertiaryinfotech.com
- training3@tertiaryinfotech.com
- training4@tertiaryinfotech.com
- training5@tertiaryinfotech.com
- PW: Tertiary@888
- https://skillbuilder.aws/
- Builder ID Login:
- training6@tertiaryinfotech.com
- training7@tertiaryinfotech.com
- training8@tertiaryinfotech.com

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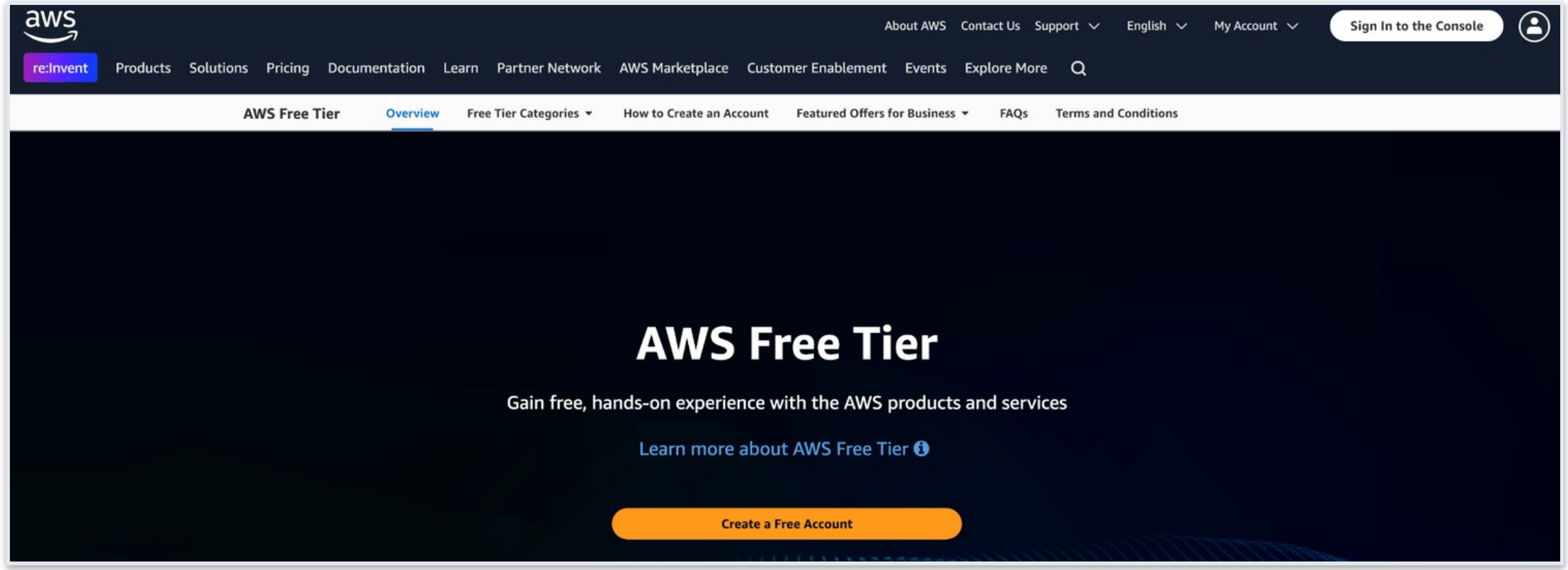
Sign Up Free Trial AWS Account



https://aws.amazon.com/free/

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Download Sample Codes

- Download the sample codes use for this course below
- https://drive.google.com/drive/folders/15Ol6kzinWR7dUIXG3t3rOUOFBeGUWNwq

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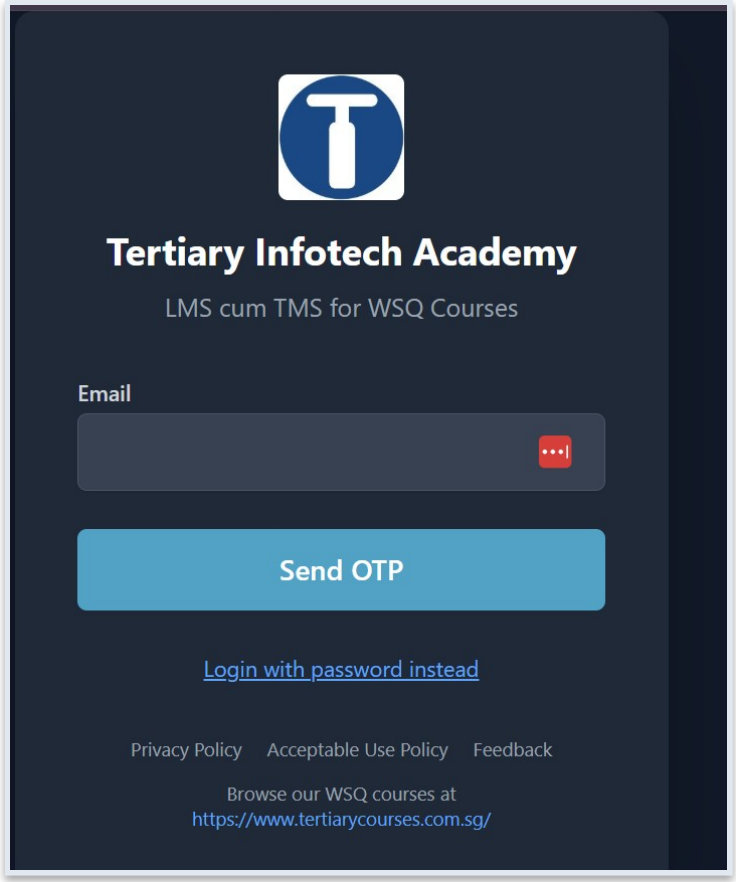
LMS/TMS



https://lms-tms.tertiaryinfotech.com

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TOPIC 1

EC2 for SysOps

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TOPIC 1

Reference

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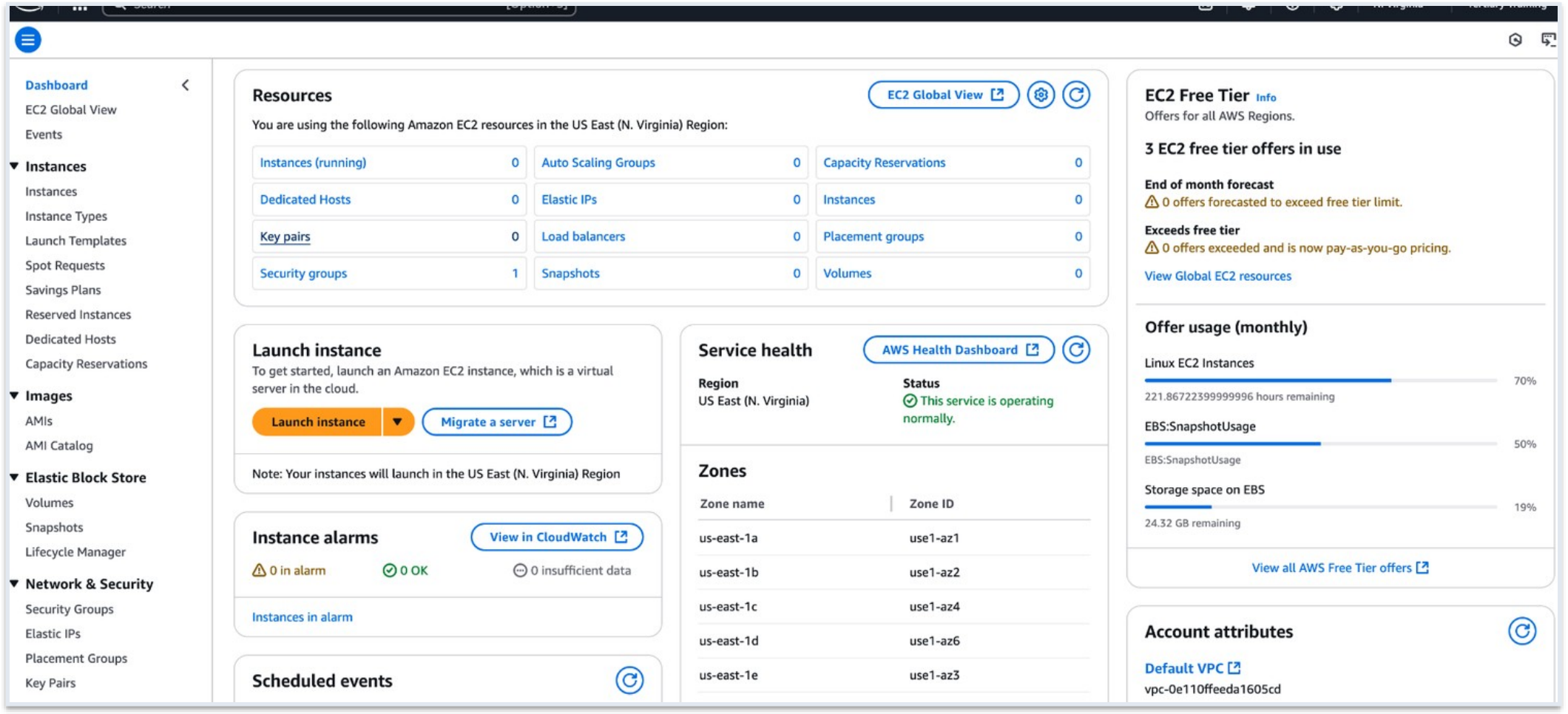
TOPIC 1

EC2 Interface



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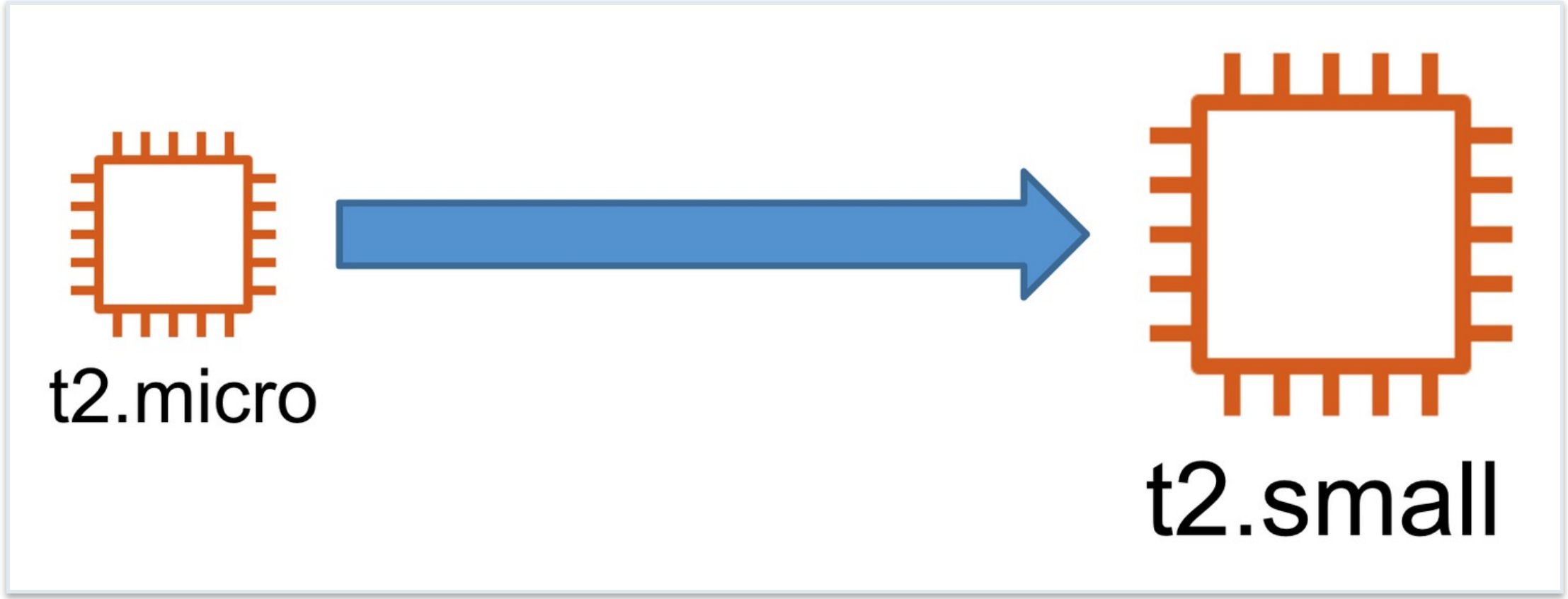
EC2 Changing Instance Type



This only works for EBS backed instances

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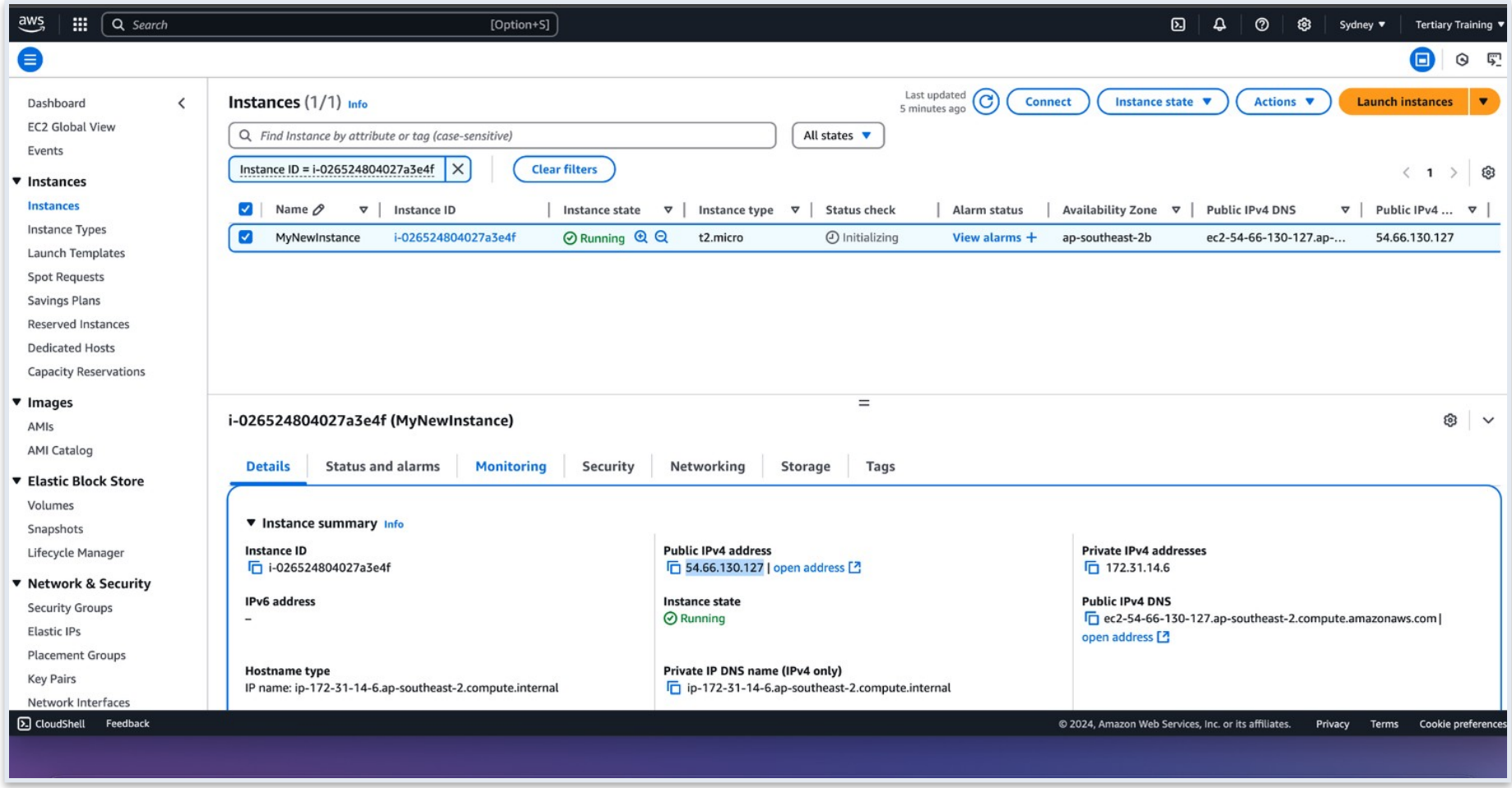
Activity: Launching a EC2 Instance



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Enhanced Networking

- EC2 Enhanced Networking (SR-IOV)
- Higher bandwidth, higher PPS (packet per second), lower latency
- Option 1: Elastic Network Adapter (ENA) up to 100 Gbps
- Option 2: Intel 82599 VF up to 10 Gbps – LEGACY
- Works for newer generation EC2 Instances
- Elastic Fabric Adapter (EFA)
- Improved ENA for HPC, only works for Linux
- Great for inter-node communications, tightly coupled workloads.
- Leverages Message Passing Interface (MPI) standard
- Bypasses the underlying Linux OS to provide low-latency reliable transport

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Placement Groups

- Sometimes you want control over the EC2 Instance placement strategy
- That strategy can be defined using placement groups
- When you create a placement group, you specify one of the following strategies for the group:
- Cluster—clusters instances into a low-latency group in a single Availability Zone
- Spread—spreads instances across underlying hardware (max 7 instances per group per AZ) – critical applications
- Partition—spreads instances across many different partitions (which rely on different sets of racks) within an AZ. Scales to 100s of EC2 instances per group (Hadoop, Cassandra, Kafka)

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Same AZ

- Placement Groups Cluster
- Placement group Cluster
- Low latency
- 10 Gbps network
- Pros: Great network (10 Gbps bandwidth between instances with Enhanced Networking enabled - recommended)
- Cons: If the AZ fails, all instances fails at the same time
- Use case:
- Big Data job that needs to complete fast
- Application that needs extremely low latency and high network throughput

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Us-east-1a

- Hardware 1
- Placement Groups Spread
- EC2
- Hardware 2
- EC2
- Us-east-1b
- Hardware 3
- EC2
- Hardware 4
- EC2
- Us-east-1c
- Hardware 5
- EC2

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TOPIC 1

Placements Groups Partition

- us-east-1a
- Partition 1
- EC2
- EC2
- EC2
- EC2
- Partition 2
- EC2
- EC2
- EC2
- EC2
- EC2
- Partition 3
- EC2

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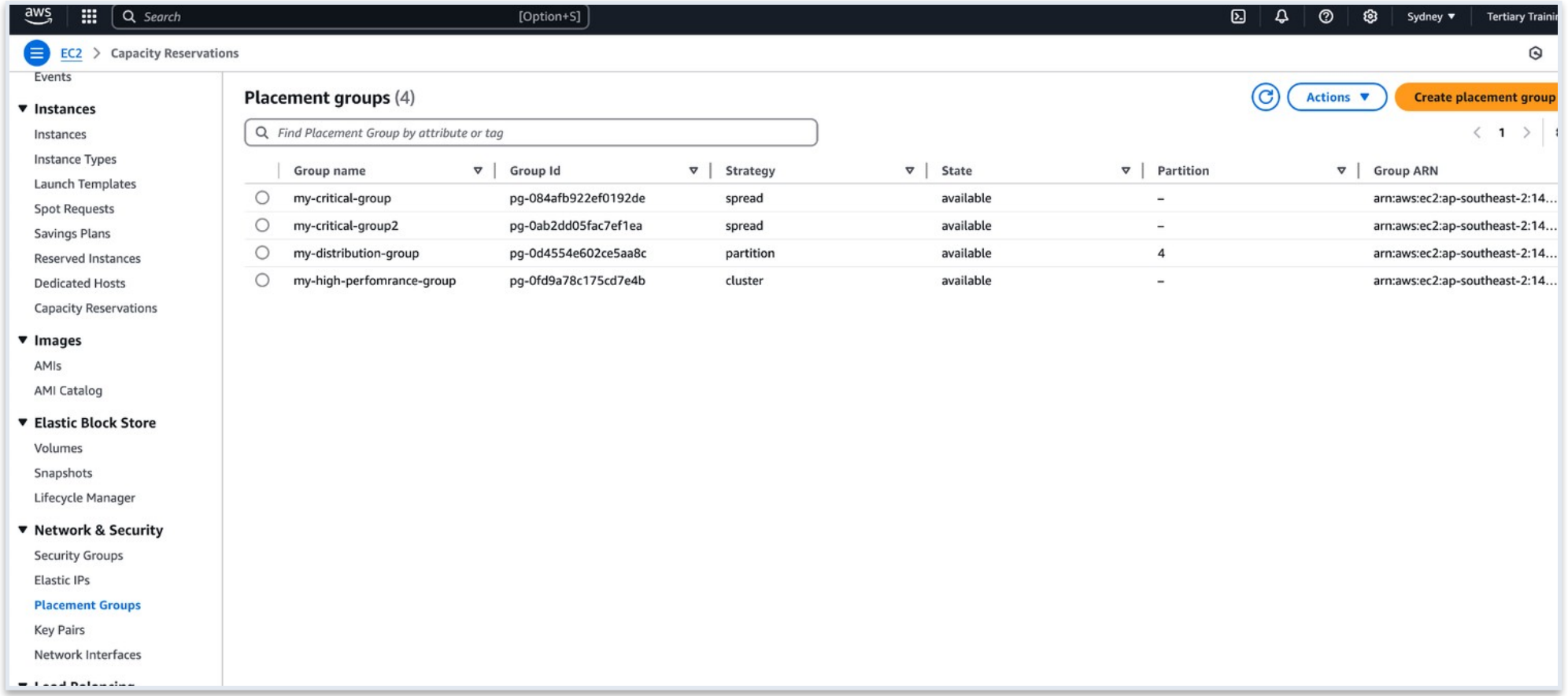
Activity: Placement Group



Goto Placement Group

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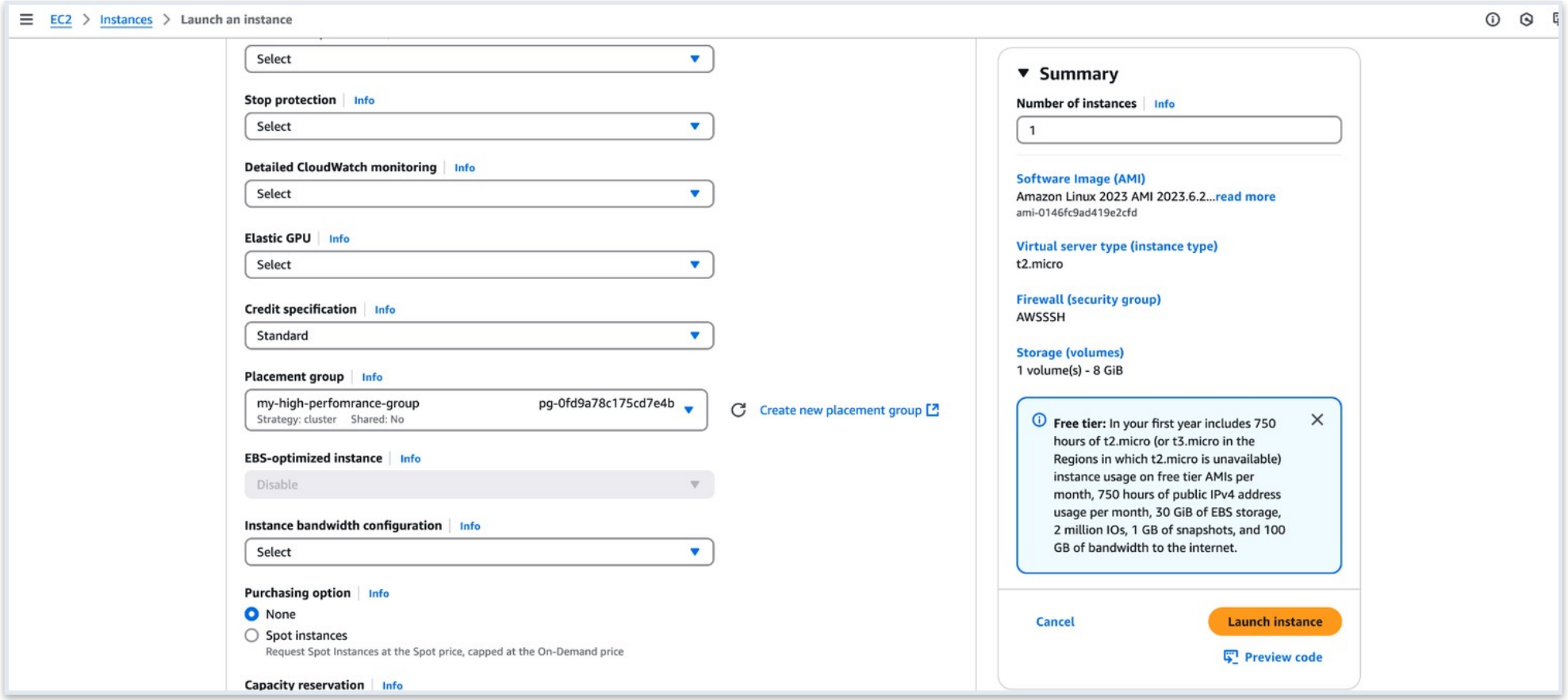
Activity: Placement Group



Launch a new t2.micro instance

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TOPIC 1

Shutdown Behaviour: How should the instance react when shutdown is done using the OS?

- Stop (default)
- Terminate
- This is not applicable when shutting down from AWS console.
- CLI Attribute:
- InstanceInitiatedShutdownBehavior
- Shutdown Behavior
- EC2 Instance (running)
- OS shutdown
- Shutdown Behavior
- Stop
- Terminate
- EC2 Instance (stopped)
- $ shutdown

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Termination Protection

- Enable termination protection:
- To protect against accidental termination in AWS Console or CLI
- Exam Tip:
- We have an instance where shutdown behavior = terminate and enable terminate protection is ticked
- We shutdown the instance from the OS, what will happen? The instance will still be terminated!

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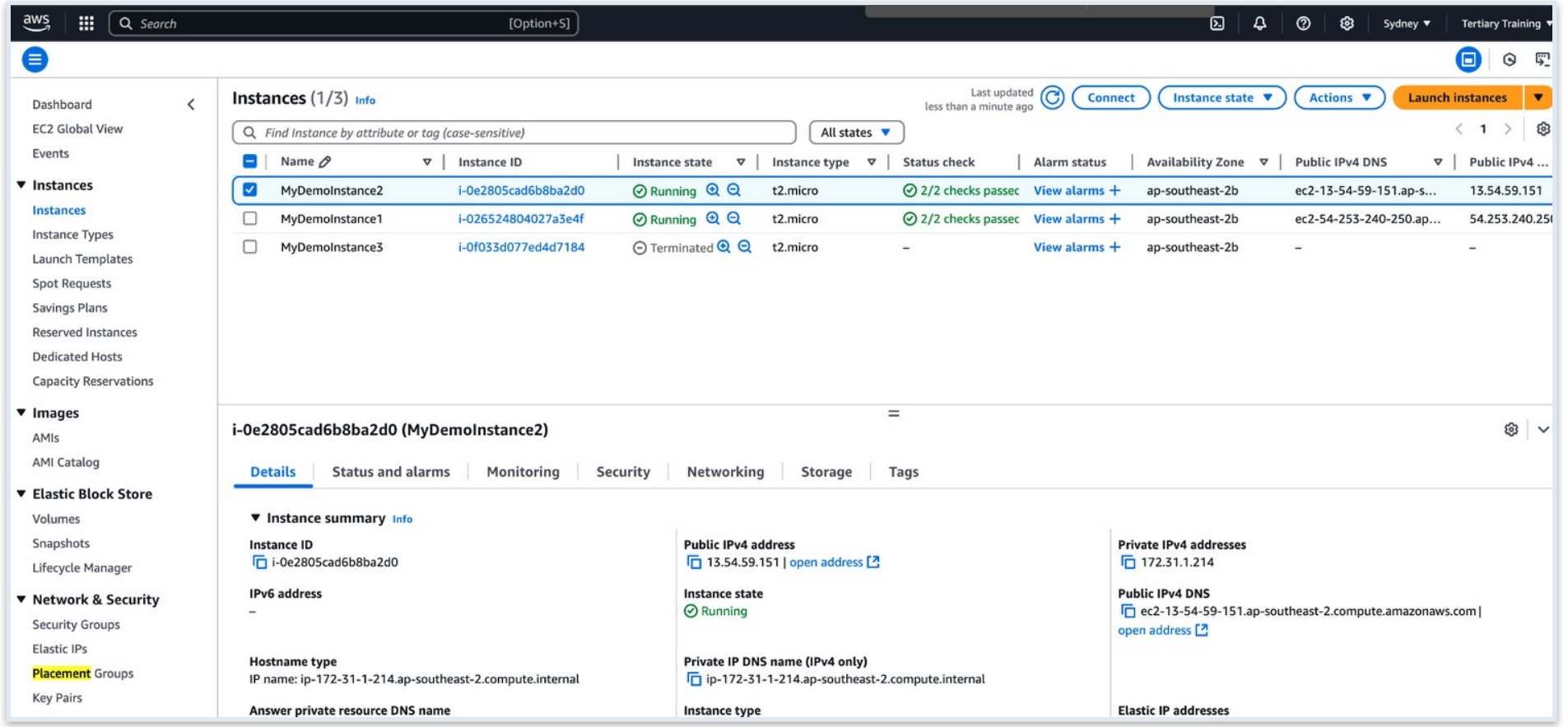
TOPIC 1

Activity: Shutdown Behavior



Launch a new t2.micro instance

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TOPIC 1

EC2 Launch Troubleshooting

- # InstanceLimitExceeded if you get this error, it means that you have reached your limit of max number of vCPUs per region
- On-Demand instance limits are set on a per-region basis
- Example: If you run on-demand (A, C, D, H, I, M, R,T, Z) instance types you'll have 64 vCPUs (default)
- Resolution: Either launch the instance in a different region or request AWS to increase your limit of the region
- NOTE: vCPU-based limits only apply to running On-Demand instances and Spot instances

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EC2 Launch Troubleshooting

- # InsufficientInstanceCapacity: if you get this error, it means AWS does not have that enough On-Demand capacity in the particular AZ where the instance is launched.
- Resolution :
- Wait for few mins before requesting again.
- If requesting more than 1 requests, break down the requests. If you need 5 instances, rather than a single request of 5, request one by one.
- If urgent, submit a request for a different instance type now, which can be resized later.
- Also, can request the EC2 instance in a different AZ

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EC2 Launch Troubleshooting

- # Instance Terminates Immediately (goes from pending to terminated)
- You've reached your EBS volume limit.
- An EBS snapshot is corrupt.
- The root EBS volume is encrypted and you do not have permissions to access the KMS key for decryption.
- The instance store-backed AMI that you used to launch the instance is missing a required part (an image.part.xx file).
- To find the exact reason, check out the EC2 console of AWS - instances - Description tab, note the reason next to the State transition reason label.

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EC2 SSH troubleshooting

- Make sure the private key (pem file) on your linux machine has 400 permissions, else you will get "Unprotected private key file" error
- Make sure the username for the OS is given correctly when logging via SSH,
- else you will get "Host key not found, "Permission denied", or "Connection closed by [instance] port 22" error
- Possible reasons for "Connection timed out" to EC2 instance via SSH:
- SG is not configured correctly
- NACL is not configured correctly
- Check the route table for the subnet (routes traffic destined outside VPC to IGW)
- Instance doesn’t have a public IPv4
- CPU load of the instance is high

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SSH vs. EC2 Instance Connect

- Connect using SSH
- Inbound Rules
- User
- (IPv4: 1.2.3.4)
- User
- (IPv4: 5.6.7.8)
- blocked
- User
- (IPv4: 1.2.3.4)
- {
- "ip_prefix": "18.206.107.24/29", "region": "us-east-1",
- "service": "EC2_INSTANCE_CONNECT",
- "network_border_group": "us-east-1"

Connect using EC2 Instance Connect

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EC2 Instances Purchasing Options

- On-Demand Instances - short workload, predictable pricing, pay by second
- Reserved (1 & 3 years)
- Reversed Instances - long workloads
- Convertible Reserved Instances - long workloads with flexible instances
- Savings Plans (1 & 3 years) - commitment to an amount of usage, long workload
- Spot Instances - short workloads, cheap, can lose instances (less reliable)
- Dedicated Hosts - book an entire physical server, control instance placement
- Dedicated Instances - no other customers will share your hardware
- Capacity reservations - reserve capacity in a specific AZ for any duration

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EC2 On Demand

- Pay for what you use:
- Linux or Windows - billing per second, after the first minute
- All other operating systems - billing per hour
- Has the highest cost but no upfront payment
- No long-term commitment
- Recommended for short term and un-interrupted workloads, where you can't predict how the application will behave

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Up to 72% discount compared to On-demand

- You reserve a specific instance attributes (Instance Type, Region, Tenancy, OS)
- Reservation Period - 1 year (+discount) or 3 years (+++discount)
- Payment options - No Upfront (+), Partial Upfront (++), All Upfront (+++)
- Reserved Instance's Scope - Regional or Zonal (reserve capacity in an AZ)
- Recommended for steady-state usage applications (think database)
- You can buy and sell in the Reserved Instance Marketplace
- Convertible Reserved Instance
- Can change the EC2 instance type, instance family, OS, scope and tenancy
- Up to 66% discount
- EC2 Reserved Instances
- Note: the % discounts are different from the video as AWS change them over time – the exact numbers are not needed for the exam. This is just for illustrative purposes ©

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EC2 Savings Plans

- Get a discount based on long-term usage (up to 72% - same as RIs)
- Commit to a certain type of usage ($10/hour for 1 or 3 years)
- Usage beyond EC2 Savings Plans is billed at the On-Demand price
- Locked to a specific instance family & AWS region (e.g., M5 in us-east-1)
- Flexible across:
- Instance Size (e.g., m5.xlarge, m5.2xlarge)
- OS (e.g., Linux, Windows)
- Tenancy (Host, Dedicated, Default)

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EC2 Spot Instances

- Can get a discount of up to 90% compared to On-demand
- Instances that you can “lose” at any point of time if your max price is less than the current spot price
- The MOST cost-efficient instances in AWS
- Useful for workloads that are resilient to failure
- Batch jobs
- Data analysis
- Image processing
- Any distributed workloads
- Workloads with a flexible start and end time
- Not suitable for critical jobs or databases

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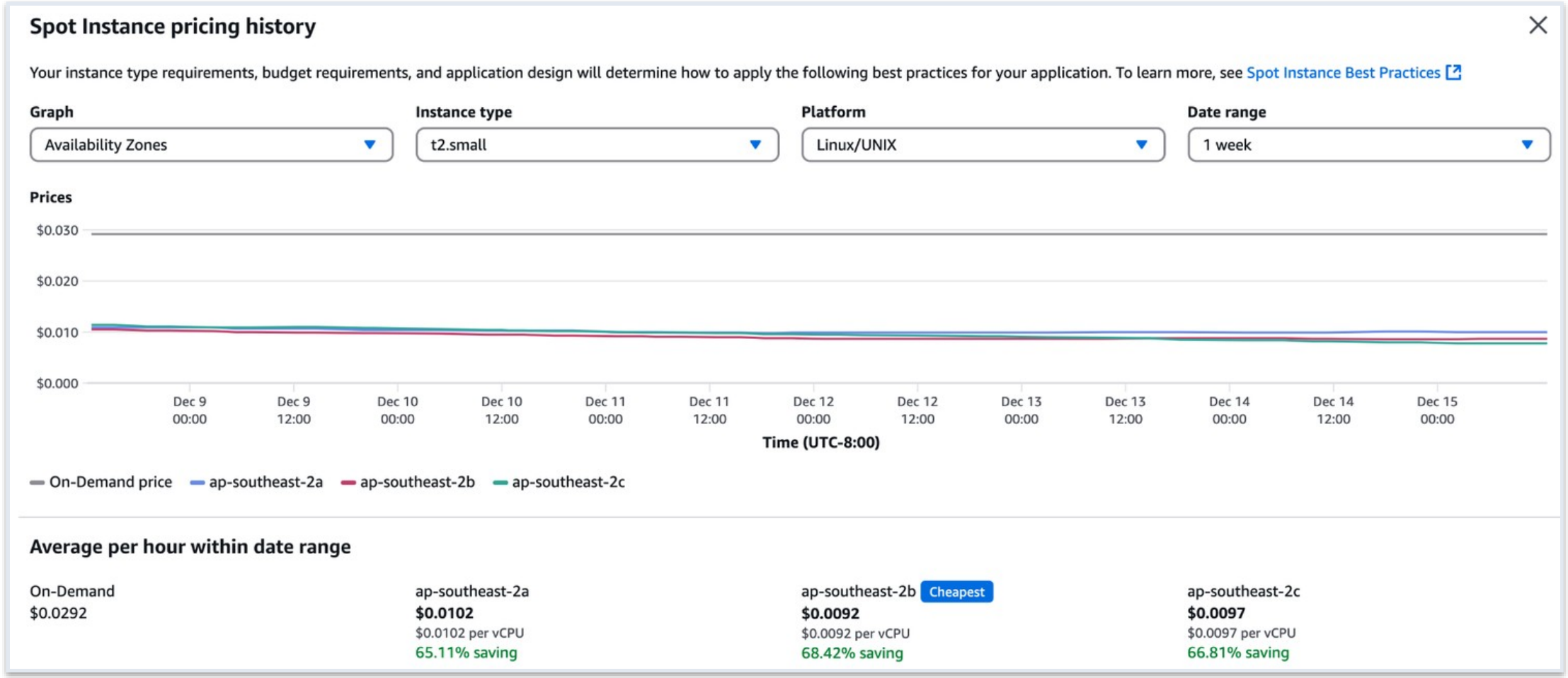
TOPIC 1

EC2 Spot Instance Pricing History



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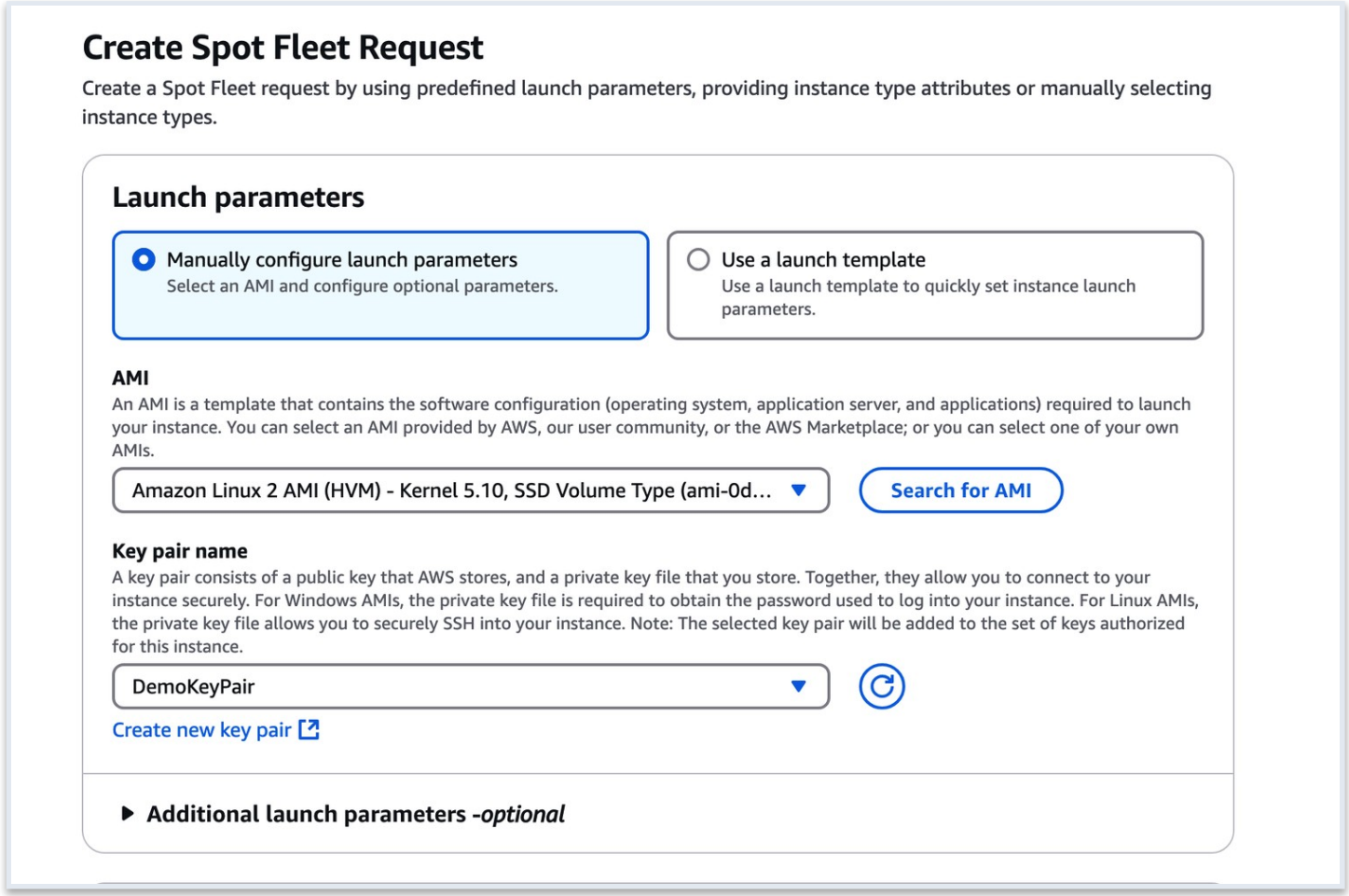
Activity: Request Spot Instance



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EC2 Dedicated Hosts

- A physical server with EC2 instance capacity fully dedicated to your use
- Allows you address compliance requirements and use your existing server - bound software licenses (per-socket, per-core, pe—VM software licenses)
- Purchasing Options:
- On-demand - pay per second for active Dedicated Host
- Reserved - 1 or 3 years (No Upfront, Partial Upfront, All Upfront) The most expensive option
- Useful for software that have complicated licensing model (BYOL – Bring Your Own License)
- Or for companies that have strong regulatory or compliance needs

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EC2 Dedicated Instances

Characteristic	Dedicated Instances	Dedicated Hosts
Enables the use of dedicated physical servers	X	X
Per instance billing (subject to a $2 per region fee)	X	
Per host billing		X
Visibility of sockets, cores, host ID		X
Affinity between a host and instance		X
Targeted instance placement		X
Automatic instance placement	X	X
Add capacity using an allocation request		X

Instances run on hardware that’s dedicated to you

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EC2 Capacity Reservations

- Reserve on-Demand instances capacity in a specific AZ for any duration
- You always have access to EC2 capacity when you need it
- No time commitment (create/cancel anytime), no billing discounts Combine with Regional Reserved Instances and Savings Plans to benefit from billing discounts
- You’re charged at On-Demand rate whether you run instances or not
- Suitable for short-term, uninterrupted workloads that needs to be in a specific AZ

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TOPIC 1

Which purchasingoption is right for me?

On demand: coming and staying in resort whenever we like, we pay the full price

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Price Comparison

- Example – m4.large – us-east-1

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AWS charges for IPv4 addresses

- IPv4 created in your account
- $0.005 per hour of Public IPv4 (~ $3.6 per month)
- For new accounts in AWS, you have a free tier for the EC2 service: 750 hours of Public IPv4 per month for the first 12 months
- For all other services there is no free tier
- Public IP
- Starting February 1st 2024, there’s a charge for all Public
- EC2 Instance
- EC2 Instance
- Public IP
- Free until 750 hours/month of usage
- Load Balancer
- One Public IPv4

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TOPIC 1

AWS charges for IPv4 addresses

- What about IPv6?
- Unfortunately, many Internet Service Provider (ISP) around the world don't support IPv6, so the course would not work for some of you
- You can test IPv6 by going to https://test-ipv6.com/
- If you use IPv6 in this course, you're on your own (security groups, networking…) but you can do it!
- How to troubleshoot charges?
- Go into your AWS Bill
- Look into the AWS Public IP Insights service
- Nice article here:
- https://repost.aws/articles/ARknH_OR0cTvqoTfJrVGaB8A/why-am-i-seeing-charges-for-public-ipv4-addresses-when-i-am-under-the-aws-free-tier

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EC2 Spot Instance Requests

- Can get a discount of up to 90% compared to On-demand
- Define max spot price and get the instance while current spot price < max
- The hourly spot price varies based on offer and capacity
- If the current spot price > your max price you can choose to stop or terminate your instance with a 2 minutes grace period.
- Other strategy: Spot Block
- “block” spot instance during a specified time frame (1 to 6 hours) without interruptions
- In rare situations, the instance may be reclaimed
- Used for batch jobs, data analysis, or workloads that are resilient to failures.
- Not great for critical jobs or databases

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TOPIC 1

EC2 Spot Instances Pricing

1. https://console.aws.amazon.com/ec2sp/v1/spot/home?region=us-east-1#

2. User-defined max price

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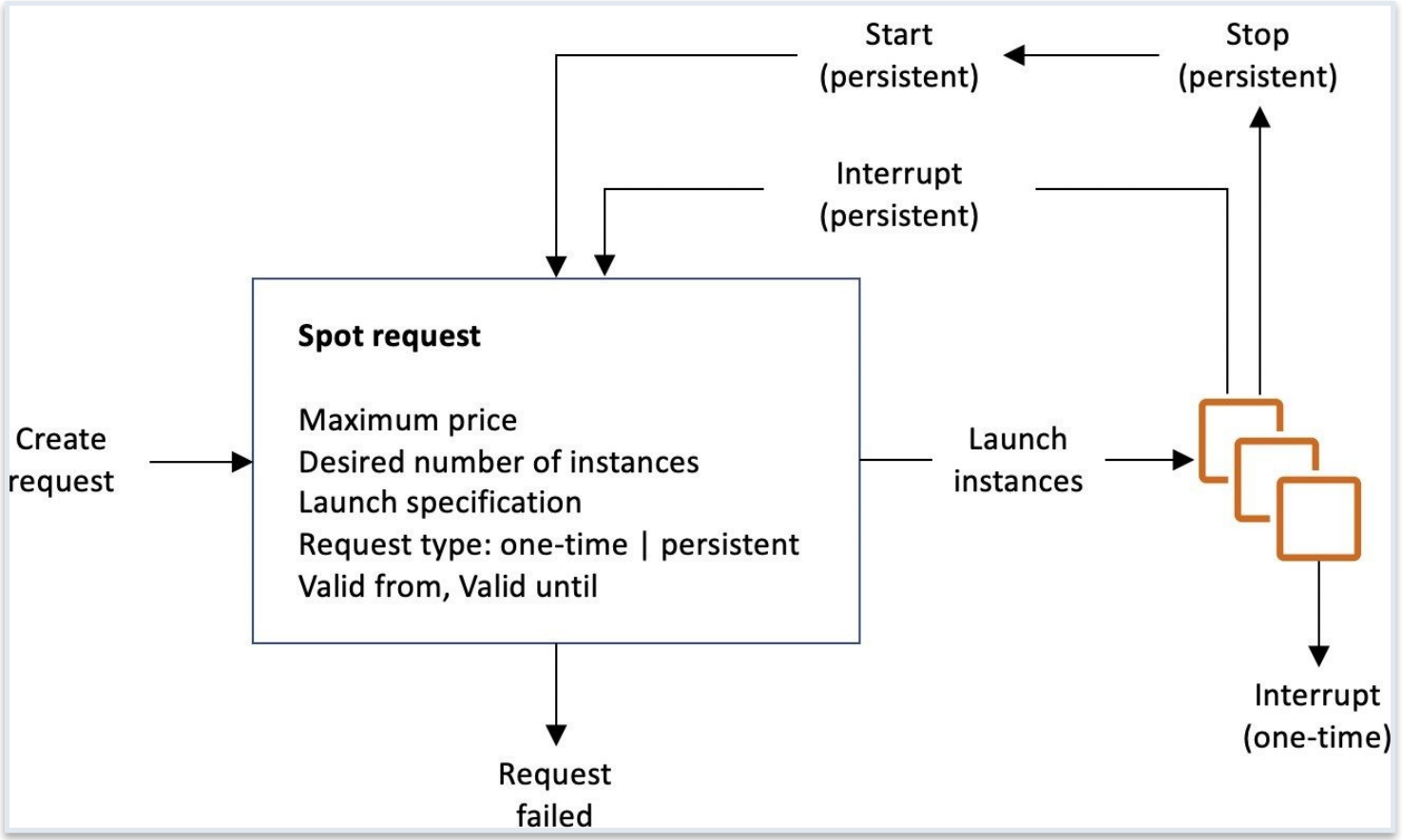
How to terminate Spot Instances?



https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/spot-requests.html

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Spot Fleets

- Spot Fleets = set of Spot Instances + (optional) On-Demand Instances
- The Spot Fleet will try to meet the target capacity with price constraints
- Define possible launch pools: instance type (m5.large), OS, Availability Zone
- Can have multiple launch pools, so that the fleet can choose
- Spot Fleet stops launching instances when reaching capacity or max cost
- Strategies to allocate Spot Instances:
- lowestPrice: rom the pool with the lowest price (cost optimization, short workload)
- diversified: distributed across all pools (great for availability, long workloads)
- capacityOptimized: pool with the optimal capacity for the number of instances
- priceCapacityOptimized (recommended): pools with highe
- st capacity available, then select the pool with the lowest price (best choice for most workloads)
- Spot Fleets allow us to automatically request Spot Instances with the lowest price

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TOPIC 1

Burstable Instances (T2/T3)

- AWS has the concept of burstable instances (T2/T3 machines)
- Burst means that overall, the instance has OK CPU performance.
- When the machine needs to process something unexpected (a spike in load for example), it can burst, and CPU can be VERY good.
- If the machine bursts, it utilizes “burst credits”
- If all the credits are gone, the CPU becomes BAD
- If the machine stops bursting, credits are accumulated over time

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TOPIC 1

Burstable Instances (T2/T3)

- Burstable instances can be amazing to handle unexpected traffic and getting the insurance that it will be handled correctly
- If your instance consistently runs low on credit, you need to move to a different kind of non-burstable instance
- Credit usage
- Credit balance

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TOPIC 1

CPU Credits

- CPU Credits

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What happens when credit ar exhausted?

1. Experiment: run a CPU stress command (to peak at 100%)
2. After the credits are exhausted, the measured CPU utilization drops
3. 100% CPU Utilization
4. Credit Usage
5. Credit exhaustion
6. Lowered CPU Utilization

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T2/T3 Unlimited

- It is possible to have an “unlimited burst credit balance”
- You pay extra money if you go over your credit balance, but you don’t lose in performance
- If average CPU usage over a 24-hour period exceeds the baseline, the instance is billed for additional usage per vCPU/hour
- Be careful, costs could go high if you’re not monitoring the CPU health of your instances
- https://aws.amazon.com/blogs/aws/new-t2-unlimited-going-beyond-the-burst-with-high-performance/

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TOPIC 1

Public subnet

- Elastic IPs
- When you stop and then start an EC2 instance, it changes its public IP
- If you need to have a fixed public IP, you need an Elastic IP
- An Elastic IP is a public IPv4 you own as long as you don’t delete it
- You can attach it to one instance at a time
- You can remap it across instances\
- You don’t pay for the Elastic IP if it’s attached to a server
- You pay for the Elastic IP if it’s not attached to a server
- VPC
- EC2 Instance
- EIP
- (IPv4: 1.2.3.4)
- http://1.2.3.4

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TOPIC 1

Elastic IPs

- With an Elastic IP address, you can mask the failure of an instance or software by rapidly remapping the address to another instance in your account.
- You can only have 5 Elastic IP in your account (you can ask AWS to increase that).
- How you can avoid using Elastic IP:
- Always think if other alternatives are available to you
- You could use a random public IP and register a DNS name to it
- Or use a Load Balancer with a static hostname

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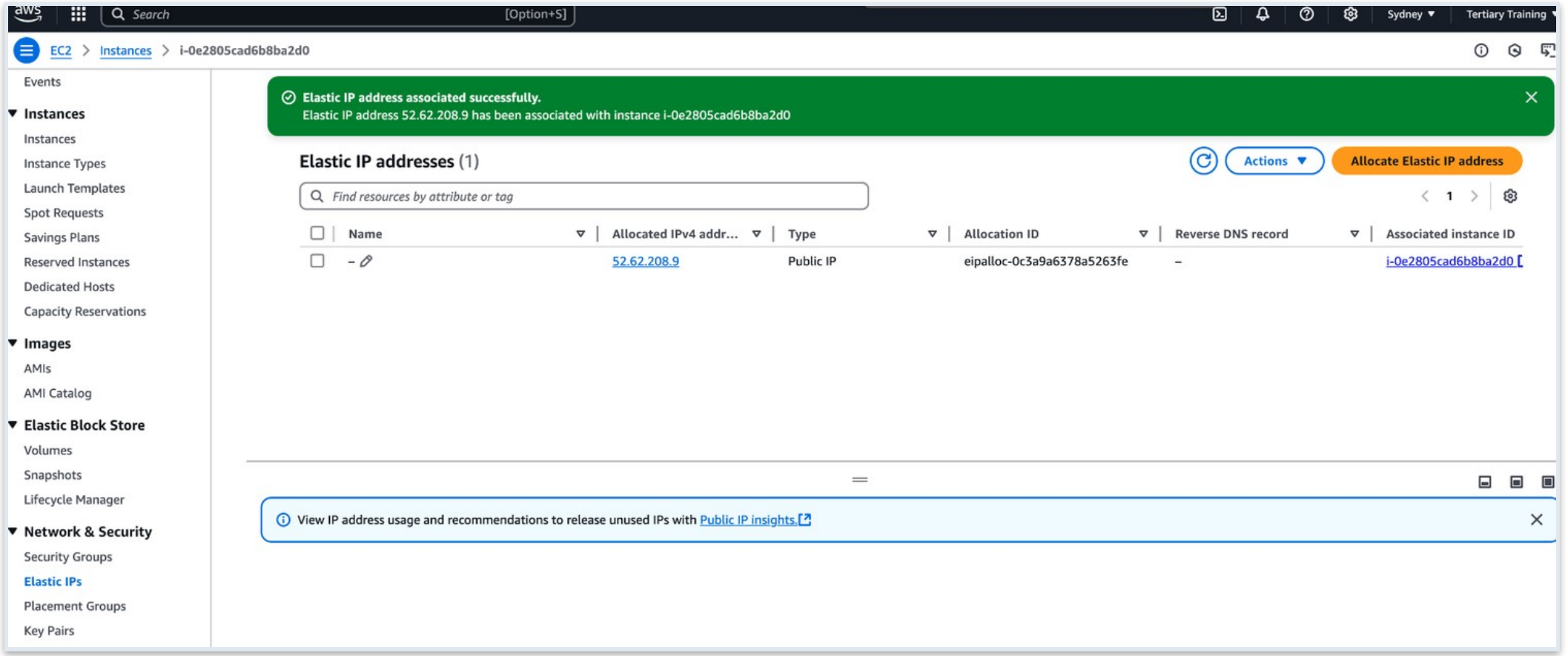
TOPIC 1

Activity: Elastic IPs



Create two instances.

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TOPIC 1

CloudWatch Metrics for EC2

- AWS Provided metrics (AWS pushes them):
- Basic Monitoring (default): metrics are collected at a 5 minute internal
- Detailed Monitoring (paid): metrics are collected at a 1 minute interval
- Includes CPU, Network, Disk and Status Check Metrics
- Custom metric (yours to push):
- Basic Resolution: 1 minute resolution
- High Resolution: all the way to 1 second resolution
- Include RAM, application level metrics
- Make sure the IAM permissions on the EC2 instance role are correct !

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TOPIC 1

EC2 included metrics

- CPU: CPU Utilization + Credit Usage / Balance
- Network: Network In / Out
- Status Check:
- Instance status = check the EC2 VM
- System status = check the underlying hardware
- Attached EBS status = check attached EBS volumes
- Disk: Read / Write for Ops / Bytes (only for instance store)
- RAM is NOT included in the AWS EC2 metrics

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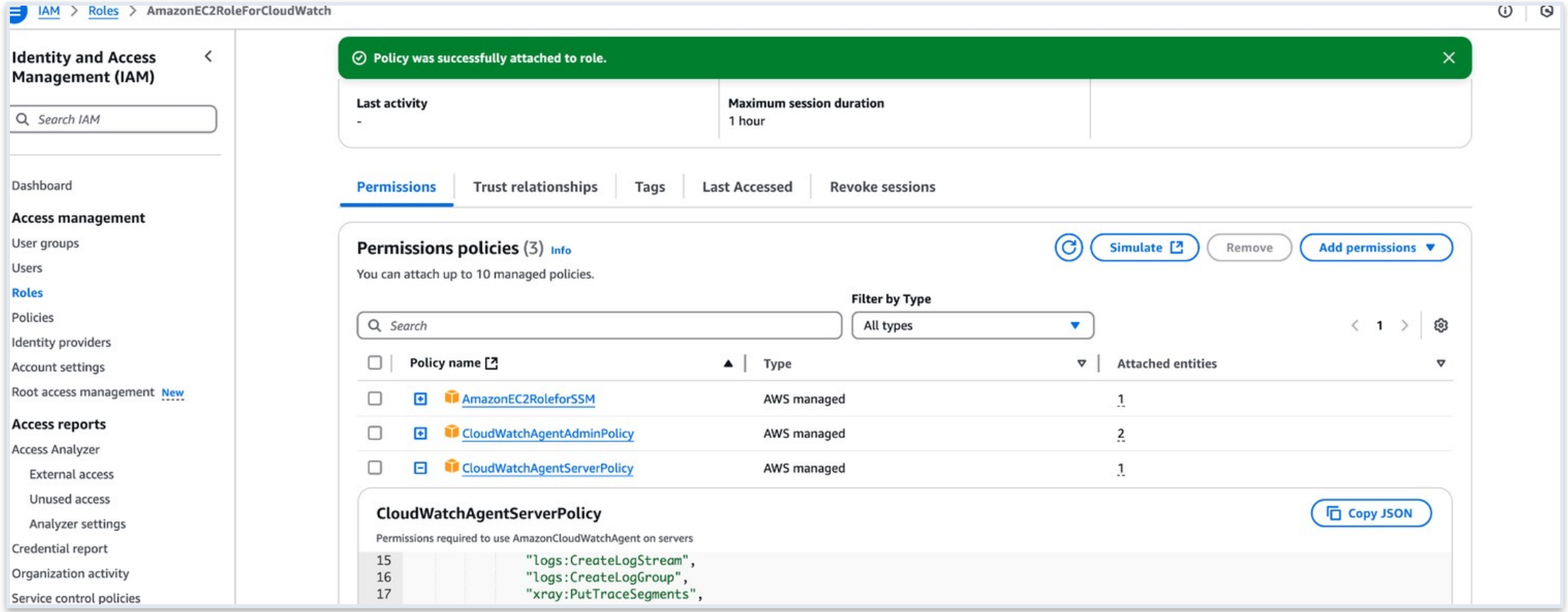
TOPIC 1

Activity: Create IAM Role For Cloudwatch



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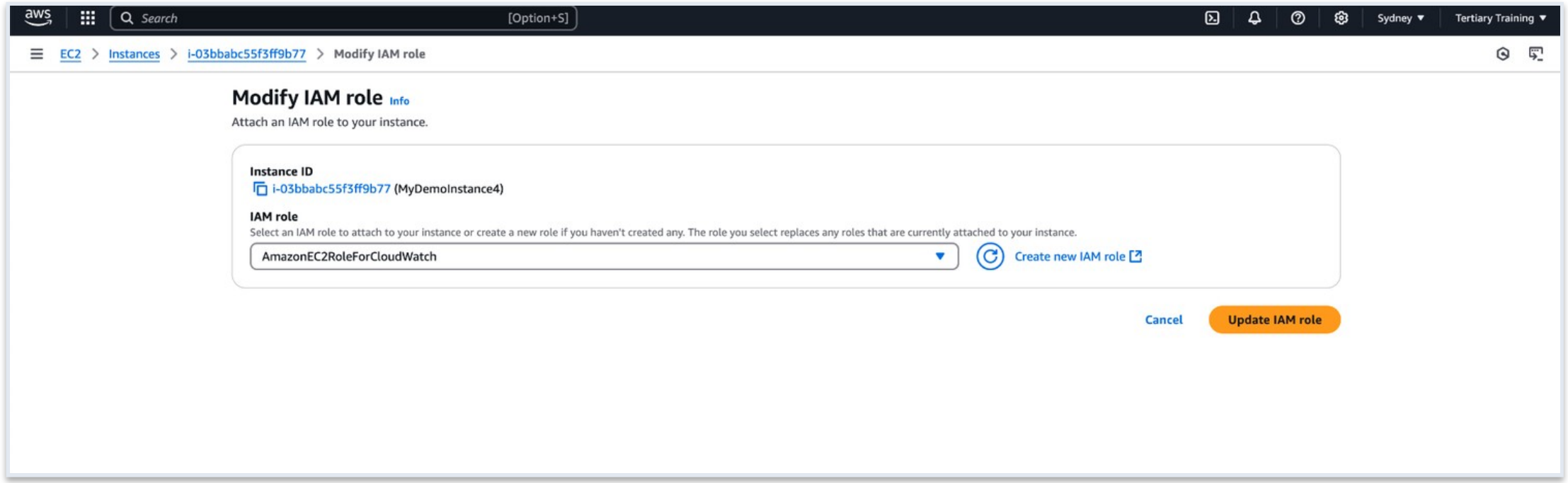
Activity: Assign IAM Role To Instance



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TOPIC 1

Unified CloudWatch Agent

- For virtual servers (EC2 instances, on-premises servers, …)
- Collect additional system-level metrics such as RAM, processes, used disk space, etc.
- Collect logs to send to CloudWatch Logs
- No logs from inside your EC2 instance will be sent to CloudWatch Logs without using an agent
- Centralized configuration using SSM Parameter Store
- Make sure IAM permissions are correct
- Default namespace for metrics collected by the Unified CloudWatch agent is CWAgent (can be configured/changed)
- Corporate Data Center
- CloudWatch
- Server with Unified CW Agent
- EC2 Instance with Unified CW Agent
- push logs

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TOPIC 1

Unified CloudWatch Agent – procstat Plugin

- Collect metrics and monitor system utilization of individual processes
- Supports both Linux and Windows servers
- Example: amount of time the process uses CPU, amount of memory the process uses, …
- Select which processes to monitor by
- pid_file: name of process identification number (PID) files they create
- exe: process name that match string you specify (RegEx)
- pattern: command lines used to start the processes (RegEx)
- Metrics collected by procstat plugin begins with "procstat" prefix (e.g., procstat_cpu_time, procstat_cpu_usage, …)

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TOPIC 1

Status Checks

- Automated checks to identify hardware and software issues
- System status Checks
- Monitors problems with AWS systems (software/hardware issues on the physical host, loss of system power, …)
- Check Personal Health Dashboard for any scheduled critical maintenance by AWS to your instance's host
- Resolution: stop and start the instance (instance migrated to a new host)
- Instance Status Checks
- Monitors software/network configuration of your instance (invalid network configuration, exhausted memory, …)
- Resolution: reboot the instance or change instance configuration
- Attached EBS Status Checks
- Monitors EBS volumes attached to your instance (reachable & complete I/O Operations)
- Resolution: reboot the instance or replace affected EBS volumes

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Status Checks - CW Metrics & Recovery

- CloudWatch Metrics (1 minute interval)
- StatusCheckFailed_System
- StatusCheckFailed_Instance
- StatusCheckFailed_AttachedEBS
- StatusCheckFailed (for any)
- Option 1: CloudWatch Alarm
- Recover EC2 instance with the same private/public IP, EIP, metadata, and Placement Group
- Send notifications using SNS
- Option 2: Auto Scaling Group
- Set min/max/desired 1 to recover an instance but won't keep the same private and elastic IP
- EC2 Instance
- CloudWatch
- monitor

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TOPIC 1

EC2 Hibernate

- We know we can stop, terminate instances
- Stop - the data on disk (EBS) is kept intact in the next start
- Terminate - any EBS volumes (root) also set-up to be destroyed is lost
- On start, the following happens:
- First start: the OS boots & the EC2 User Data script is run
- Following starts: the OS boots up
- Then your application starts, caches get warmed up, and that can take time!

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TOPIC 1

EC2 Hibernate

- Introducing EC2 Hibernate:
- The in-memory (RAM) state is preserved
- The instance boot is much faster! (the OS is not stopped / restarted)
- Under the hood: the RAM state is written to a file in the root EBS volume
- The root EBS volume must be encrypted
- Use cases:
- Long-running processing
- Saving the RAM state
- Services that take time to initialize
- RAM
- EC2 Instance
- Root EBS
- Volume

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TOPIC 1

EC2 Hibernate – Good to know

- Supported Instance Families - C3, C4, C5, I3, M3, M4, R3, R4,T2,T3, …
- instance RAM Size - must be less than 150 GB.
- Instance Size - not supposed for bare metal instances.
- AMI - Amazon Linux 2, Linux AMI, Ubuntu, RHEL, CentOS & Windows…
- Root Volume - must be EBS, encrypted, not instance store, and large.
- Available for On-Demand, Reserved and Spot Instances.
- An instance can NOT be hibernated for more than 60 days.

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TOPIC 2

Amazon Machine Image (AMI)

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TOPIC 2

AMI Overview

- AMI = Amazon Machine Image
- AMI is a customization of an EC2 instance
- You add your own software, configuration, operating system, monitoring…
- Faster boot / configuration time because all your software is pre-packaged
- AMI are built for a specific region (and can be copied across regions)
- You can launch EC2 instances from:
- A Public AMI: AWS provided
- Your own AMI: you make and maintain them yourself
- An AWS Marketplace AMI: an AMI someone else made (and potentially sells)

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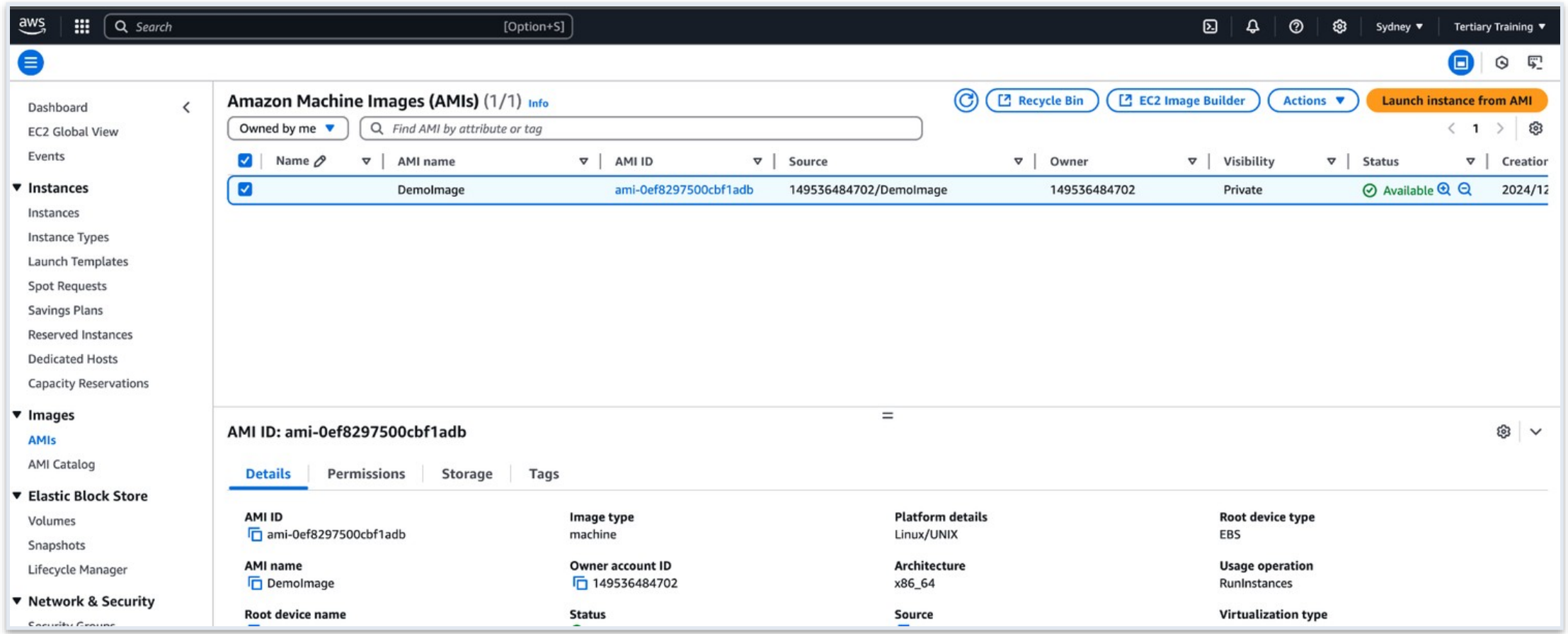
TOPIC 2

Activity: Create AMI



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TOPIC 2

AMI Process (from an EC2 instance)

- Start an EC2 instance and customize it
- Stop the instance (for data integrity)
- Build an AMI – this will also create EBS snapshots
- Launch instances from other AMIs
- US-EAST-1A
- US-EAST-1B
- Custom AMI
- Create AMI
- Launch from AMI

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TOPIC 2

AMI No-Reboot Option

- Enables you to create an AMI without shutting down your instance
- By default, it’s not selected (AWS will shut down the instance before creating an AMI to maintain the file system integrity)
- With No-Reboot Disabled (default)
- EC2 Instance (running)
- Attached EBS Volume
- EBS Snapshot
- AMI
- EC2 Instance (stopped)
- Attached
- EBS Volume
- shut down
- create image

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TOPIC 2

AWS Backup Plans to create AMI

- AWS Backup doesn't reboot the instances while taking EBS snapshots (no-reboot behavior)
- This won't help you to create an AMI that guarantees file system integrity since you need to reboot the instance
- To maintain integrity you need to provide the reboot parameter while taking images (EventBridge
- + Lambda + CreateImage API with reboot)
- AMI
- Create AMI
- --no-reboot
- Amazon EC2 (rebooting)
- AMI
- Create AMI
- --reboot
- EventBridge Rule (schedule)

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TOPIC 2

EC2 Instance Migration between AZ

- us-east-1
- us-east-1a
- us-east-1b
- EC2 Instance
- EBS Volume
- EC2 Instance
- EBS Volume
- create image launch/restore
- AMI

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TOPIC 2

Cross-Account AMI Sharing

- You can share an AMI with another AWS account
- Sharing an AMI does not affect the ownership of the AMI
- You can only share AMIs that have unencrypted volumes and volumes that are encrypted with a customer managed key
- If you share an AMI with encrypted volumes, you must also share any customer managed keys used to encrypt them.
- Account A
- Source AMI
- Account B
- Share with Account B
- EC2 Instance
- (AMI: Source AMI)

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TOPIC 2

AMI Sharing with KMS Encryption

- Account A
- Custom AMI
- us-east-1
- EBS
- Snapshot
- (encrypted)
- CMK - A
- Source
- Account B
- Target
- IAM Permissions
- kms:DescribeKey kms:CreateGrant kms:Decrypt kms:GenerateDataKey kms:ReEncrypt
- share

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TOPIC 2

Cross-Account AMI Copy

- If you copy an AMI that has been shared with your account, you are the owner of the target AMI in your account
- The owner of the source AMI must grant you read permissions for the storage that backs the AMI (EBS Snapshot)
- If the shared AMI has encrypted snapshots, the owner must share the key or keys with you as well
- Can encrypt the AMI with your own CMK while copying
- Account A
- Source AMI
- EBS Snapshot
- Account B
- Copy of Source AMI
- EBS Snapshot
- IAM Permissions (read snapshot)

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TOPIC 2

AMI Copy with KMS Encryption

- Custom AMI
- EBS
- Snapshot
- (encrypted)
- CMK - A
- Custom AMI
- EBS
- Snapshot
- (encrypted)
- IAM Permissions
- kms:DescribeKey kms:CreateGrant kms:Decrypt
- CMK - B
- share

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TOPIC 2

EC2 Image Builder

- Used to automate the creation of Virtual Machines or container images
- => Automate the creation, maintain, validate and test EC2 AMIs
- Can be run on a schedule (weekly, whenever packages are updated, etc…)
- Free service (only pay for the underlying resources)
- Builder EC2 Instance
- create
- Build Components applied (customize software on instance)
- New AMI
- create
- Test EC2 Instance
- Test suite is run
- (is the AMI working, secure?)
- AMI is distributed

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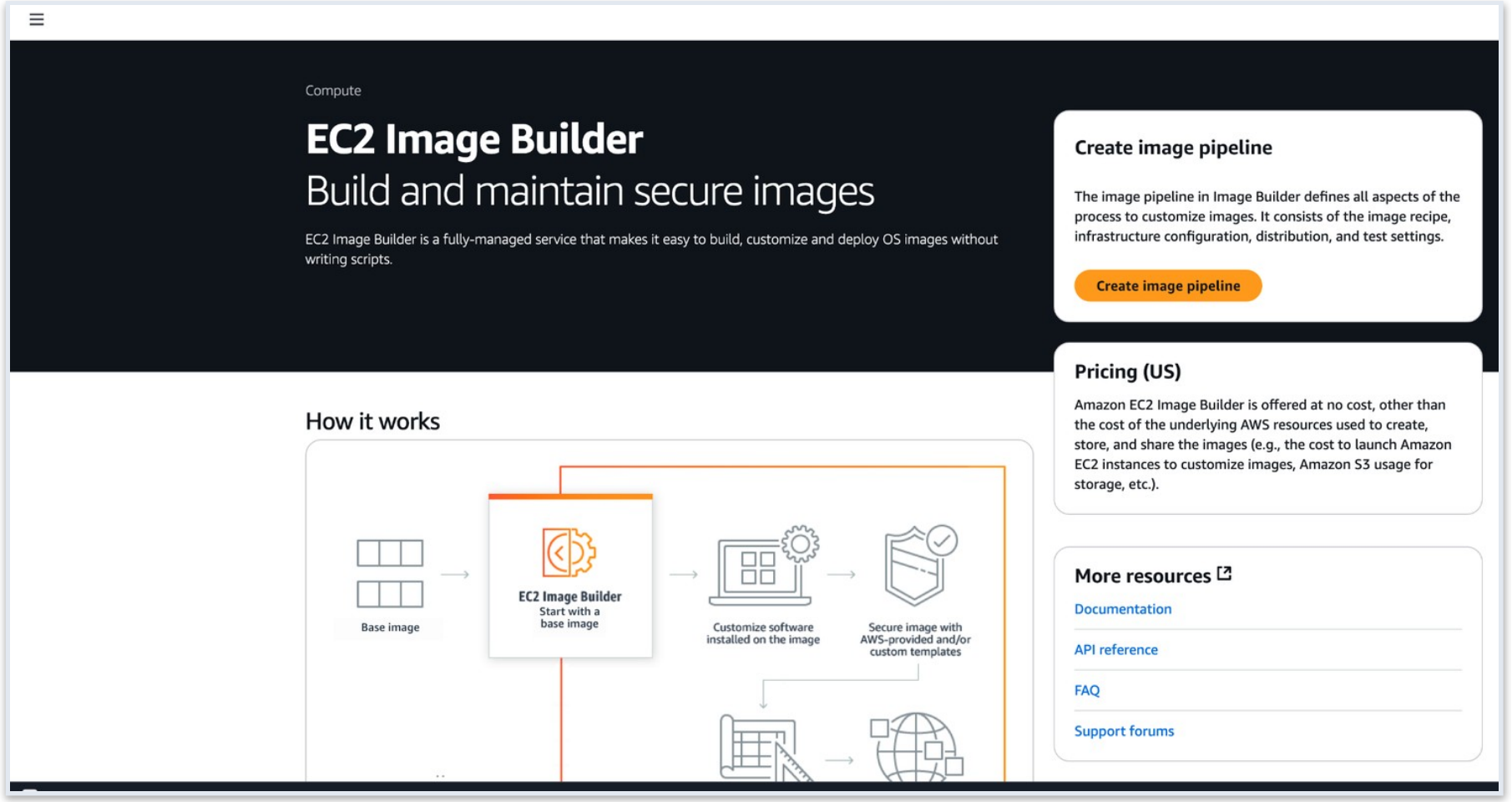
TOPIC 2

Image Builder



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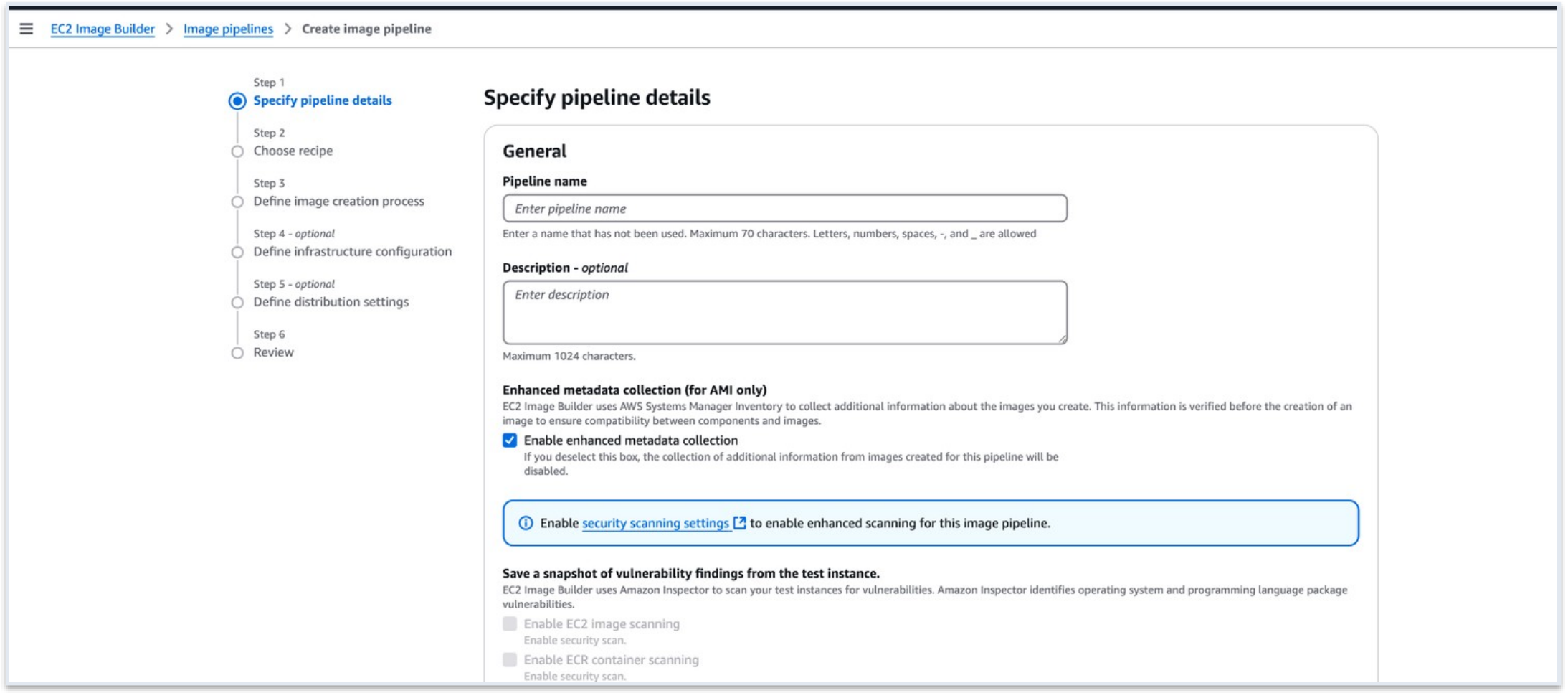
TOPIC 2

Activity: Image Builder



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AMI in Production

- You can force users to only launch EC2 instances from pre-approved AMIs (AMIs tagged with specific tags) using IAM policies
- Combine with AWS Config to find non- compliant EC2 instance (instances launched with non-approved AMIs)
- AMI
- (not approved)
- AMI
- (approved)
- launch
- IAM Permissions
- {
- … "Condition": {
- "StringEquals": {

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TOPIC 3

Management of EC2 at Scale Systems Manager

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TOPIC 3

AWS Systems Manager Overview

- Helps you manage your EC2 and On-Premises systems at scale
- Get operational insights about the state of your infrastructure
- Easily detect problems
- Patching automation for enhanced compliance
- Works for both Windows and Linux OS
- Integrated with CloudWatch metrics / dashboards
- Integrated with AWS Config
- Free service

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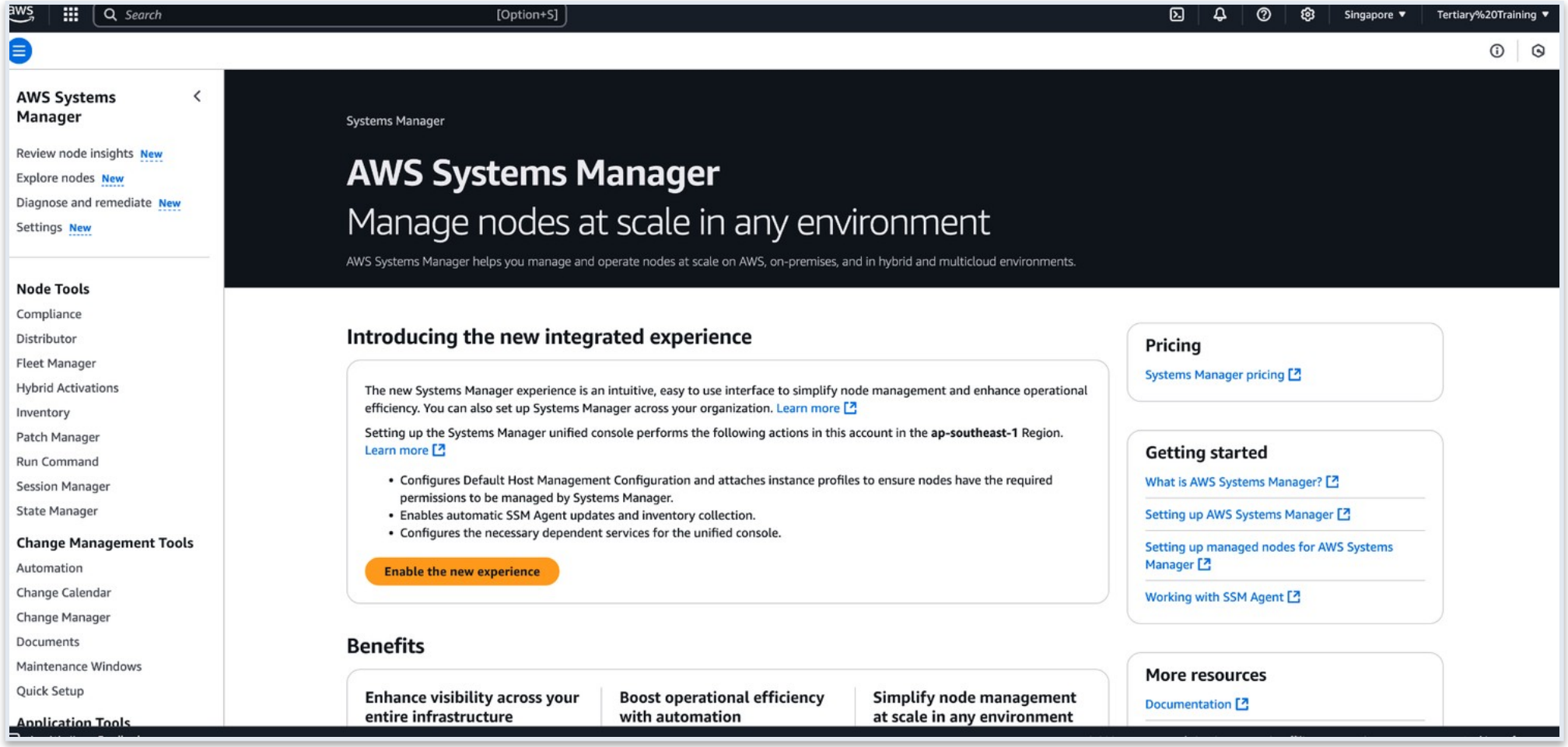
TOPIC 3

AWS Systems Manager



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TOPIC 3

AWS Systems Manager Features

- Resource Groups
- Operations Management
- Explorer
- OpsCenter
- CloudWatch Dashboard
- PHD
- Incident Manager
- Shared Resources
- Documents
- Change Management
- Change Manager
- Automation
- Change Calendar

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TOPIC 3

How Systems Manager works

- We need to install the SSM agent onto the systems we control
- Installed by default on Amazon Linux 2 AMI & some Ubuntu AMI
- If an instance can’t be controlled with SSM, it’s probably an issue with the SSM agent!
- Make sure the EC2 instances have a proper IAM role to allow SSM actions
- EC2 Instance
- (SSM Agent)
- AWS Systems Manager
- (SSM Agent)
- EC2 Instance On-premise
- Server/VM
- (SSM Agent)
- IAM Permissions

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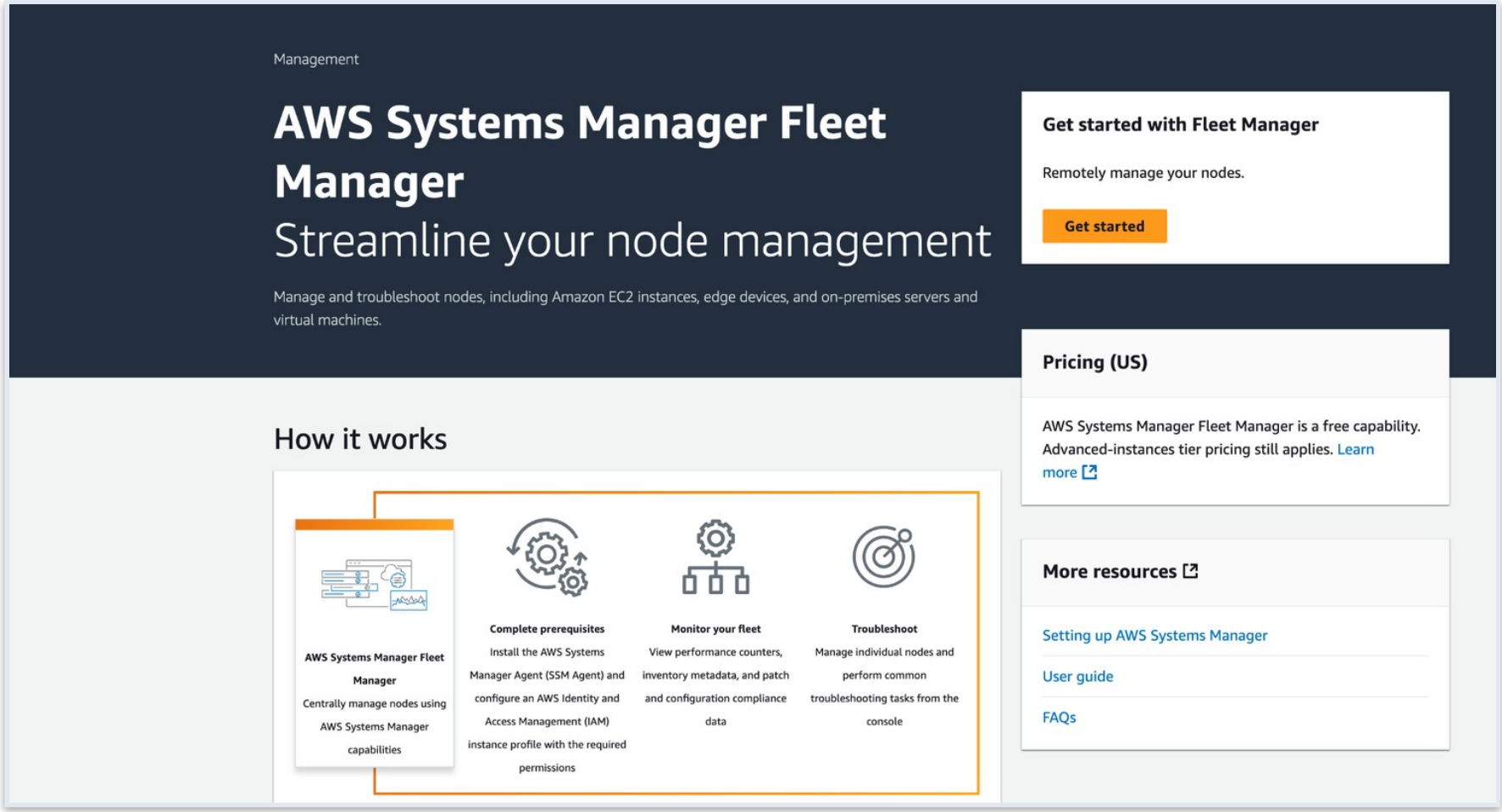
TOPIC 3

Fleet Manager



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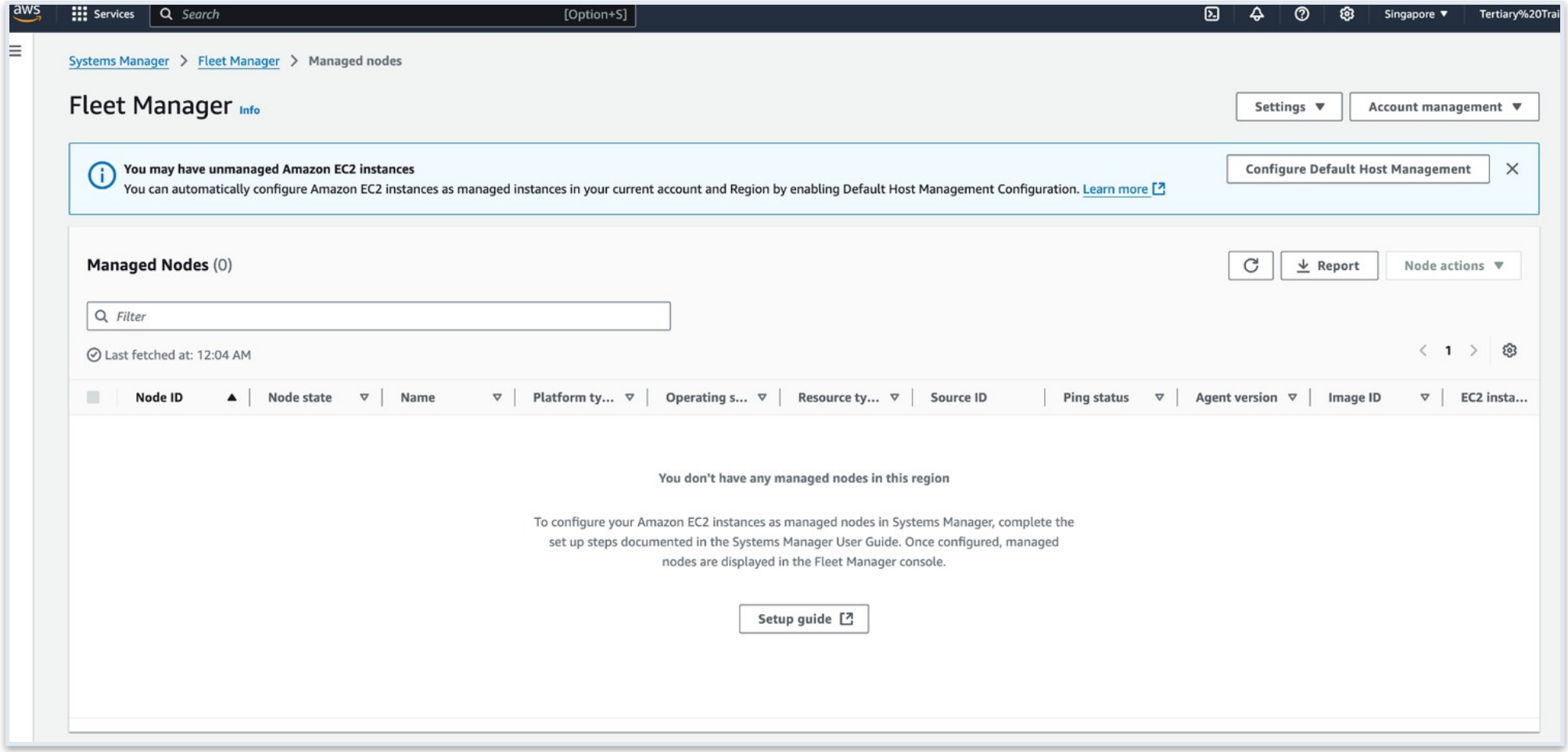
Activity: Fleet Manager



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AWS Tags

- You can add text key-value pairs called Tags to many AWS resources
- Commonly used in EC2
- Free naming, common tags are Name, Environment,Team …
- They’re used for
- Resource grouping
- Automation
- Cost allocation
- Better to have too many tags than too few!

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Resource Groups

- Create, view or manage logical group of resources thanks to tags
- Allows creation of logical groups of resources such as
- Applications
- Different layers of an application stack
- Production versus development environments
- Regional service
- Works with EC2, S3, DynamoDB, Lambda, etc…
- EC2 Instances
- Resource Group Environment = Dev

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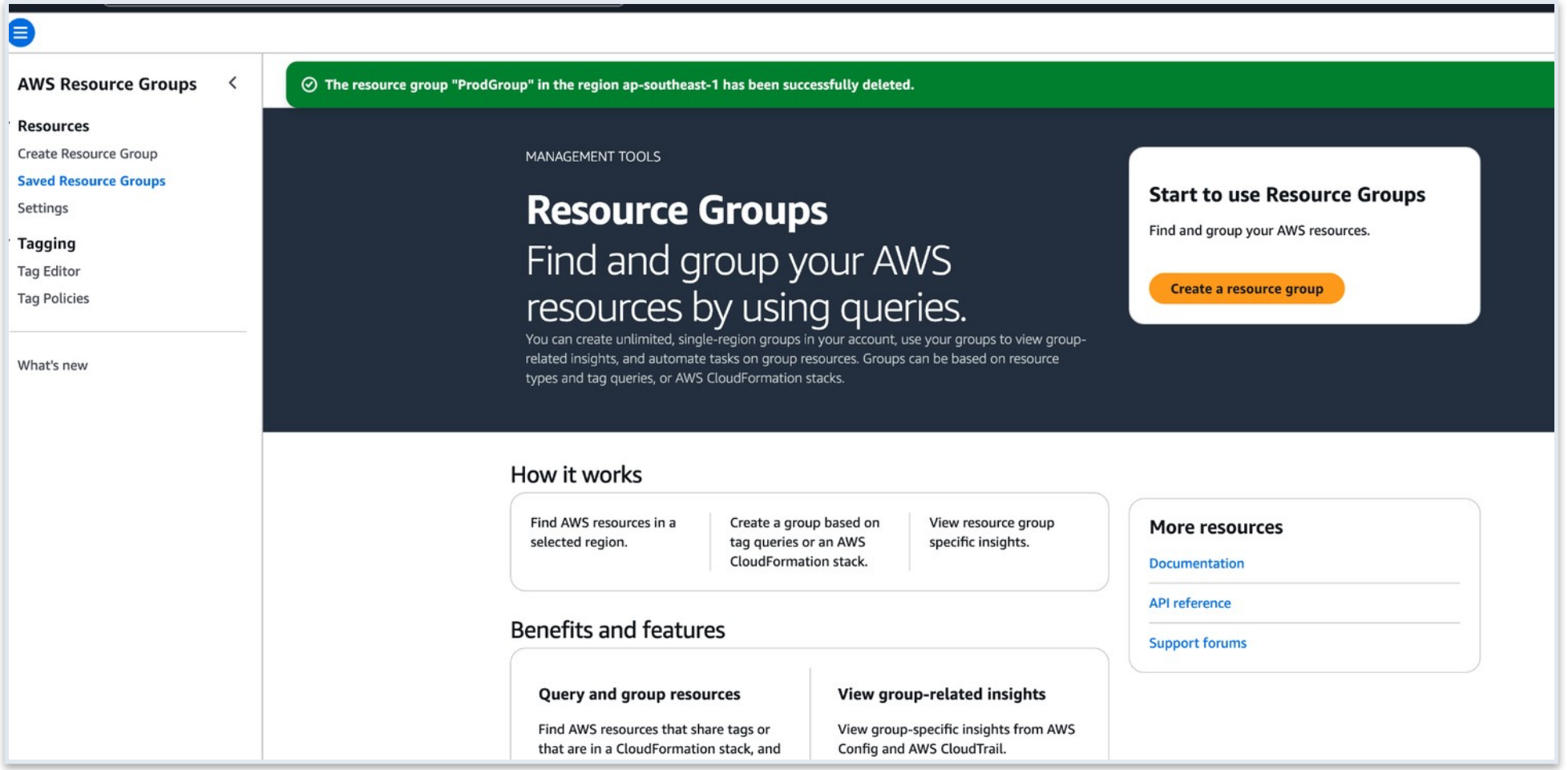
Resource Groups



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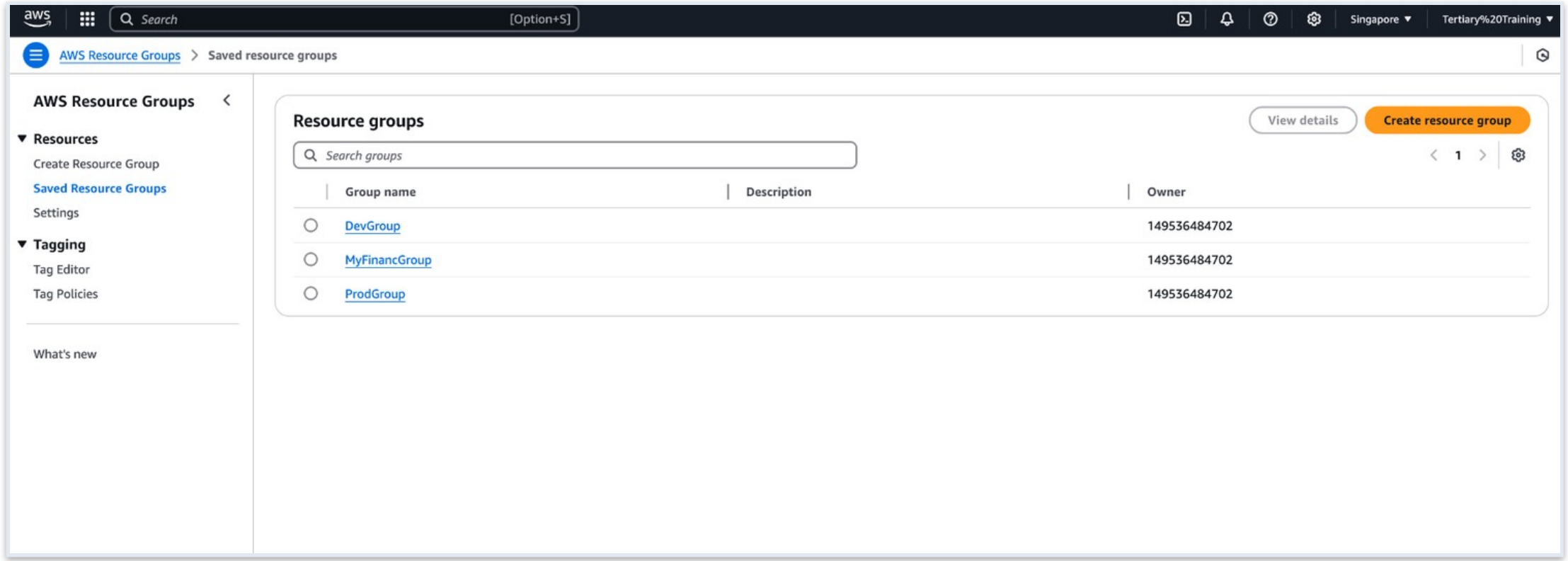
Activity: Resource Groups



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TOPIC 3

SSM - Documents

```yaml
---
schemaVersion: '2.2'
description: State Manager Bootstrap Example
parameters: {}
mainSteps:
  - action: aws:runShellScript
    name: configureServer
    inputs:
      runCommand:
        - sudo yum install -y httpd
        - sudo yum --enablerepo=epel install -y clamav
```

Documents can be in JSON or YAML

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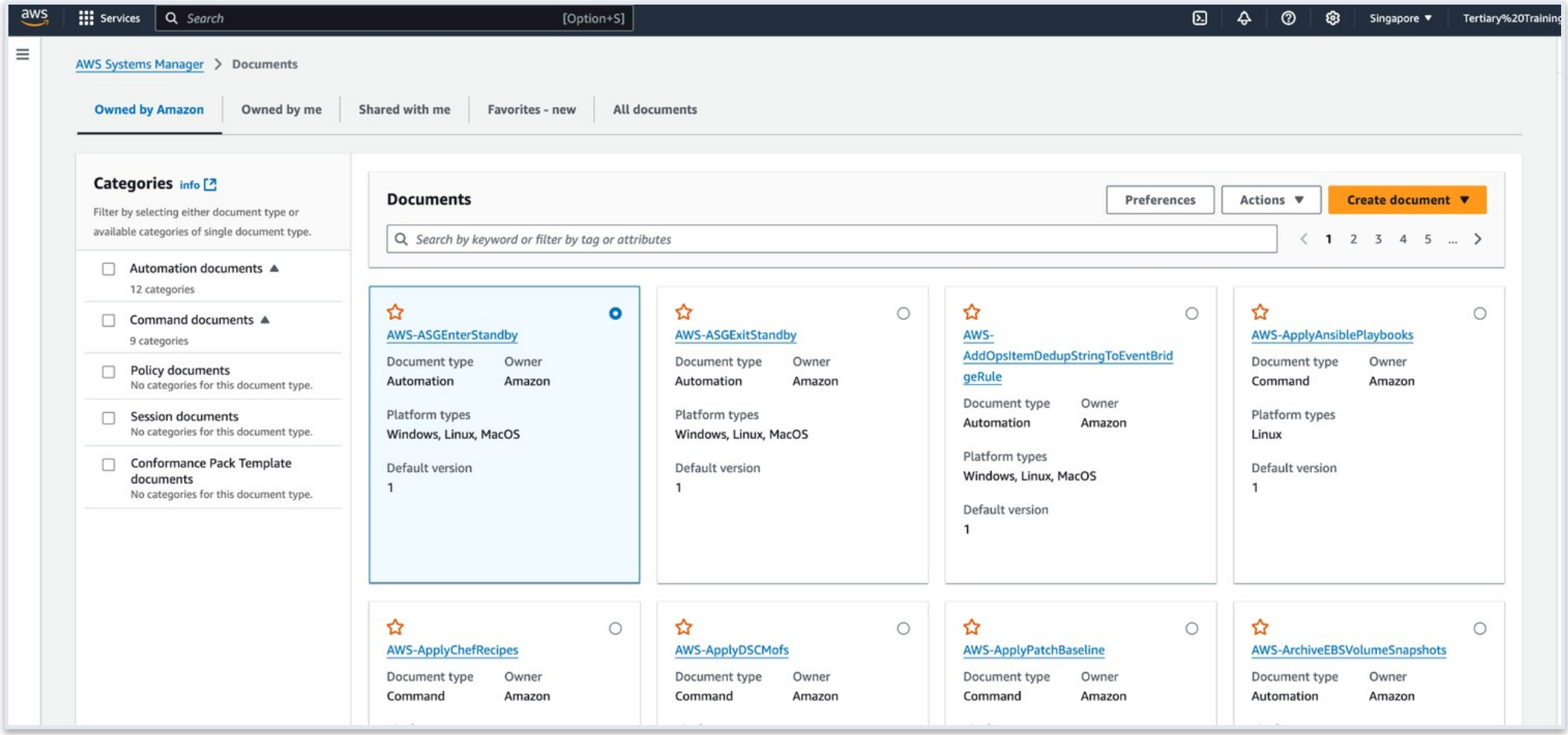
Activity: Document



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TOPIC 3

SSM – Run Command

- Execute a document (= script) or just run a command
- Run command across multiple instances (using resource groups)
- Rate Control / Error Control
- Integrated with IAM & CloudTrail
- No need for SSH
- Command Output can be shown in the Console, sent to S3 bucket or CloudWatch Logs
- Send notifications to SNS about command statues (In progress, Success, Failed, …)
- Can be invoked using EventBridge
- EventBridge
- Event
- Run Command
- Amazon S3
- CloudWatch Logs

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TOPIC 3

SSM - Automation

- Simplifies common maintenance and deployment tasks of EC2 instances and other AWS resources
- Example: restart instances, create an AMI, EBS snapshot
- Automation snapbook
- SSM Documents of type Automation
- Defines actions performed on your EC2 instances or AWS resources
- Pre-defined runbooks (AWS) or create custom runbooks
- Can be triggered
- Manually using AWS Console, AWS CLI or SDK
- By Amazon EventBridge
- On a schedule using Maintenance Windows
- By AWS Config for rules remediations
- SSM Automation
- Runbooks (automation documents)

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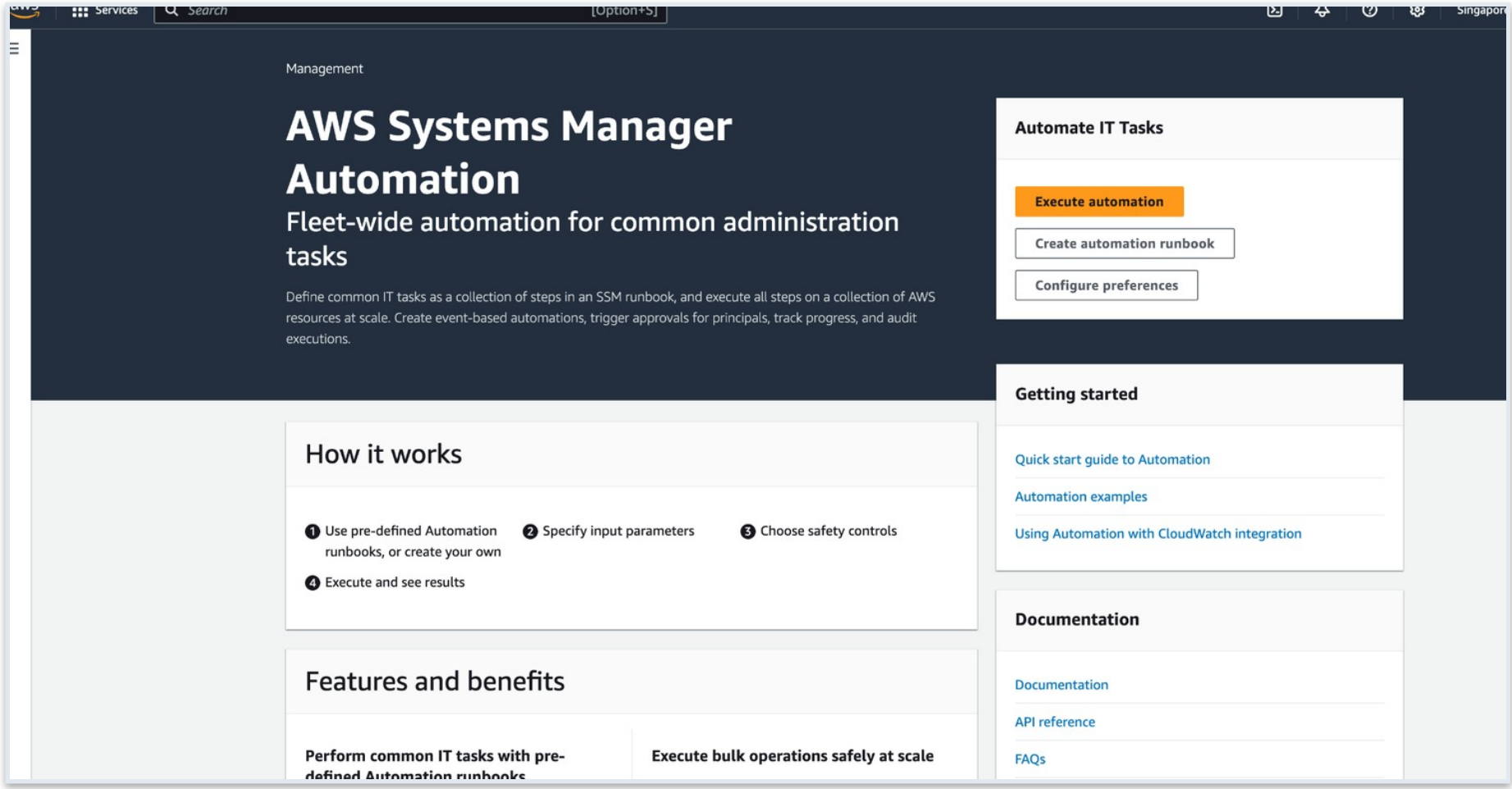
SSM - Automation



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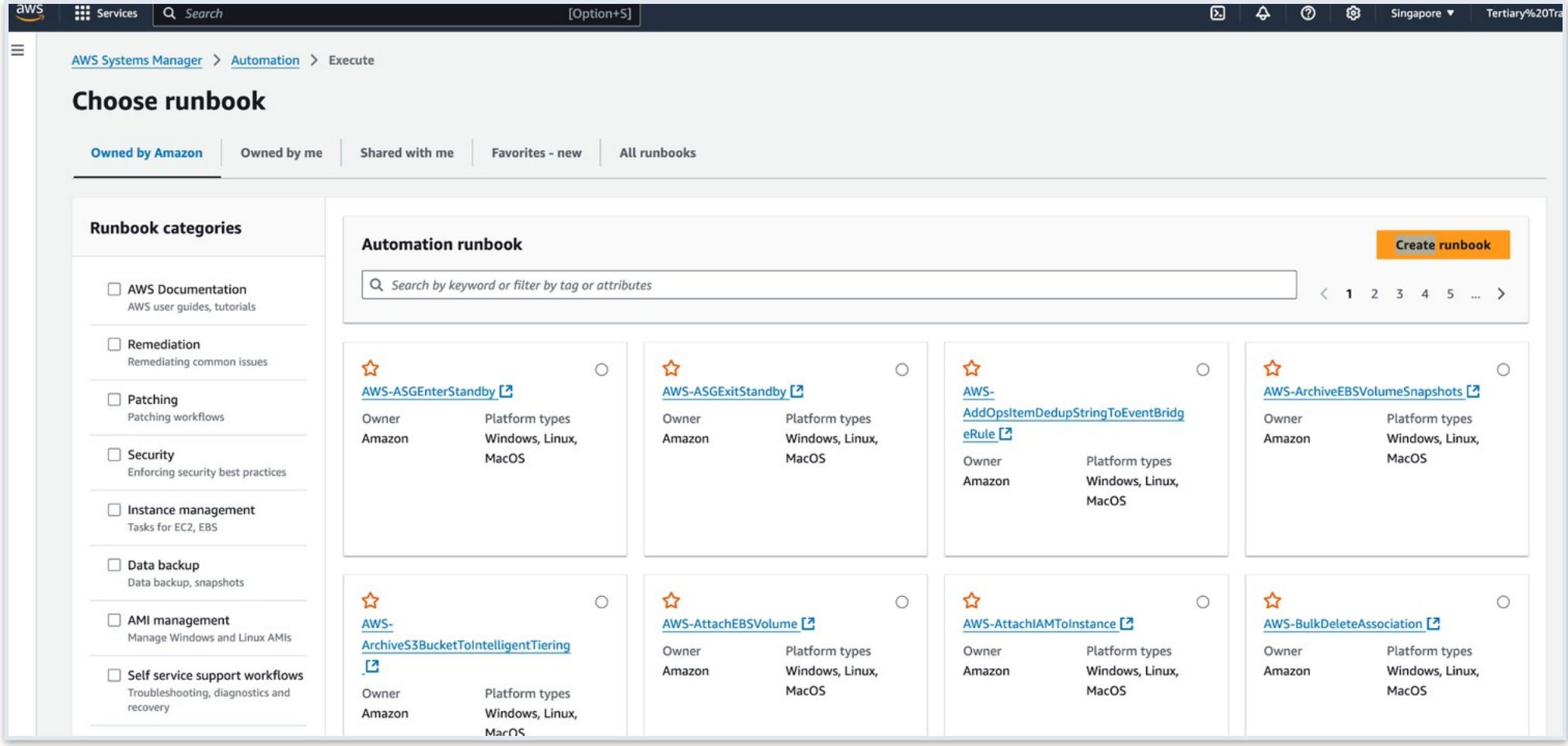
TOPIC 3

Activity: Automation



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TOPIC 3

SSM Parameter Store

- Secure storage for configuration and secrets
- Optional Seamless Encryption using KMS
- Serverless, scalable, durable, easy SDK
- Version tracking of configurations / secrets
- Security through IAM
- Notifications with Amazon EventBridge
- Integration with CloudFormation
- SSM Parameter Store
- Applications
- Plaintext configuration
- Encrypted configuration
- Check IAM permissions
- Decryption Service

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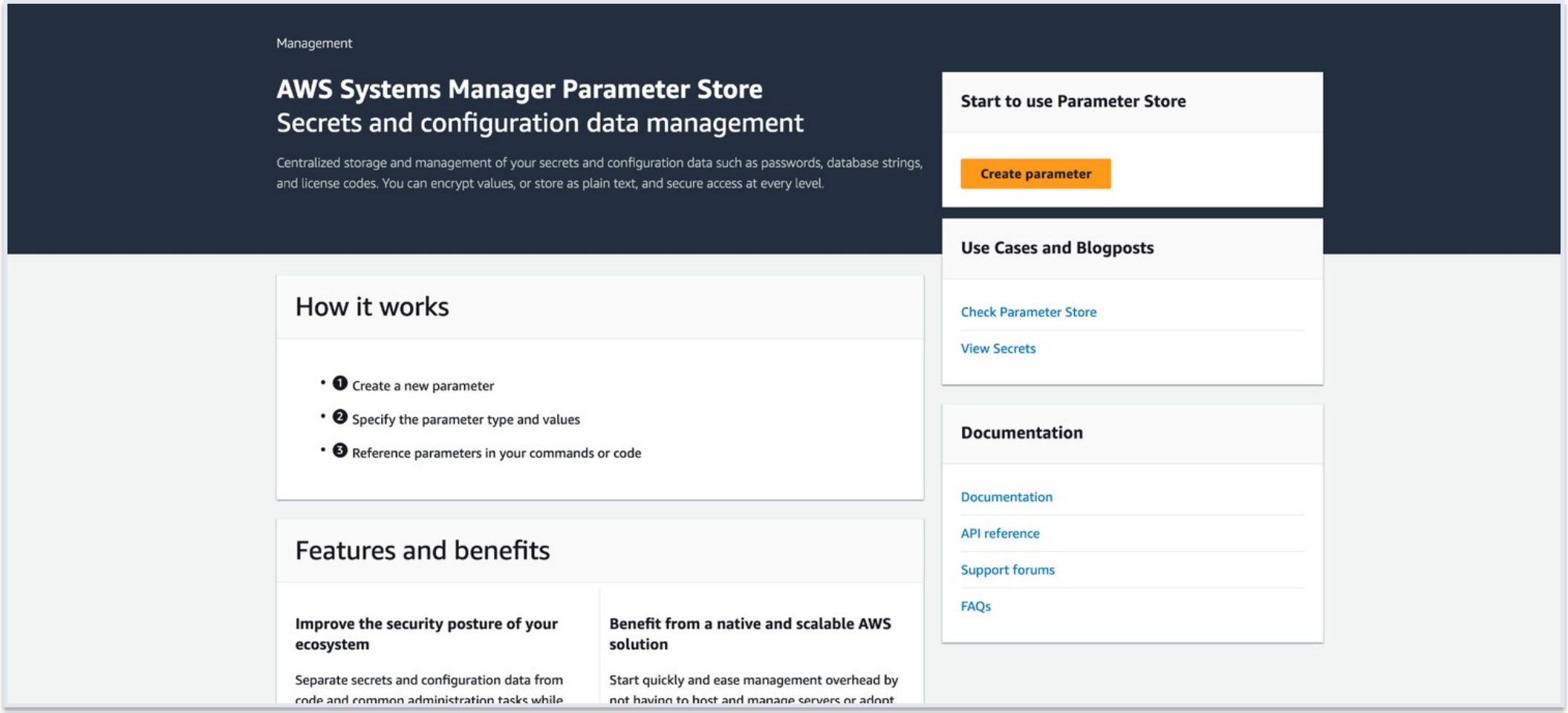
TOPIC 3

SSM Parameter Store



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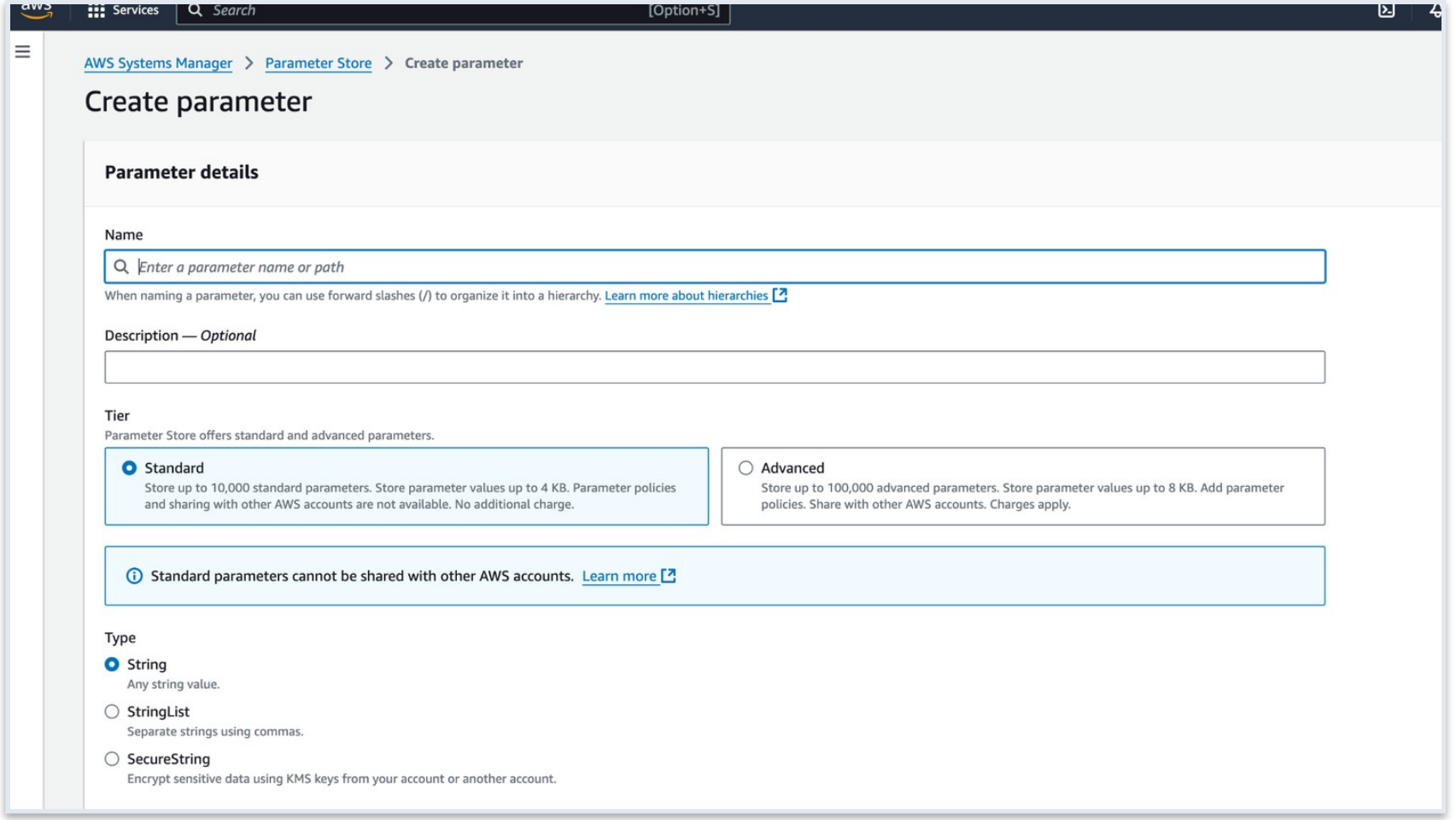
TOPIC 3

Activity: Parameter Store



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TOPIC 3

SSM Parameter Store Hierarchy

- /my-department/
- my-app/
- dev/
- db-url
- db-password
- prod/
- db-url
- db-password
- other-app/
- /other-department/
- /aws/reference/secretsmanager/secret_ID_in_Secrets_Manager
- /aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2 (public)
- Dev Lambda Function

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TOPIC 3

Standard and advanced parameter tiers

- Standard and advanced parameter tiers

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TOPIC 3

Parameters Policies (for advanced parameters)

- Allow to assign a TTL to a parameter (expiration date) to force updating or deleting sensitive data such as passwords
- Can assign multiple policies at a time
- Expiration (to delete a parameter)
- ExpirationNotification (EventBridge)
- NoChangeNotification (EventBridge)

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TOPIC 3

SSM - Inventory

- Collect metadata from your managed instances (EC2/On-premises)
- Metadata includes installed software, OS drivers, configurations, installed updates, running services …
- View data in AWS Console or store in S3 and query and analyze using Athena and QuickSight
- Specify metadata collection interval (minutes, hours, days)
- Query data from multiple AWS accounts and regions
- Create Custom Inventory for your custom metadata (e.g., rack location of each managed instance)

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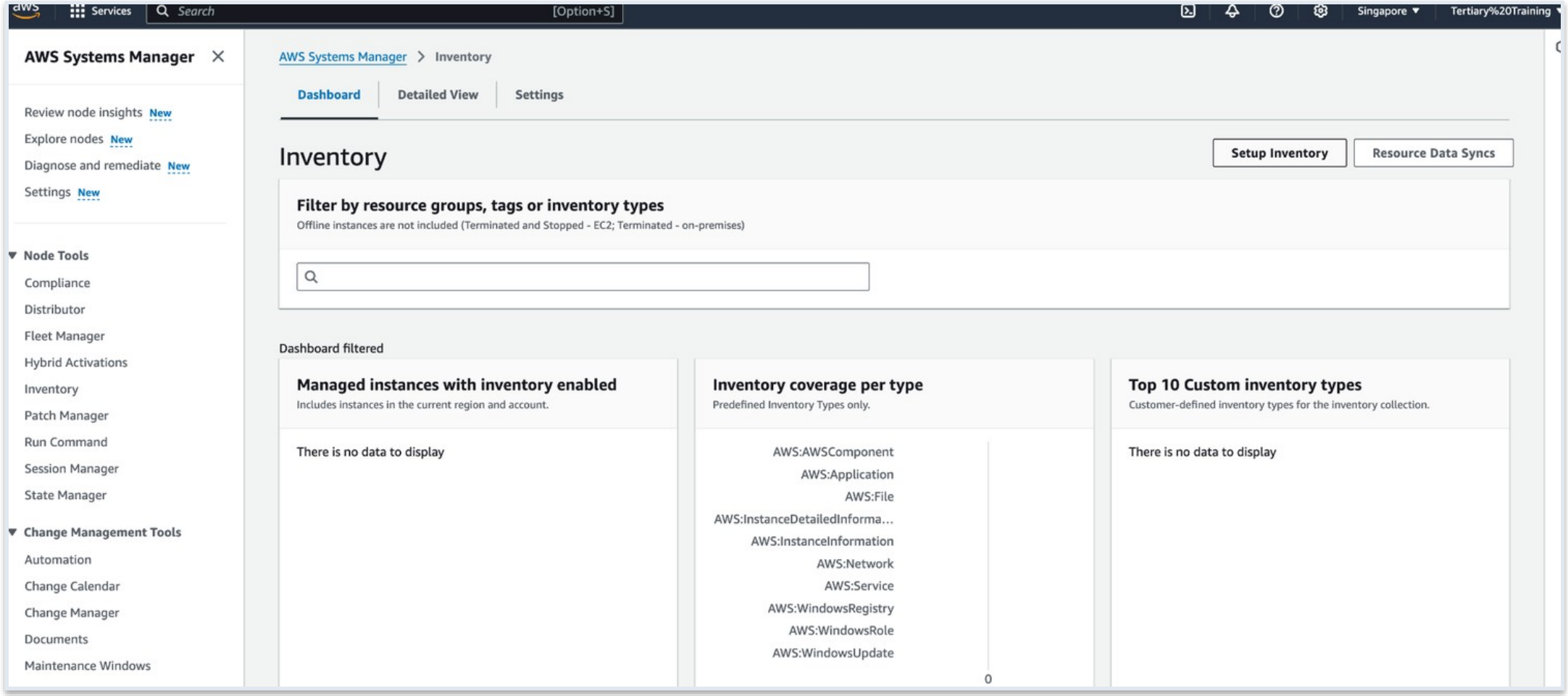
TOPIC 3

Activity: Inventory



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TOPIC 3

SSM - State Manager

- Automate the process of keeping your managed instances
- (EC2/On- premises) in a state that you define
- Use cases: bootstrap instances with software, patch OS/software updates on a schedule …
- State Manager Association:
- Defines the state that you want to maintain to your managed instances
- Example: port 22 must be closed, antivirus must be installed …
- Specify a schedule when this configuration is applied
- Uses SSM Documents to create an Association (e.g., SSM Document to configure CW Agent)

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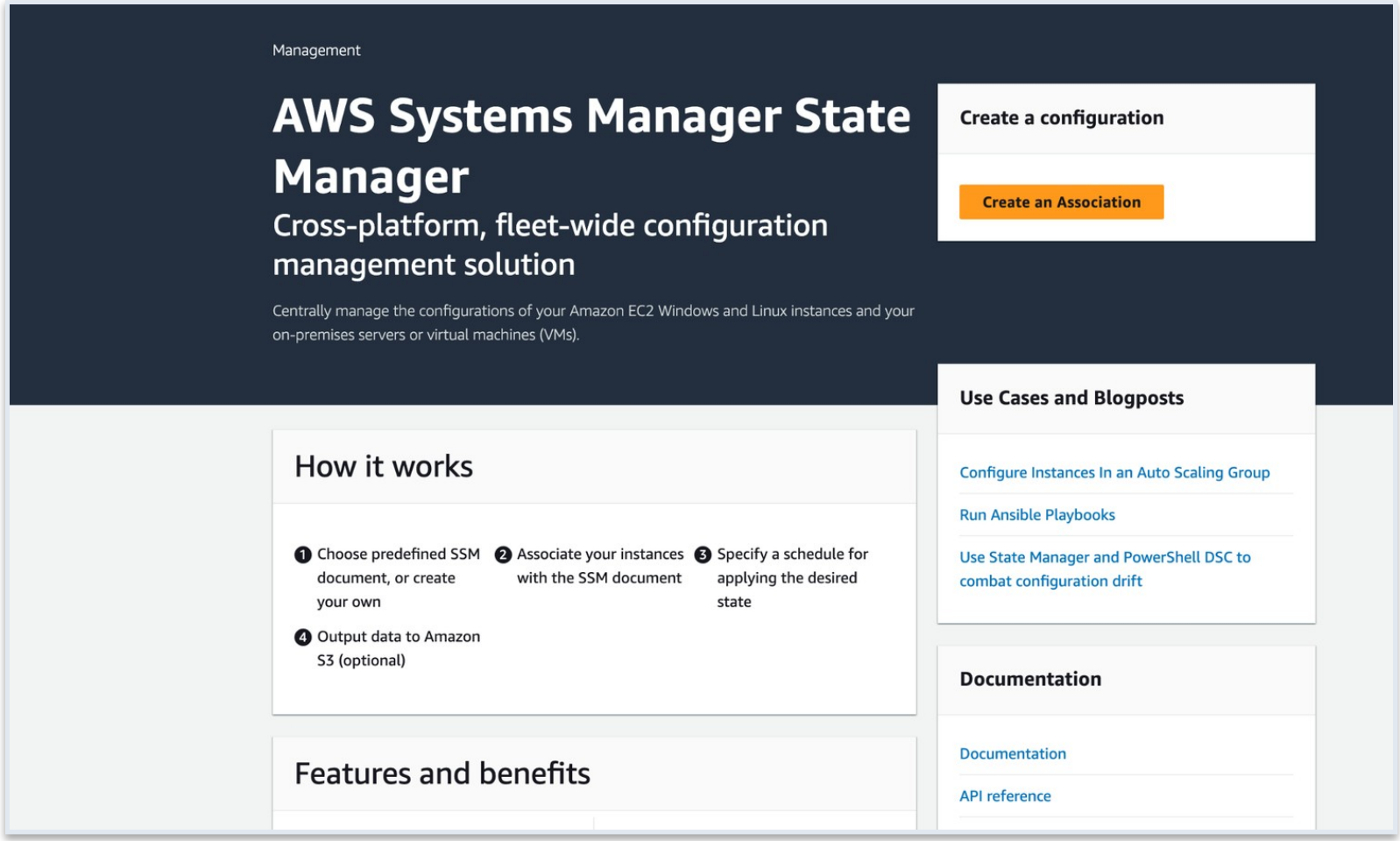
TOPIC 3

SSM - State Manager



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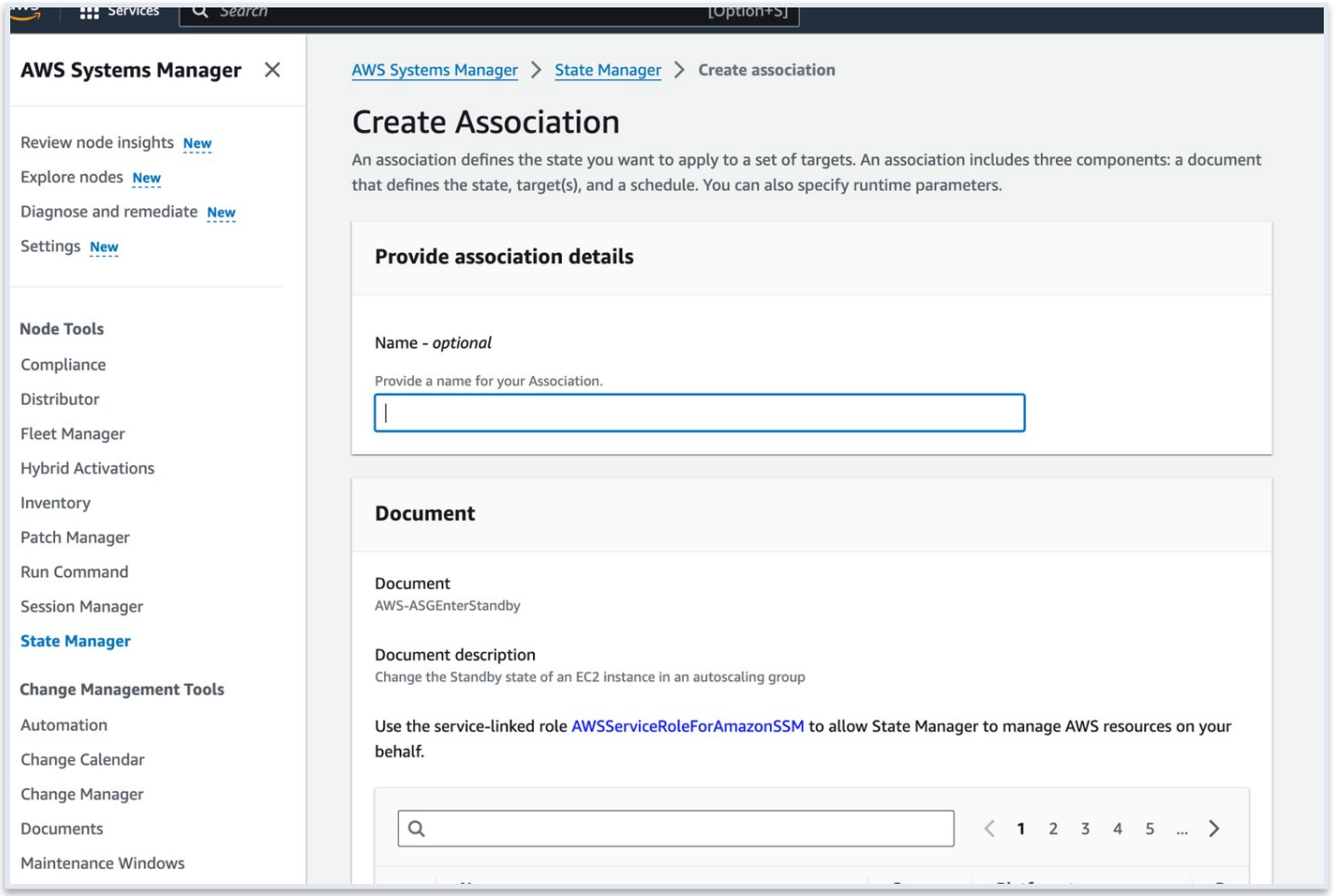
Activity: State Manager



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TOPIC 3

SSM - Patch Manager

- Automates the process of patching managed instances
- OS updates, applications updates, security updates, …
- Supports both EC2 instances and on-premises servers
- Supports Linux, macOS, and Windows
- Patch on-demand or on a schedule using Maintenance Windows
- Scan instances and generate patch compliance report (missing patches)
- Patch compliance report can be sent to S3

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TOPIC 3

SSM - Patch Manager

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TOPIC 3

SSM - Patch Manager

- Patch Baseline
- Defines which patches should and shouldn’t be installed on your instances
- Ability to create custom Patch Baselines (specify approved/rejected patches)
- Patches can be auto-approved within days of their release
- By default, install only critical patches and patches related to security
- Patch Group
- Associate a set of instances with a specific Patch Baseline
- Example: create Patch Groups for different environments (dev, test, prod)
- Instances should be defined with the tag key Patch Group
- An instance can only be in one Patch Group
- Patch Group can be registered with only one Patch Baseline

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SSM - Patch Manager Patch Baselines

- Pre-Defined Patch Baseline
- Managed by AWS for different Operating Systems (can’t be modified)
- AWS-RunPatchBaseline (SSM Document) - apply both operating system and application patches (Linux, macOS, Windows Server)
- Custom Patch Baseline
- Create your own Patch Baseline and choose which patches to auto-approve
- Operating System, allowed patches, rejected patches, …
- Ability to specify custom and alternative patch repositories

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TOPIC 3

SSM - Patch Manager

- Run Command
- EC2 Instances (with SSM Agent)
- Tags
- EC2 Instances (with SSM Agent)
- Tags
- EC2 Instances (with SSM Agent)
- Patch Manager
- Default
- Yes
- No
- Patch Baselines
- Patch Baseline ID Patch Group
- pb-0123456789abcdef0

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TOPIC 3

SSM - Maintenance Windows

- Defines a schedule for when to perform actions on your instances
- Example: OS patching, updating drivers, installing software, …
- Maintenance Window contains
- Schedule
- Duration
- Set of registered instances
- Set of registered tasks

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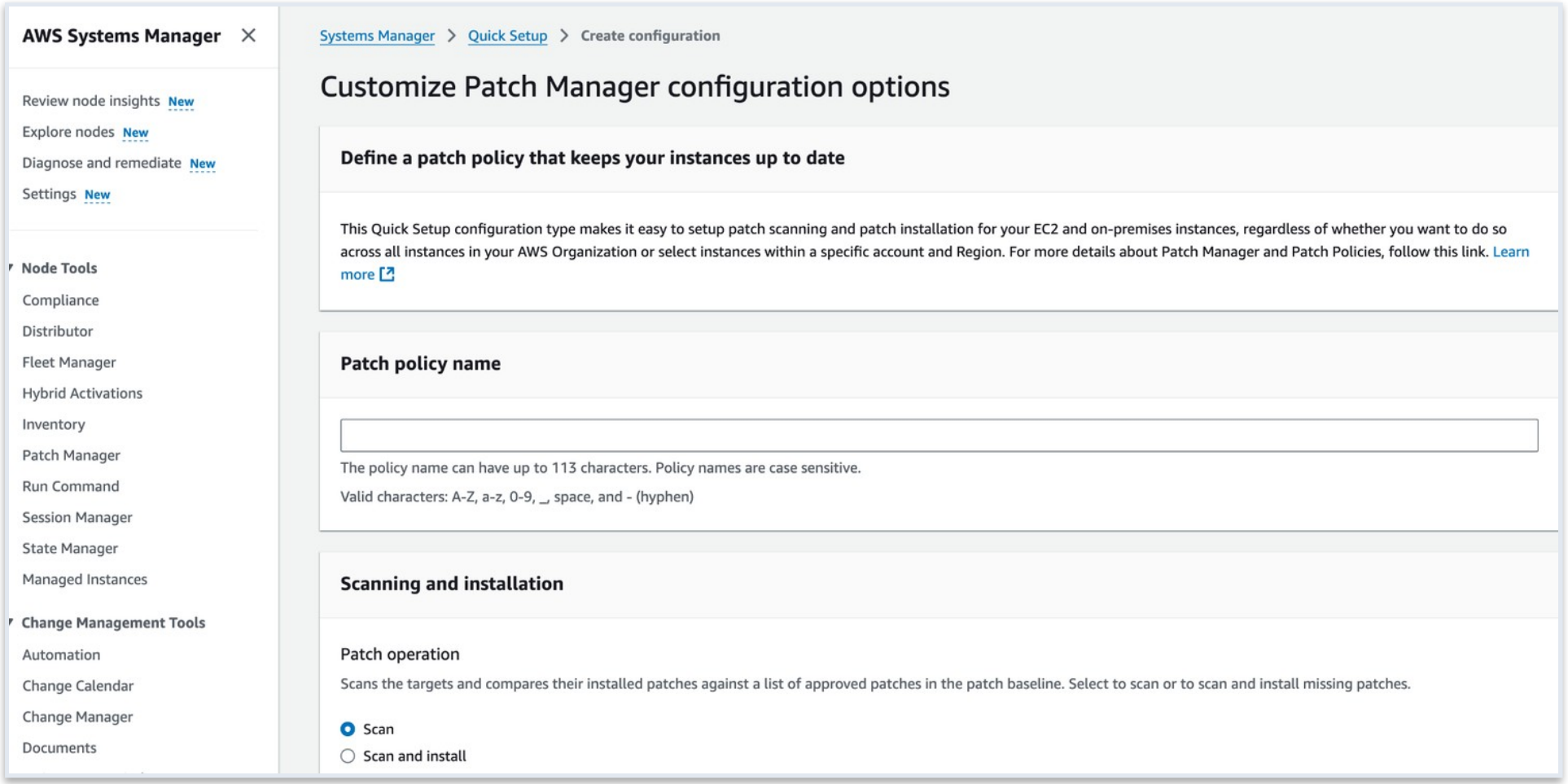
Activity: Patch Manager



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TOPIC 3

SSM - Session Manager

- Allows you to start a secure shell on your EC2 and on-premises servers
- Access through AWS Console, AWS CLI, or Session Manager SDK
- Does not need SSH access, bastion hosts, or SSH keys
- Supports Linux, macOS, and Windows
- Log connections to your instances and executed commands
- Session log data can be sent to S3 or CloudWatch Logs
- CloudTrail can intercept startsession events
- EC2 Instance
- (SSM Agent)
- User
- IAM
- Permissions
- Execute commands

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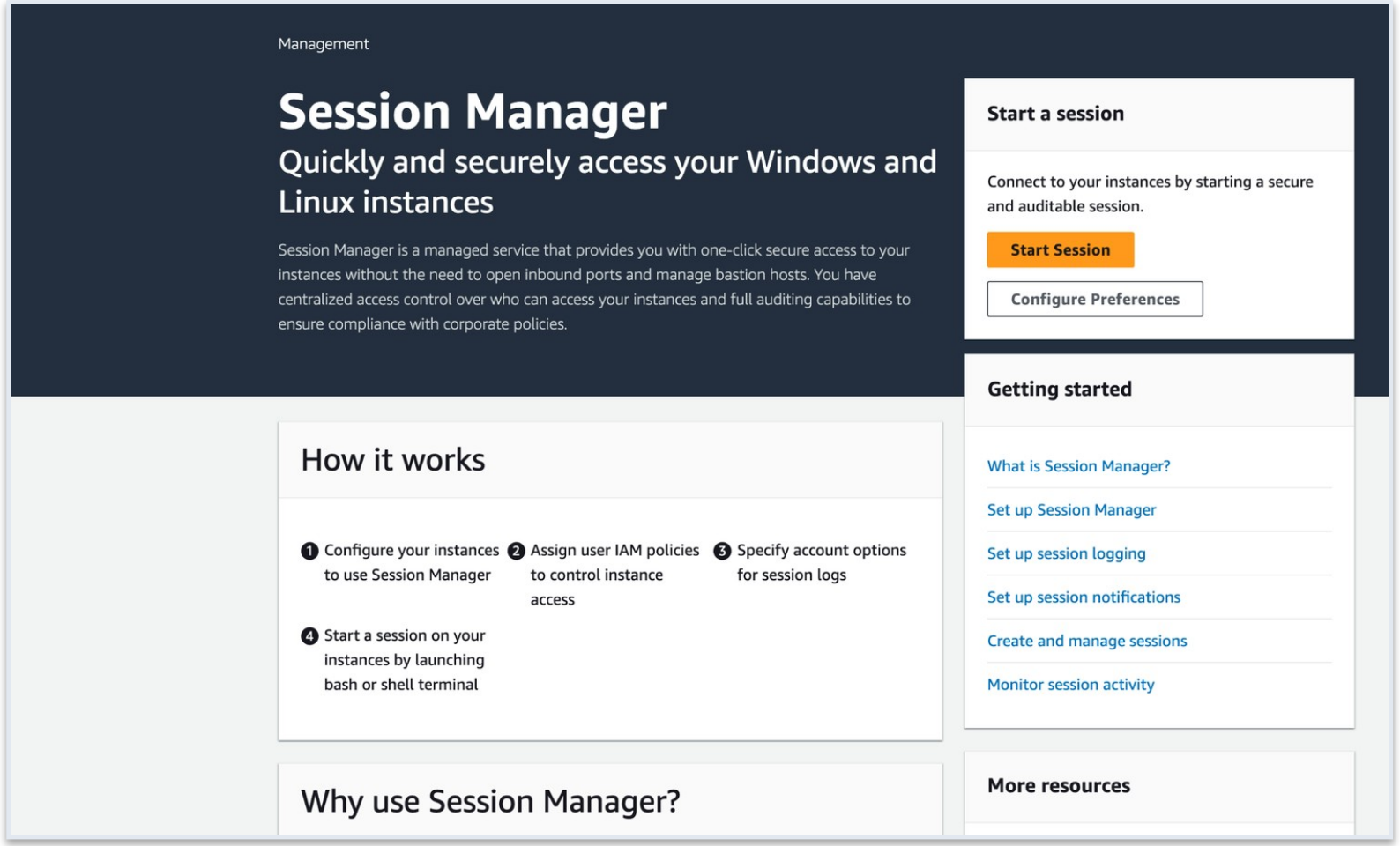
TOPIC 3

SSM - Session Manager



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SSM - Session Manager

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "ssm:StartSession",
            "Resource": "arn:aws:ec2:us-east-1:123456789012:instance/*",
            "Condition": {
                "StringLike": {
                    "ssm:resourceTag/Environment": ["Dev"]
                }
            }
        }
    ]
}
```

IAM Permissions

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SSH vs. SSM Session Manager

- Inbound Rules
- User
- (IPv4: 1.2.3.4)
- EC2 Instance
- (SSM Agent)
- Inbound Rules
- User
- Session Manager
- AWS Console, AWS CLI,
- or Session Manager SDK
- Security Group
- IAM Instance Profile

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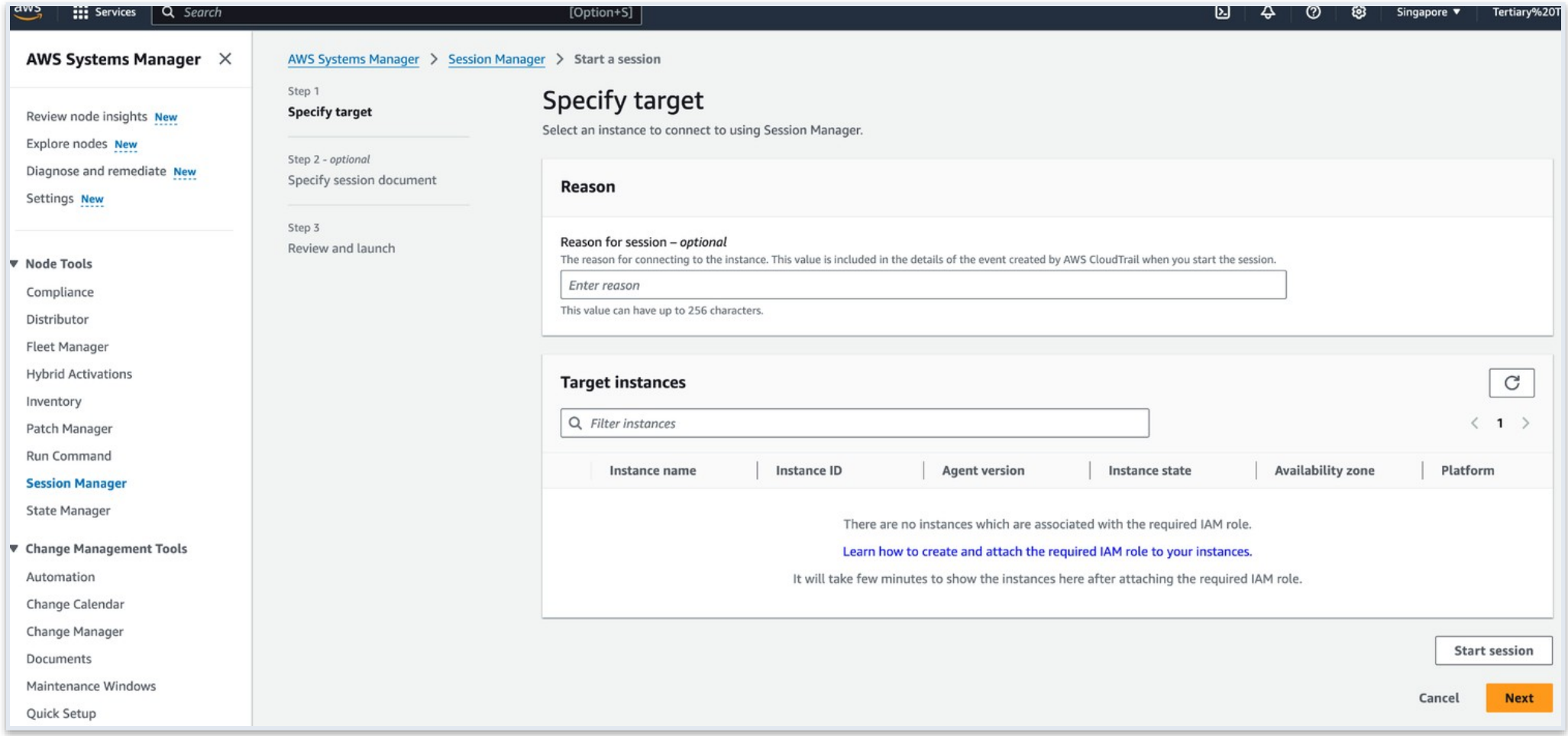
- Connect using SSH Connect using SSM Session Manager

TOPIC 3

Activity: Session Manager



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TOPIC 4

High Availability & Scalability

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TOPIC 4

Scalability and High Availability Section

- Load Balancers:
- Troubleshooting
- Advanced options and logging
- CloudWatch integrations
- Auto Scaling
- Troubleshooting
- Advanced options and logging
- CloudWatch integrations

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Scalability & High Availability

- Scalability means that an application / system can handle greater loads by adapting.
- There are two kinds of scalability:
- Vertical Scalability
- Horizontal Scalability (= elasticity)
- Scalability is linked but different to High Availability
- Let’s deep dive into the distinction, using a call center as an example

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Vertical Scalability

- Vertically scalability means increasing the size of the instance
- For example, your application runs on a t2.micro
- Scaling that application vertically means running it on a t2.large
- Vertical scalability is very common for non distributed systems, such as a database.
- RDS, ElastiCache are services that can scale vertically.
- There’s usually a limit to how much you can vertically scale (hardware limit)
- junior operator
- senior operator

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Horizontal Scalability

- Horizontal Scalability means increasing the number of instances / systems for your application
- Horizontal scaling implies distributed systems.
- This is very common for web applications / modern applications
- It’s easy to horizontally scale thanks the cloud offerings such as Amazon EC2
- operator
- operator
- operator
- operator
- operator
- operator
- operator

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second building in San Francisco

- first building in New York
- High Availability
- High Availability usually goes hand in hand with horizontal scaling
- High availability means running your
- application / system in at least 2 data centers (== Availability Zones)
- The goal of high availability is to survive a data center loss
- The high availability can be passive (for RDS Multi AZ for example)
- The high availability can be active (for horizontal scaling)

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High Availability & Scalability For EC2

- Vertical Scaling: Increase instance size (= scale up / down)
- From: t2.nano - 0.5G of RAM, 1 vCPU
- To: u-12tb1.metal – 12.3 TB of RAM, 448 vCPUs
- Horizontal Scaling: Increase number of instances (= scale out / in)
- Auto Scaling Group
- Load Balancer
- High Availability: Run instances for the same application across multi-AZ
- Auto Scaling Group multi-AZ
- Load Balancer multi-AZ

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TOPIC 4

Whatis load balancing?

- Elastic Load Balancer
- EC2 Instance
- EC2 Instance
- EC2 Instance
- Load Balances are servers that forward traffic to multiple servers (e.g., EC2 instances) downstream

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TOPIC 4

Why use aload balancer?

- Spread load across multiple downstream instances
- Expose a single point of access (DNS) to your application
- Seamlessly handle failures of downstream instances
- Do regular health checks to your instances
- Provide SSL termination (HTTPS) for your websites
- Enforce stickiness with cookies
- High availability across zones
- Separate public traffic from private traffic

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Why use an Elastic Load Balancer?

- An Elastic Load Balancer is a managed load balancer
- AWS guarantees that it will be working
- AWS takes care of upgrades, maintenance, high availability
- AWS provides only a few configuration knobs
- It costs less to setup your own load balancer but it will be a lot more effort on your end
- It is integrated with many AWS offerings / services
- EC2, EC2 Auto Scaling Groups, Amazon ECS
- AWS Certificate Manager (ACM), CloudWatch
- Route 53, AWS WAF, AWS Global Accelerator

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Health Checks

- Health Checks are crucial for Load Balancers
- They enable the load balancer to know if instances it forwards traffic to are available to reply to requests
- The health check is done on a port and a route (/health is common)
- If the response is not 200 (OK), then the instance is unhealthy
- Elastic Load Balancer
- EC2 Instance
- Health Checks
- Protocol: HTTP Port: 4567 Endpoint: /health

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Types of load balancer on AWS

- AWS has 4 kinds of managed Load Balancers
- Classic Load Balancers (v1 - old generation) - 2009 - CLB
- HTTP, HTTPS,TCP, SSL (secure TCP)
- Application Load Balancer v2 - new generation) - 2016 - ALB
- HTTP, HTTPS, WebSocket
- Network Load Balancer (v2 - new generation) - 2017 - NLB
- TCP, TLS (secure TCP), UDP
- Gateway Load Balancer - 2020 - GWLB
- Operates at layer 3 (Network layer) - IP Protocol
- Overall, it is recommended to use the newer generation load balancers as they provide more features
- Some load balancers can be setup as internal (private) or external (public) ELB

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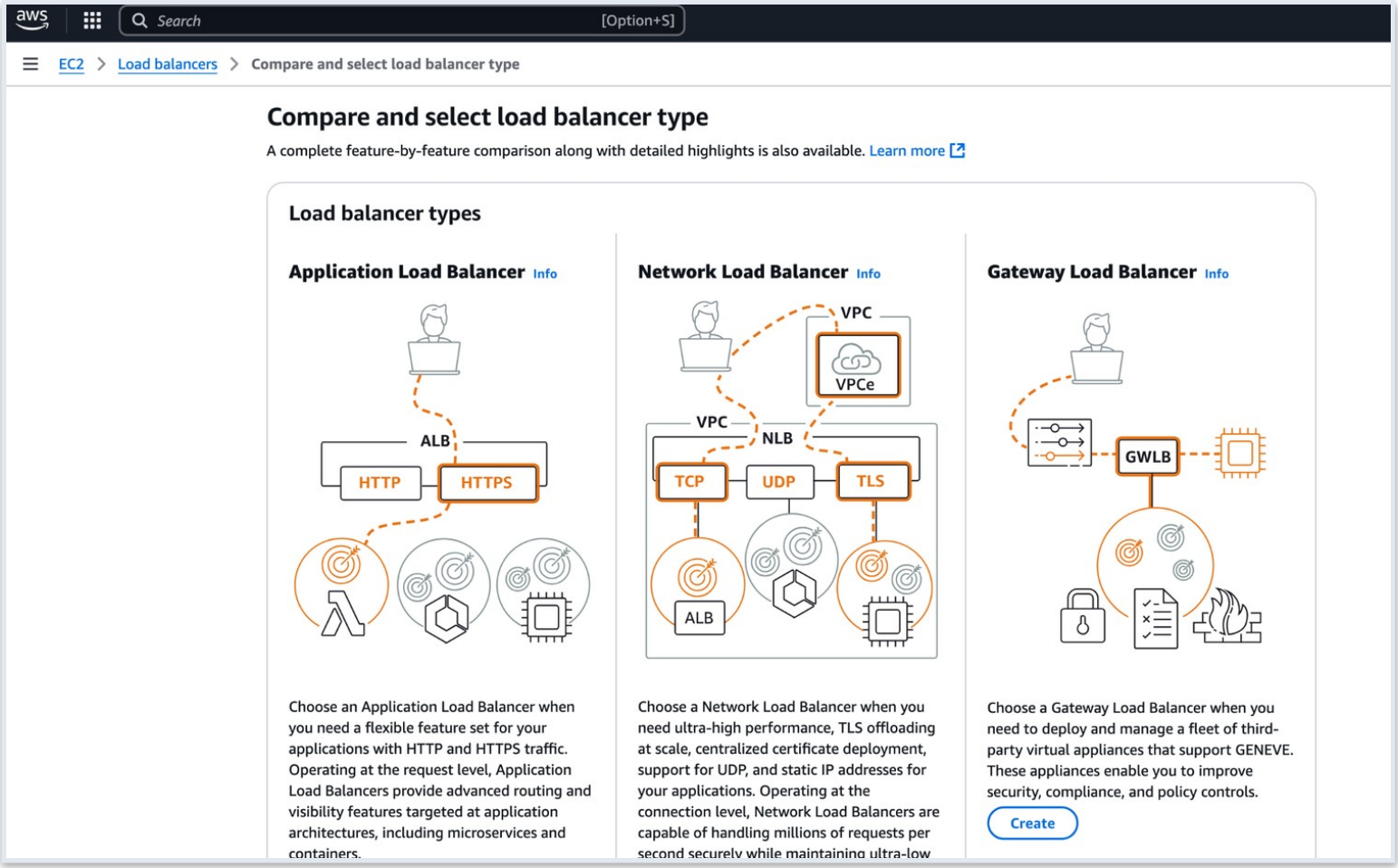
Types of load balancer on AWS



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Users

Type	Protocol	Port Range	Source	Description
HTTP	TCP	80	0.0.0.0/0	Allow HTTP from an...
HTTPS	TCP	443	0.0.0.0/0	Allow HTTPS from a...

HTTPS / HTTP

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Classic Load Balancers (v1)

- Client
- CLB
- EC2
- listener
- internal
- Supports TCP (Layer 4), HTTP & HTTPS (Layer 7)
- Health checks are TCP or HTTP based
- Fixed hostname XXX.region.elb.amazonaws.com

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Application Load Balancer (v2)

- Application load balancers is Layer 7 (HTTP)
- Load balancing to multiple HTTP applications across machines (target groups)
- Load balancing to multiple applications on the same machine (ex: containers)
- Support for HTTP/2 and WebSocket
- Support redirects (from HTTP to HTTPS for example)

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Application Load Balancer (v2)

- Routing tables to different target groups:
- Routing based on path in URL (example.com/users & example.com/posts)
- Routing based on hostname in URL (& other.example.com)
- Routing based on Query String, Headers (example.com/users? id=123=false)
- ALB are a great fit for micro services & container-based application (example: Docker & Amazon ECS)
- Has a port mapping feature to redirect to a dynamic port in ECS
- In comparison, we'd need multiple Classic Load Balancer per application

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Target Group for Users application

- Application Load Balancer (v2) HTTP Based Traffic
- External Application Load Balancer (v2)
- HTTP
- WWW
- Route /user
- Target Group for Search application
- HTTP
- WWW
- Route /search
- Health Check
- Health Check

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Application Load Balancer (v2) Target Groups

1. EC2 instances (can be managed by an Auto Scaling Group) - HTTP
2. ECS tasks (managed by ECS itself) - HTTP
3. Lambda functions - HTTP request is translated into a JSON event
4. IP Addresses - must be private IPs
5. ALB can route to multiple target groups
6. Health checks are at the target group level

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Target Group 1

- AWS – EC2 based
- Application Load Balancer (v2) Query Strings/Parameters Routing
- External Application Load Balancer (v2)
- ?Platform=Mobile
- WWW
- Requests
- Target Group 2
- On-premises – Private IP routing
- ?Platform=Desktop

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Application Load Balancer (v2) Good to Know

- Fixed hostname (XXX.region.elb.amazonaws.com)
- The application servers don’t see the IP of the client directly
- The true IP of the client is inserted in the header X-Forwarded-For
- We can also get Port (X-Forwarded-Port) and proto (X-Forwarded-Proto)
- EC2
- Instance
- Client IP 12.34.56.78
- Load Balancer IP (Private IP)
- Connection termination

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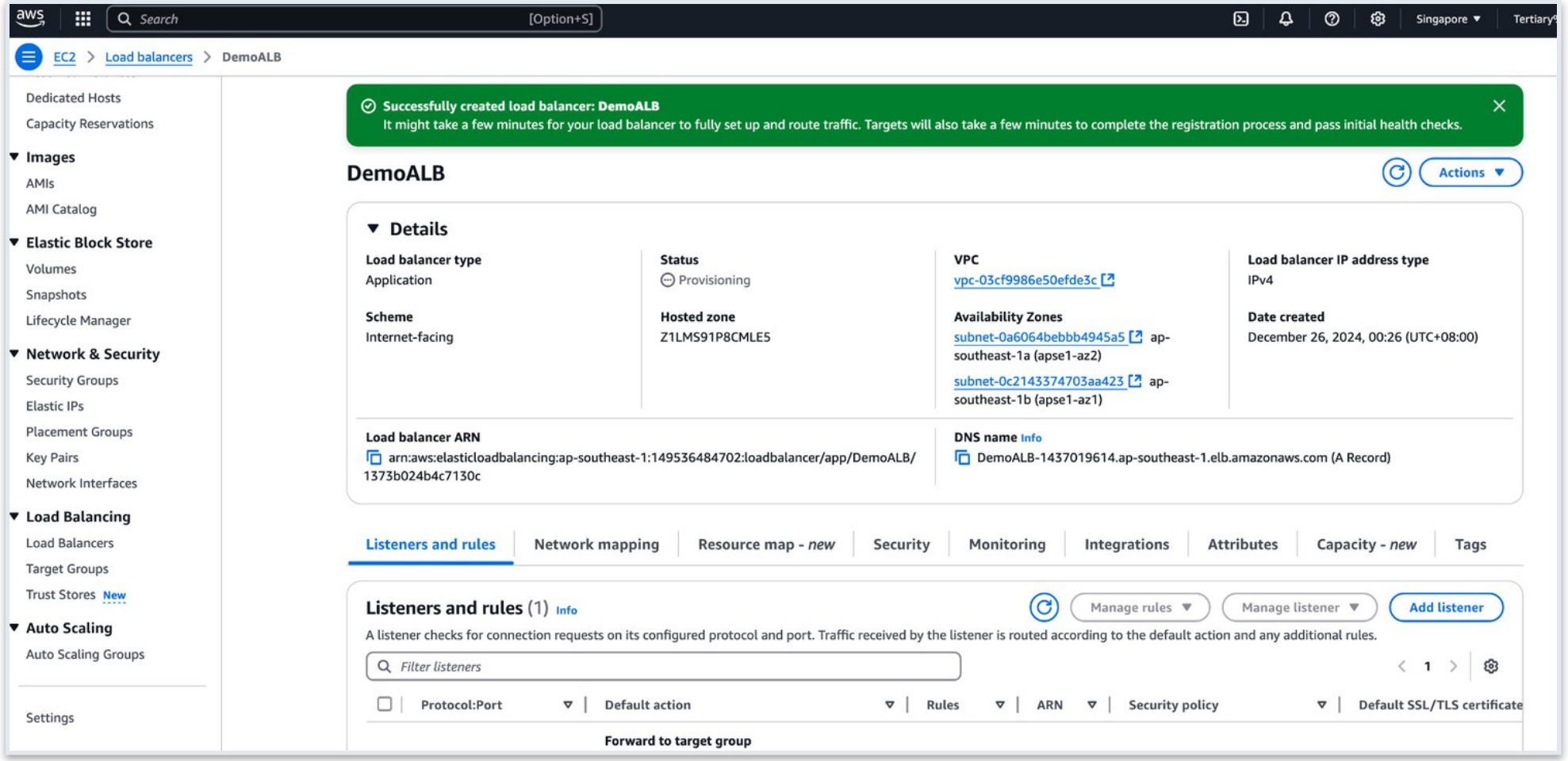
Activity: Application Load Balancer



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Network Load Balancer (v2)

- Network load balancers (Layer 4) allow to:
- Forward TCP & UDP traffic to your instances
- Handle millions of request per seconds
- Ultra-low latency
- NLB has one static IP per AZ and supports assigning Elastic IP
- (helpful for whitelisting specific IP)
- NLB are used for extreme performance,TCP or UDP traffic
- Not included in the AWS free tier

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TOPIC 4

Network Load Balancer (v2) TCP (Layer 4) Based Traffic

- Target Group for Users application
- External Network Load Balancer (v2)
- TCP
- WWW
- TCP + Rules
- Target Group for Search application
- HTTP
- WWW
- TCP + Rules
- Health Check
- Health Check

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TOPIC 4

Network Load Balancer - Target Groups

- EC2 Instances
- IP Addresses - must be private IPs
- Application Load Balancer
- Health Checks support the TCP, HTTP, HTTPS Protocols
- i-1234567890abcdef0 i-1234567890abcdef0
- Target Group
- (EC2 Instances)
- Network
- Load Balancer
- Target Group
- (IP Addresses)
- 192.168.1.118
- 10.0.4.21

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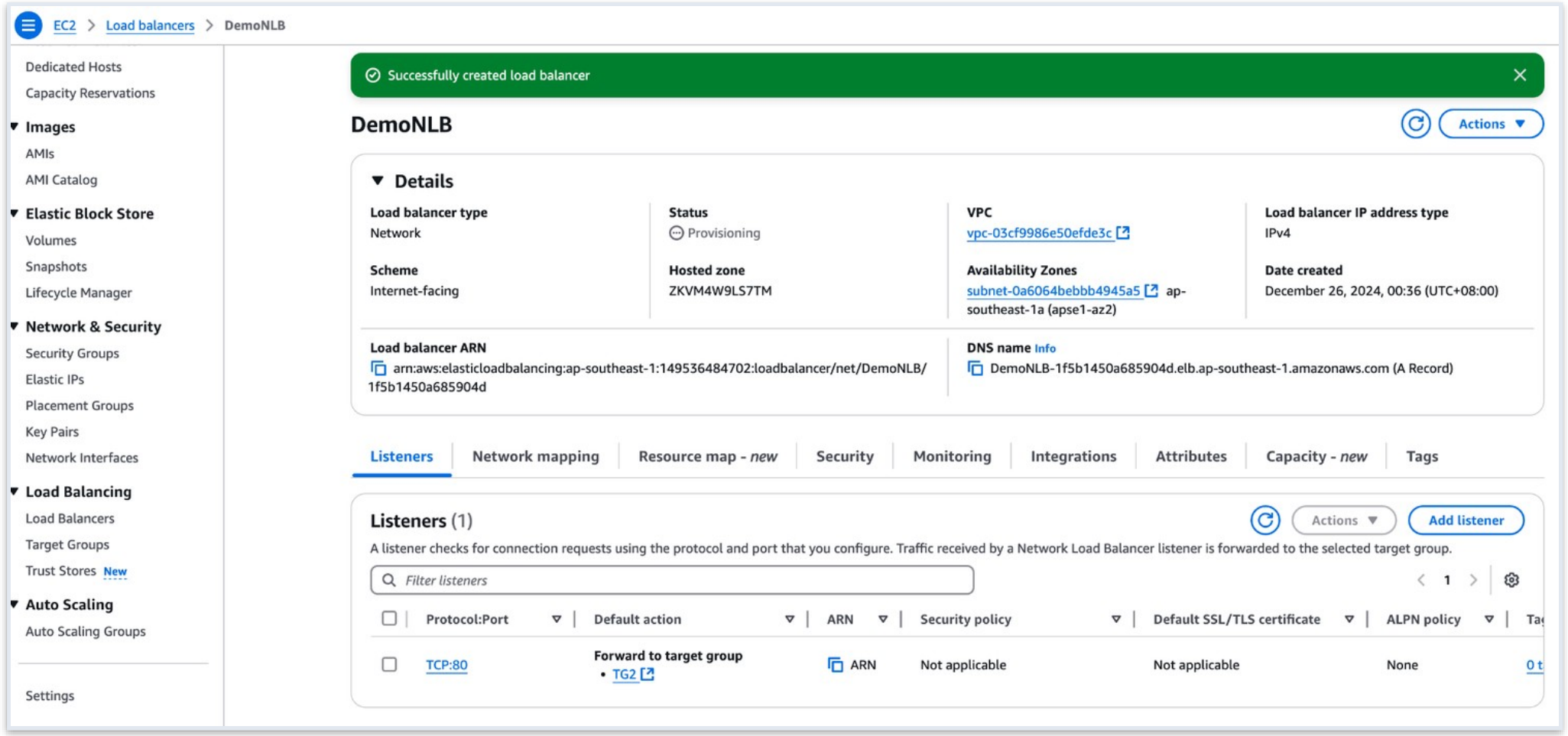
Activity: Network Load Balancer



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Gateway Load Balancer

- Deploy, scale, and manage a fleet of 3rd party network virtual appliances in AWS
- Example: Firewalls, Intrusion Detection and
- Prevention Systems, Deep Packet Inspection Systems, payload manipulation, …
- Operates at Layer 3 (Network Layer) – IP Packets
- Combines the following functions:
- Transparent Network Gateway - single entry/exit for all traffic
- Load Balancer - distributes traffic to your virtual appliances
- Uses the GENEVE protocol on port 60
- Users
- (source)
- Gateway Load Balancer
- Application
- (destination)

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TOPIC 4

Gateway Load Balancer - Target Groups

- EC2 Instances
- IP Addresses - must be private IPs
- i-1234567890abcdef0 i-1234567890abcdef0
- Target Group
- (EC2 Instances)
- Gateway Load Balancer
- Target Group
- (IP Addresses)
- 192.168.1.118
- 10.0.4.21
- Gateway Load Balancer

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TOPIC 4

Sticky Sessions (Session Affinity)

- It is possible to implement stickiness so that the
- same client is always redirected to the same instance behind a load balancer
- This works for Classic Load Balancer
- Application Load Balancer, and Network Load Balancer
- For both CLB & ALB, the “cookie” used for stickiness has an expiration date you control
- Use case: make sure the user doesn’t lose his session data
- Enabling stickiness may bring imbalance to the load over the backend EC2 instances
- EC2 Instance
- EC2 Instance
- Client 1
- Client 2
- Client 3

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TOPIC 4

Sticky Sessions – Cookie Names

- Application based cookies
- Custom cookie
- Generated by the target
- Can include any custom attributes required by the application
- Cookie name must be specified individually for each target group
- Don’t use AWSALB, AWSALBAPP or AWSALBTG reserved for use by the ELB)
- Application cookie
- Generated by the load balancer
- Cookie name is AWSALBAPP
- Duration based cookies
- Cookie generated by the load balancer
- Cookie name is AWSALB for ALB, AWSELB for CLB

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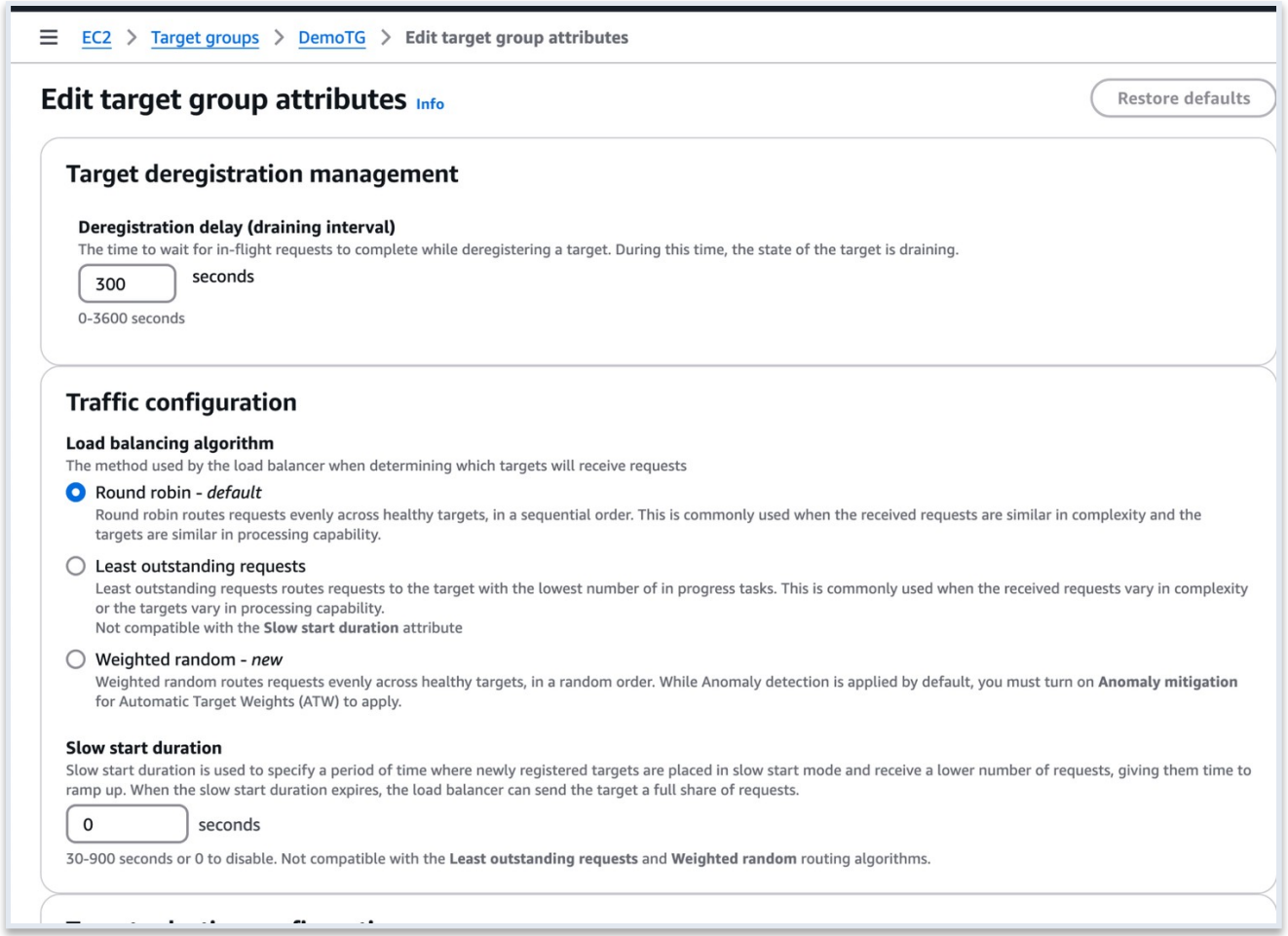
Activity: Sticky Session



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Cross-Zone Load Balancing

- Availability Zone 1
- Availability Zone 2
- 10
- 10
- 10
- 10
- 10
- 10
- 10
- 10
- 10
- 10
- Availability Zone 1

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Cross-Zone Load Balancing

- Application Load Balancer
- Enabled by default (can be disabled at the Target Group level)
- No charges for inter AZ data
- Network Load Balancer $ Gateway Load Balancer
- Disabled by default
- You pay charges ($) for inter AZ data if enabled
- Classic Load Balancer
- Disabled by default
- No charges for inter AZ data if enabled

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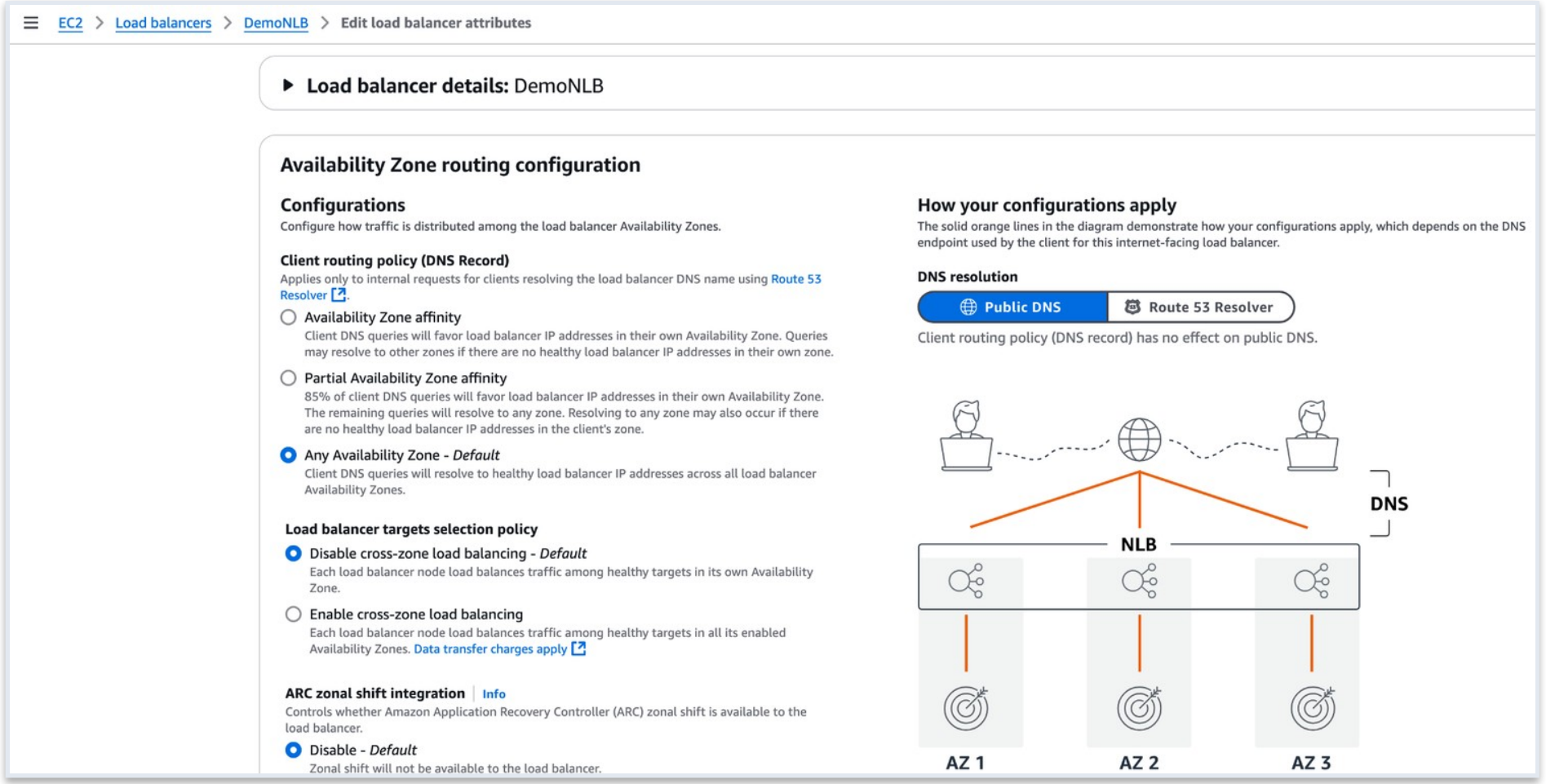
Activity: Cross Zone Load Balancing



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SSL/TLS - Basics

- An SSL Certificate allows traffic between your clients and your load balancer to be encrypted in transit (in-flight encryption)
- SSL refers to Secure Sockets Layer, used to encrypt connections
- TLS refers to Transport Layer Security, which is a newer version
- Nowadays, TLS certificates are mainly used, but people still refer as SSL
- Public SSL certificates are issued by Certificate Authorities (CA)
- Comodo, Symantec, GoDaddy, GlobalSign, Digicert, Letsencrypt, etc…
- SSL certificates have an expiration date (you set) and must be renewed

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The load balancer uses an X.509 certificate (SSL/TLS server certificate)

- You can manage certificates using ACM (AWS Certificate Manager)
- You can create upload your own certificates alternatively
- HTTPS listener:
- You must specify a default certificate
- You can add an optional list of certs to support multiple domains
- Cilents can use SNI (Server Name Indication) to specify the hostname they reach
- Ability to specify a security policy to support older versions of SSL / TLS (legacy clients)
- EC2
- Instance
- Users
- HTTPS (encrypted) Over www
- HTTP
- Over private VPC

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TOPIC 4

SSL - Server Name Indication (SNI)

- SNI solves the problem of loading multiple SSL certificates onto multiple web servers (to serve multiple websites)
- It’s a “newer” protocol, and requires the client
- to indicate the hostname of the target server in the initial SSL handshake
- The server will then find the correct certificate, or return the default one
- Note:
- Only works for ALB & NLB (newer generation), CloudFront
- Does not work for CLB (older gen)
- Client
- ALB
- SSL Cert: Domain1.example.com
- SSL Cert: www.mycorp.com
- I would like www.mycorp.com

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TOPIC 4

Elastic Load Balancers - SSL Certificates

- Classic Load Balancer (v1)
- Support only one SSL certificate
- Must use multiple CLB for multiple hostname with multiple SSL certificates
- Application Load Balancer (v2)
- Supports multiple listeners with multiple SSL certificates
- Uses Server Name Indication (SNI) to make it work
- Network Load Balancer (v2)
- Supports multiple listeners with multiple SSL certificates
- Uses Server Name Indication (SNI) to make it work

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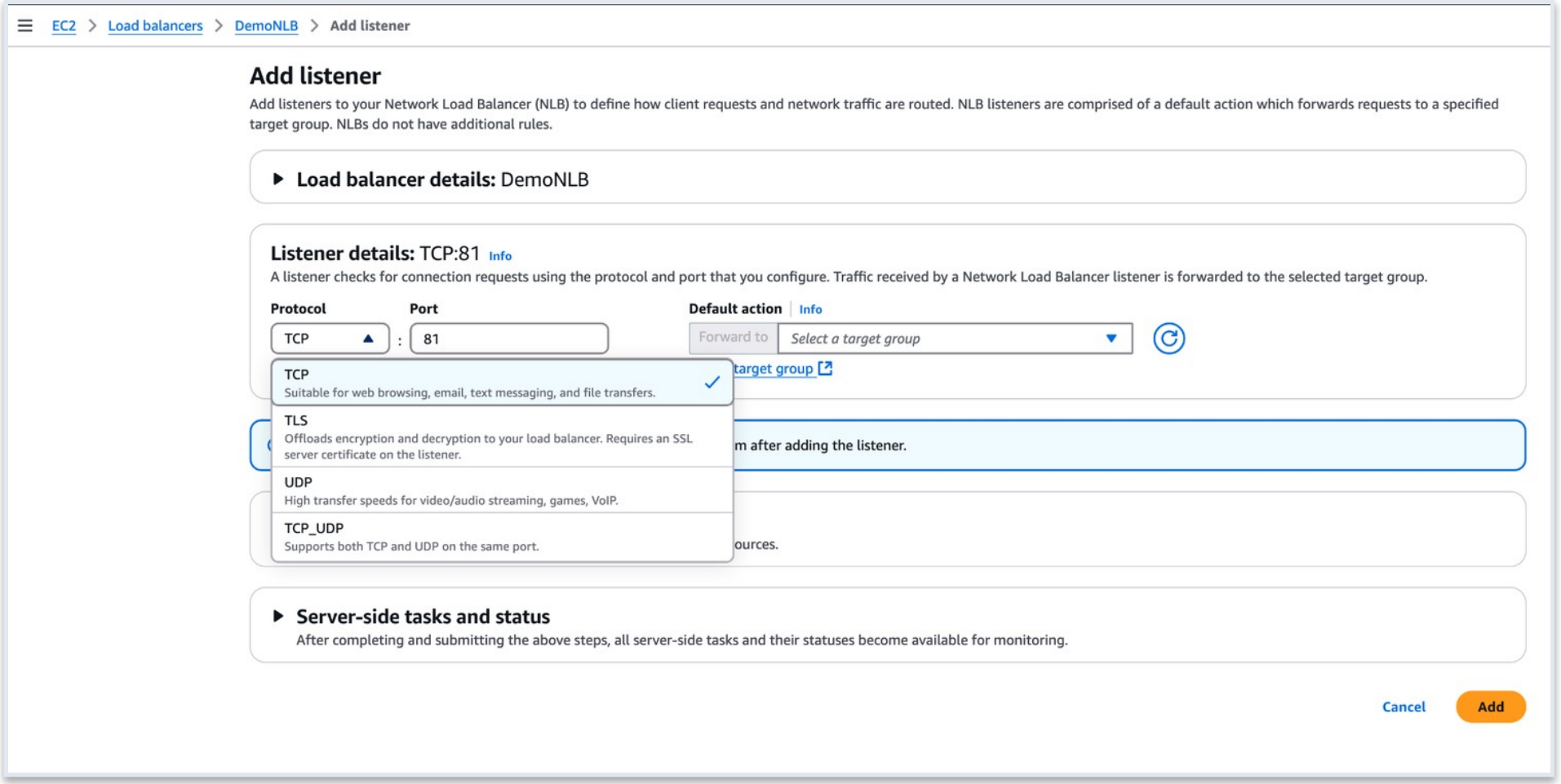
Activity: SSL/TLS



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TOPIC 4

Connection Draining

- Feature naming
- Connection Draining – for CLB
- Deregistration Delay – for ALB & NLB
- Time to complete “in-flight requests” while the instance is de-registering or unhealthy
- Stops sending new requests to the EC2 instance which is de-registering
- Between 1 to 3600 seconds (default: 300 seconds)
- Can be disabled (set value to 0)
- Set to a low value if your requests are short
- ELB
- waiting for existing
- connections to complete
- new connections
- established to all other instances

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ELB Health Checks

- Target Health Status
- Initial: registering the target
- Healthy
- Unhealthy
- Unused: target is not registered
- Draining: de-registering the target
- unavailable: health checks disabled
- If a group contains only unhealthy targets, ELB routes requests across its unhealthy targets
- Target Group
- Application Load Balancer
- Health Check
- Setting
- Value

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TOPIC 4

Load Balancer Error codes

- Successful request : Code 200.
- Unsuccessful at client side : 4XX code.
- Error 400 : Bad Request
- Error 401 : Unauthorized
- Error 403 : Forbidden
- Error 460 : Client closed connection.
- Error 463 : X-Forwarded For header with >30 IP (Similar to malformed request).
- Unsuccessful at server side : 5xx code.
- An error 500 / Internal server error would mean some error on the ELB itself.
- Error 502 : Bad Gateway
- An error 503/ Service Unavailable
- Error 504 / Gateway timeout : probably an issue within the server.
- Error 561 : Unauthorized

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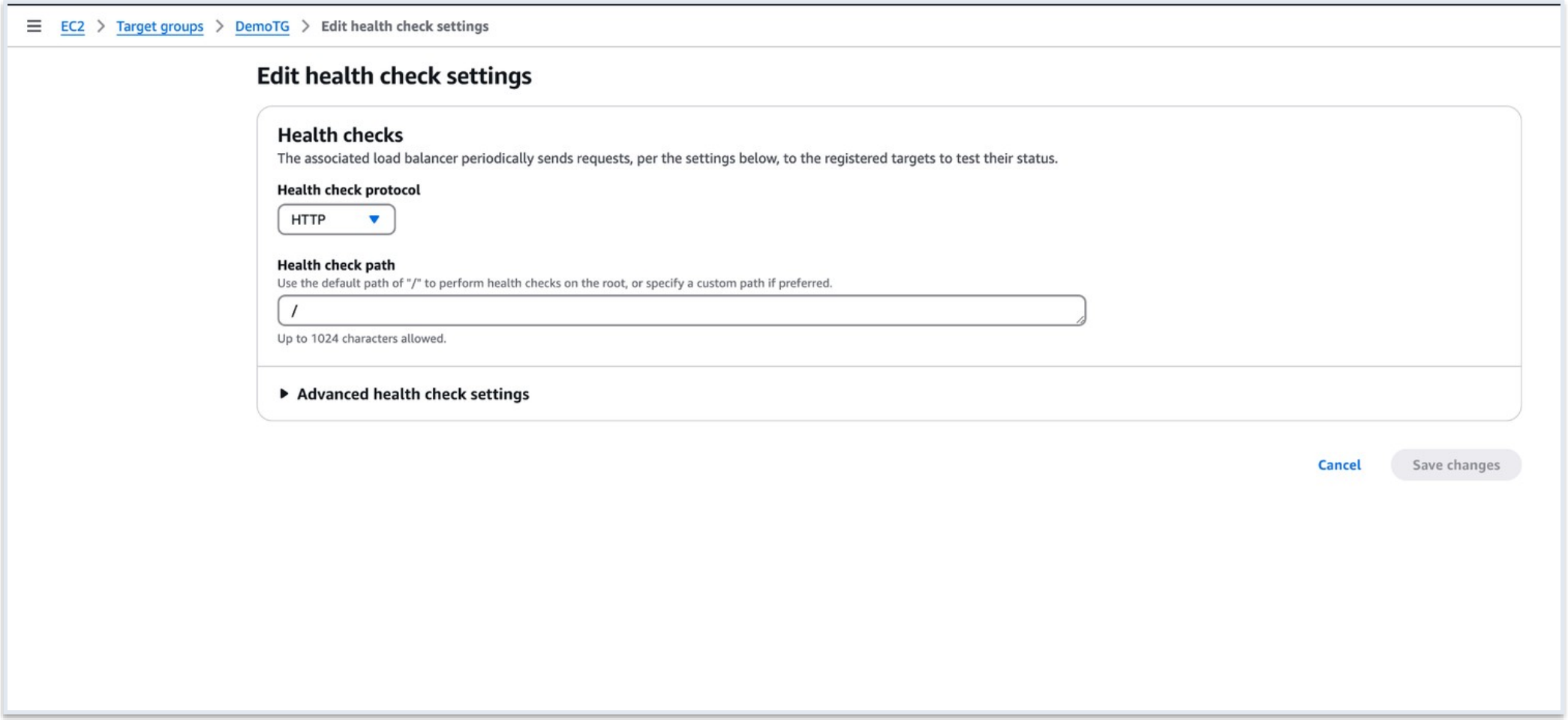
Activity: Health Check



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Load Balancers Monitoring

- All Load Balancer metrics are directly pushed to CloudWatch metrics
- BackendConnectionErrors
- HealthyHostCount / UnHealthyHostCount
- HTTPCode_Backend_2XX: Successful request.
- HTTPCode_Backend_3XX, redirected request
- HTTPCode_ELB_4XX: Client error codes
- HTTPCode_ELB_5XX:
- Server error codes generated by the load balancer.
- Latency
- RequestCount
- ReuqestCountPerTarget:
- SurgeQueueLength: the total
- number of requests (HTTP listener) or connections (TCP

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Load Balancer troubleshooting using metrics

- HTTP 400: BAD_REQUEST=> The client sent a malformed request that does not meet HTTP specifications.
- HTTP 503: Service Unavailable => Ensure that you have healthy instances in
- every Availability Zone that your load balancer is configured to respond in. Look for HealthyHostCount in CloudWatch
- HTTP 504: Gateway Timeout => Check if keep-alive settings on your EC2
- instances are enabled and make sure that the keep-alive timeout is greater than the idle timeout settings of load balancer.
- Set alarms & look at the documentation for troubleshooting: https://docs.aws.amazon.com/elasticloadbalancing/latest/classic/ts-elb-error- message.html

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Load Balancers Access Logs

- Access logs from Load Balancers can be stored in S3 and contain:
- Time
- Client IP address
- Latencies
- Request paths
- Server response
- Trace Id
- Only pay for the S3 storage
- Helpful for compliance reason
- Helpful for keeping access data even after ELB or EC2 instances are terminated
- Access Logs are already encrypted

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Application Load Balancer Request Tracing

- Request tracing – Each HTTP request has an added custom header
- ‘X-Amzn-Trace-Id’
- Example:
- X-Amzn-Trace-Id: Root=1-67891233-abcdef012345678912345678
- This is useful in logs / distributed tracing platform to track a single request
- Application Load Balancer is not (yet) integrated with X-Ray

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TOPIC 4

Target Groups Settings

- Deregistration_delay.timeout_seconds time the load balancer waits before deregistering a target
- slow_start.duration_seconds: (see next slide)
- load_balancing.algorithm.type: ow the load balancer selects targets when routing requests (Round Robin, Least Outstanding Requests)
- Stickiness.enabled
- Stickiness.type: application-based or duration-based cookie
- Stickiness.app_cookie.cookie_name: name of the application cookie
- Stickiness.app_cookie.duration_seconds: application-based cookie expiration period
- Stickiness.lb_cookie.duration_seconds: duration-based cookie expiration period

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Slow Start Mode

- By default, a target receives its full share of requests once it’s registered with the target group
- slow start mode gives healthy targets time to warm-up before the load balancer sends them a full share of requests
- The load balancer linearly increases the number of requests that it sends to the target
- A target exits Slow Start Mode when:
- The duration period elapses
- The target becomes unhealthy
- To disable, set slow start duration value to 0
- Client
- ALB
- Slow Start duration
- without Slow Start mode
- with Slow Start mode

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Request Routing Algorithms - Least Outstanding Requests

- The next instance to receive the request is the instance that has the lowest number of pending/unfinished requests
- Works with Application Load Balancer and Classic Load Balancer (HTTP/HTTPS)
- EC2 Instance
- EC2 Instance
- EC2 Instance
- Registered Instances
- ALB
- CLB
- (HTTP/HTTPS Listener)

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Request Routing Algorithms - Round Robin

- Equally choose the targets from the target group
- Works with Application Load Balancer and Classic Load Balancer (TCP)
- ALB
- --- OR ---
- CLB (TCP Listener)
- EC2 Instance
- EC2 Instance
- EC2 Instance
- Target Group (ALB) / Registered Instances (CLB)

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Request Routing Algorithms - Flow Hash

- Selects a target based on the protocol, source/destination IP address, source/destination port, and TCP sequence number
- Each TCP/UDP connection is routed to a single target for the life of the connection
- Works with Network Load Balancer
- Network Load Balancer
- Flow Hash Algorithm
- Protocol
- Source & destination IP Source & destination Port TCP sequence no.
- Hash 8743b…
- EC2 Instance
- EC2 Instance
- EC2 Instance
- Target Group

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ALB - Listener Rules

- Processed in order (with Default Rule)
- Supported Actions (forward, redirect, fixed-response)
- Rule Conditions:
- host-header
- http-request-method
- path-pattern
- source-ip
- http-header
- query-string
- Rule #1
- Application Load Balancer
- Target
- Rule #2

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Target Group Weighting

- Specify weight for each Target Group on a single Rule
- Example: multiple versions of your app, blue/green deployment
- Allows you to control the distribution of the traffic to your applications
- Target Group 1 (Blue)
- Target Group 2 (Green)
- Application Load Balancer
- Target Group
- Weight
- Target Group 1
- 8
- Target Group 2
- 2
- Users

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TOPIC 4

What’s an Auto Scaling Group?

- In real-life, the load on your websites and application can change
- In the cloud, you can create and get rid of servers very quickly
- The goal of an Auto Scaling Group (ASG) is to:
- Scale out (add EC2 instances) to match an increased load
- Scale in (remove EC2 instances) to match a decreased load
- Ensure we have a minimum and a maximum number of EC2 instances running
- Automatically register new instances to a load balancer
- Re-create an EC2 instance in case a previous one is terminated (ex: if unhealthy)
- ASG are free (you only pay for the underlying EC2 instances)

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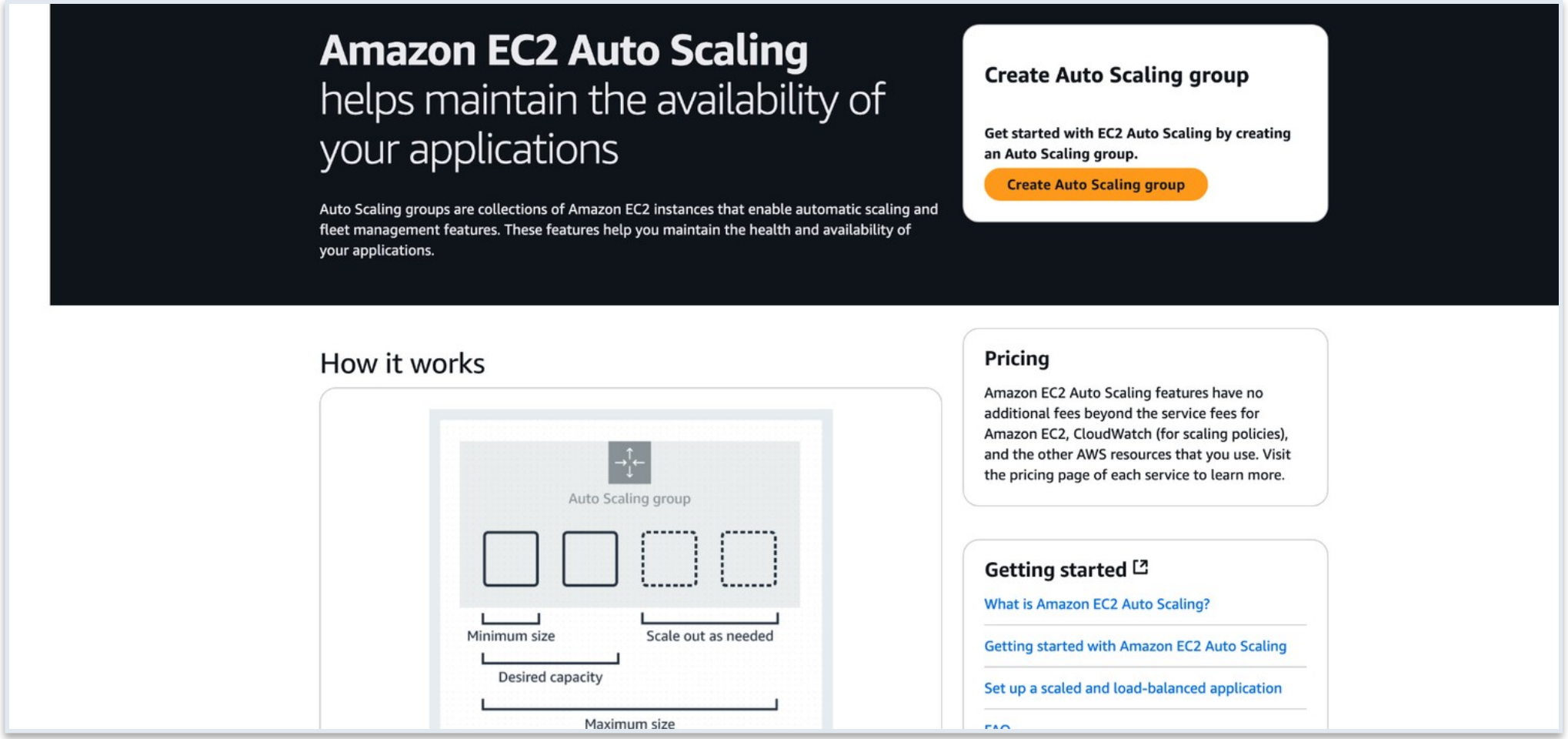
What's an Auto Scaling Group?



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Auto Scaling Group in AWS

1. Auto Scaling Group
2. Minimum Capacity
3. Desired Capacity
4. Maximum Capacity
5. Scale Out as Needed

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Auto Scaling Group in AWS With Load Balancer

1. Users
2. Elastic Load Balancer
3. ELB can check the health of your EC2 instances!
4. Auto Scaling Group

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Auto Scaling Group Attributes

- A Launch Template (older “Launch Configurations” are deprecated)
- AMI + Instance Type
- EC2 User Data
- EBS Volumes
- Security Groups
- SSH Key Pair
- IAM Roles for your EC2 Instances
- Network + Subnets Information
- Load Balancer Information
- Min Size / Max Size / Initial Capacity
- Scaling Policies
- ASG Launch Template
- AMI

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Auto Scaling - CloudWatch Alarms & Scaling

- It is possible to scale an ASG based on CloudWatch alarms
- An alarm monitors a metric (such as Average CPU or a custom metric)
- Metrics such as Average CPU are computed for the overall ASG instances
- Based on the alarm:
- We can create scale-out policies (increase the number of instances)
- We can create scale-in policies (decrease the number of instances)
- EC2
- Instance
- EC2
- Instance
- EC2
- Instance
- EC2

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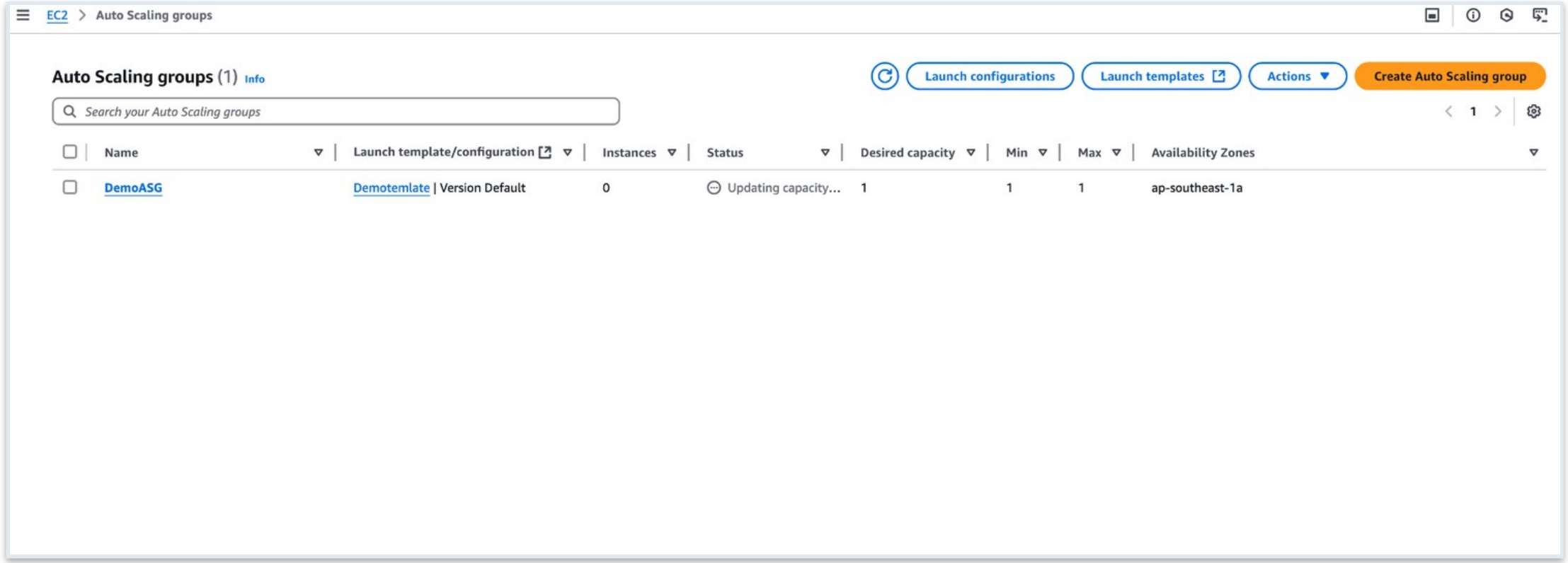
Activity: Auto Scaling Group



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Auto Scaling Groups - Scaling Policies

- Dynamic Scaling
- Target Tracking Scaling
- Simple to set-up
- Example: I want the average ASG CPU to stay at around 40%
- Simple / Step Scaling
- When a CloudWatch alarm is triggered (example CPU > 70%), then add 2 units
- When a CloudWatch alarm is triggered (example CPU < 30%), then remove 1
- Scheduled Scaling
- Anticipate a scaling based on known usage patterns
- Example: increase the min capacity to 10 at 5 pm on Fridays

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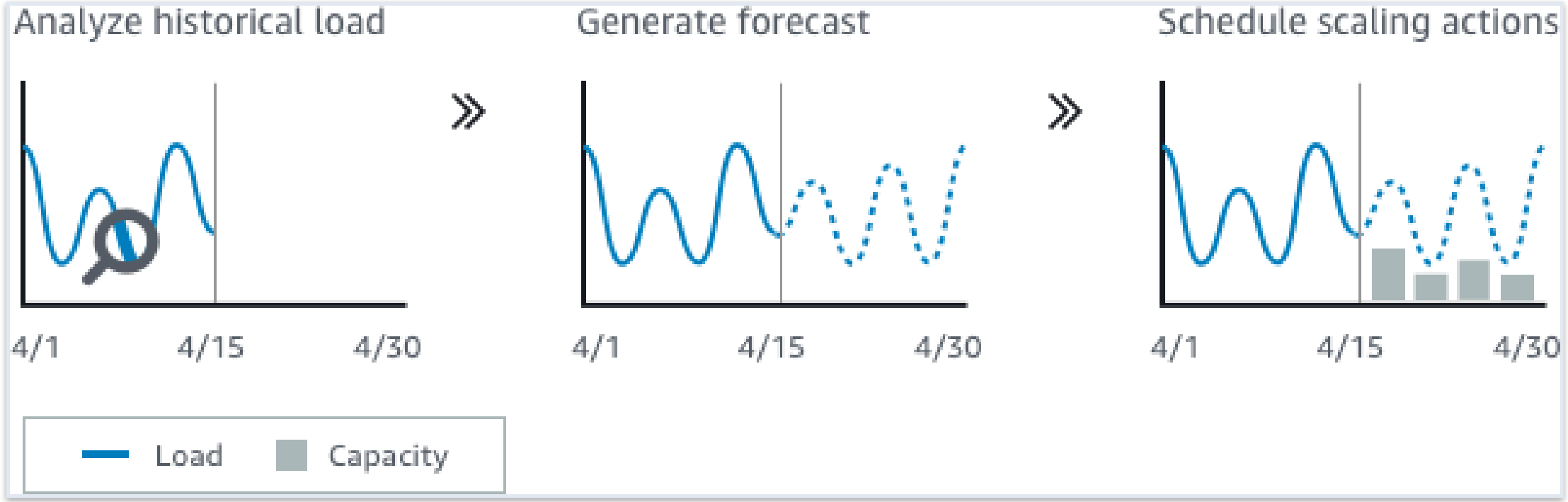
Auto Scaling Groups - Scaling Policies



Predictive scaling: continuously forecast load and schedule scaling ahead

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Good metrics to scale on

- CPUUtilization: Average CPU utilization across your instances
- RequestCountPerTarget: to make sure the number of requests per EC2 instances is stable
- Average Network In / Out: if you’re application is network bound)
- Any custom metric (that you push using CloudWatch)
- Auto Scaling group
- Users
- Application Load Balancer
- RequestCountPerTarget
- Target Value: 3

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Auto Scaling Groups - Scaling Cooldowns

- After a scaling activity happens, you are in the cooldown period (default 300 seconds)
- During the cooldown period, the ASG will not launch or terminate additional instances (to allow for metrics to stabilize)
- Advice: Use a ready-to-use AMI to reduce configuration time in order to be serving request fasters and reduce the cooldown period
- Scaling Action Occurs
- Default Cooldown in effect?
- Ignore Action
- Launch or Teminate Instance
- No
- Yes

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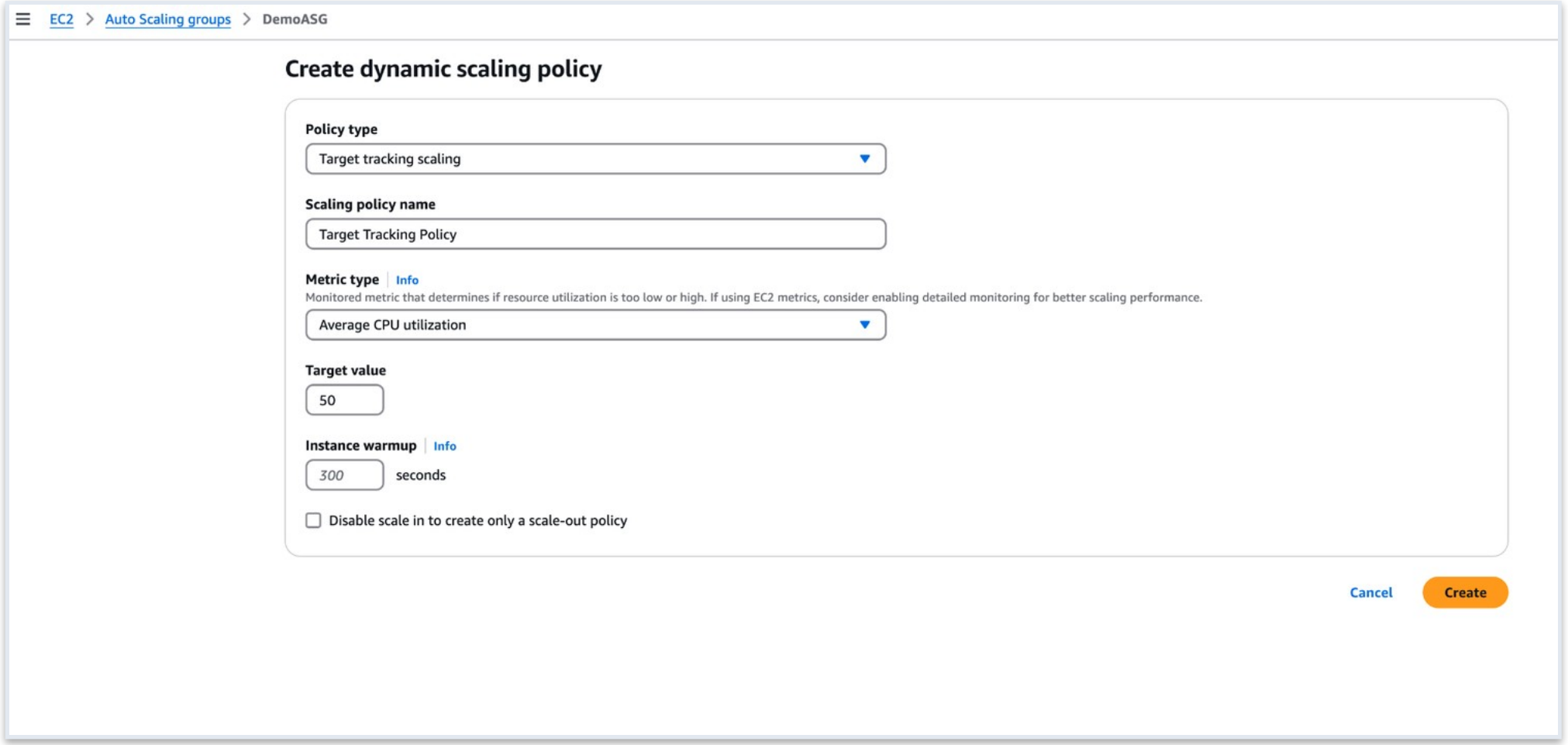
Activity: Scaling Policy



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ASG - Lifecycle Hooks

- By default, as soon as an instance is launched in an ASG it’s in service
- You can perform extra steps before
- the instance goes in service (Pending state)
- Define a script to run on the instances as they start
- You can perform some actions before
- the instance is terminated (Terminating state)
- Pause the instances before they’re terminated for troubleshooting
- Use cases: cleanup, log extraction, special health checks
- Integration with EventBridge, SNS, and SQS
- Pending
- Lifecycle Hook
- (EC2_Instance_Launching)
- Pending:Wait

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Launch Configuration vs. Launch Template

- Both:
- ID of the Amazon Machine Image (AMI), the instance type, a key pair, security groups, and the other parameters that you use to launch EC2 instances (tags, EC2 user-data...)
- You can’t edit both Launch Configurations and Launch Templates
- Launch configuration (legacy):
- must be re-created everytime
- Launch Template (newer):
- Can have multiple versions
- Create parameters subsets (partial configuration for re-use and inheritance)
- Provision using both On-Demand and Spot instances (or a mix)
- Supports Placement Groups, Capacity Reservations, Dedicated hosts, multiple instance types
- Can use T2 unlimited burst feature
- Recommended by AWS going forward

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SQS with Auto Scaling Group (ASG)

- SQS Queue
- CloudWatch Metric - Queue Length
- ApproximateNumberOfMessages
- EC2 Instances
- Poll for messages
- Auto Scaling Group
- scale
- CloudWatch Alarm
- Alarm for breach

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ASG Health Checks

- To make sure you have high availability, means you have least 2 instances running across 2 AZ in your ASG (must configure multi-AZ ASG)
- Health checks available:
- EC2 Status Checks
- ELB Health Checks
- Custom Health Checks: send instance's health to ASG using AWS CLI or AWS SDK
- ASG will launch a new instance after terminating an unhealthy one
- ASG will not reboot unhealthy hosts for you
- Good to know CLI:
- set-instance-health (use with Custom Health Checks)
- terminate-instance-in-auto-scaling-group

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Troubleshooting ASG issues

- <number of instances> instance(s) are already running. Launching EC2 instance failed.
- The Auto Scaling group has reached the limit set by the MaximumCapacity parameter. Update your Auto Scaling group by providing a new value for the maximum capacity.
- Launching EC2 instances is failing:
- The security group does not exist. SG might have been deleted
- The key pair does not exist. The key pair might have been deleted
- If the ASG fails to launch an instance for over 24 hours, it will automatically suspend the processes (administration suspension)

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CloudWatch Metrics for ASG

- Metrics are collected every 1 minute
- ASG-level metrics: (opt-in)
- GroupMinSize, GroupMaxSize, GroupDesiredCapacity
- GroupInServiceInstances, GroupPendingInstances, GroupStandbyInstances
- GroupTerminatingInstances, GroupTotalInstances
- You should enable metric collection to see these metrics
- EC2-level metrics (enabled): CPU Utilization, etc…
- Basic monitoring: 5 minutes granularity
- Detailed monitoring: 1 minute granularity

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AWS Auto Scaling

- Backbone service of auto scaling for scalable resources in AWS:
- Amazon EC2 Auto Scaling groups: Launch or terminate EC2 instances
- Amazon EC2 Spot Fleet requests: Launch or terminate instances from a Spot Fleet request, or automatically replace instances that get interrupted for price or capacity reasons.
- Amazon ECS: Adjust the ECS service desired count up or down
- Amazon DynamoDB (table or global secondary index): WCU & RCU
- Amazon Aurora: Dynamic Read Replicas Auto Scaling

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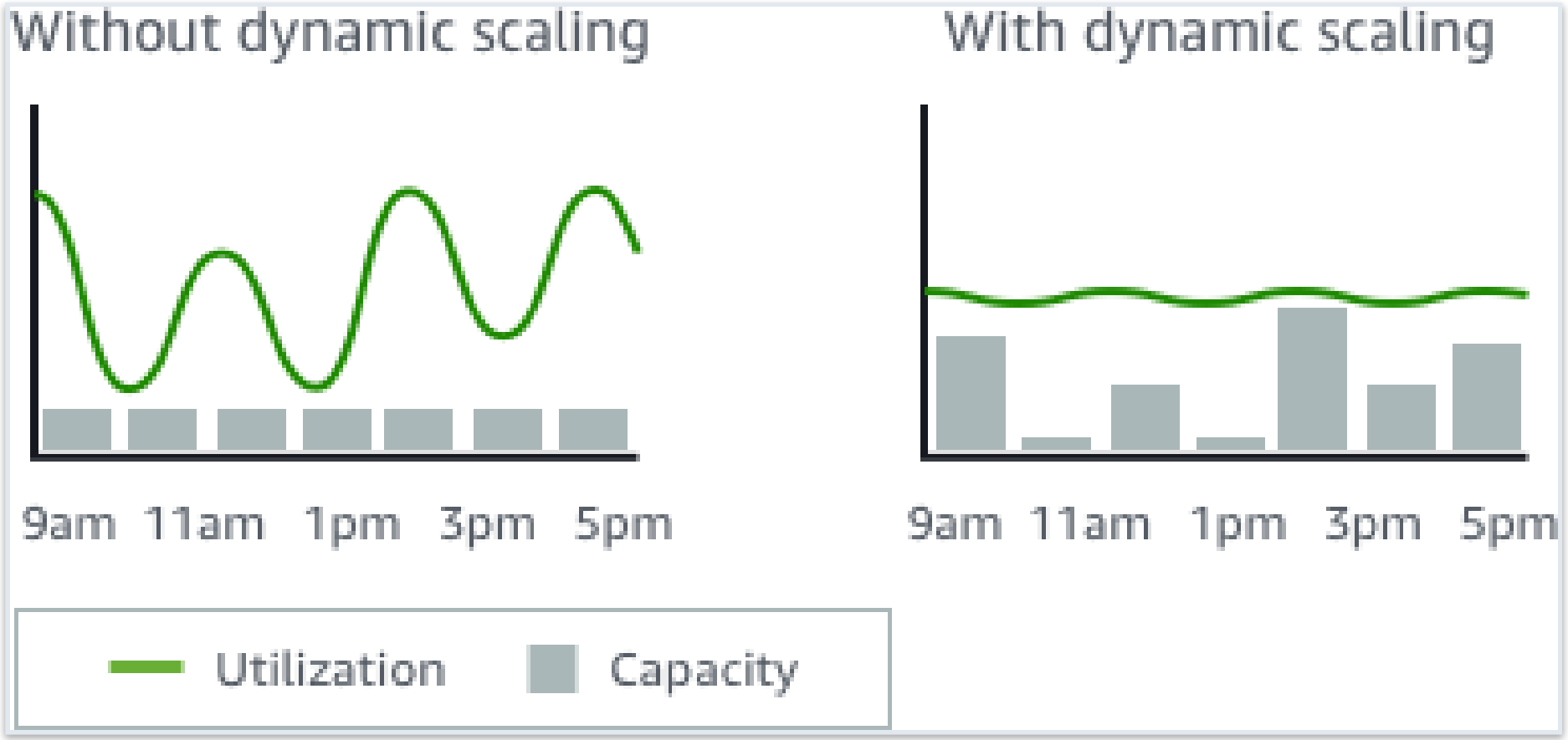
AWS Auto Scaling - Scaling Plans



Dynamic scaling: creates a target tracking scaling policy

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TOPIC 5

AWS Elastic Beanstalk

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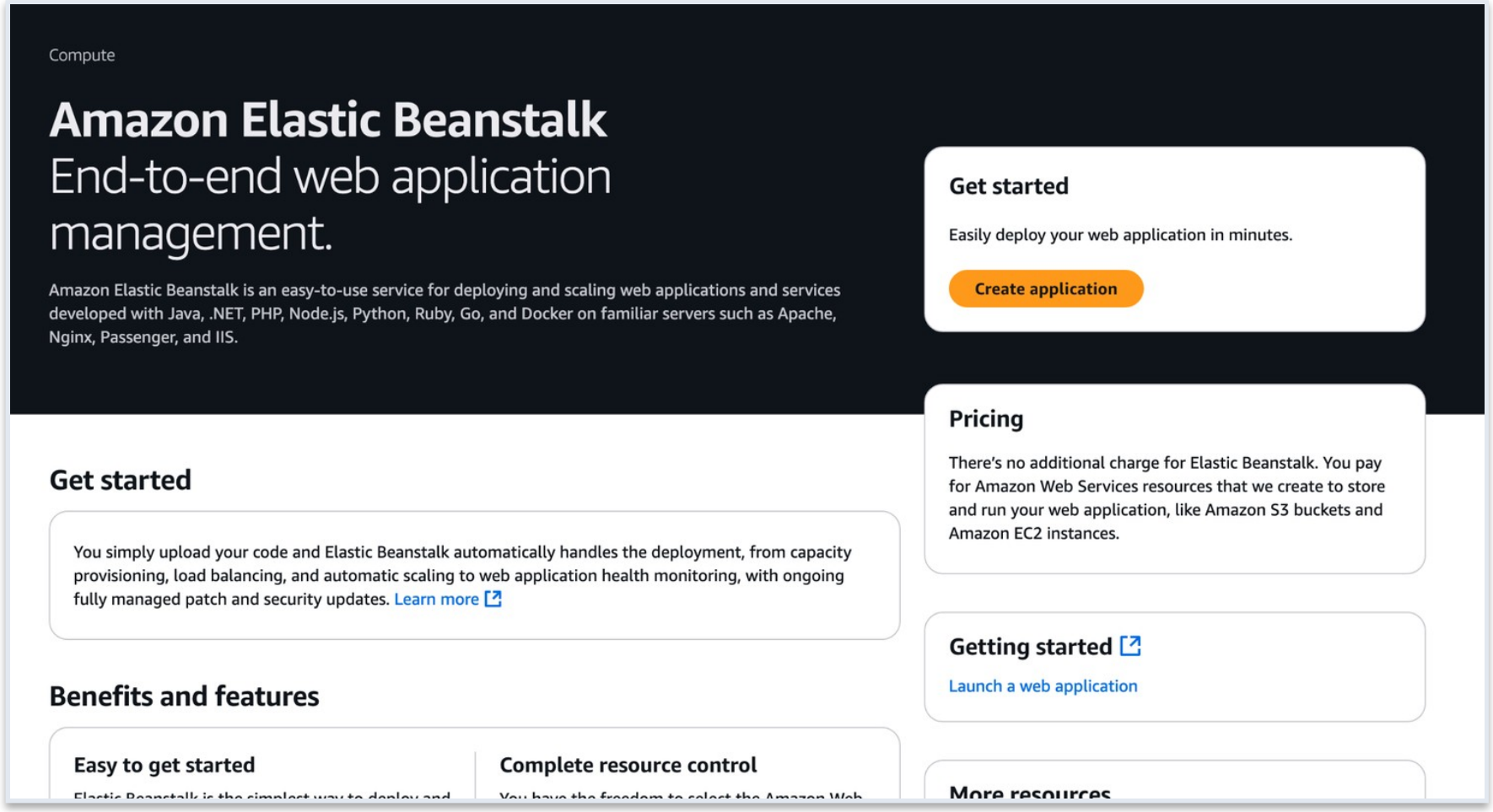
TOPIC 5

Elastic Beanstalk - Overview



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Elastic Beanstalk - Overview

- Elastic Beanstalk is a developer centric view of deploying an application on AWS
- It uses all the component's we've seen before: EC2, ASG, ELB, RDS, …
- Managed service
- Automatically handles capacity provisioning, load balancing, scaling, application health monitoring, instance configuration, …
- Just the application code is the responsibility of the developer
- We still have full control over the configuration
- Beanstalk is free but you pay for the underlying instances

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Multi AZ

- Availability zone 1
- Auto Scaling group
- Availability zone 2
- Availability zone 3
- ElastiCache
- Store / retrieve
- session data
- + Cached data
- PUBLIC SUBNET
- PRIVATE SUBNET
- Amazon RDS Read / write data
- DATA SUBNET
- ELB

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Developer problems on AWS

- Managing infrastructure
- Deploying Code
- Configuring all the databases, load balancers, etc
- Scaling concerns
- Most web apps have the same architecture (ALB + ASG)
- All the developers want is for their code to run!
- Possibly, consistently across different applications and environments

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Elastic Beanstalk - Components

- Application: collection of Elastic Beanstalk components (environments, versions, configurations, …)
- Application Version: an iteration of your application code
- Environment
- Collection of AWS resources running an application version (only one application version at a time)
- Tiers: Web Server Environment Tier & Worker Environment Tier
- You can create multiple environments (dev, test, prod, …)
- update version
- Create Application
- Upload Version
- Launch Environment
- Manage Environment
- deploy new version

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Elastic Beanstalk - Supported Platforms

- Go
- Java SE
- Java with Tomcat
- .NET Core on Linux
- .NET on Windows Server
- Node.js
- PHP
- Python
- Ruby
- Packer Builder
- Single Container Docker
- Multi-container Docker
- Preconfigured Docker

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Web Server Tier vs Worker Tier

- Web Environment
- (myapp.us-east-1.elasticbeanstalk.com)
- Auto Scaling group
- Worker Environment
- Availability Zone 1
- Availability Zone 2
- SQS Queue
- SQS message
- pull SQS message
- messages
- EC2 Instance (Worker)
- EC2 Instance (Worker)
- Scale based on the number of SQS messages

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Elastic Beanstalk Deployment Modes

- Single Instance Great for dev
- High Availability with Load Balancer Great for prod
- Availability Zone 1
- Auto Scaling Group
- Availability Zone 2
- RDS Master
- RDS Standby
- EC2 Instance
- EC2 Instance
- ALB
- Availability Zone 1
- Elastic IP
- EC2 Instance

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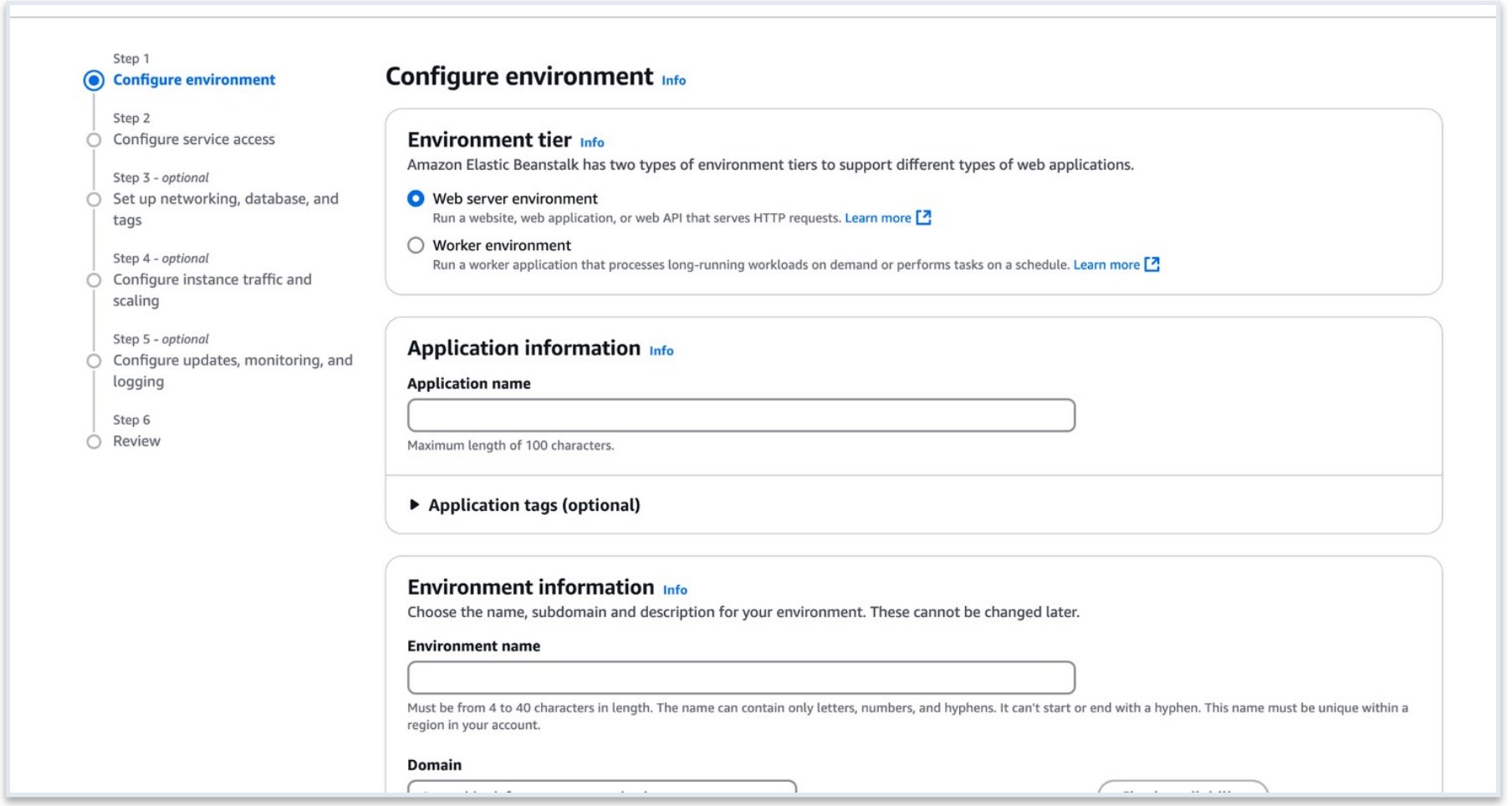
Activity: Elastic Beamstalk



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TOPIC 6

AWS CloudFormation for SysOps

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TOPIC 6

AWS CloudFormation

- CloudFormation is a declarative way of outlining your AWS Infrastructure, for any resources (most of them are supported)
- For example, within a CloudFormation template, you say:
- I want a security group
- I want two EC2 instances using this security group
- I want two Elastic IPs for these EC2 instances
- I want an S3 bucket
- I want a load balancer (ELB) in front of these EC2 instances
- Then CloudFormation creates those for you, in the right order, with the exact configuration that you specify

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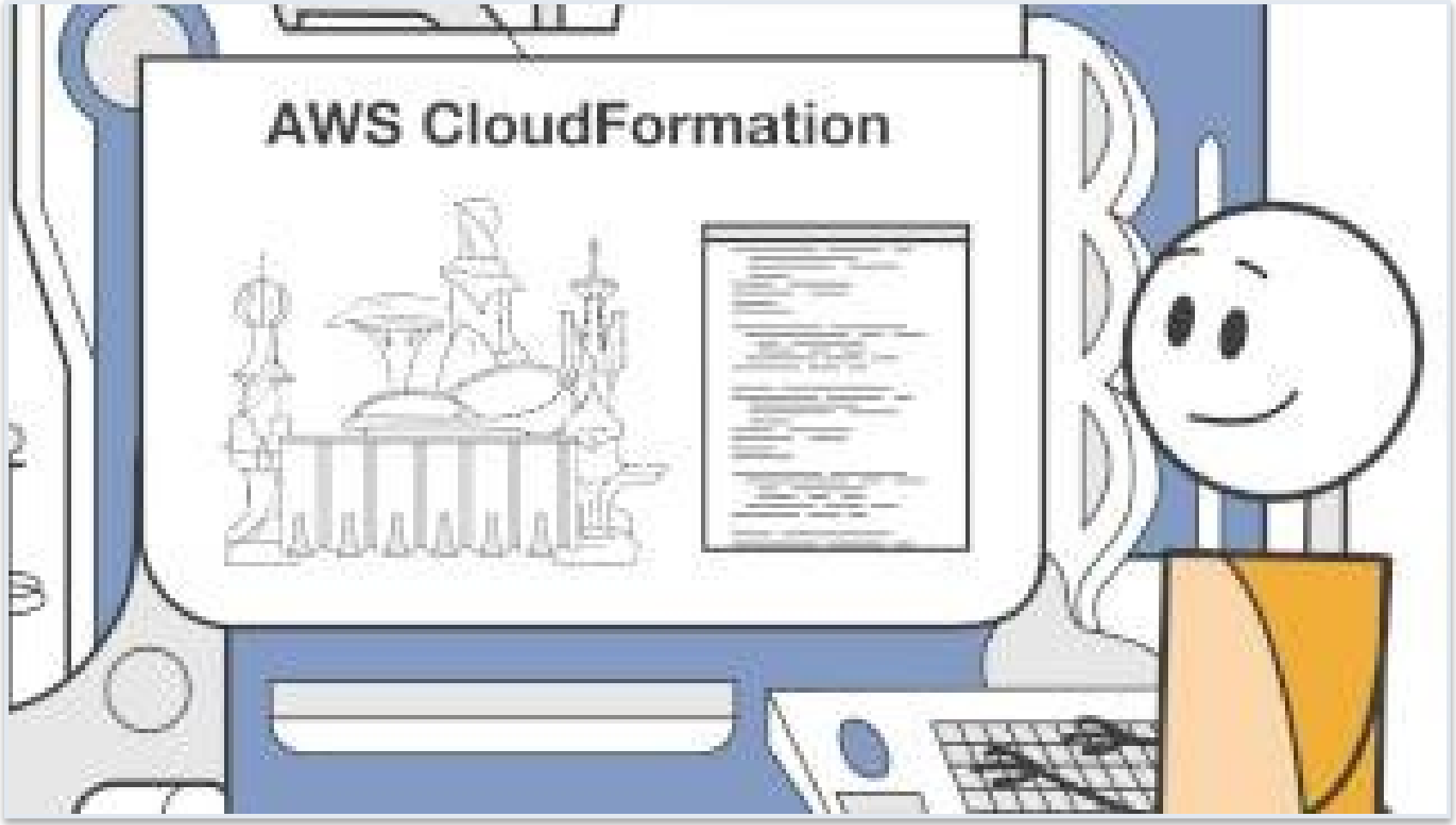
Reference



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CloudFormation - Template Example

```
AWSTemplateFormatVersion: '2010-09-09'
Description: 'AWS CloudFormation Sample Template LAMP_Multi_AZ: C
Parameters: ⋯
Mappings: ⋯
Resources:
  ApplicationLoadBalancer:
    Type: AWS::ElasticLoadBalancingV2::LoadBalancer
    Properties:
      Subnets: !Ref Subnets
  ALBListener:
    Type: AWS::ElasticLoadBalancingV2::Listener
    Properties:
      DefaultActions:
        - Type: forward
          TargetGroupArn: !Ref ALBTargetGroup
      LoadBalancerArn: !Ref ApplicationLoadBalancer
      Port: '80'
      Protocol: HTTP
  ALBTargetGroup:
    Type: AWS::ElasticLoadBalancingV2::TargetGroup
    Properties:
      HealthCheckIntervalSeconds: 10
      HealthCheckTimeoutSeconds: 5
      HealthyThresholdCount: 2
      Port: 80
      Protocol: HTTP
      UnhealthyThresholdCount: 5
      VpcId: !Ref VpcId
      TargetGroupAttributes:
        - Key: stickiness.enabled
          Value: 'true'
```

Application Composer

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Benefits of AWS CloudFormation (1/2)

- Infrastructure as code
- No resources are manually created, which is excellent for control
- The code can be version controlled for example using Git
- Changes to the infrastructure are reviewed through code
- Cost
- Each resources within the stack is tagged with an identifier so you can easily see how much a stack costs you
- You can estimate the costs of your resources using the CloudFormation template
- Savings strategy: In Dev, you could automation deletion of templates at 5 PM and recreated at 8 AM, safely

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Benefits of AWS CloudFormation (2/2)

- Productivity
- Ability to destroy and re-create an infrastructure on the cloud on the fly
- Automated generation of Diagram for your templates!
- Declarative programming (no need to figure out ordering and orchestration)
- Separation of concern: create many stacks for many apps and many layers. Eg.
- VPC stacks
- Network stacks
- App stacks
- Don't re-invent the wheel
- Leverage existing templates on the web!
- Leverage the documentation

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How CloudFormation Works

- Templates must be uploaded in S3 and then referenced in CloudFormation
- To update a template, we can’t edit previous ones. We have to re- upload a new version of the template to AWS
- Stacks are identified by a name
- Deleting a stack deletes every single artifact that was created by CloudFormation.
- Template
- S3 bucket
- AWS CloudFormation
- Stack
- upload
- reference
- create
- Create

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Deploying CloudFormation Templates

- Manual way
- Editing templates in Application Composer or code editor
- Using the console to input parameters, etc…
- We'll mostly do this way in the course for learning purposes
- Automated way
- Editing templates in a YAML file
- Using the AWS CLI (Command Line Interface) to deploy the templates, or using a Continuous Delivery (CD) tool
- Recommended way when you fully want to automate your flow
- CloudFormation Console
- Template
- Application Composer / Code Editor
- Stack

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CloudFormation - Building Blocks

- Template's Components
- AWSTemplateFormatVersion - identifies the capabilities of the template “2010-09-09”
- Description - comments about the template
- Resources (MANDATORY) - your AWS resources declared in the template
- Parameters - the dynamic inputs for your template
- Mappings - the static variables for your template
- Outputs - references to what has been created
- Conditionals - list of conditions to perform resource creation
- Template's Helpers
- Reference
- Functions

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Introductory Example

- We’re going to create a simple EC2 instance
- And we’re going to add security group to it
- For now, forget about the code syntax
- We’ll look at the structure of the files later
- We’ll see how in no-time, we are able to get started with CloudFormation!
- Web Server
- WebServerSecurityGroup
- Port: 22
- Port: 80

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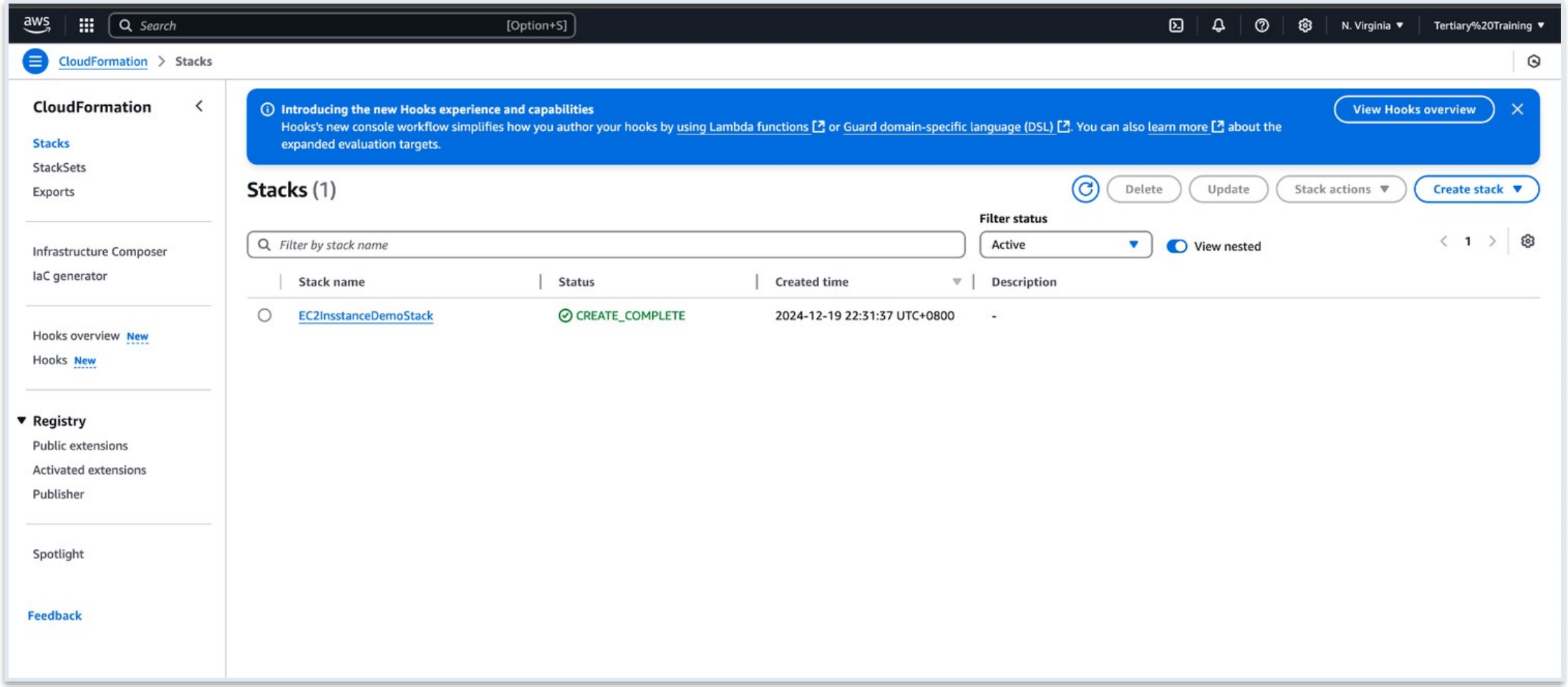
Activity: CloudFormation



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YAML Crash Course

```yaml
invoice: 34843
date: 2001-01-23
bill-to:
  given: Chris
  family: Dumars
  address:
    lines: |
      458 Walkman Dr.
      Suite #292
    city: Royal Oak
    state: MI
    postal: 48046
products:
  - sku: BL394D
    quantity: 4
    description: Basketball
    price: 450.00
  - sku: BL4438H
    quantity: 1
    description: Super Hoop
    price: 2392.00
```

YAML and JSON are the languages you can use for CloudFormation

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CloudFormation - Resources

- Resources are the core of your CloudFormation template (MANDATORY)
- They represent the different AWS Components that will be created and configured
- Resources are declared and can reference each other
- AWS figures out creation, updates and deletes of resources for us
- There are over 700 types of resources (!)
- Resource types identifiers are of the form:
- service-provider::service-name::data-type-name

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How do I find Resources documentation?

- I can’t teach you all the 700+ resources, but I can teach you how to learn how to use them
- All the resources can be found here: https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/aws-template-resource-type-ref.html
- Then, we just read the docs ©
- Example here (for an EC2 instance): https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/aws-resource-ec2-instance.html

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Analysis of CloudFormation Template

- Going back to the example of the introductory lecture, let's learn why it was written this way.
- Relevant documentation can be found here:
- https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/aws-resource-ec2-instance.html
- https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/aws-resource-ec2-securitygroup.html
- http://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/aws-resource-ec2-eip.html

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CloudFormation - Resources FAQ

1. Can I create a dynamic number of resources?

2. Yes, you can by using CloudFormation Macros and Transform

3. It is not in the scope of this course

4. Is every AWS Service supported?

5. Almost. Only a select few niches are not there yet

6. You can work around that using CloudFormation Custom Resources

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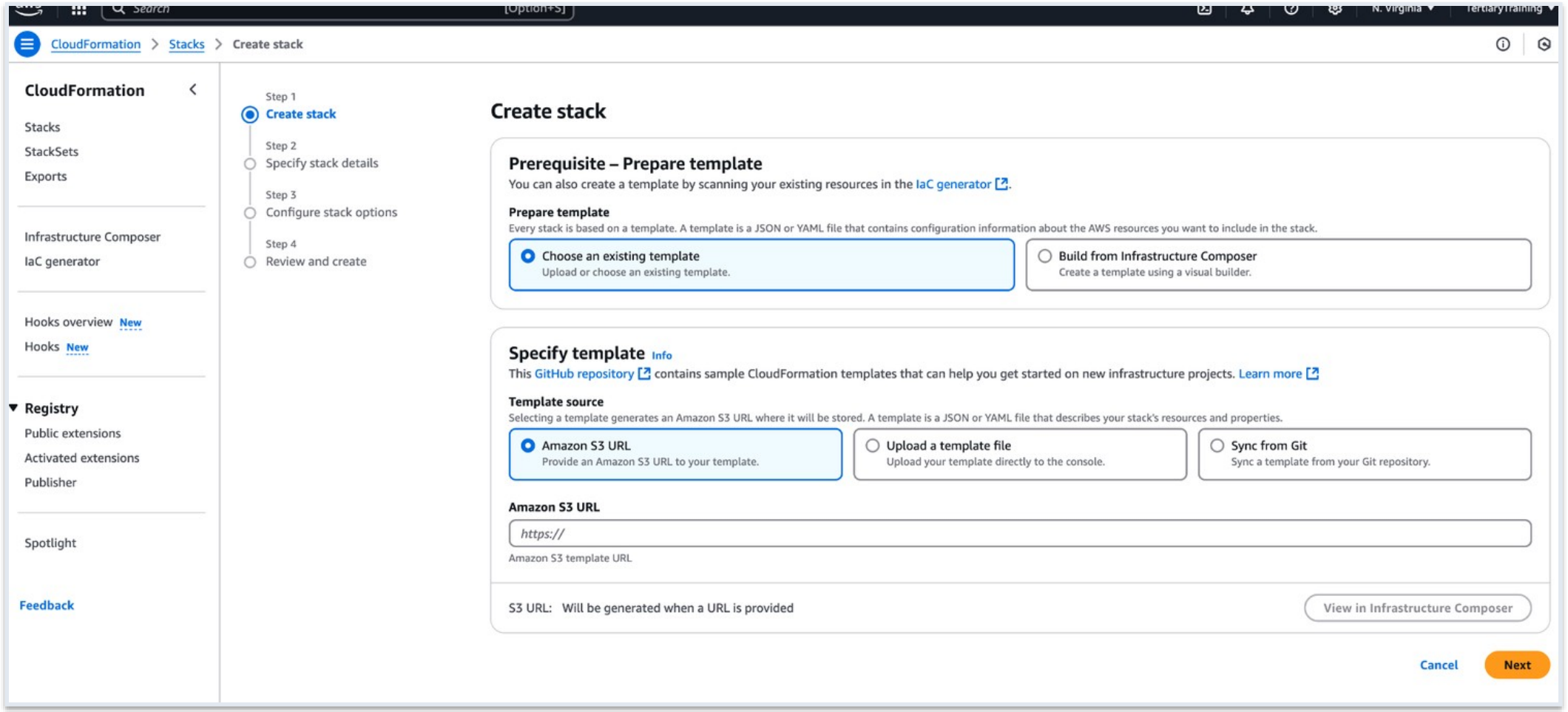
Activity: Cloud Formation



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CloudFormation - Parameters

- Parameters are a way to provide inputs to your AWS CloudFormation template
- They’re important to know about if:
- You want to reuse your templates across the company
- Some inputs can not be determined ahead of time
- Parameters are extremely powerful, controlled, and can prevent errors from happening in your templates, thanks to types
- AWS CloudFormation
- User
- upload template
- +
- provide parameter values
- Parameter1: value1
- + Parameter2: valu

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When should you use a Parameter?

```
Parameters:
  SecurityGroupDescription:
    Description: Security Group Description
    Type: String
```

Ask yourself this:

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CloudFormation - Parameters Settings

- Type:
- String
- Number
- CommaDelimitedList
- List<Number>
- AWS-Specific Parameter
- (to help catch invalid values - match against existing values in the AWS account)
- List<AWS-Specific Parameter>
- SSM Parameter (get parameter value from SSM Parameter store)
- Description
- Constraint Description (String)
- Min/MaxLength
- Min/MaxValue

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TOPIC 6

CloudFormation - Parameters Example

- AllowedValues
- NoEcho

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TOPIC 6

How to Reference a Parameter?

1. The Fn::Ref function can be leveraged to reference parameters
2. Parameters can be used anywhere in a template
3. The shorthand for this in YAML is !Ref
4. The function can also reference other elements within the template

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TOPIC 6

CloudFormation - Pseudo Parameters

1. AWS offers us Pseudo Parameters in any CloudFormation template

2. These can be used at any time and are enabled by default

3. Important pseudo parameters:

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TOPIC 6

CloudFormation - Mappings

- Mappings are fixed variables within your CloudFormation template
- They’re very handy to differentiate between different environments (dev vs prod), regions (AWS regions), AMI types…
- All the values are hardcoded within the template

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TOPIC 6

Accessing Mapping Values (Fn::FindInMap)

1. We use Fn::FindInMap to return a named value from a specific key

2. !FindInMap [MapName, TopLevelKey, SecondLevelKey]

3. Mappings work great for AMIs Because AMIs are region-specific!

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TOPIC 6

When would you use Mappings vs Parameters?

- Mappings are great when you know in advance all the values that can be taken and that they can be deduced from variables such as
- Region
- Availability Zone
- AWS Account
- Environment (dev vs prod)
- etc…
- They allow safer control over the template
- Use parameters when the values are really user specific

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TOPIC 6

CloudFormation - Outputs

- The Outputs section declares optional outputs values that we can import into other stacks (if you export them first)!
- You can also view the outputs in the AWS Console or in using the AWS CLI
- They’re very useful for example if you define a network CloudFormation, and output the variables such as VPC ID and your Subnet IDs
- It’s the best way to perform some collaboration cross stack, as you let expert handle their own part of the stack
- Application Stack
- Network Stack
- VPC ID: vpc-2f09a348
- Exported Output
- reference

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CloudFormation - Outputs

1. Creating a SSH Security Group as part of one template

2. We create an output that references that security group

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CloudFormation - Outputs Cross-Stack Reference

- We then create a second template that leverages that security group
- For this, we use the Fn::ImportValue function
- You can’t delete the underlying stack until all the references are deleted

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CloudFormation - Conditions

- Conditions are used to control the creation of resources or outputs based on a condition
- Conditions can be whatever you want them to be, but common ones are:
- Environment (dev / test / prod)
- AWS Region
- Any parameter value
- Each condition can reference another condition, parameter value or mapping
- User
- EC2 Instance
- Dev Stack
- EC2 Instance
- EBS Volume
- Prod Stack
- Parameters

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TOPIC 6

How to define a Condition

```
Conditions:
  CreateProdResources: !Equals [ !Ref EnvType, prod ]
```

The logical ID is for you to choose. It's how you name condition

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TOPIC 6

How to use a Condition

- Conditions can be applied to resources / outputs / etc…

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TOPIC 6

CloudFormation - Intrinsic Functions

- Ref
- Fn::GetAtt
- Fn::FindInMap
- Fn::ImportValue
- Fn::Join
- Fn::Sub
- Fn::ForEach
- Fn::ToJsonString
- Condition Functions (Fn::If, Fn::Not, Fn::Equals, etc…)
- Fn::Base64
- Fn::Cidr
- Fn::GetAZs
- Fn::Select

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TOPIC 6

Intrinsic Functions - Fn::Ref

- The Fn::Ref function can be leveraged to reference
- Parameters - returns the value of the parameter
- Resources - returns the physical ID of the underlying resource (e.g., EC2 ID)
- The shorthand for this in YAML is !Ref

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TOPIC 6

Intrinsic Functions - Fn::GetAtt

- Attributes are attached to any resources you create
- To know the attributes of your resources, the best place to look at is the documentation
- Example: the AZ of an EC2 instance!

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TOPIC 6

Intrinsic Functions - Fn::FindInMap

1. We use Fn::FindInMap to return a named value from a specific key

2. !FindInMap [MapName, TopLevelKey, SecondLevelKey]

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TOPIC 6

Intrinsic Functions - Fn::ImportValue

1. Import values that are exported in other stacks

2. For this, we use the Fn::ImportValue function

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TOPIC 6

Intrinsic Functions - Fn::Base64

1. Convert String to it's Base64 representation

2. Example: pass encoded data to EC2 Instance's UserData property

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TOPIC 6

Intrinsic Functions -Condition Functions

```
Conditions:
  CreateProdResources: !Equals [ !Ref EnvType, prod ]
```

The logical ID is for you to choose. It's how you name condition

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CloudFormation - Rollbacks

- Stack Creation Fails:
- Default: everything rolls back (gets deleted). We can look at the log
- Option to disable rollback and troubleshoot what happened
- Stack Update Fails:
- The stack automatically rolls back to the previous known working state
- Ability to see in the log what happened and error messages
- Rollback Failure? Fix resources manually then issue
- ContinueUpdateRollback API from Console
- Or from the CLI using continue-update-rollback API call20

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CloudFormation - Service Role

- IAM role that allows CloudFormation to create/update/delete stack resources on your behalf
- Give ability to users to create/update/delete the
- stack resources even if they don’t have permissions to work with the resources in the stack
- Use cases:
- You want to achieve the least privilege principle
- But you don’t want to give the user all the required permissions to create the stack resources
- User must have iam:PassRole permissions
- cloudformation:*
- iam:PassRole
- Permissions
- s3:*Bucket
- Service Role
- User

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CloudFormation Capabilities

- CAPABILITY_NAMED_IAM and CAPABILITY_IAM Necessary to enable when you CloudFormation template is creating or updating IAM resources (IAM User, Role, Group, Policy, Access Keys, Instance Profile…)
- Specify CAPABILITY_NAMED_IAM if the resources are named
- CAPABILITY_AUTO_EXPAND
- Necessary when your CloudFormation template includes Macros or Nested Stacks (stacks within stacks) to perform dynamic transformations
- You’re acknowledging that your template may change before deploying
- InsufficientCapabilitiesException
- Exception that will be thrown by CloudFormation if the capabilities haven’t been acknowledged when deploying a template (security measure)

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CloudFormation - DeletionPolicy Delete

- DeletionPolicy:
- Control what happens when the CloudFormation template is deleted or when a resource is removed from a CloudFormation template
- Extra safety measure to preserve and backup resources
- Default DeletionPolicy=Delete
- Delete won’t work on an S3 bucket if the bucket is not empty

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CloudFormation - DeletionPolicy Retain

1. DeletionPolicy=Retain:
2. Specify on resources to preserve in case of CloudFormation deletes
3. Works with any resources

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CloudFormation - DeletionPolicy Snapshot

- DeletionPolicy=Snapshot
- Create one final snapshot before deleting the resource
- Examples of supported resources:
- EBS Volume, ElastiCache Cluster, ElastiCache ReplicationGroup
- RDS DBInstance, RDS DBCluster, Redshift Cluster, Neptune DBCluster, DocumentDB DBCluster0

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CloudFormation - Stack Policies

```
{
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "Update:*",
            "Principal": "*",
            "Resource": "*"
        },
        {
            "Effect": "Deny",
            "Action": "Update:*",
            "Principal": "*",
            "Resource": "LogicalResourceId/ProductionDatabase"
        }
    ]
}
```

During a CloudFormation Stack update, all

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CloudFormation - Termination Protection

- To prevent accidental deletes of CloudFormation Stacks, use TerminationProtection

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CloudFormation – Custom Resources

- Used to
- define resources not yet supported by CloudFormation
- define custom provisioning logic for resources can that be outside of CloudFormation (on-premises resources, 3rd party resources…)
- have custom scripts run during create / update / delete through Lambda functions (running a Lambda function to empty an S3 bucket before being deleted)
- Defined in the template using
- AWS::CloudFormation::CustomResource or Custom::MyCustomResourceTypeName (recommended)
- Backed by a Lambda function (most common) or an SNS topic

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How to define a Custom Resource?

- Service token specifies where CloudFormation sends requests to, such as Lambda ARN or SNS ARN (required & must be in the same region)
- Input data parameters (optional)

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Use Case - Delete content from an S3 bucket

- You can't delete a non-empty S3 bucket
- To delete a non-empty S3 bucket, you must first delete all the objects inside it
- We can use a custom resource to empty an S3 bucket before it gets deleted by CloudFormation
- Custom resource Lambda function
- delete stack
- Stack
- empty bucket
- S3 bucket
- CloudFormation
- User

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CloudFormation - Dynamic References

- Reference external values stored in Systems Manager Parameter Store and Secrets Manager within CloudFormation templates
- CloudFormation retrieves the value of the specified reference during create/update/delete operations
- For example: retrieve RDS DB Instance master password from Secrets Manager
- Supports
- ssm - for plaintext values stored in SSM Parameter Store
- ssm-secure - for secure strings stored in SSM Parameter Store
- secretsmanager - for secret values stored in Secrets Manager
- ‘{{resolve:service-name:reference-key}}’
- CloudFormation
- create/update
- Template
- SSM Parameter Store

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TOPIC 6

CloudFormation - Dynamic References

```
Resources:
  S3Bucket:
    Type: AWS::S3::Bucket
    Properties:
      AccessControl: '{{resolve:ssm:S3AccessControl:2}}'
```

SSM

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TOPIC 6

CloudFormation, Secrets Manager & RDS Option 1 - ManageMasterUserPassword

- ManageMasterUserPassword - creates admin secret implicitly
- RDS, Aurora will manage the secret in Secrets Manager and its rotation
- User
- create Stack
- CloudFormation
- Secrets Manager
- create Secret
- RDS
- create DB and
- configure Username/Password

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2. Reference secret in RDS DB instance

1. 3. link the secret to RDS DB instance (for rotation)

2. 1. secret is generated

3. CloudFormation, Secrets Manager & RDS Option 2 - Dynamic Reference

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User Data in EC2 for CloudFormation

- We can have user data at EC2 instance launch through the console
- Let’s learn how to write the same EC2 user-data script in our CloudFormation template
- The important thing to pass is the entire script through the function Fn::Base64
- Good to know, user data script log is in /var/log/cloud-init-output.log
- Let’s see how to do this in CloudFormation!

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The Problems with EC2 User Data

- What if we want to have a very large instance configuration?
- What if we want to evolve the state of the EC2 instance without terminating it and creating a new one?
- How do we make EC2 user-data more readable?
- How do we know or signal that our EC2 user-data script completed successfully?
- Enter CloudFormation Helper Scripts!
- Python scripts, that come directly on Amazon Linux AMIs, or can be installed using yum
- or dnf on non-Amazon Linux AMIs
- cfn-init, cfn-signal, cfn-get-metadata, cfn-hup

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AWS::CloudFormation::Init

```
Resources:
  EC2Instance:
    Type: AWS::EC2::Instance
    Properties:
      ...
    Metadata:
      AWS::CloudFormation::Init:
        config:
          packages:
            ...
          groups:
            ...
          users:
            ...
          sources:
            ...
          files:
            ...
          commands:
            ...
          services:
            ...
```

A config contains the following and is executed in that order

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CloudFormation - cfn-init

- Used to retrieve and interpret the resource metadata, installing packages, creating files and starting services
- With the cfn-init script, it helps make complex EC2 configurations readable
- The EC2 instance will query the CloudFormation service to get init data
- AWS::CloudFormation::Init must be in the Metadata of a resource
- Logs go to /var/log/cfn-init.log
- launch
- retrieve init data
- run cfn-init
- CloudFormation
- EC2 Instance

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TOPIC 6

CloudFormation - cfn-signal & Wait Conditions

- We still don’t know how to tell CloudFormation that the EC2 instance got properly configured after a cfn-init
- For this, we can use the cfn-signal script!
- We run cfn-signal right after cfn-init
- Tell CloudFormation service that the resource creation success/fail to keep on going or fail
- We need to define WaitCondition:
- Block the template until it receives a signal from cfn-signal
- We attach a CreationPolicy (also works on EC2, ASG)
- We can define a Count > 1 (in case you need more than 1 signal)
- launch
- run cfn-init
- CloudFormation
- EC2 Instance

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TOPIC 6

Wait Condition Didn't Receive the Required

- Number of Signals from an Amazon EC2 Instance
- Ensure that the AMI you're using has the AWS CloudFormation helper scripts
- installed. If the AMI doesn't include the helper scripts, you can also download them to your instance
- Verify that the cfn-init & cfn-signal command was successfully run on the instance.
- You can view logs, such as /var/log/cloud-init.log or /var/log/cfn-init.log, to help you debug the instance launch
- You can retrieve the logs by logging in to your instance, but you must disable
- rollback on failure or else AWS CloudFormation deletes the instance after your stack fails to create
- Verify that the instance has a connection to the Internet. If the instance is in a VPC,
- the instance should be able to connect to the Internet through a NAT device if it's is in a private subnet or through an Internet gateway if it's in a public subnet
- For example, run: curl -I https://aws.amazon.com

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TOPIC 6

CloudFormation - Nested Stacks

- Nested stacks are stacks as part of other stacks
- They allow you to isolate repeated patterns / common components in separate stacks and call them from other stacks
- Example:
- Load Balancer configuration that is re-used
- Security Group that is re-used
- Nested stacks are considered best practice
- To update a nested stack, always update the parent (root stack)
- Nested stacks can have nested stacks themselves!
- Stack B
- Stack A (root)
- Stack C
- Stack D

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TOPIC 6

Cross Stacks vs Nested Stacks

- Cross Stacks
- Helpful when stacks have different lifecycles
- Use Outputs Export and Fn::ImportValue
- When you need to pass export values to many stacks (VPC Id…)
- Nested Snacks
- Helpful when components must be re-used
- Example: re-use how to properly configure an Application Load Balancer
- The nested stack only is important to the higher-level stack (it’s not shared)
- VPC Stack
- App 1 Stack
- App 2 Stack
- App 3 Stack
- RDS Stack

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TOPIC 6

CloudFormation - DependsOn

- Specify that the creation of a specific resource follows another
- When added to a resource, that resource is created only after the creation of the resource specified in the DependsOn attribute
- Applied automatically when using !Ref and
- !GetAtt
- Use with any resource

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TOPIC 6

CloudFormation - StackSets

- Create, update, or delete stacks across
- multiple accounts and regions with a single operation/template
- Administrator account to create StackSets
- Target accounts to create, update, delete stack instances from StackSets
- When you update a stack
- set, all associated stack instances are updated throughout all accounts and regions
- Can be applied into all accounts of an AWS organizations
- Administrator account
- Template
- Create StackSet
- StackSet
- Region 1
- Target account A

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TOPIC 6

CloudFormation - StackSets Permission Models

- Self-managed Permissions
- Create the IAM roles (with established trusted relationship) in both administrator and target accounts
- Deploy to any target account in which you have permissions to create IAM role
- Service-managed Permissions
- Deploy to accounts managed by AWS Organizations
- StackSets create the IAM roles on your behalf (Enable trusted access with AWS Organizations)
- Must enable all features in AWS Organizations
- Ability to deploy to accounts added to your organization in the future (Automatic Deployments)
- Administrator account
- Target accounts
- AWSCloudFormagonStackSetAdministrationRole
- AWSCloudFormationStackSetExecutionRole
- Trust relationship

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TOPIC 6

StackSets with AWS Organizations

- Ability to automatically deploy Stack instances to new Accounts in an Organization
- Can delegate StackSets administration to member accounts in AWS Organization
- Trusted access with AWS Organizations must be enabled before delegated administrators can deploy to accounts managed by Organizations

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TOPIC 6

StackSets with AWS Organizations

- OU (Prod)
- Member Account - A
- Member Account - B
- OU (Dev)
- Member Account - C
- New Account
- AWS Organization
- (Delegated) Administrator Account
- StackSet
- manage
- Stack Instance
- Stack Instance
- Stack Instance

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TOPIC 6

Hands-On: StackSets

- We’ll use StackSets to enable AWS Config across AWS regions with a single click
- Let’s see how this works!
- Region 1
- Administrator/Target account
- Template
- Create StackSet
- StackSet
- AWS Config
- Region 2
- AWS Config
- Region 3
- AWS Config
- enable

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TOPIC 6

CloudFormation - Troubleshooting

- DELETE_FAILED
- Some resources must be emptied before deleting, such as S3 buckets
- Use Custom Resources with Lambda functions to automate some actions
- Security Groups cannot be deleted until all EC2 instances in the group are gone
- Think about using DeletionPolicy=Retain to skip deletions
- UPDATE_ROLLBACK_FAILED
- Can be caused by resources changed outside of CloudFormation, insufficient permissions, Auto Scaling Group that doesn't receive enough signals…
- Manually fix the error and then ContinueUpdateRollback

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TOPIC 6

CloudFormation - Stack Set Troubleshooting

- A stack operation failed, and the stack instance status is OUTDATED.
- Insufficient permissions in a target account for creating resources that are specified in your template.
- The template could be trying to create global resources that must be unique but aren't, such as S3 buckets
- The administrator account does not have a trust relationship with the target account
- Reached a limit or a quota in the target account (too many resources)

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AWS LAMBDA

It’s a Serverless world

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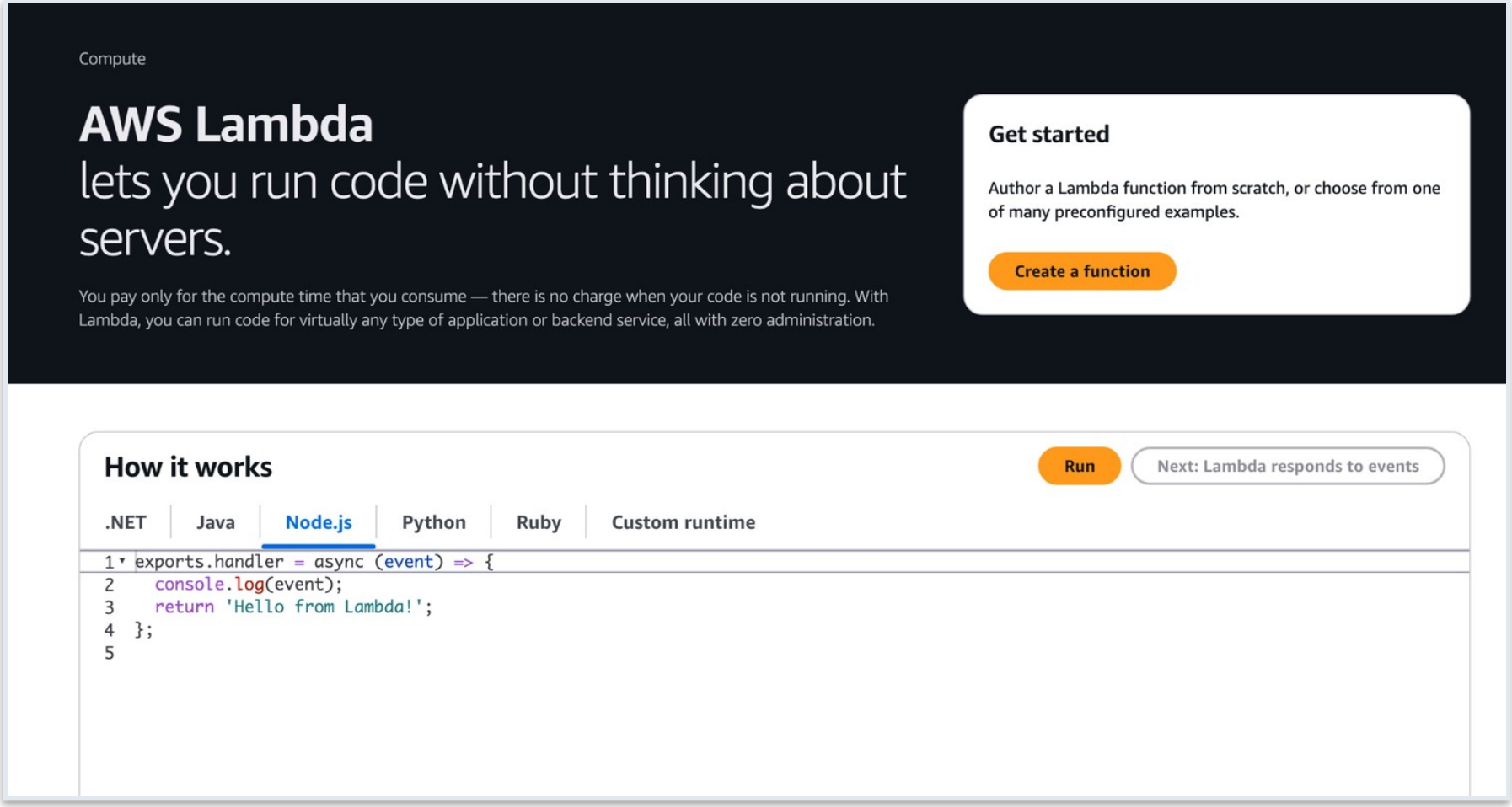
AWS Lambda



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Reference

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Reference

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Why AWS Lambda

- Virtual Servers in the Cloud
- Limited by RAM and CPU
- Continuously running
- Scaling means intervention to add / remove servers
- Amazon Lambda
- Amazon EC2
- Virtual functions – no servers to manage!
- Limited by time - short executions
- Run on-demand
- Scaling is automated!

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Benefits of AWS Lambda

- Easy Pricing:
- Pay per request and compute time
- Free tier of 1,000,000 AWS Lambda requests and 400,000 GBs of compute time
- Integrated with the whole AWS suite of services
- Integrated with many programming languages
- Easy monitoring through AWS CloudWatch
- Easy to get more resources per functions (up to 10GB of RAM!)
- Increasing RAM will also improve CPU and network!

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AWS Lambda language support

- Node.js (JavaScript)
- Python
- Java
- C# (.NET Core) / Powershell
- Ruby
- Custom Runtime API (community supported, example Rust or Golang)
- Lambda Container Image
- The container image must implement the Lambda Runtime API
- ECS / Fargate is preferred for running arbitrary Docker images

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AWS Lambda Integrations Main ones

- CloudWatch Logs
- SNS
- Cognito
- SQS
- S3
- Kinesis
- API Gateway
- DynamoDB
- CloudFront
- CloudWatch
- Events
- EventBridge

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Example: Serverless Thumbnail creation

- New image in S3
- AWS Lambda
- Function Creates a
- Thumbnail
- trigger
- push
- New thumbnail in S3
- Metadata in DynamoDB
- push
- Image name Image size Creation date etc…

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Example: Serverless CRON Job

- CloudWatch
- Events
- EventBridge
- AWS Lambda
- Function Perform
- a task
- Trigger Every 1 hour

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AWS Lambda Pricing: example

- You can find overall pricing information here: https://aws.amazon.com/lambda/pricing/
- Pay per calls:
- First 1,000,000 requests are free
- $0.20 per 1 million requests thereafter ($0.0000002 per request)
- Pay per duration: (in increment of 1 ms)
- 400,000 GB-seconds of compute time per month for FREE
- == 400,000 seconds if function is 1GB RAM
- == 3,200,000 seconds if function is 128 MB RAM
- After that $1.00 for 600,000 GB-seconds
- It is usually very cheap to run AWS Lambda so it’s very popular

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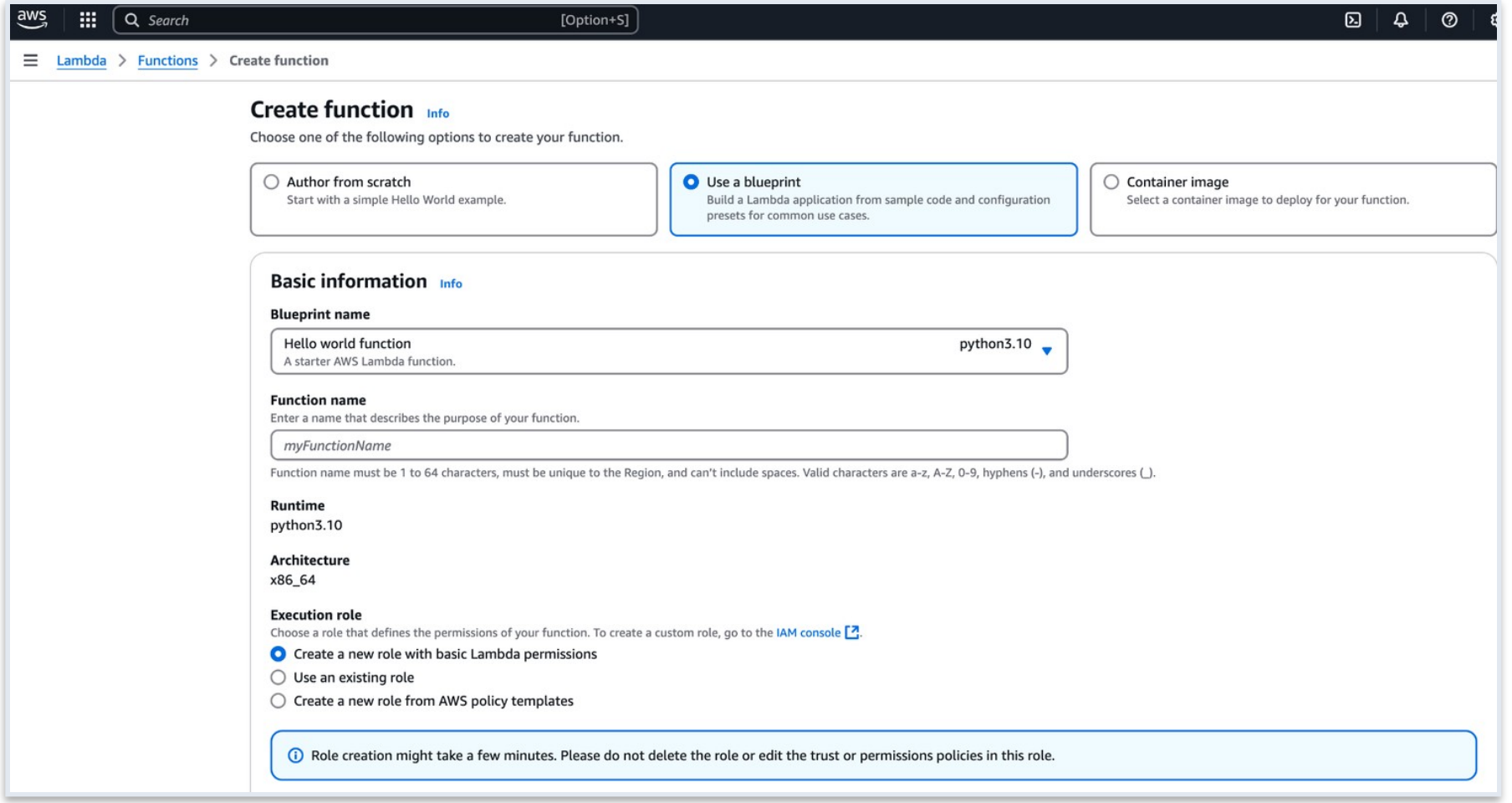
Activity: Create a Lambda



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CloudWatch Events / EventBridge

- CRON or Rate EventBridge Rule
- AWS Lambda
- Function Perform
- a task
- Trigger Every 1 hour
- CodePipeline EventBridge Rule
- AWS Lambda
- Function Perform
- a task
- Trigger on
- State Changes

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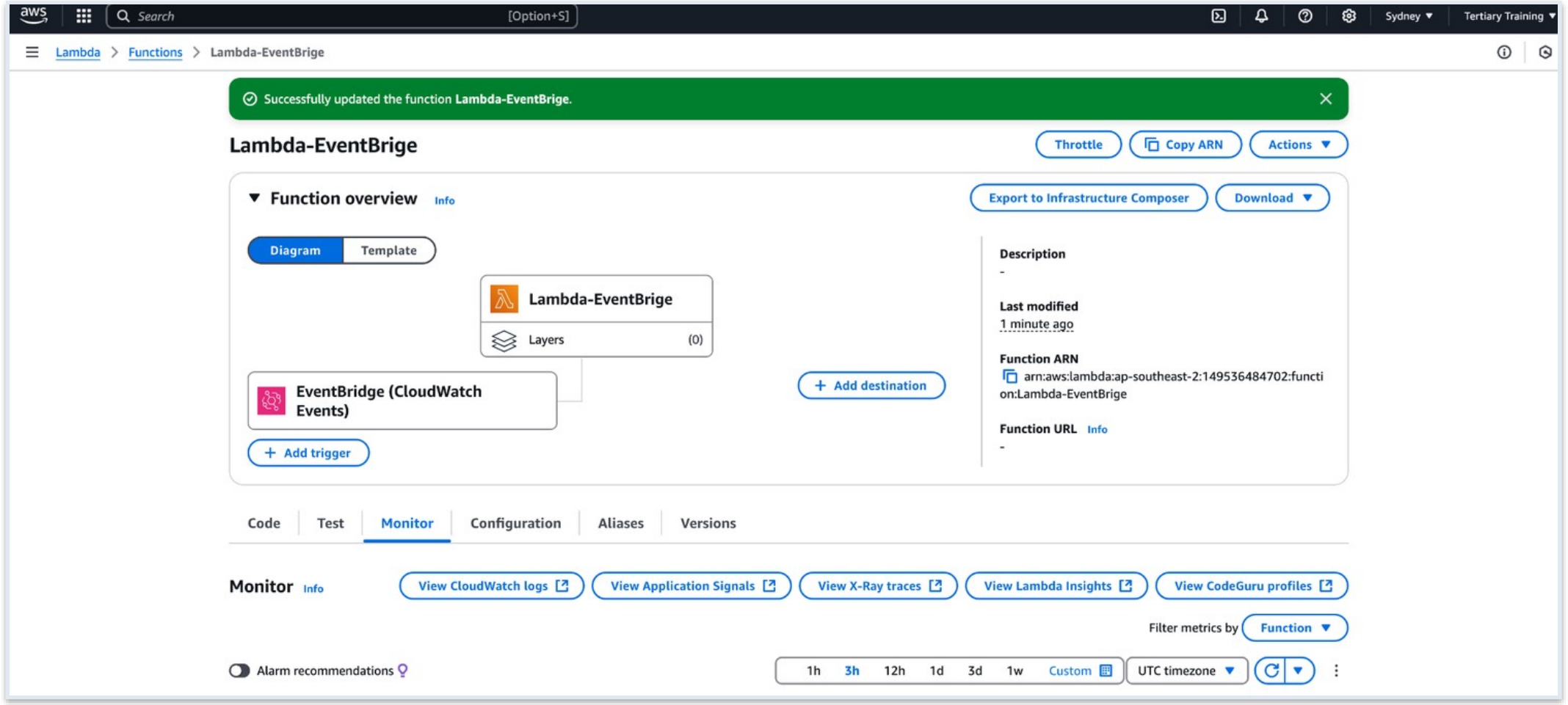
Activity: Lambda Triggered by Eventbridge



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S3 Events Notifications

- S3:ObjectCreated, S3:ObjectRemoved,
- S3:ObjectRestore, S3:Replication…
- Object name filtering possible (*.jpg)
- Use case: generate thumbnails of images uploaded to S3
- S3 event notifications typically deliver events in seconds but can sometimes take a minute or longer
- If two writes are made to a single non- versioned object at the same time, it is possible that only a single event notification will be sent
- If you want to ensure that an event notification is sent for every successful write, you can enable versioning on your bucket.
- Amazon S3
- events
- Lambda Function
- SQS

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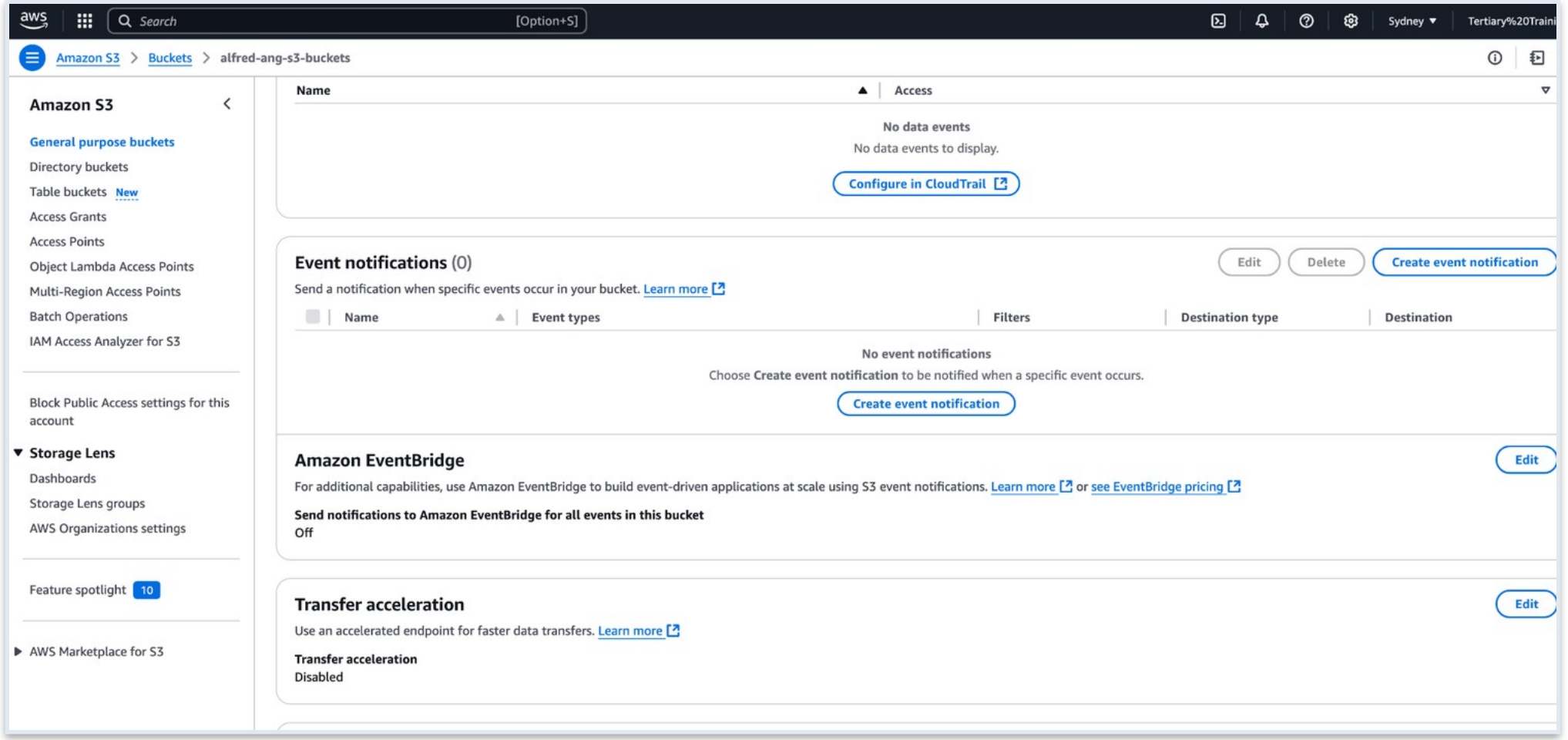
Activity: S3 Event Notification



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.com

1. Simple S3 Event Pattern - Metadata Sync
2. S3 bucket
3. Table in RDS
4. New file event
5. Update metadata table
6. DynamoDB Table

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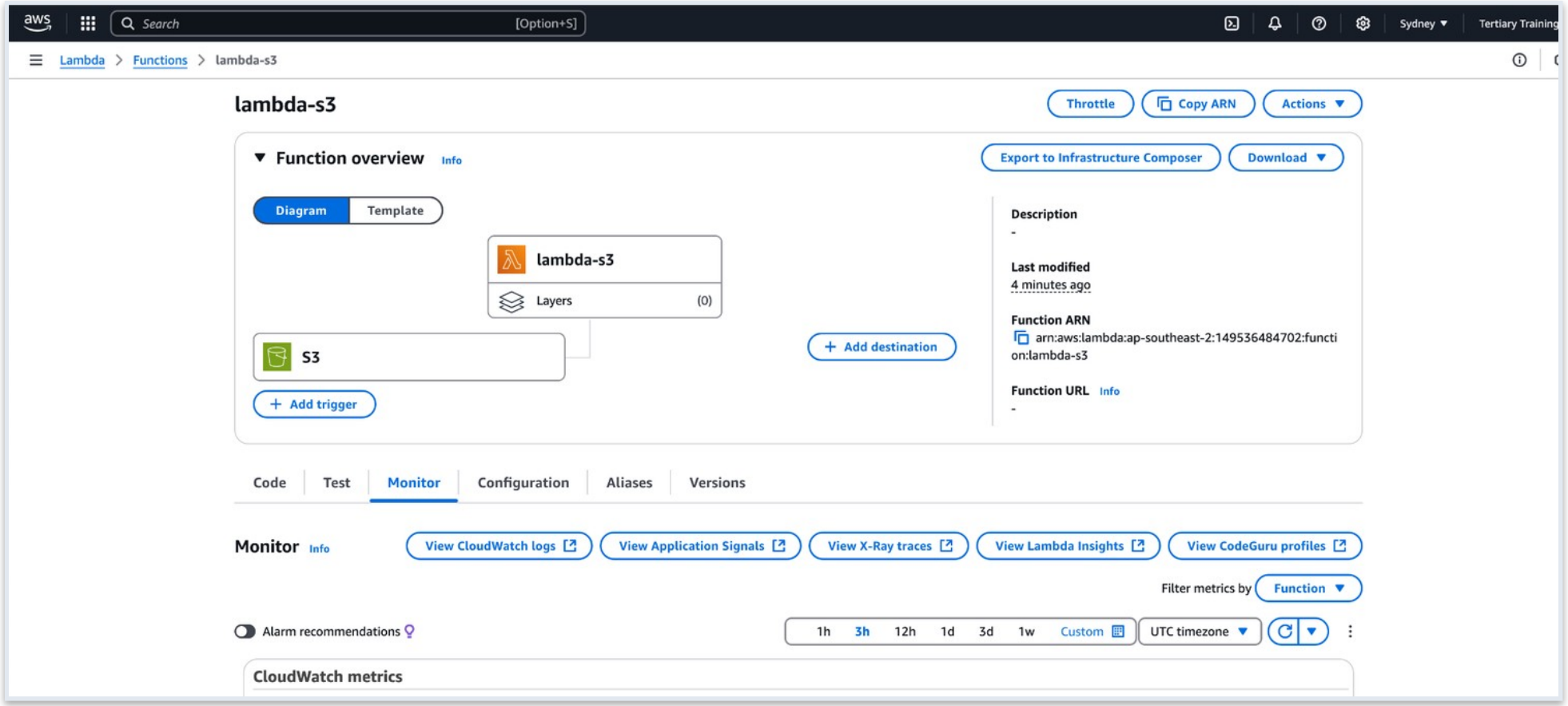
Activity: Lambda Triggered by S3



Upload coffee.jpg to S3

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Lambda Execution Role (IAM Role)

- Grants the Lambda function permissions to AWS services / resources
- Sample managed policies for Lambda:
- AWSLambdaBasicExecutionRole - Upload logs to CloudWatch.
- AWSLambdaKinesisExecutionRole - Read from Kinesis
- AWSLambdaDynamoDBExecutionRole - Read from DynamoDB Streams
- AWSLambdaSQSQueueExecutionRole - Read from SQS
- AWSLambdaVPCAccessExecutionRole - Deploy Lambda function in VPC
- AWSXRayDaemonWriteAccess - Upload trace data to X-Ray.
- When you use an event source mapping to invoke your function, Lambda uses the execution role to read event data.
- Best practice: create one Lambda Execution Role per function

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Lambda Resource Based Policies

- Use resource-based policies to give other accounts and AWS services permission to use your Lambda resources
- Similar to S3 bucket policies for S3 bucket
- An IAM principal can access Lambda:
- if the IAM policy attached to the principal authorizes it (e.g. user access)
- OR if the resource-based policy authorizes (e.g. service access)
- When an AWS service like Amazon S3 calls your Lambda function, the resource-based policy gives it access.

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Lambda Logging & Monitoring

- CloudWatch Logs:
- AWS Lambda execution logs are stored in AWS CloudWatch Logs
- Make sure your AWS Lambda function has an execution role with an IAM policy that authorizes writes to CloudWatch Logs
- CloudWatch Metrics:
- AWS Lambda metrics are displayed in AWS CloudWatch Metrics
- Invocations, Durations, Concurrent Executions
- Error count, Success Rates,Throttles
- Async Delivery Failures
- Iterator Age (Kinesis & DynamoDB Streams)

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Lambda Tracing with X-Ray

- Enable in Lambda configuration (Active Tracing)
- Runs the X-Ray daemon for you
- Use AWS X-Ray SDK in Code
- Ensure Lambda Function has a correct IAM Execution Role
- The managed policy is called AWSXRayDaemonWriteAccess
- Environment variables to communicate with X-Ray
- _X_AMZN_TRACE_ID: contains the tracing header
- AWS_XRAY_CONTEXT_MISSING: by default, LOG_ERROR
- AWS_XRAY_DAEMON_ADDRESS: the X-Ray Daemon IP_ADDRESS:PORT

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Lambda Function Configuration

- RAM:
- From 128MB to 10GB in 1MB increments
- The more RAM you add, the more vCPU credits you get
- At 1,792 MB, a function has the equivalent of one full vCPU
- After 1,792 MB, you get more than one CPU, and need to use multi-threading in your code to benefit from it (up to 6 vCPU)
- If your application is CPU-bound (computation heavy), increase RAM
- Timeout: default 3 seconds, maximum is 900 seconds (15 minutes)

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Lambda Execution Context

- The execution context is a temporary runtime environment that initializes any external dependencies of your lambda code
- Great for database connections, HTTP clients, SDK clients
- The execution context is maintained for some time in anticipation of another Lambda function invocation
- The next function invocation can “re-use” the context to execution time and save time in initializing connections objects
- The execution context includes the /tmp directory

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Initialize outside the handler

1. GOOD!
2. BAD!
3. The DB connection is established At every function invocation
4. The DB connection is established once And re-used across invocations

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Lambda Functions /tmp space

- If your Lambda function needs to download a big file to work…
- If your Lambda function needs disk space to perform operations…
- You can use the /tmp directory
- Max size is 10GB
- The directory content remains when the execution context is frozen, providing transient cache that can be used for multiple invocations (helpful to checkpoint your work)
- For permanent persistence of object (non temporary), use S3
- To encrypt content on /tmp, you must generate KMS Data Keys

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Lambda Concurrency and Throttling

- Concurrency limit: up to 1000 concurrent executions
- Can set a "reserved concurrency" at the function level (=limit)
- Each invocation over the concurrency limit will trigger a “Throttle”
- Throttle behavior:
- If synchronous invocation => return ThrottleError - 429
- If asynchronous invocation => retry automatically and then go to DLQ
- If you need a higher limit, open a support ticket

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Lambda Concurrency Issue

- If you don’t reserve (=limit) concurrency, the following can happen:
- Application Load Balancer
- SDK / CLI
- 1000 concurrent executions
- THROTTLE!
- API Gateway
- THROTTLE!
- Many users
- Few users

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Concurrency and Asynchronous Invocations

- S3 bucket
- New file event
- New file event
- New file event
- If the function doesn't have enough concurrency available to process all events, additional requests are throttled.
- For throttling errors (429) and system errors (500-series), Lambda returns the event to the queue and attempts to run the function again for up to 6 hours.
- The retry interval increases exponentially from 1 second after the first attempt to a maximum of 5 minutes.

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Cold Starts & Provisioned Concurrency

- Cold Start:
- New instance => code is loaded and code outside the handler run (init)
- If the init is large (code, dependencies, SDK…) this process can take some time.
- First request served by new instances has higher latency than the rest
- Provisioned Concurrency:
- Concurrency is allocated before the function is invoked (in advance)
- So the cold start never happens and all invocations have low latency
- Application Auto Scaling can manage concurrency (schedule or target utilization)
- Note:
- Note: cold starts in VPC have been dramatically reduced in Oct & Nov 2019
- https://aws.amazon.com/blogs/compute/announcing-improved-vpc-networking-for-aws-lambda-functions/

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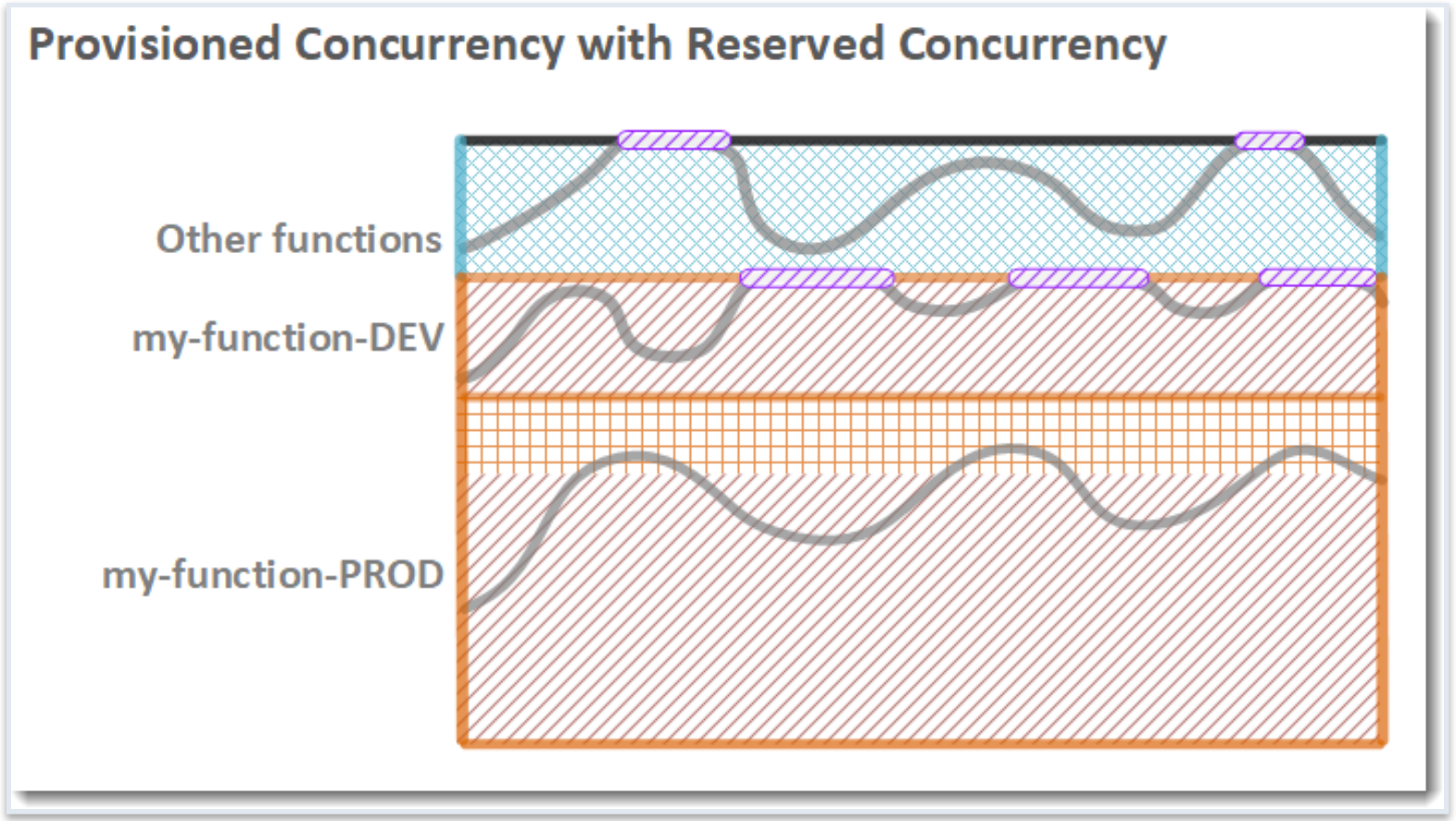
Reserved and Provisioned Concurrency



https://docs.aws.amazon.com/lambda/latest/dg/configuration-concurrency.html

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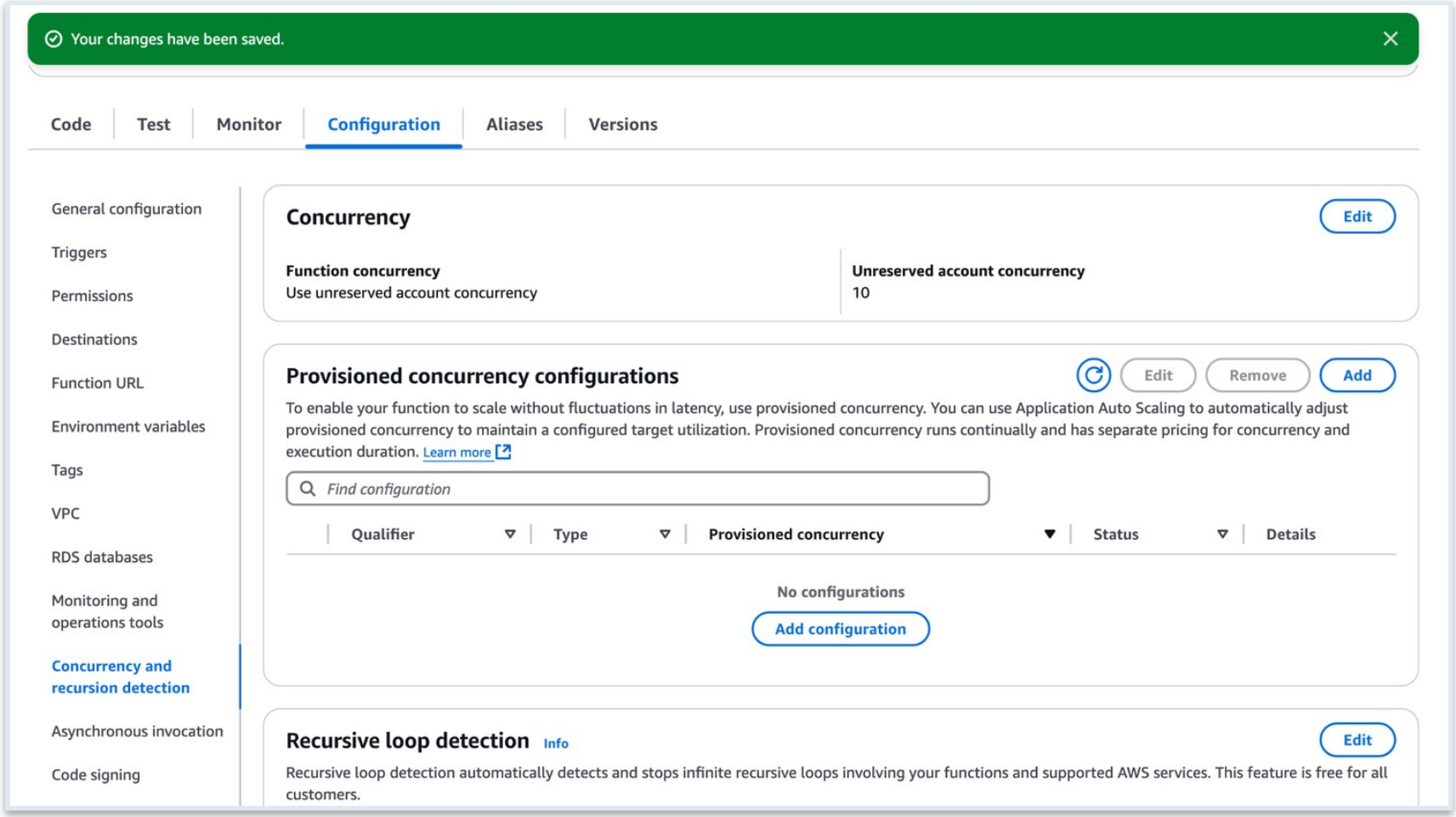
Activity: Lambda Concurrency



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Lambda Monitoring - CloudWatch Metrics

- Invocations - number of times your function is invoked (success/failure)
- Duration - amount of time your function spends processing an event
- Errors - number of invocations that result in a function error
- Throttles - number of invocation requests that are throttled (no concurrency available)
- DeadLetterErrors - number of times Lambda failed to send an event to a DLQ (async invocations)
- LteratorAge - time between when a Stream receives a record and when the Event
- Source Mapping sends the event to the function (for Event Source Mapping that reads from Stream)
- ConcurrentExecutions - number of function instances that are processing events

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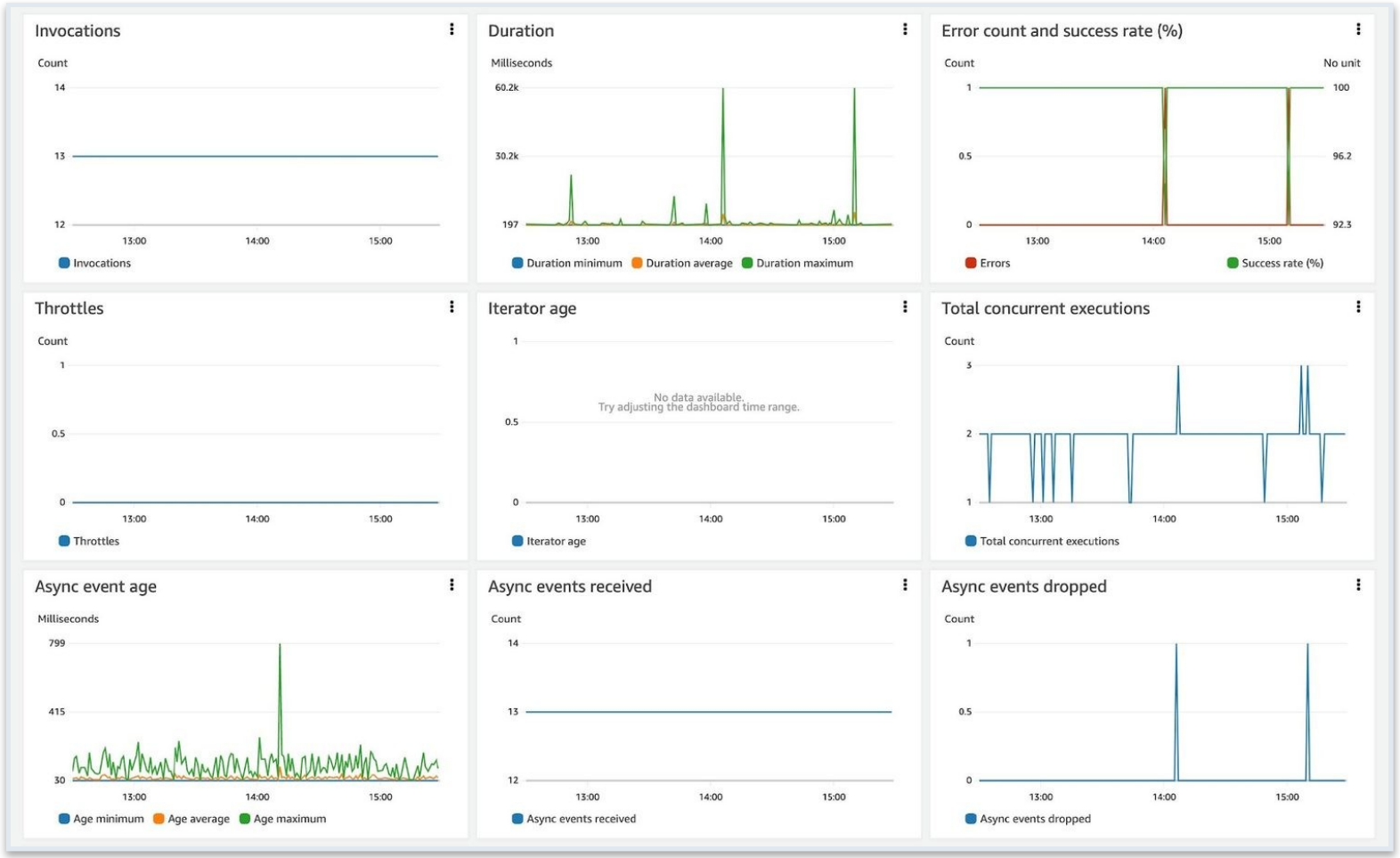
Example Lambda Metrics Dashboard



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Lambda Monitoring - CloudWatch Alarms

- Example 1 - No Lambda Invocations in the last hour using Invocations CloudWatch Metric
- Example 2 - When error > 0 using Errors CloudWatch Metric
- Example 3 - When throttles > 0 using Throttles CloudWatch Metric

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Lambda Monitoring - CloudWatch Logs

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "logs:CreateLogGroup",
                "logs:CreateLogStream",
                "logs:PutLogEvents",
                "logs:DescribeLogStreams"
            ],
            "Resource": [
                "arn:aws:logs:*:*:*"
            ]
        }
    ]
}
```

Lambda Function

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Lambda Monitoring - CloudWatch Logs Insights

```
1   fields @timestamp, @message
2   | filter @message like /ERROR/
3   | sort @timestamp desc
```

Allows you to search through all your Lambda functions logs

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Lambda Monitoring - CloudWatch Logs Insights

- Example CloudWatch Logs Insights Queries

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Lambda Monitoring – Lambda Insights

- Collects, aggregates, and summarizes:
- System-level Metrics - CPU time, memory, disk, network
- Diagnostic Information - old starts, Lambda worker shutdowns
- Helps you isolate issues with your Lambda functions and resolve them quickly
- Uses a CloudWatch Lambda Extension (provided as a Lambda layer)
- CloudWatch Logs
- /aws/lambda-insights/..
- Lambda function

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Amazon EC2 Storage

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What’s an EBS Volume?

- An EBS (Elastic Block Store) Volume is a network drive you can attach to your instances while they run
- It allows your instances to persist data, even after their termination
- They can only be mounted to one instance at a time (at the CCP level)
- They are bound to a specific availability zone
- Analogy:Think of them as a “network USB stick”
- Free tier: 30 GB of free EBS storage of type General Purpose (SSD) or Magnetic per month

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EBS Volume

- It’s a network drive (i.e. not a physical drive)
- It uses the network to communicate the instance, which means there might be a bit of latency
- It can be detached from an EC2 instance and attached to another one quickly
- It’s locked to an Availability Zone (AZ)
- An EBS Volume in us-east-1a cannot be attached to us-east-1b
- To move a volume across, you first need to snapshot it
- Have a provisioned capacity (size in GBs, and IOPS)
- You get billed for all the provisioned capacity
- You can increase the capacity of the drive over time

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US-EAST-1B

- US-EAST-1A
- EBS Volume - Example
- EBS (10 GB)
- EBS (100 GB)
- EBS (50 GB)
- EBS (50 GB)
- EBS (10 GB)
- unattached

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EC2 Instance Store

- EBS volumes are network drives with good but “limited” performance
- If you need a high-performance hardware disk, use EC2 Instance store
- Better I/O performance
- EC2 Instance Store lose their storage if they’re stopped (ephemeral)
- Good for buffer / cache / scratch data / temporary content
- Risk of data loss if hardware fails
- Backups and Replication are your responsibility

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1. Local EC2 Instance Store
2. Very high IOPS

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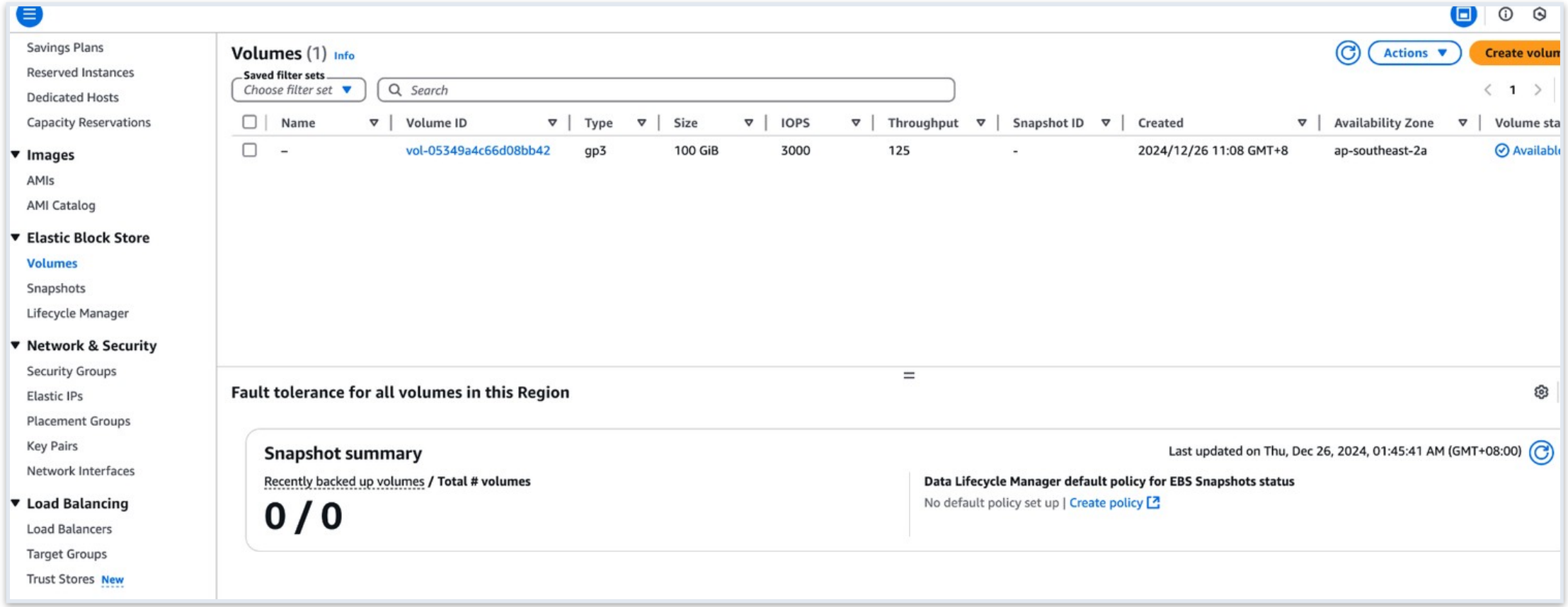

Activity: EBS

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EBS Volume Types

- EBS Volumes come in 6 types
- gp2 / gp3 (SSD): General purpose SSD volume that balances price and performance for a wide variety of workloads
- io1 / io2 Block Express (SSD): Highest-performance SSD volume for mission-critical low-latency or high-throughput workloads
- st1 (HDD): Low cost HDD volume designed for frequently accessed, throughput- intensive workloads
- sc1 (HDD): Lowest cost HDD volume designed for less frequently accessed workloads
- EBS Volumes are characterized in Size | Throughput | IOPS (I/O Ops Per Sec)
- When in doubt always consult the AWS documentation – it's good!
- Only gp2/gp3 and io1/io2 Block Express can be used as boot volumes

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- EBS Volume Types Use cases General Purpose SSD
- Cost effective storage, low-latency
- System boot volumes,Virtual desktops, Development and test environments
- 1 GiB - 16 TiB
- gp3:
- Baseline of 3,000 IOPS and throughput of 125 MiB/s
- Can increase IOPS up to 16,000 and throughput up to 1000 MiB/s independently
- gp2:
- Small gp2 volumes can burst IOPS to 3,000
- Size of the volume and IOPS are linked, max IOPS is 16,000
- 3 IOPS per GB, means at 5,334 GB we are at the max IOPS

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EBS Volume Types Use cases Provisioned IOPS (PIOPS) SSD

- Critical business applications with sustained IOPS performance
- Or applications that need more than 16,000 IOPS
- Great for databases workloads (sensitive to storage perf and consistency)
- io1 (4 GiB - 16 TiB):
- Max PIOPS: 64,000 for Nitro EC2 instances & 32,000 for other
- Can increase PIOPS independently from storage size
- io2 Block Express (4 GiB – 64 TiB):
- Sub-millisecond latency
- Max PIOPS: 256,000 with an IOPS:GiB ratio of 1,000:1
- Supports EBS Multi-attach

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EBS Volume Types Use cases

- Hard Disk Drives (HDD)
- Cannot be a boot volume
- 125 GiB to 16 TiB
- Throughput Optimized HDD (st1)
- Big Data, Data Warehouses, Log Processing
- Max through put500 MiB/s – max IOPS 500
- Cold HDD (sc1):
- For data that is infrequently accessed
- Scenarios where lowest cost is important
- Max throughput 250 MiB/s – max IOPS 250

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EBS - Volume Types Summary

	General Purpose SSD volumes		Provisioned IOPS SSD volumes	
Volume type	gp3	gp2	io2 Block Express [3]	io1
Durability	99.8% - 99.9% durability (0.1% - 0.2% annual failure rate)		99.999% durability (0.001% annual failure rate)	99.8% - 99.9% durability (0.1% - 0.2% annual failure rate)
Use cases	• Transactional workloads • Virtual desktops • Medium-sized, single-instance databases • Low-latency interactive applications • Boot volumes • Development and test environments		Workloads that require: • Sub-millisecond latency • Sustained IOPS performance • More than 64,000 IOPS or 1,000 MiB/s of throughput	• Workloads that require sustained IOPS performance or more than 16,000 IOPS • I/O-intensive database workloads
Volume size	1 GiB - 16 TiB		4 GiB - 64 TiB [4]	4 GiB - 16 TiB
Max IOPS per volume (16 KiB I/O)	16,000		256,000 [5]	64,000
Max throughput per volume	1,000 MiB/s	250 MiB/s [1]	4,000 MiB/s	1,000 MiB/s [2]
Amazon EBS Multi-attach	Not supported		Supported	
NVMe reservations	Not supported		Supported	Not supported
Boot volume	Supported			

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-volume-types.html#solid-state-drives

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EBS Multi-Attach - io1/io2 family

- Attach the same EBS volume to multiple EC2 instances in the same AZ
- Each instance has full read & write permissions to the high-performance volume
- Use case:
- Achieve higher application availability in clustered Linux applications (ex:Teradata)
- Applications must manage concurrent write operations
- Up to 16 EC2 Instances at a time
- Must use a file system that’s cluster-aware (not XFS, EXT4, etc…)
- Availability Zone 1
- io2 volume with Multi-Attach

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EBS Volume Resizing

- You can only increase the EBS volumes:
- Size (any volume type)
- IOPS (only in IO1)
- After resizing an EBS volume, you need to repartition your drive
- After increasing the size, it’s possible for the volume to be in a long time in the “optimisation” phase.The volume is still usable
- You can’t decrease the size of your EBS volume (create another smaller volume then migrate data)
- EBS Volume
- (250 GiB)
- EBS Volume
- (1 TiB)
- Modifying
- Optimizing

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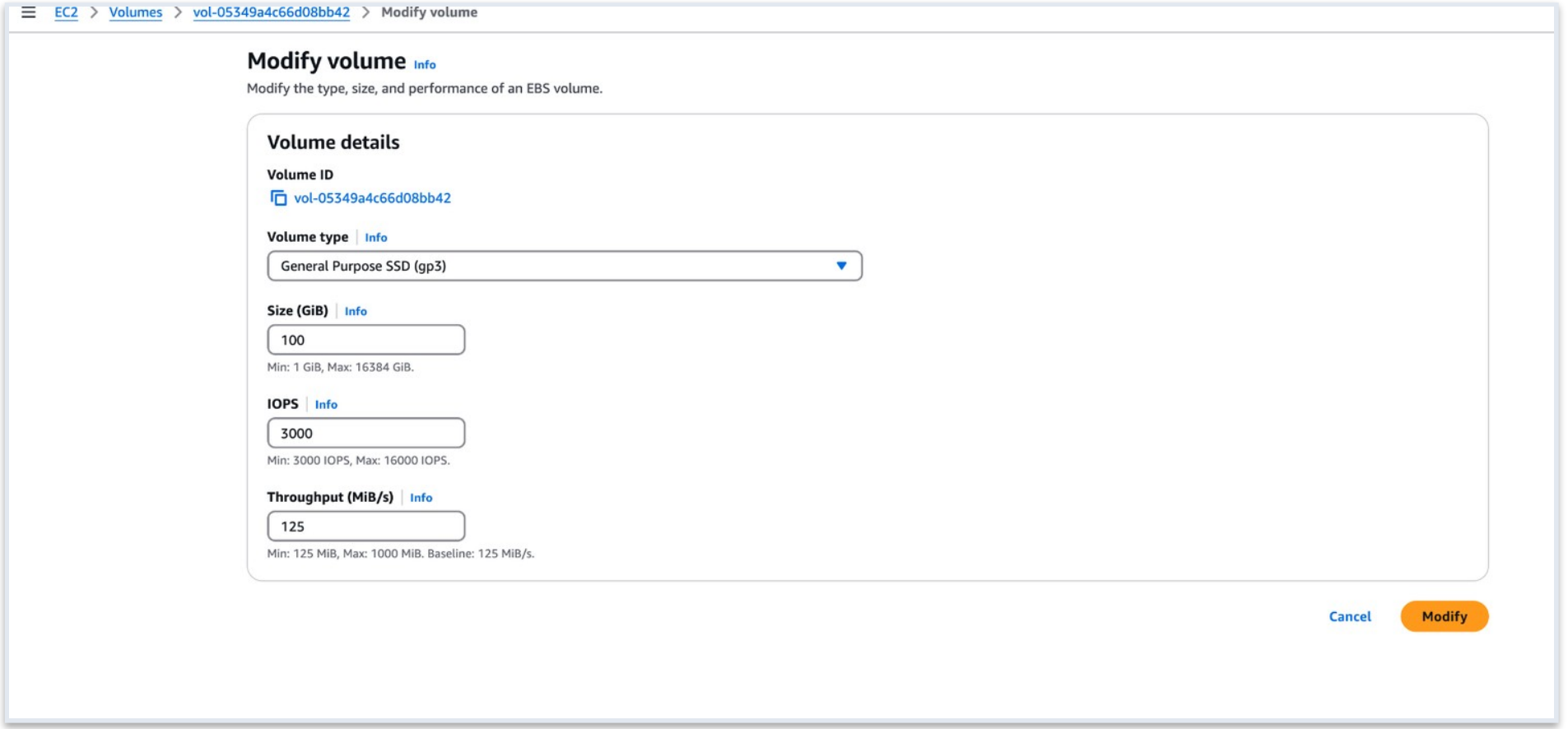

Activity: Modify EBS

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EBS Snapshots

- Make a backup (snapshot) of your EBS volume at a point in time
- Not necessary to detach volume to do snapshot, but recommended
- Can copy snapshots across AZ or Region
- US-EAST-1A
- EBS (50 GB)
- US-EAST-1B
- EBS (50 GB)
- EBS Snapshot
- snapshot
- restore

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Amazon Data Lifecycle Manager

- Automate the creation, retention, and deletion of EBS snapshots and EBS-backed AMIs
- Schedule backups, cross-account snapshot copies, delete outdated backups, …
- Uses resource tags to identify the resources (EC2 instances, EBS volumes)
- Can’t be used to manage snapshots/AMIs created outside DLM
- Can’t be used to manage instance-store backed AMIs
- Amazon Data Lifecycle Manager
- EBS Volume
- EBS Volumes
- EC2 Instance
- Tags
- Tags
- Snapshots
- AMI

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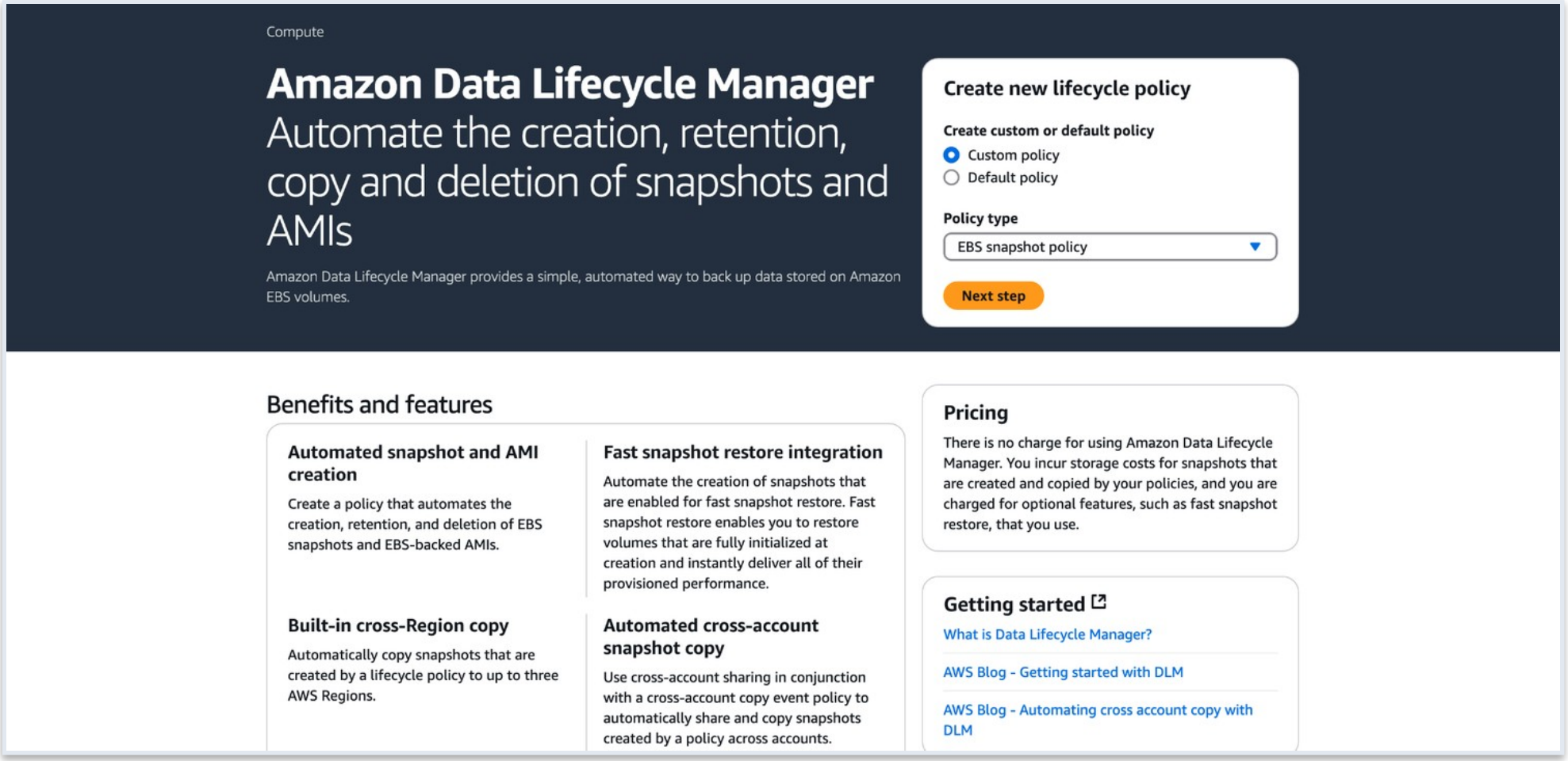
Amazon Data Lifecycle Manager



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EBS Snapshots - Fast Snapshot Restore (FSR)

- EBS Snapshots stored in S3
- By default, there's a latency of I/O operations the first time each block is accessed (block must be pulled from S3)
- Solution: force the initialization of the entire
- volume (using the dd or fio command), or you can enable FSR
- FSR helps you to create a volume from a snapshot that is fully initialized at creation (no I/O latency)
- Enabled for a snapshot in a particular AZ (billed per minute – very expensive $$$)
- Can be enabled on snapshots created by Data Lifecycle Manager
- Amazon S3
- EBS Snapshot
- EBS Snapshot
- (FSR enabled)
- EBS Volume

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EBS Snapshots Features

- EBS Snapshot Archive
- Move a Snapshot to an ”archive tier” that is 75% cheaper
- Takes within 24 to 72 hours for restoring the archive
- Recycle Bin for EBS Snapshots
- Setup rules to retain deleted snapshots so you can recover them after an accidental deletion
- Specify retention (from 1 day to 1 year)
- EBS Snapshot
- EBS Snapshot
- Archive
- archive
- EBS Snapshot
- Recycle Bin
- delete

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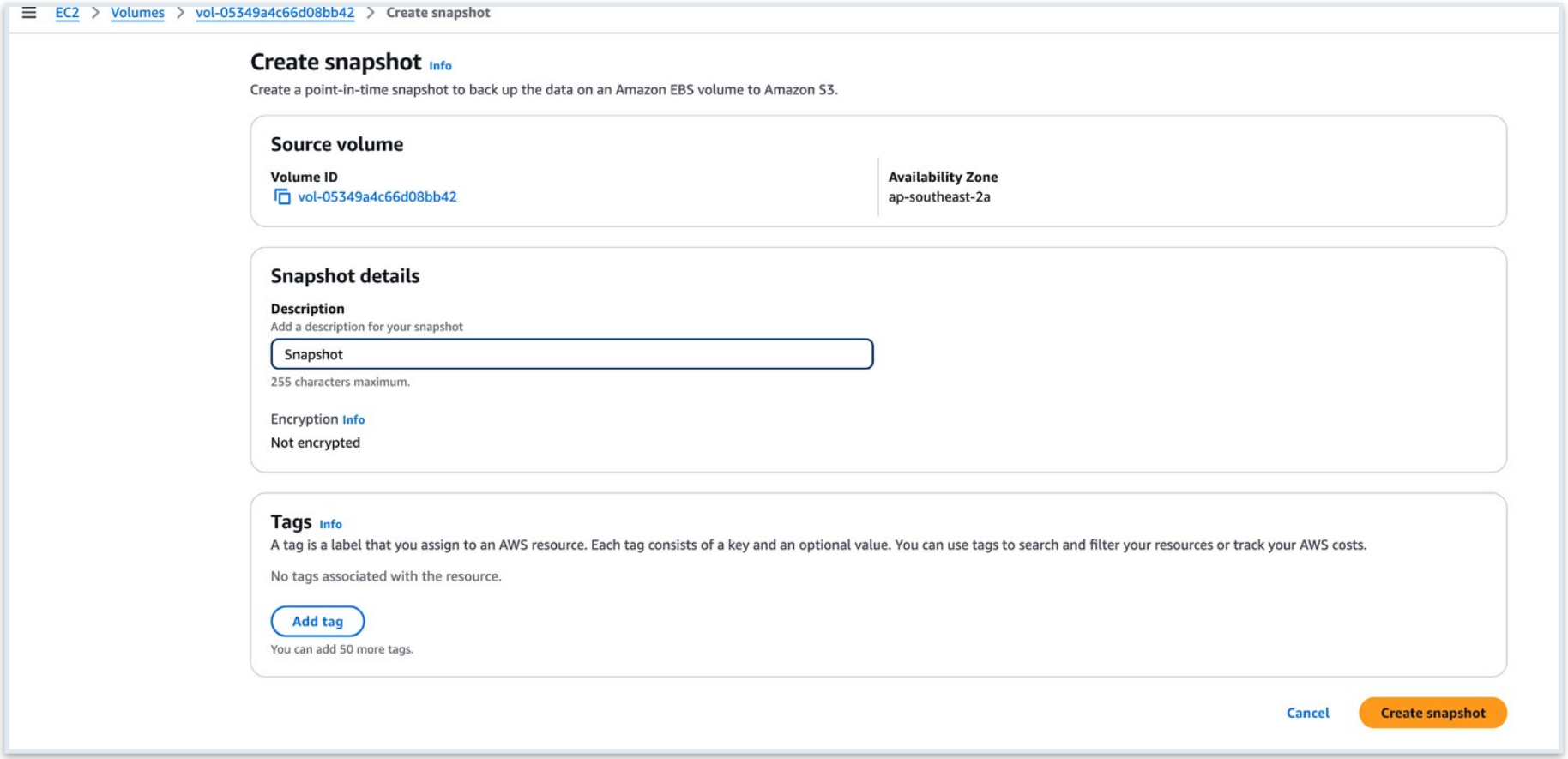

Activity: EBS Snapshot

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EBS Migration

1. EBS Volumes are only locked to a specific AZ
2. To migrate it to a different AZ (or region):
3. Snapshot the volume
4. (optional) Copy the volume to a different region
5. Create a volume from the snapshot in the AZ of your choice
6. Let's practice!

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EBS Encryption

- When you create an encrypted EBS volume, you get the following:
- Data at rest is encrypted inside the volume
- All the data in flight moving between the instance and the volume is encrypted
- All snapshots are encrypted
- All volumes created from the snapshot
- Encryption and decryption are handled transparently (you have nothing to do)
- Encryption has a minimal impact on latency
- EBS Encryption leverages keys from KMS (AES-256)
- Copying an unencrypted snapshot allows encryption
- Snapshots of encrypted volumes are encrypted

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Encryption: encrypt an unencrypted EBS volume

- Create an EBS snapshot of the volume
- Encrypt the EBS snapshot (using copy)
- Create new ebs volume from the snapshot (the volume will also be encrypted)
- Now you can attach the encrypted volume to the original instance

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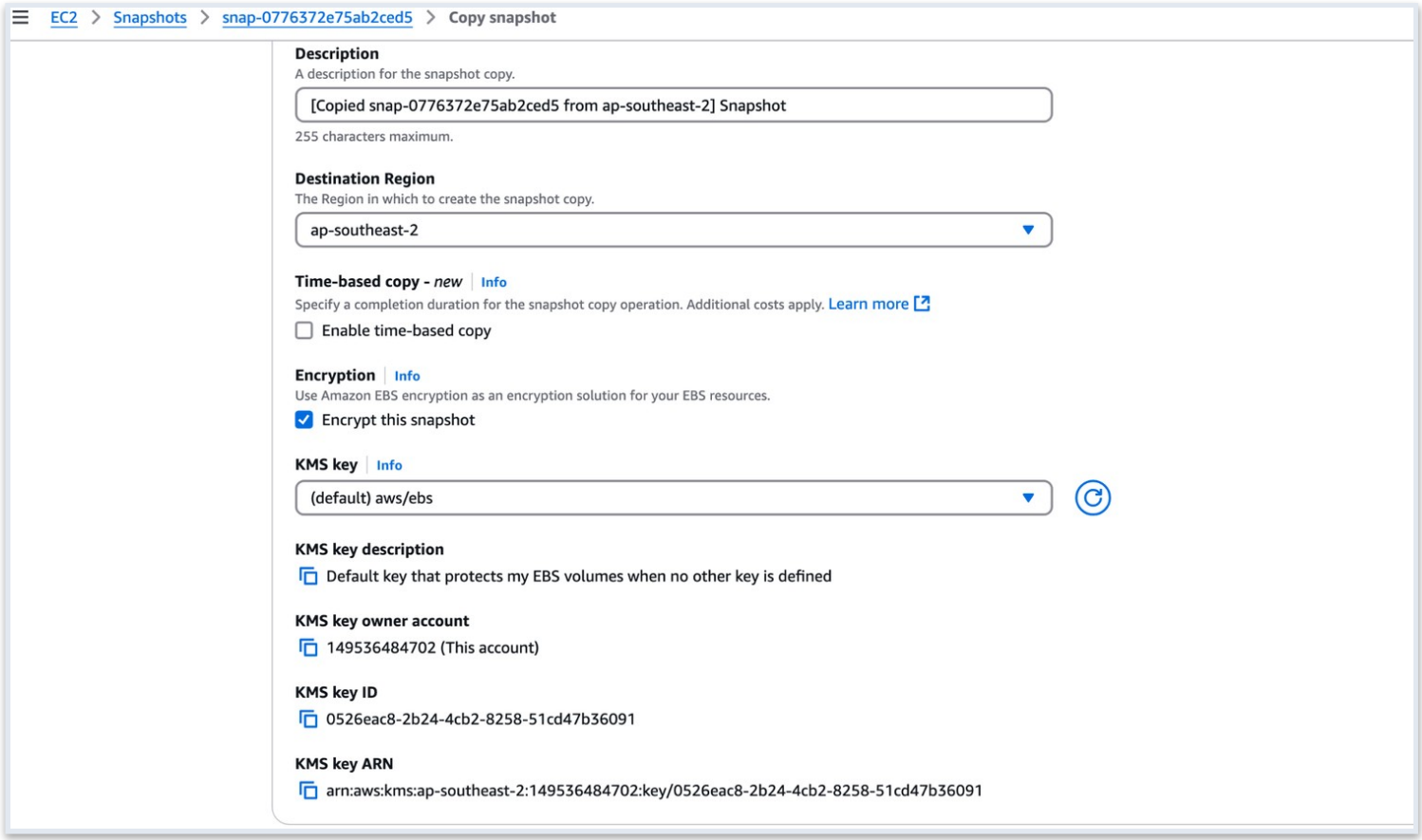

Activity: Encrypt EBS

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Amazon EFS - Elastic File System

- Managed NFS (network file system) that can be mounted on many EC2
- EFS works with EC2 instances in multi-AZ
- Highly available, scalable, expensive (3x gp2), pay per use
- us-east-1a
- EC2 Instances
- us-east-1b
- EC2 Instances
- us-east-1c
- EC2 Instances
- Security Group
- EFS FileSystem

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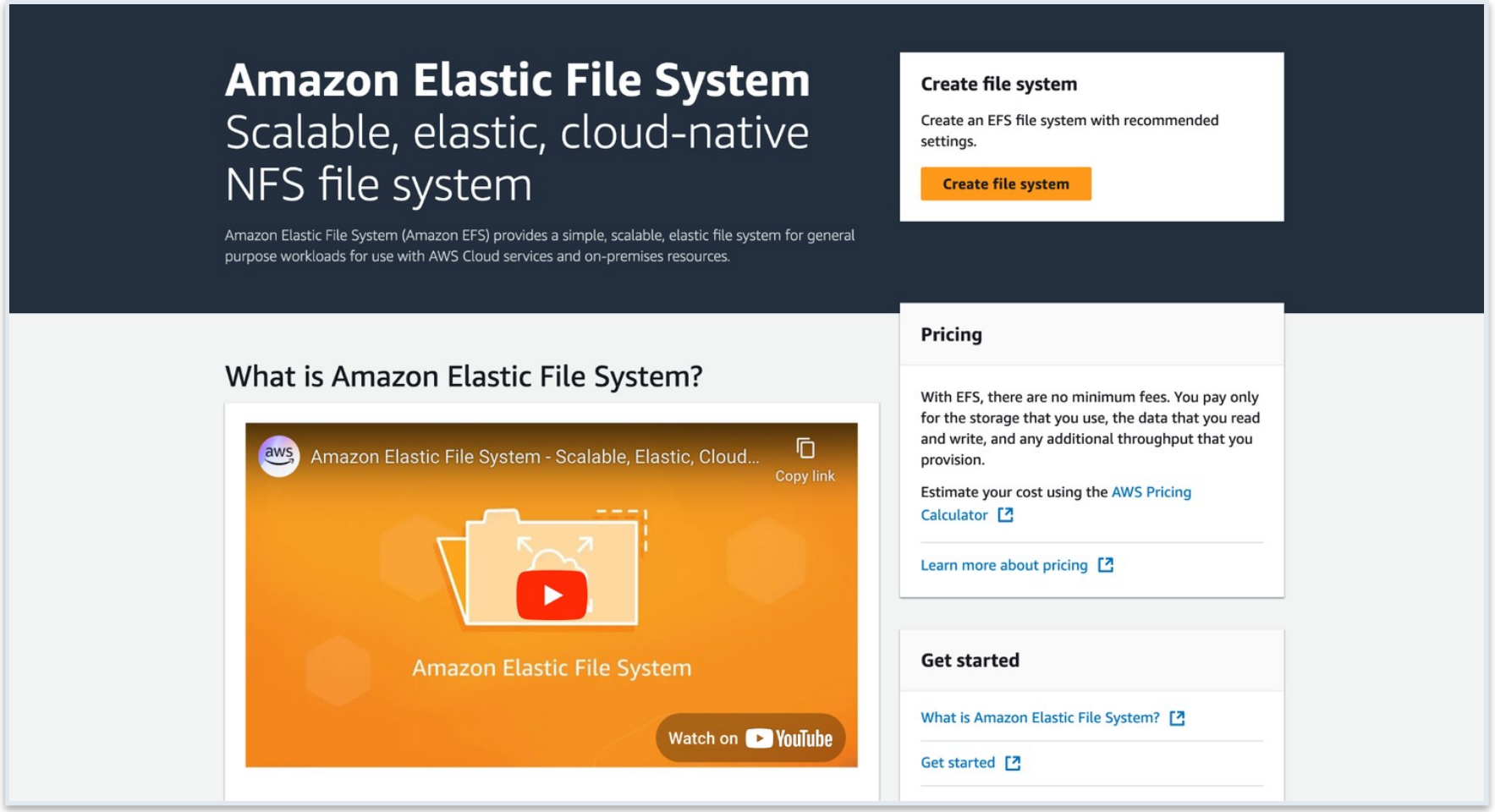
Amazon EFS - Elastic File System



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Amazon EFS - Elastic File System

- Use cases: content management, web serving, data sharing, Wordpress
- Uses NFSv4.1 protocol
- Uses security group to control access to EFS
- Compatible with Linux based AMI (not windows)
- Encryption at rest using KMS
- POSIX file system (~Linux) that has a standard file API
- File system scales automatically, pay-per-use, no capacity planning!

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EFS - Performance & Storage Classes

- EFS Scale
- 1000s of concurrent NFS clients, 10 GB+ /s throughput
- Grow to Petabyte-scale network file system, automatically
- Performance Mode (set at EFS creation time)
- General Purpose (default) - latency-sensitive use cases (web server, CMS, etc…)
- Max I/O - higher latecy, throughput, highly parallel (big data, media processing)
- Throughput Mode
- Bursting - 1TB = 50MiB/s + burst of up to 100MiB/s
- Provisioned - set your throughput regardless of storage size, ex: 1 GiB/s for 1 TB storage
- Elastic - automatically scales throughput up or down based on your workloads
- Up to 3GiB/s for reads and 1GiB/s for writes
- Used for unpredictable workloads

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EFS - Storage Classes

- Storage Tiers (lifecycle management feature move file after N days)
- Standard: for frequently accessed files
- Infrequent access (EFS-IA): cost to retrieve files, lower price to store.
- Archive: rarely accessed data (few times each year), 50% cheaper
- Implement Lifecycle policies to move files between storage tiers
- Availability and durability
- Standard: Multi-AZ, great for prod
- One Zone: One AZ, great for dev, backup enabled by default, compatible with IA (EFS One Zone-IA)
- Over 90% in cost savings
- Amazon EFS File System
- EFS IA
- Lifecycle Policy
- no access for 60 days

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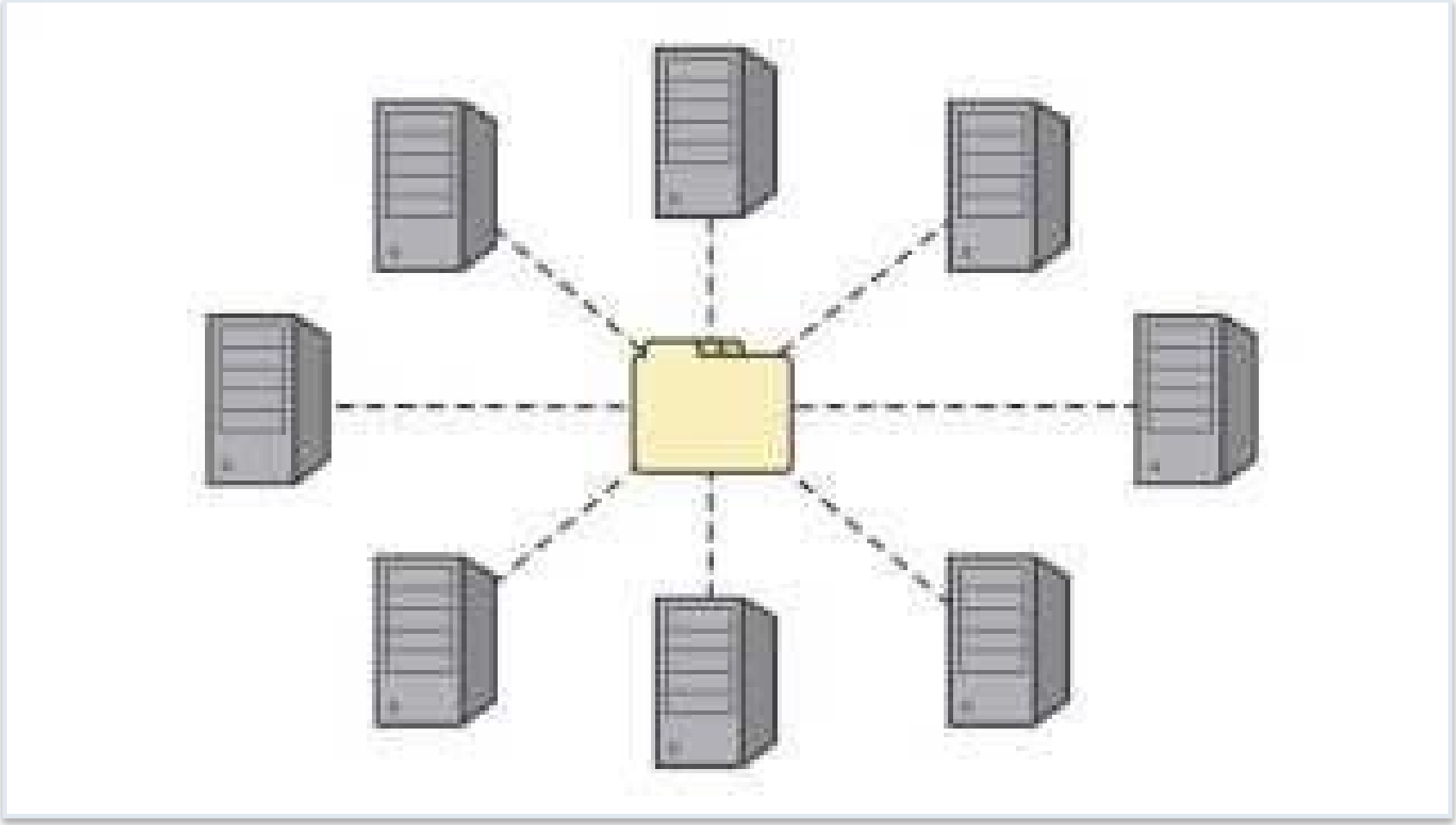

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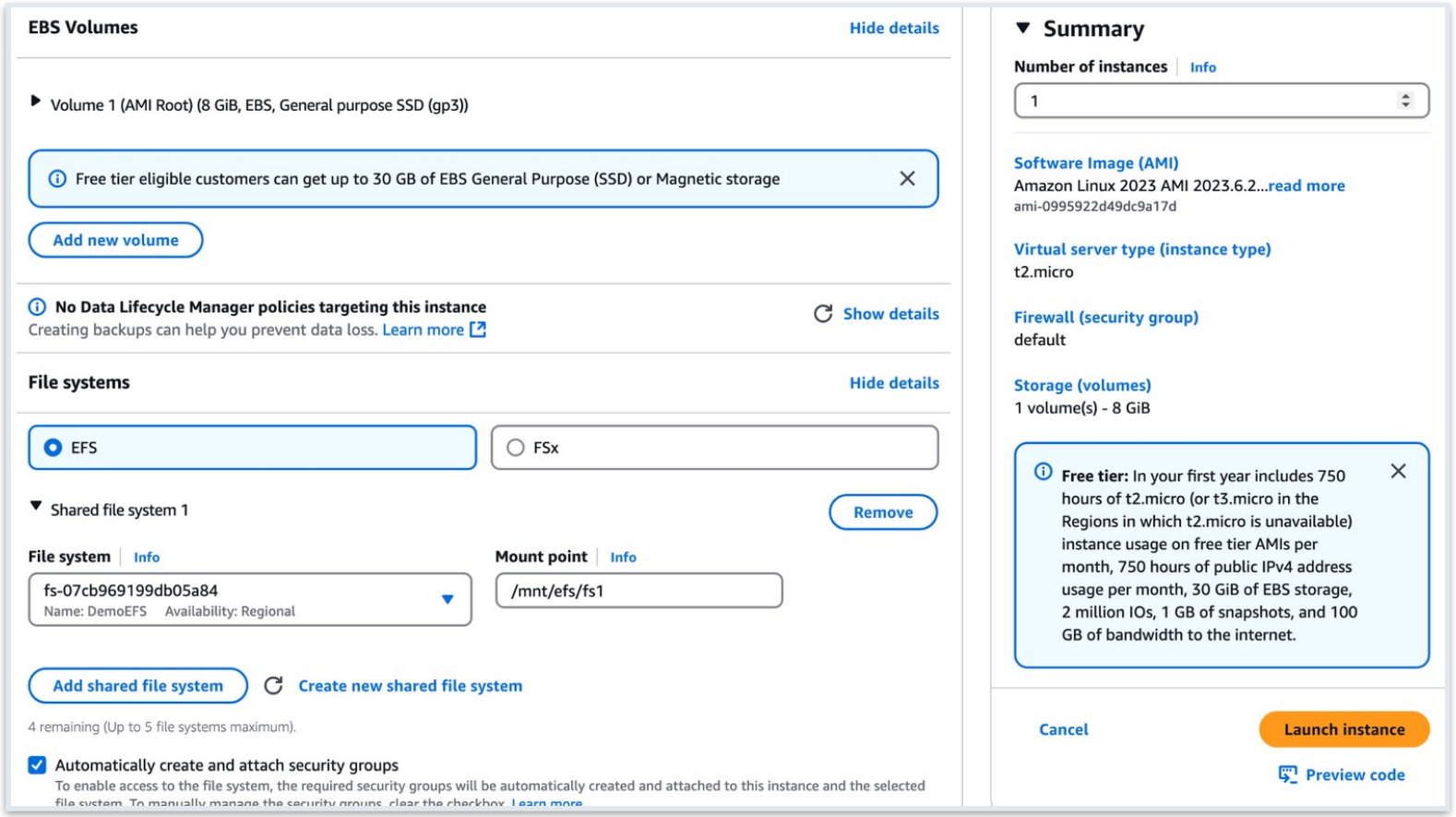

Activity: EFS

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EBS vs EFS - Elastic Block Storage

- EBS volumes…
- one instance (except multi-attach io1/io2)
- are locked at the Availability Zone (AZ) level
- gp2: IO increases if the disk size increases
- gp3 & io1: can increase IO independently
- To migrate an EBS volume across AZ
- Take a snapshot
- Restore the snapshot to another AZ
- EBS backups use IO and you shouldn’t run
- them while your application is handling a lot of traffic
- Root EBS Volumes of instances get terminated by default if the EC2 instance gets terminated. (you can disable that)
- snapshot

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EBS vs EFS - Elastic File System

- Mounting 100s of instances across AZ
- EFS share website files (WordPress)
- Only for Linux Instances (POSIX)
- EFS has a higher price point than EBS
- Can leverage Storage Tiers for cost savings
- Remember: EFS vs EBS vs Instance Store
- Availability Zone 1
- EFS
- Availability Zone 2
- EFS
- Mount
- Target
- Linux

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EFS - Access Points

- Easily manage applications access to NFS environments
- Enforce a POSIX user and group to use when accessing the file system
- Restrict access to a directory within the file system and optionally specify a different root directory
- Can restrict access from NFS clients using IAM policies
- EFS File System
- /
- /data
- /secret
- /config
- Access Point 1
- UID: 1001
- GID: 1001
- Path: /config

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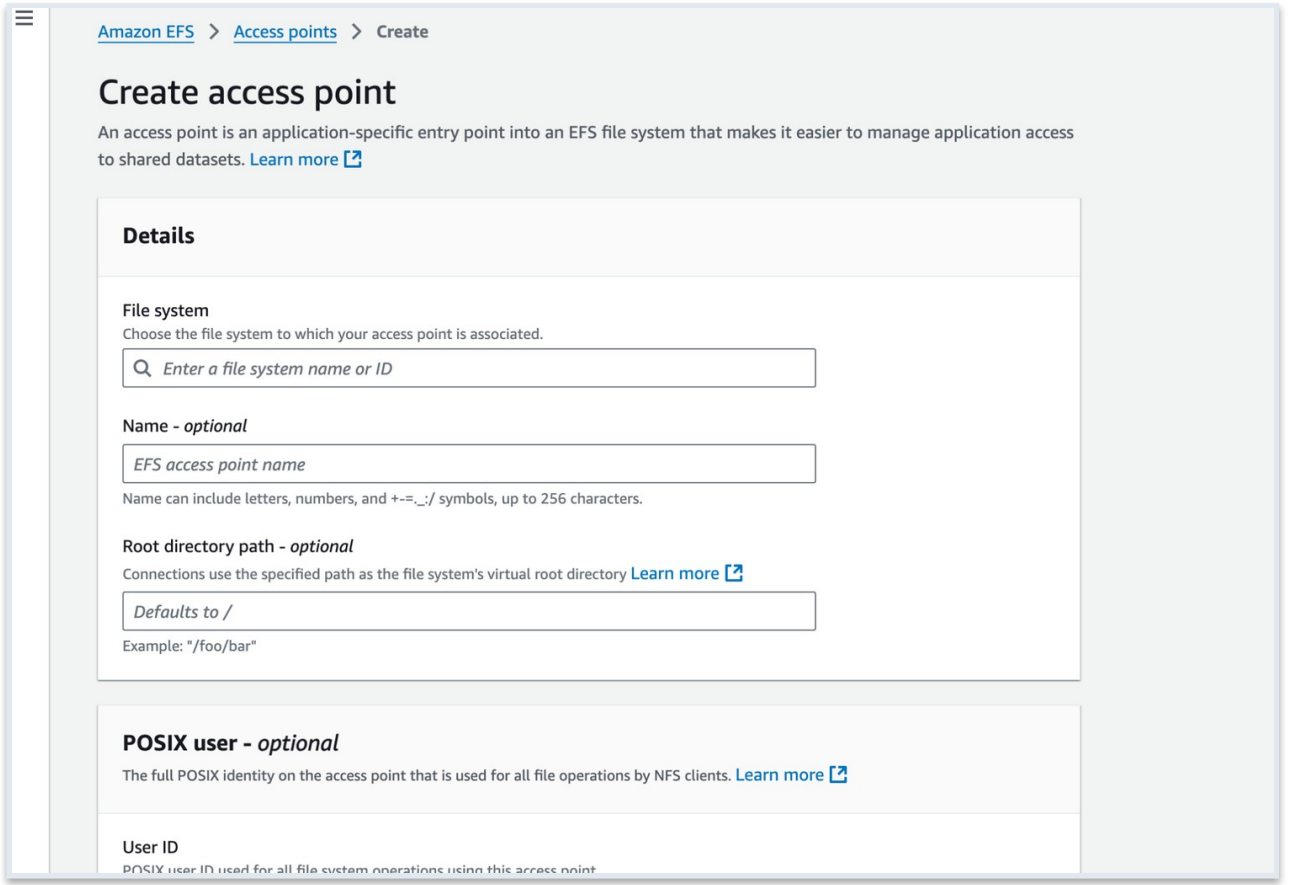

Activity: EFS Access Point

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EFS - Operations

- Operations that can be done in place:
- Lifecycle Policy (enable IA or change IA settings)
- Throughput Mode and Provisioned Throughput Numbers
- EFS Access Points
- Operations that require a migration using DataSync (replicates all file attributes and metadata)
- Migration to encrypted EFS
- Performance Mode (e.g. Max IO)
- AWS DataSync
- EFS File System
- (source)
- EFS File System
- (destination - encrypted)

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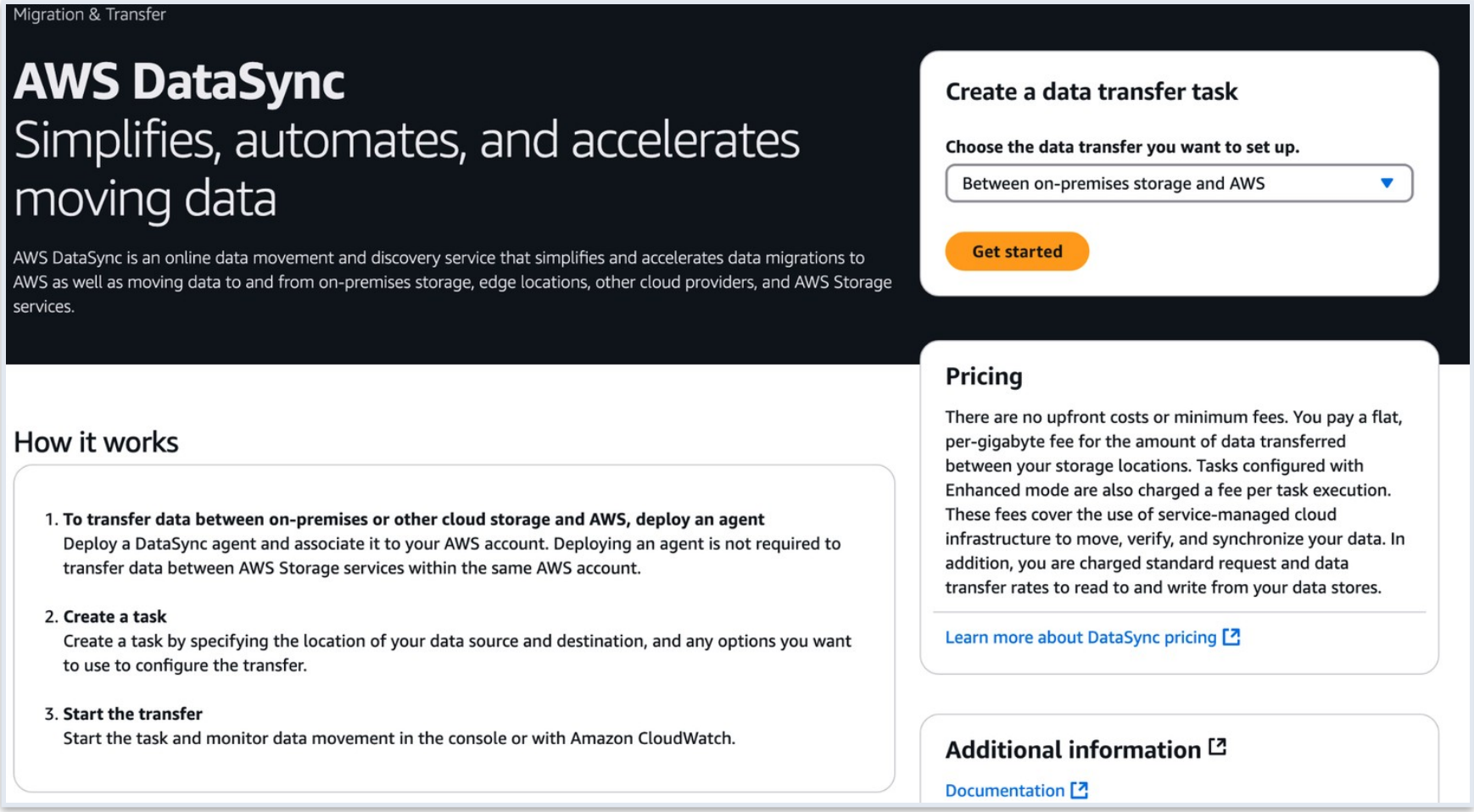
AWS DataSync



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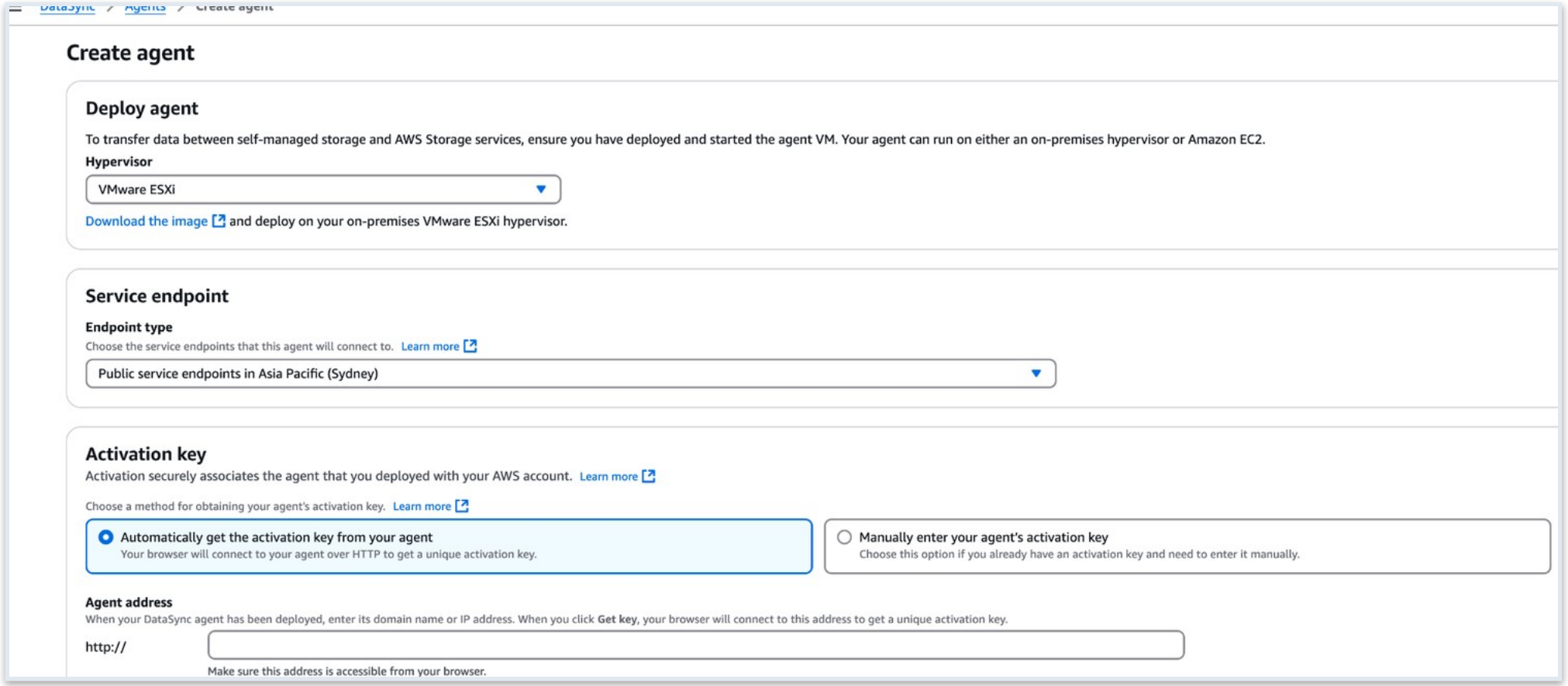

Activity: AWS DataSync

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EFS - CloudWatch Metrics

PercentIOLimit

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Amazon S3

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Amazon S3

1. Amazon S3 is one of the main building blocks of AWS
2. It’s advertised as ”infinitely scaling” storage
3. Many websites use Amazon S3 as a backbone
4. Many AWS services use Amazon S3 as an integration as well
5. We’ll have a step-by-step approach to S3

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Reference

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Amazon S3 Use cases

Backup and storage

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- Amazon S3 - Buckets
- Amazon S3 allows people to store objects (files) in “buckets” (directories)
- Buckets must have a globally unique name (across all regions all accounts)
- Buckets are defined at the region level
- S3 looks like a global service but buckets are created in a region
- Naming convention
- No uppercase, No underscore
- 3-63 characters long
- Not an IP
- Must start with lowercase letter or number
- Must NOT start with the prefix xn--
- Must NOT end with the suffix -s3alias
- S3 Bucket

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Amazon S3 - Objects

- Object (files) have a Key
- The key is the FULL path:
- s3://my-bucket/my_file.txt
- s3://my-bucket/my_folder1/another_folder/my_file.txt
- They key is composed of prefix + object name
- s3://my-bucket/my_folder1/another_folder/my_file.txt
- There’s no concept of “directories” within buckets (although the UI will trick you to think otherwise)
- Just keys with very long names that contain slashes (“/”)
- S3 Bucket with Objects
- Object

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Amazon S3 - Objects (cont.)

- Object values are the content of the body:
- Max. Object Size is 5TB (5000GB)
- If uploading more than 5GB, must use “multi-part upload”
- Metadata (list of text key / value pairs – system or user metadata)
- Tags (Unicode key / value pair – up to 10) – useful for security / lifecycle
- Version ID (if versioning is enabled)

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Amazon S3 - Security

- User-Based
- IAM Policies - which API calls should be allowed for a specific user from IAM
- Resource-Based
- Bucket policies - bucket wide rules from the S3 console - allows cross account
- Object Access Control List (ACL) - finer grain (can be disabled)
- Bucket Access Control List (ACL) - less common (can be disabled)
- Note: an IAM principal can access an S3 object if
- The user IAM permissions ALLOW it OR the resource policy ALLOWS it
- AND there’s no explicit DENY
- Encryption: encrypt objects in Amazon S3 using encryption keys

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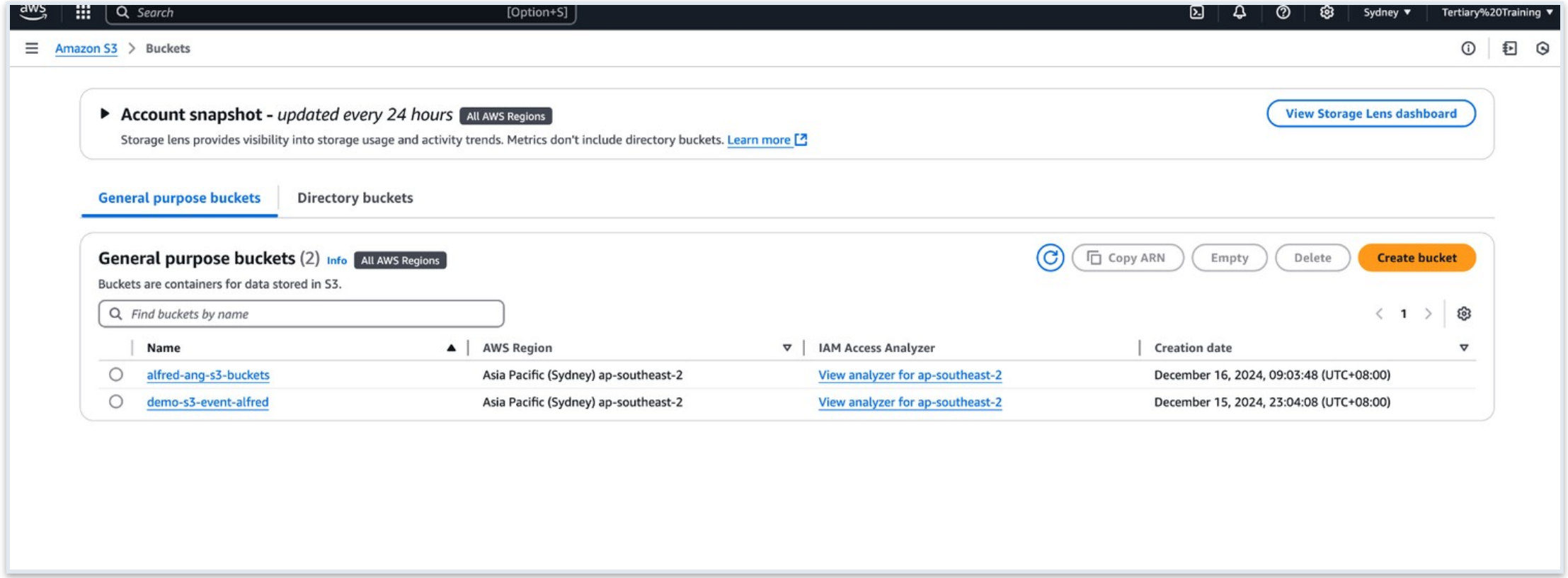
Activity - Amazon S3



Create bucket and upload images to S3

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S3 Bucket Policies

- JSON based policies
- Resources: buckets and objects
- Effect: Allow / Deny
- Actions: Set of API to Allow or Deny
- Principal: The account or user to apply the policy to
- Use S3 bucket for policy to:
- Grant public access to the bucket
- Force objects to be encrypted at upload
- Grant access to another account (Cross Account)

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Example: Public Access - Use Bucket Policy

1. S3 Bucket
2. Anonymous www website visitor
3. S3 Bucket Policy Allows Public Access

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Example: User Access to S3 - IAM permissions

1. S3 Bucket
2. IAM Policy
3. IAM User

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1. Example: EC2 instance access - Use IAM Roles
2. S3 Bucket
3. EC2 Instance Role
4. IAM permissions
5. EC2 Instance

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Advanced: Cross-Account Access - Use Bucket Policy

1. S3 Bucket
2. IAM User
3. Other AWS account
4. S3 Bucket Policy Allows Cross-Account

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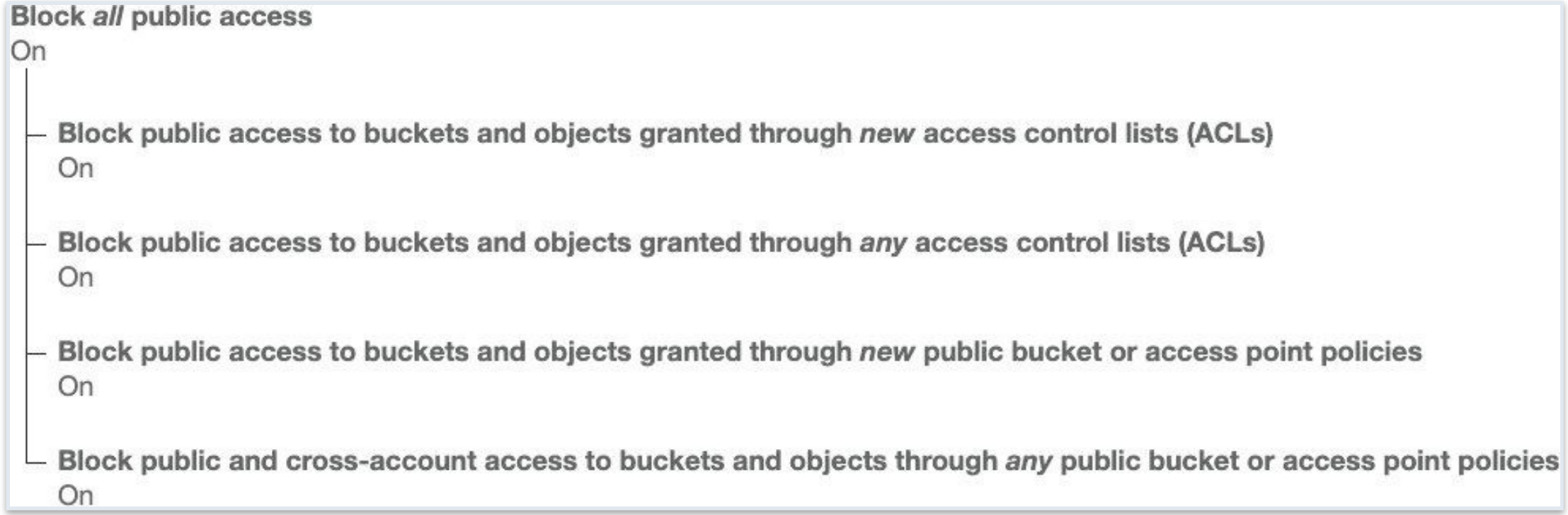
Bucket settings for Block Public Access



These settings were created to prevent company data leaks

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S3 Bucket Policies

- Use S3 bucket for policy to:
- Grant public access to the bucket
- Force objects to be encrypted at upload
- Grant access to another account (Cross account)
- Optional Conditions on:
- Public IP or Elastic IP (not on Private IP)
- Source VPC or Source VPC Endpoint – only works with VPC Endpoints
- CloudFront Origin Identity
- MFA
- Examples here: https://docs.aws.amazon.com/AmazonS3/latest/dev/example- bucket-policies.html

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Bucket Policies Examples

- {
- "Version": "2012-10-17",
- "Id": "Policy1734311494060",
- "Statement": [
- {
- "Sid": "Stmt1734311424063",
- "Effect": "Allow",
- "Principal": "*",
- "Action": "s3:GetObject",
- "Resource": "arn:aws:s3:::alfred-ang-s3-buckets/*"
- }
-]
- }

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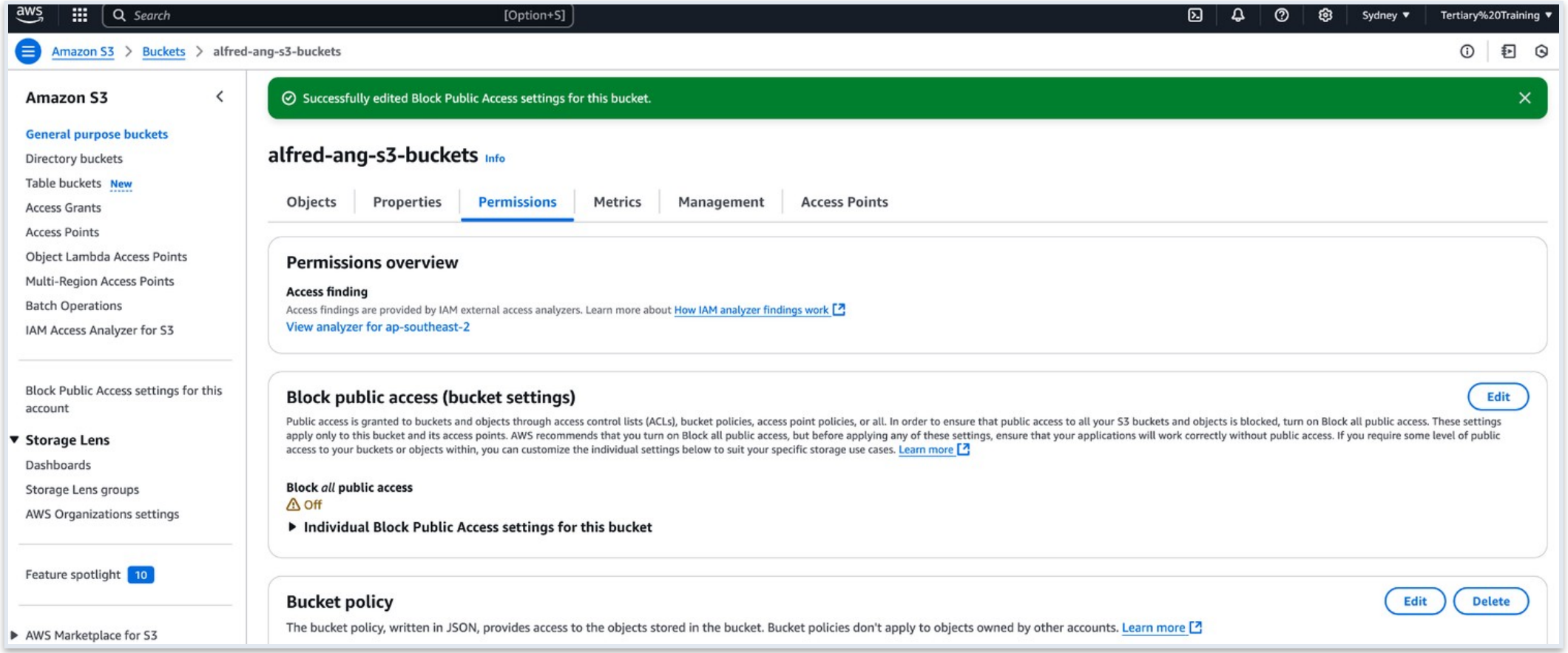
Activity - Amazon S3 Bucket Policies



Allow public to access the images on S3

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Amazon S3 - Static Website Hosting

- S3 can host static websites and have them accessible on the Internet
- http://bucket-name.s3-website-aws-region.amazonaws.com/
- OR
- http://bucket-name.s3-website/
- If you get a 403 Forbidden error, make sure the bucket policy allows public reads!
- S3 Bucket
- (demo-bucket)
- us-west-2
- User
- http://demo-bucket.s3-website-us-west-2.amazonaws.com http://demo-bucket.s3-website.us-west-2.amazonaws.com
- The website URL will be (depending on the region)

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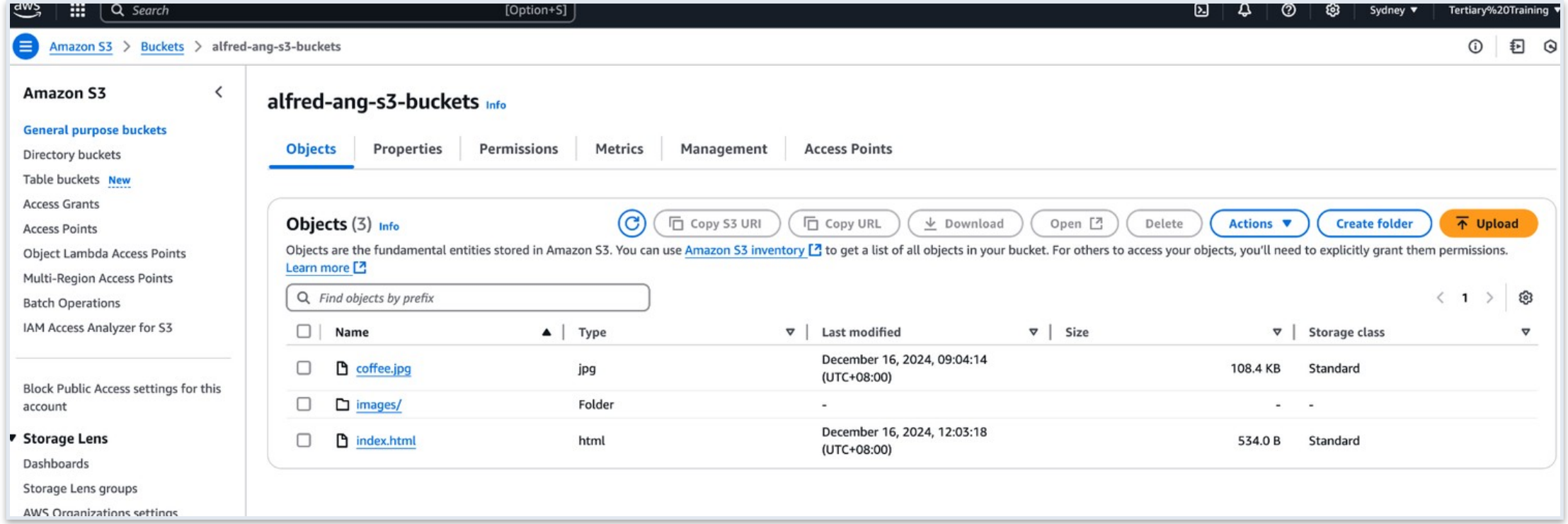
Activity - Website Hosting



Enable Website hosting on S3

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Amazon S3 - Versioning

- You can version your files in Amazon S3
- It is enabled at the bucket level
- Same key overwrite will change the “version”: 1, 2, 3….
- It is best practice to version your buckets
- Protect against unintended deletes (ability to restore a version)
- Easy roll back to previous version
- Notes:
- Any file that is not versioned prior to enabling versioning will have version “null”
- Suspending versioning does not delete the previous versions
- S3 Bucket
- (my-bucket)
- s3://my-bucket/my-file.docx
- User

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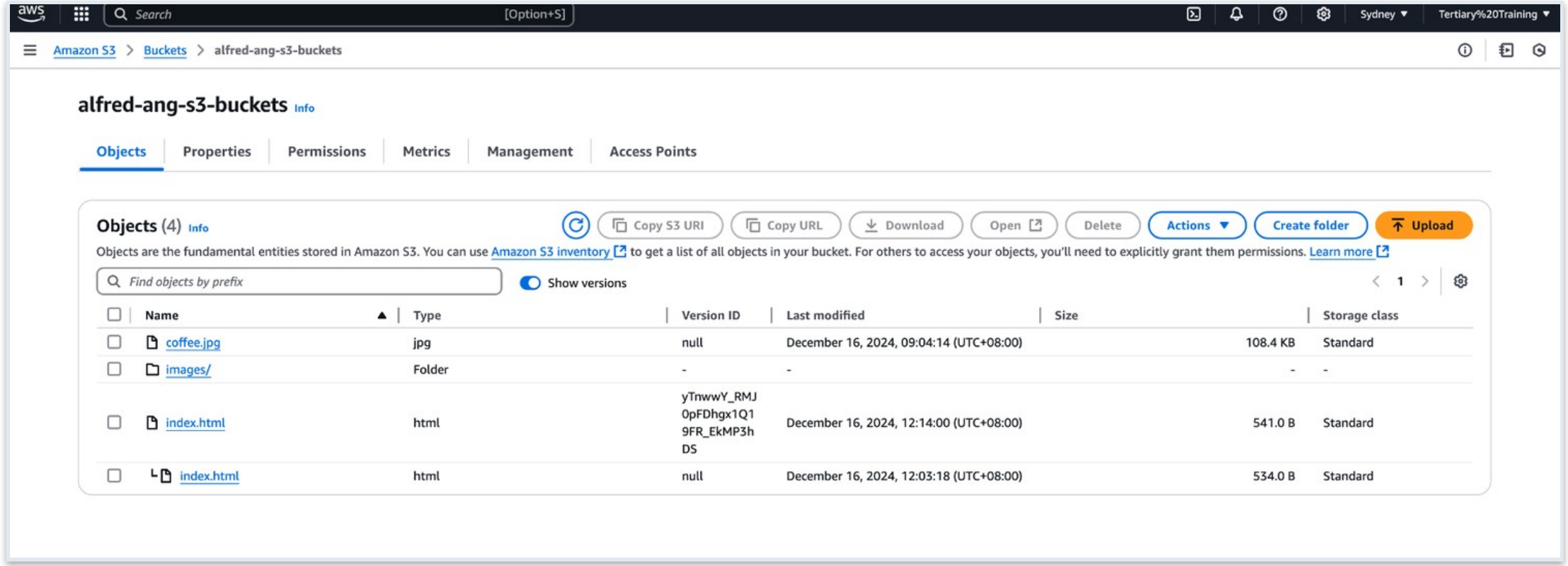
Activity - S3 Versioning



Allow public to access the images on S3

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Amazon S3 - Replication (CRR & SRR)

- Must enable Versioning in source and destination buckets
- Cross-Region Replication (CRR)
- Same-Region Replication (SRR)
- Buckets can be in different AWS accounts
- Copying is asynchronous
- Must give proper IAM permissions to S3
- Use cases:
- CRR - compliance, lower latency access, replication across accounts
- SRR - log aggregation, live replication between production and test accounts
- S3 Bucket
- (eu-west-1)
- S3 Bucket
- (us-east-2)

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Amazon S3 - Replication (Notes)

- After you enable Replication, only new objects are replicated
- Optionally, you can replicate existing objects using S3 Batch Replication
- Replicates existing objects and objects that failed replication
- For DELETE operations
- Can replicate delete markers from source to target (optional setting)
- Deletions with a version ID are not replicated (to avoid malicious deletes)
- There is no "chaining" of replication
- If bucket 1 has replication into bucket 2, which has replication into bucket 3
- Then objects created in bucket 1 are not replicated to bucket 3

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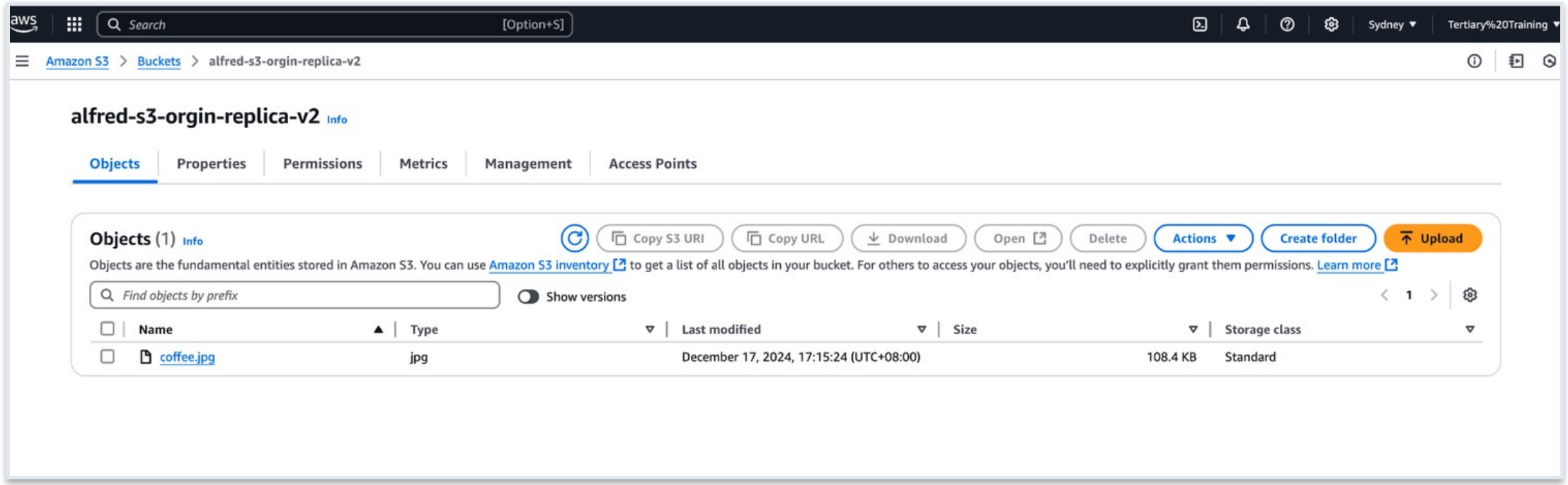
Activity - S3 Replication



Create a Replication Rule between two buckets

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S3 Storage Classes

- Amazon S3 Standard - General Purpose
- Amazon S3 Standard-Infrequent Access (IA)
- Amazon S3 One Zone-Infrequent Access
- Amazon S3 Glacier Instant Retrieval
- Amazon S3 Glacier Flexible Retrieval
- Amazon S3 Glacier Deep Archive
- Amazon S3 Intelligent Tiering
- Can move between classes manually or using S3 Lifecycle configurations

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S3 Durability and Availability

- Durability:
- High durability (99.999999999%, 11 9's) of objects across multiple AZ
- If you store 10,000,000 objects with Amazon S3, you can on average expect to incur a loss of a single object once every 10,000 years
- Same for all storage classes
- Availability:
- Measures how readily available a service is
- Varies depending on storage class
- Example: S3 standard has 99.99% availability = not available 53 minutes a year

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S3 Standard - General Purpose

- 99.99% Availability
- Used for frequently accessed data
- Low latency and high throughput
- Sustain 2 concurrent facility failures
- Use Cases: Big Data analytics, mobile & gaming applications, content distribution…

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S3 Storage Classes - Infrequent Access

- For data that is less frequently accessed, but requires rapid access when needed
- Lower cost than S3 Standard
- Amazon S3 Standard-Infrequent Access (S3 Standard-IA)
- 99.9% Availability
- Use cases: Disaster Recovery, backups
- Amazon S3 One Zone-Infrequent Access (S3 Standard-IA)
- High durability (99.999999999%) in a single AZ; data lost when AZ is destroyed
- 99.5% Availability
- Use Cases: Storing secondary backup copies of on-premises data, or data you can recreate

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Amazon S3 Glacier Storage Classes

- Low-cost object storage meant for archiving / backup
- Pricing: price for storage + object retrieval cost
- Amazon S3 Glacier Instant Retrieval
- Millisecond retrieval, great for data accessed once a quarter
- Minimum storage duration of 90 days
- Amazon S3 Glacier Flexible Retrieval (formerly Amazon S3 Glacier):
- Expedited (1 to 5 minutes), Standard (3 to 5 hours), Bulk (5 to 12 hours) – free
- Minimum storage duration of 90 days
- Amazon S3 Glacier Deep Archive - for long term storage:
- Standard (12 hours), Bulk (48 hours)
- Minimum storage duration of 180 days

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S3 Intelligent-Tiering

- Small monthly monitoring and auto-tiering fee
- Moves objects automatically between Access Tiers based on usage
- There are no retrieval charges in S3 Intelligent-Tiering
- Frequent Access tier (automatic): default tier
- Infrequent Access tier (automatic): objects not accessed for 30 days
- Archive Instant Access tier (automatic): objects not accessed for 90 days
- Archive Access tier (optional): configurable from 90 days to 700+ days
- Deep Archive Access tier (optional): config. from 180 days to 700+ days

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S3 Storage Classes Comparison

- https://aws.amazon.com/s3/storage-classes/

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S3 Storage Classes - Price Comparison Example: us-east-1

- https://aws.amazon.com/s3/pricing/

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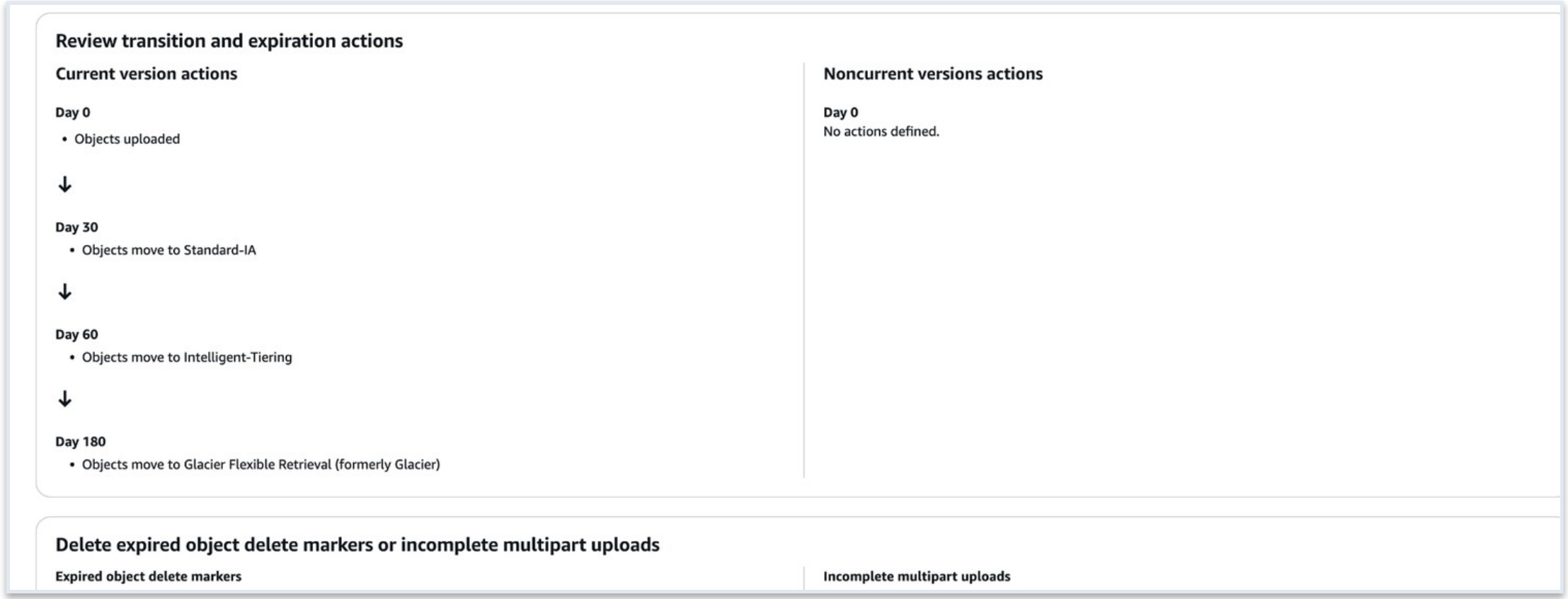
Activity - S3 Life Cycle Rule



Create a S3 life cycle rule as shown below:

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Advanced Amazon S3 and Athena

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Amazon S3 - Moving between Storage Classes

- You can transition objects between storage classes
- For infrequently accessed object, move them to Stanard IA
- For archive objects that you don’t need fast access to, move them to Glacier or Glacier Deep Archive
- Moving objects can be automated using a Lifecycle Rules
- Standard
- Standard IA
- Intelligent Tiering
- One-Zone IA
- Glacier Instant Retrieval
- Glacier Flexible Retrieval
- Glacier Deep Archive

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Amazon S3 - Lifecycle Rules

- Transition Actions - configure objects to transition to another storage class
- Move objects to Standard IA class 60 days after creation
- Move to Glacier for archiving after 6 months
- Expiration Actions - configure objects to expire (delete) after some time
- Access log files can be set to delete after a 365 days
- Can be used to delete old versions of files (if versioning is enabled)
- Can be used to delete incomplete Multi-Part uploads
- Rules can be created for a certain prefix (example: s3://mybucket/mp3/*)
- Rules can be created for certain objects Tags (example: Department: Finance)

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Amazon S3 - Lifecycle Rules (Scenario 1)

- Your application on EC2 creates images thumbnails after profile
- photos are uploaded to Amazon S3. These thumbnails can be easily recreated, and only need to be kept for 60 days. The source images should be able to be immediately retrieved for these 60 days, and afterwards, the user can wait up to 6 hours. How would you design this?
- S3 source images can be on Standard with a lifecycle configuration to trainsition them to Glacier after 60 days
- S3 thumbnails can be on One-Zone IA with a lifecycle configuration to expire them (delete them) after 60 days

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Amazon S3 - Lifecycle Rules (Scenario 2)

- A rule in your company states that you should be able to recover your deleted S3 objects immediately for 30 days, although this may happen rarely. After this time, and for up to 365 days, deleted objects should
- be recoverable within 48 hours.
- Enable S3 Versioning in order to have object versions, so that “deleted objects” are in fact hidden by a “delete marker” and can be recovered
- Transition the “noncurrent versions” of the object to Standard IA
- Transition afterwards the “noncurrent versions” to Glacier Deep Archive

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Amazon S3 Analytics - Storage Class Analysis

- Help you decide when to transition objects to the right storage class
- Recommendations for Standard and Stanard IA
- Does NOT work for One-Zone IA or Glacier
- Report is updated daily
- 24 to 48 hours to start seeing data analysis
- Good first step to put together Lifecycle Rules (or improve them)!
- S3 Bucket
- S3 Analytics
- .csv report

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S3 Event Notifications

- Amazon S3
- events
- Lambda Function
- SQS
- SNS
- S3:ObjectCreated, S3:ObjectRemoved, S3:ObjectRestore, S3:Replication…
- Object name filtering possible (*.jpg)
- Use case: generate thumbnails of images uploaded to S3
- Can create as many "S3 events" as desired
- S3 event notifications typically deliver events in seconds but can sometimes take a minute or longer

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S3 Event Notifications - IAM Permissions

```
{
  "Version": "2012-10-17",
  "Statement": {
    "Effect": "Allow",
    "Action": "SQS:SendMessage",
    "Principal": {
      "Service": "s3.amazonaws.com"
    },
    "Resource": "arn:aws:sqs:us-east-1:123456789012:MyQueue",
    "Condition": {
      "ArnLike": {
        "aws:SourceArn": "arn:aws:s3:::MyBucket"
      }
    }
  }
}
```

Lambda Resource Policy

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S3 Event Notifications with Amazon EventBridge

- Amazon S3 bucket
- events
- All events
- Amazon EventBridge
- rules
- Over 18
- AWS services
- as destinations
- Advanced filtering options with JSON rules (metadata, object size, name...)
- Multiple Destinations - ex Step Functions, Kinesis Streams / Firehose…
- EventBridge Capabilties - Archive, Replay Events, Reliable delivery

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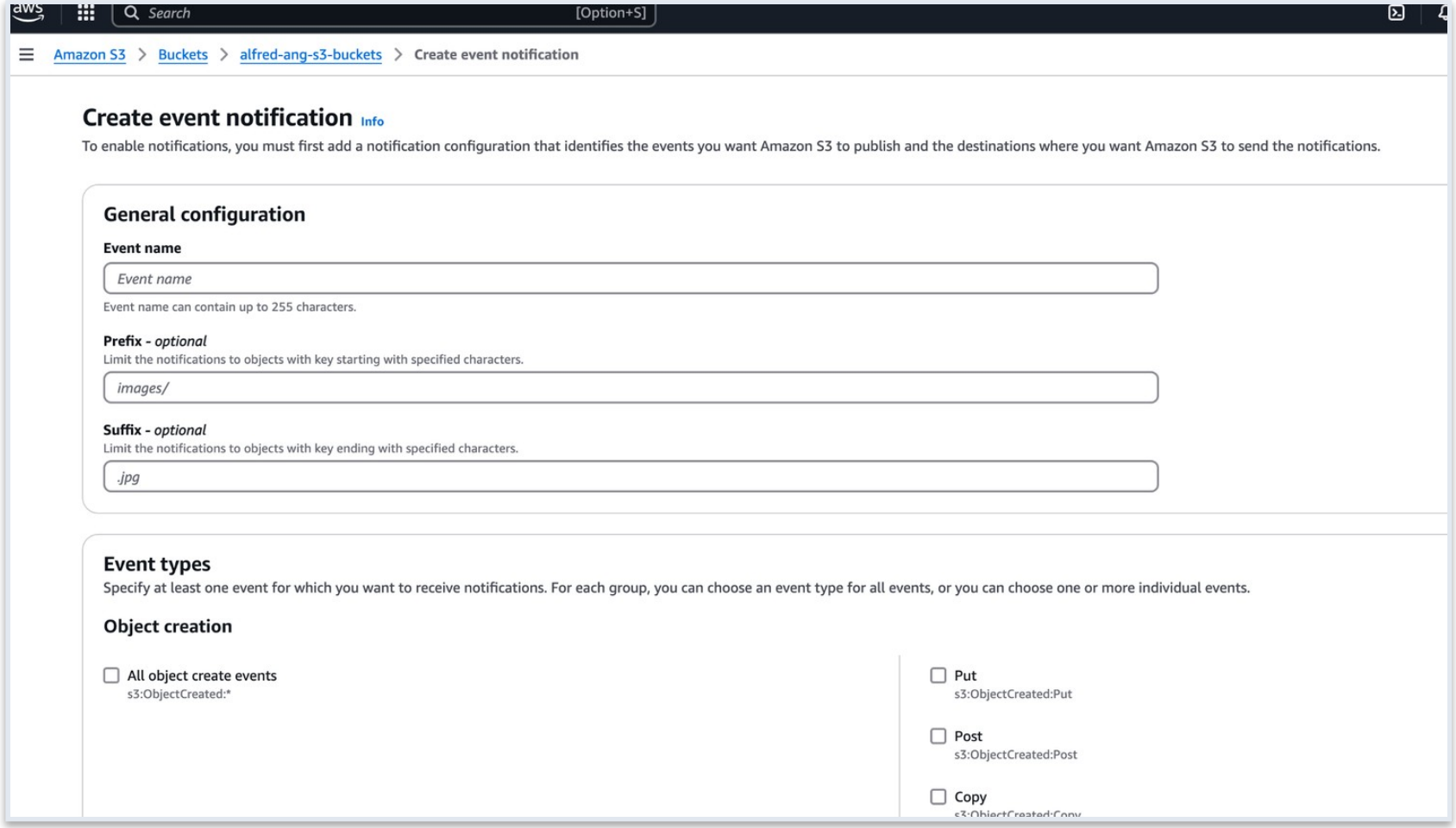
Activity: Event Notification



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S3 - Baseline Performance

- Amazon S3 automatically scales to high request rates, latency 100-200 ms
- Your application can achieve at least 3,500 PUT/COPY/POST/DELETE or 5,500 GET/HEAD requests per second per prefix in a bucket.
- There are no limits to the number of prefixes in a bucket.
- Example (object path => prefix):
- bucket/folder1/sub1/file => /folder1/sub1/
- bucket/folder1/sub2/file => /folder1/sub2/
- bucket/1/file => /1/
- bucket/2/file => /2/
- If you spread reads across all four prefixes evenly, you can achieve 22,000 requests per second for GET and HEAD

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S3 Performance

- Multi-Part upload:
- recommended for files > 100MB, must use for files > 5GB
- Can help parallelize uploads (speed up transfers)
- S3 Transfer Acceleration
- Increase transfer speed by transferring file to an AWS edge location which will forward the data to the S3 bucket in the target region
- Compatible with multi-part upload
- Amazon S3
- Parallel uploads
- Divide In parts
- BIG file
- S3 Bucket
- Australia

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S3 Performance - S3 Byte-Range Fetches

- Can be used to speed up downloads
- Parallelize GETs by requesting specific byte ranges
- Better resilience in case of failures
- Can be used to retrieve only partial
- File in S3
- Byte-range request for header (first XX bytes)
- header
- File in S3
- Part 1
- Part 2
- Part N
- …
- data (for example the head of a file)

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S3 Batch Operations

- Perform bulk operations on existing S3 objects with a single request, example:
- Modify object metadata & properties
- Copy objects between S3 buckets
- Encrypt un-encrypted objects
- Modify ACLs, tags
- Restore objects from S3 Glacier
- Invoke Lambda function to perform custom action on each object
- A job consists of a list of objects, the action to perform, and optional parameters
- S3 Batch Operations manages retries, tracks progress, sends completion notifications, generate reports …
- You can use S3 Inventory to get object list to use
- S3 Select to filter your objects
- S3 Inventory

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S3 Inventory

- List objects and their corresponding metadata (alternative to S3 List API operation)
- Usage examples:
- Audit and report on the replication and encryption status of your objects
- Get the number of objects in an S3 bucket
- Identify the total storage of previous object versions
- Generate daily or weekly reports
- Output files: CSV, ORC, or Apache Parquet
- You can query all the data using Amazon Athena, Redshift, Presto, Hive, Spark…
- You can filter generated report using S3 Select
- Use cases: Business, Compliance, Regulatory needs, …

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Amazon S3 Glacier

- Low-cost object storage meant for archiving / backup
- Data is retained for the longer term (10s of years)
- Alternative to on-premises magnetic tape storage
- Average annual durability is 99.999999999%
- Cost per storage per month ($0.004 / GB – Standard | $0.00099 / GB Deep Archive)
- Each item in Glacier is called "Archive" (up to 40TB)
- Archives are stored in "Vaults"
- By default, data encrypted at rest using AES-256 – keys managed by AWS
- Exam tip: archival from S3 after XXX days => use Glacier

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Amazon S3 Glacier Operations

- Vault Operations:
- Create & Delete - delete only when there’s no archives in it
- Retrieving Metadata - creation date, number of archives, total size of all archives, …
- Download Inventory - list of archives in the vault (archive ID, creation date, size, …)
- Glacier Operations:
- Upload- single operation or by parts (MultiPart upload) for larger archives
- Download - first initiate a retrieval job for the archive, Glacier then prepares it for download.
- User then has a limited time to download the data from staging server. (optionally, specify a range or portion of bytes to retrieve)
- Delete - use Glacier Rest API or AWS SDKs by specifying archive ID
- Restore links have an expiry date
- Retrieval Options:
- Expedited (1 to 5 minutes retrieval) - $0.03 per GB and $10 per 1000 requests

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Amazon S3 Glacier - Vault Policies & Vault Lock

- Each Vault has:
- ONE vault access policy
- ONE vault lock policy
- Vault Policies are written in JSON
- Vault Access Policy is like a bucket policy (restrict user / account permissions)
- Vault Lock Policy is a policy you lock, for regulatory and compliance requirements.
- The policy is immutable, it can never be changed (that's why it’s called a LOCK)
- Example 1: forbid deleting an archive if less than 1 year old
- Example 2: implement WORM policy (write once read many)
- Users
- Permissions
- Vault Access Policy
- Vault

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Glacier - Notifications for Restore Operations

- Vault Notification Configuration
- Configure a vault so that when a job completes a message is sent to SNS
- Optionally, specify an SNS topic when you initiate a job
- S3 Event Notifications
- S3 supports the restoration of objects archived to S3 Glacier storage classes
- s3:ObjectRestore:Post => notify when object restoration initiated
- s3:ObjectRestore:Completed => notify when object restoration completed
- Vault
- Archives
- initiate restore archive job
- SNS
- Amazon Glacier
- archive ready restore

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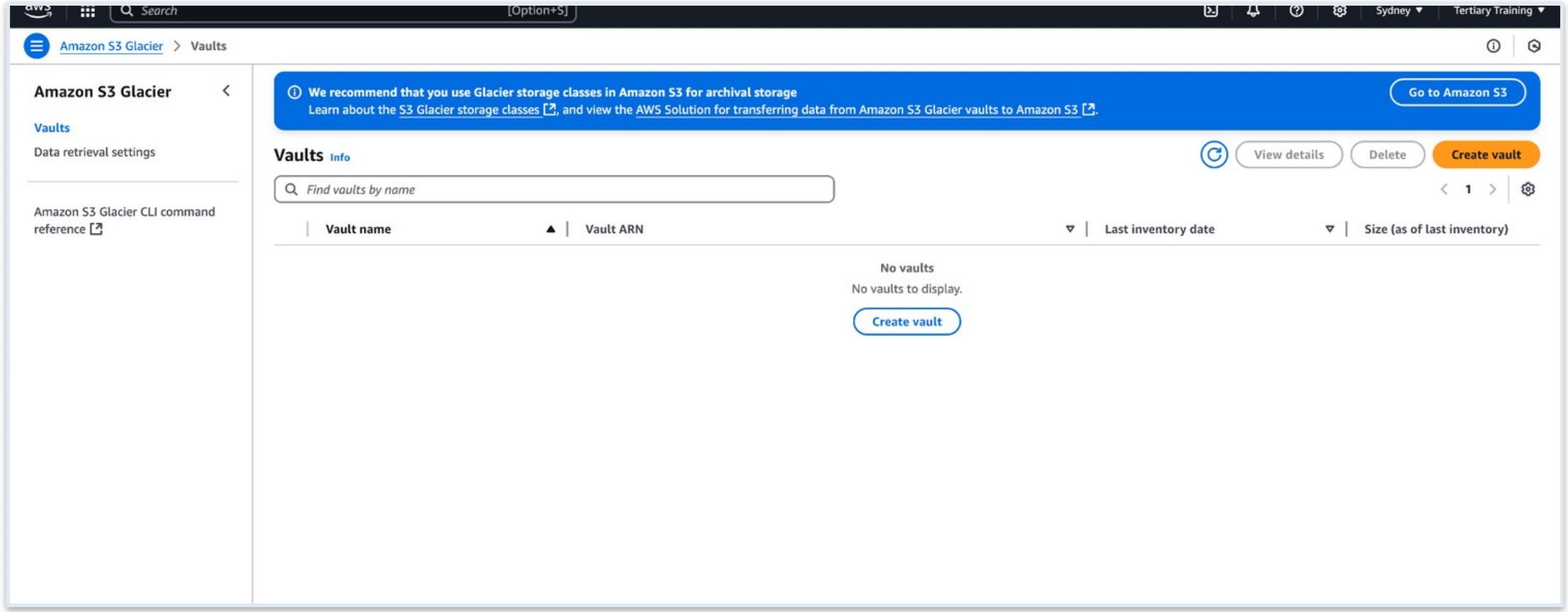
Activity: S3 Glacier



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TOPIC 9

S3 Multi Part Upload - Deep Dive

- Upload large objects in parts (any order)
- Recommended for files > 100MB, must use for files > 5GB
- Can help parallelize uploads (speed up transfers)
- Max. parts: 10,000
- Failures: restart uploading ONLY failed parts (improves performance)
- Use Lifecycle policy to automate old parts
- Deletion of unfinished upload after x days (e.g., network outage)
- Upload using AWS CLI or AWS SDK
- BIG file
- …
- parts
- (max. 10,000)
- parallel uploads

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Amazon Athena

- Serverless query service to analyze data stored in Amazon S3
- Uses standard SQL language to query the files (built on Presto)
- Supports CSV, JSON, ORC, Avro, and Parquet
- Pricing: $5.00 per TB of data scanned
- Commonly used with Amazon Quicksight for reporting/dashboards
- Use cases: Business intelligence / analytics / reporting, analyze & query VPC Flow Logs, ELB Logs, CloudTrail trails, etc...
- Exam Tip: analyze data in S3 using serverless SQL, use Athena
- S3 Bucket
- load data
- Amazon Athena
- Query & Analyze
- Amazon QuickSight

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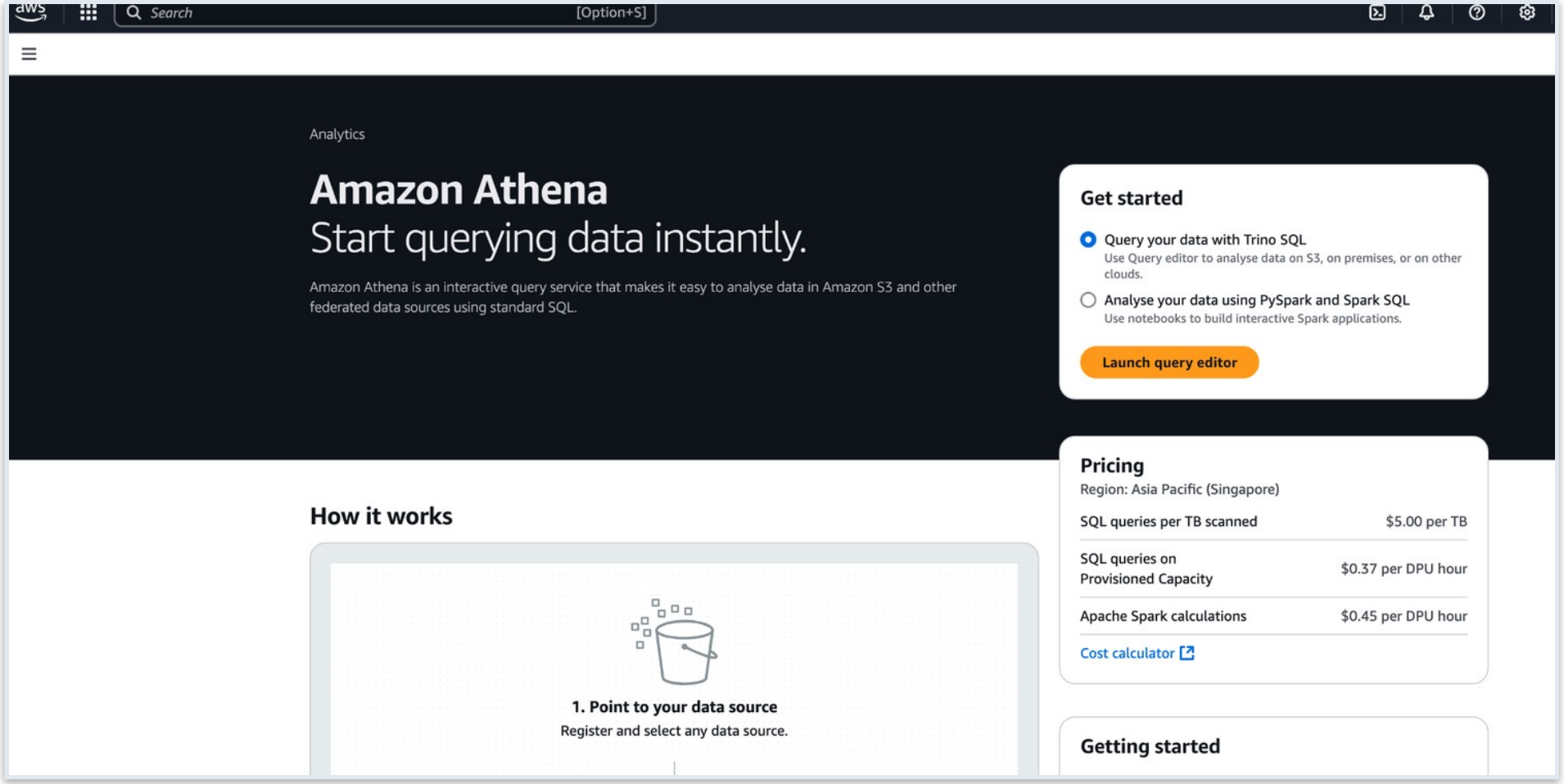
TOPIC 9

Amazon Anthena



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Amazon Athena - Performance Improvement

- Use Columnar data for cost-savings (less scan)
- Apache Parquet or ORC is recommended
- Huge performance improvement
- Use Glue to convert your data to Parquet or ORC
- Compress data for smaller retrievals (bzip2, gzip, lz4, snappy, zlip, zstd…)
- Partition datasets in S3 for easy querying on virtual columns
- s3://yourBucket/pathToTable
- /<PARTITION_COLUMN_NAME>=<VALUE>
- /<PARTITION_COLUMN_NAME>=<VALUE>
- /<PARTITION_COLUMN_NAME>=<VALUE>
- /etc…
- Example: s3://athena-examples/flight/parquet/year=1991/month=1/day=1/
- Use larger files (> 128 MB) to minimize overhead

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Amazon Athena - Federated Query

- Allows you to run SQL queries across data stored in relational, non-relational, object, and custom data sources (AWS or on-premises)
- Uses Data Source Connectors that run on AWS Lambda to run Federated Queries (e.g., CloudWatch Logs, DynamoDB, RDS, …)
- Store the results back in Amazon S3
- Amazon Athena
- Database (On-Premises)
- S3 Bucket
- Lambda
- (Data Source
- Connector)
- ElastiCache
- DocumentDB

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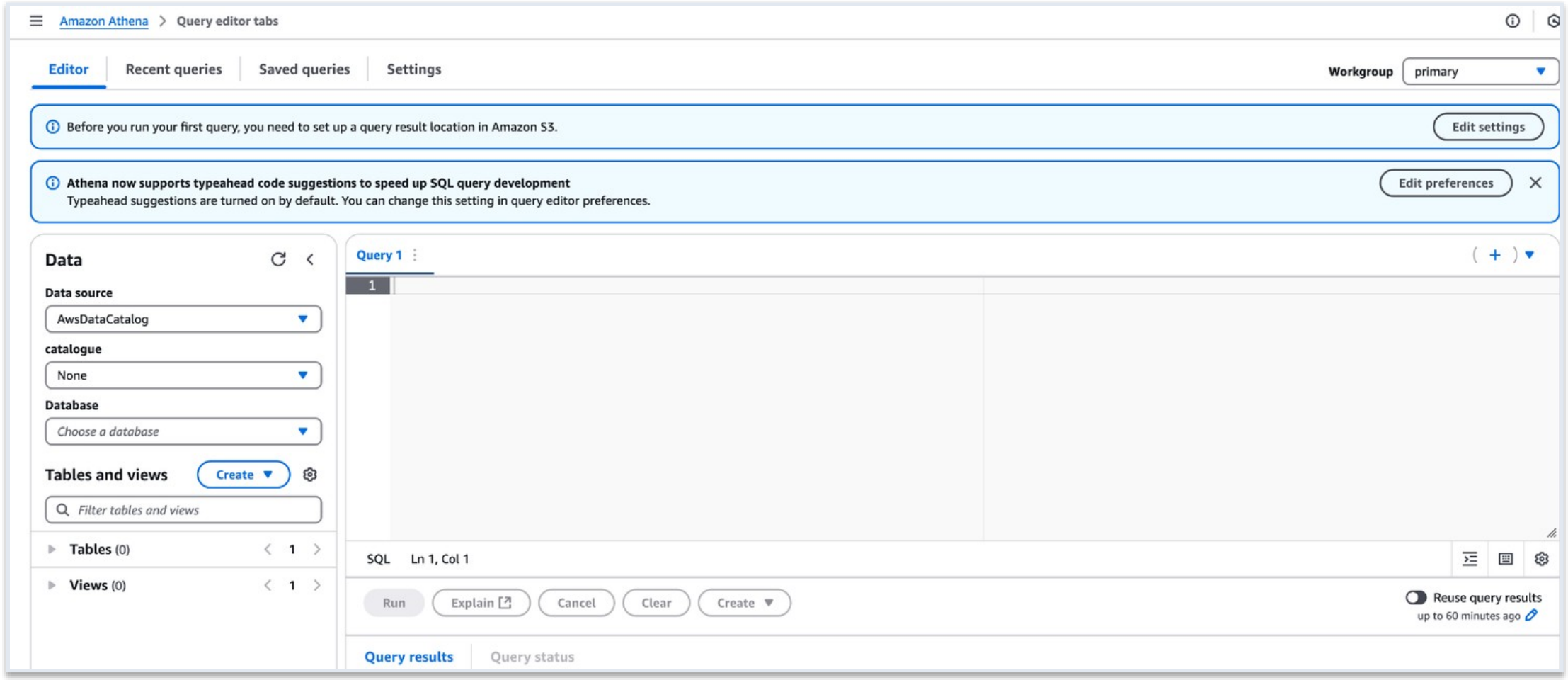
Activity: Athena



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AMAZON S3 - SECURITY

Amazon S3 - Security

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Amazon S3 - Object Encryption

- You can encrypt objects in S3 buckets using one of 4 methods
- Server-Side Encryption (SSE)
- Server-Side Encryption with Amazon S3-Managed Keys (SSE-S3) Enabled by Default
- Encrypts S3 objects using keys handled, managed, and owned by AWS
- Server-Side Encryption with KMS Keys stored in AWS KMS (SSE-KMS)
- Leverage AWS Key Management Service (AWS KMS) to manage encryption keys
- Server-Side Encryption with Customer-Provided Keys (SSE-C)
- When you want to manage your own encryption keys
- Cilent-Side Encryption
- It’s important to understand which ones are for which situation for the exam

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Amazon S3 Encryption - SSE-S3

- User
- upload
- HTTP(S) + Header
- S3 Bucket
- +
- Encryption using keys handled, managed, and owned by AWS
- Object is encrypted server-side
- Encryption type is AES-256
- Must set header "x-amz-server-side-encryption":"AES256"
- Enabled by default for new buckets & new objects
- S3 Owned Key
- Encryption
- Amazon S3

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Amazon S3 Encryption - SSE-KMS

- User
- upload
- HTTP(S) + Header
- S3 Bucket
- +
- Encryption using keys handled and managed by AWS KMS (Key Management Service)
- KMS advantages: user control + audit key usage using CloudTrail
- Object is encrypted server side
- Must set header "x-amz-server-side-encryption":"AWS:KMS"
- KMS Key
- AWS KMS
- Encryption
- Amazon S3

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SSE-KMS Limitation

- If you use SSE-KMS, you may be impacted by the KMS limits
- When you upload, it calls the
- GenerateDataKey KMS API
- When you download, it calls the Decrypt KMS API
- Count towards the KMS quota per second (5500, 10000, 30000 req/s based on region)
- You can request a quota increase using the Service Quotas Console
- KMS Key
- S3 Bucket
- Users
- Upload / download SSE-KMS
- API call

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Amazon S3 Encryption - SSE-C

- Server-Side Encryption using keys fully managed by the customer outside of AWS
- Amazon S3 does NOT store the encryption key you provide
- HTTPS must be used
- Encryption key must provided in HTTP headers, for every HTTP request made
- User
- HTTPS ONLY
- + Key in Header
- S3 Bucket
- +
- Client-Provided Key
- +
- upload
- Encryption

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Amazon S3 Encryption - Client-Side Encryption

- Amazon S3
- S3 Bucket
- Use client libraries such as Amazon S3 Cilent-Side Encryption Library
- Clients must encrypt data themselves before sending to Amazon S3
- Clients must decrypt data themselves when retrieving from Amazon S3
- Customer fully manages the keys and encryption cycle
- +
- File
- Client Key
- File
- (encrypted)
- upload
- HTTP(S)

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Amazon S3 - Encryption in transit (SSL/TLS)

- Encryption in flight is also called SSL/TLS
- Amazon S3 exposes two endpoints:
- HTTP Endpoint - non encrypted
- HTTPS Endpoint - encryption in flight
- HTTPS is recommended
- HTTPS is mandatory for SSE-C
- Most clients would use the HTTPS endpoint by default

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Amazon S3 – Force Encryption in Transit aws:SecureTransport

- Account B
- S3 Bucket
- (my-bucket)
- Bucket Policy
- User
- User
- http
- https

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Amazon S3 - Default Encryption vs. Bucket Policies

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Action": "s3:PutObject",
            "Principal": "*",
            "Resource": "arn:aws:s3:::my-bucket/*",
            "Condition": {
                "StringNotEquals": {
                    "s3:x-amz-server-side-encryption": "aws:kms"
                }
            }
        }
    ]
}
```

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Action": "s3:PutObject",
            "Principal": "*",
            "Resource": "arn:aws:s3:::my-bucket/*",
            "Condition": {
                "Null": {
                    "s3:x-amz-server-side-encryption-customer-algorithm": "true"
                }
            }
        }
    ]
}
```

SSE-S3 encryption is automatically applied to new objects stored in S3 bucket

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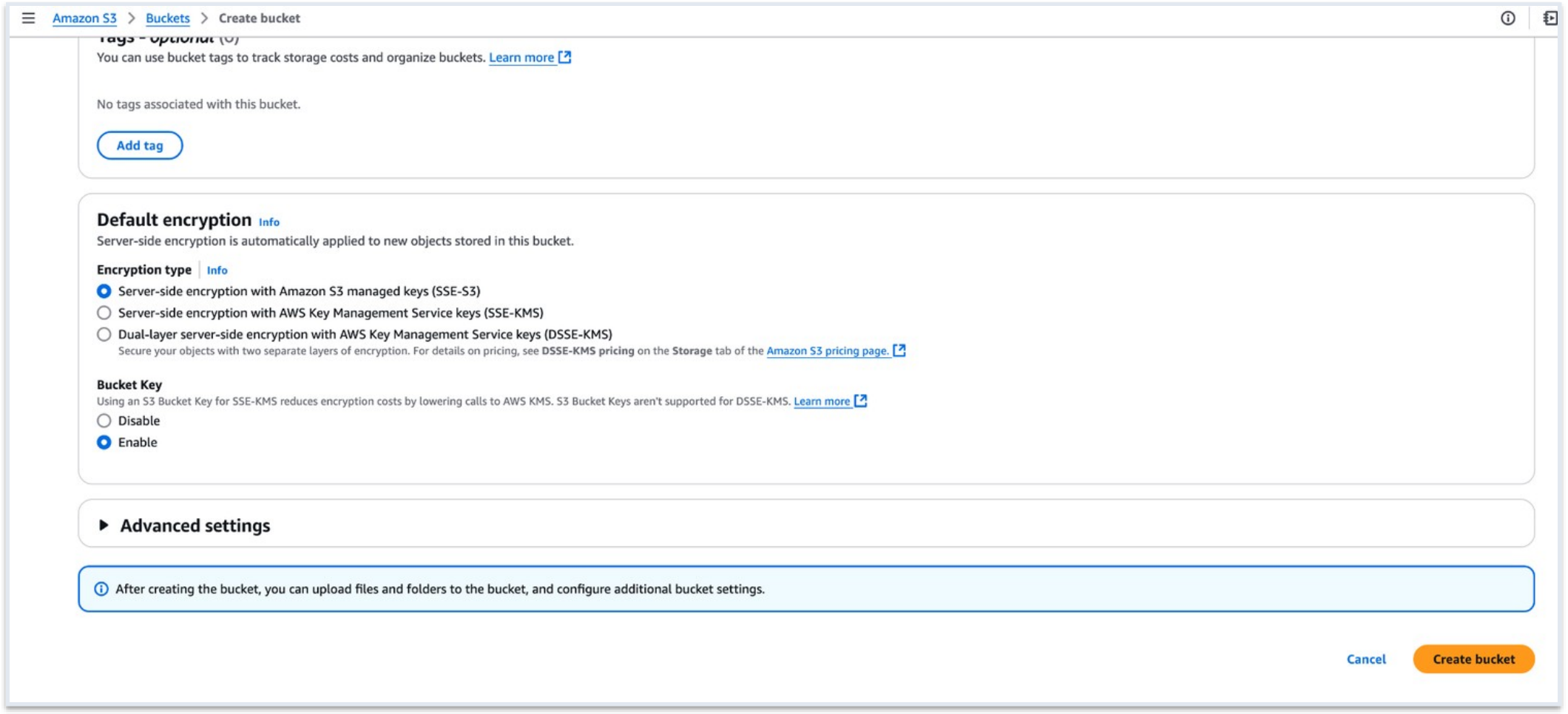
Activity: S3 Encryption



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What is CORS?

- Cross-Origin Resource Sharing (CORS)
- Origin = scheme (protocol) + host (domain) + port
- example: https://www.example.com (implied port is 443 for HTTPS, 80 for HTTP)
- Web Browser based mechanism to allow requests to other origins while visiting the main origin
- Same origin: http://example/app1 & http://example/app2
- Different origins: http://example.com & http://other.example/
- The requests won't be fulfilled unless the other origin allows for the requests, using CORS Headers (example: Access-Control-Allow-Origin)

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What is CORS?

- Web Server
- (Origin) https://www.example.com
- Web Server
- (Cross-Origin) https://www.other.com
- Web Browser
- HTTPS Request
- OPTIONS /
- Host: www.other.com
- Origin: https://www.example.com
- Preflight Request
- Access-Control-Allow-Origin: https://www.example.com Access-Control-Allow-Methods: GET, PUT, DELETE
- Preflight Response

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Amazon S3 - CORS

- If a client makes a cross-origin request on our S3 bucket, we need to enable the correct CORS headers
- It’s a popular exam question
- You can allow for a specific origin or for * (all origins)
- Web Browser
- S3 Bucket
- (my-bucket-html)
- (Static Website Enabled)
- S3 Bucket
- (my-bucket-assets)
- (Static Website Enabled)
- GET /index.html
- Host: http://my-bucket-html.s3-website.us-west-2.amazonaws.com

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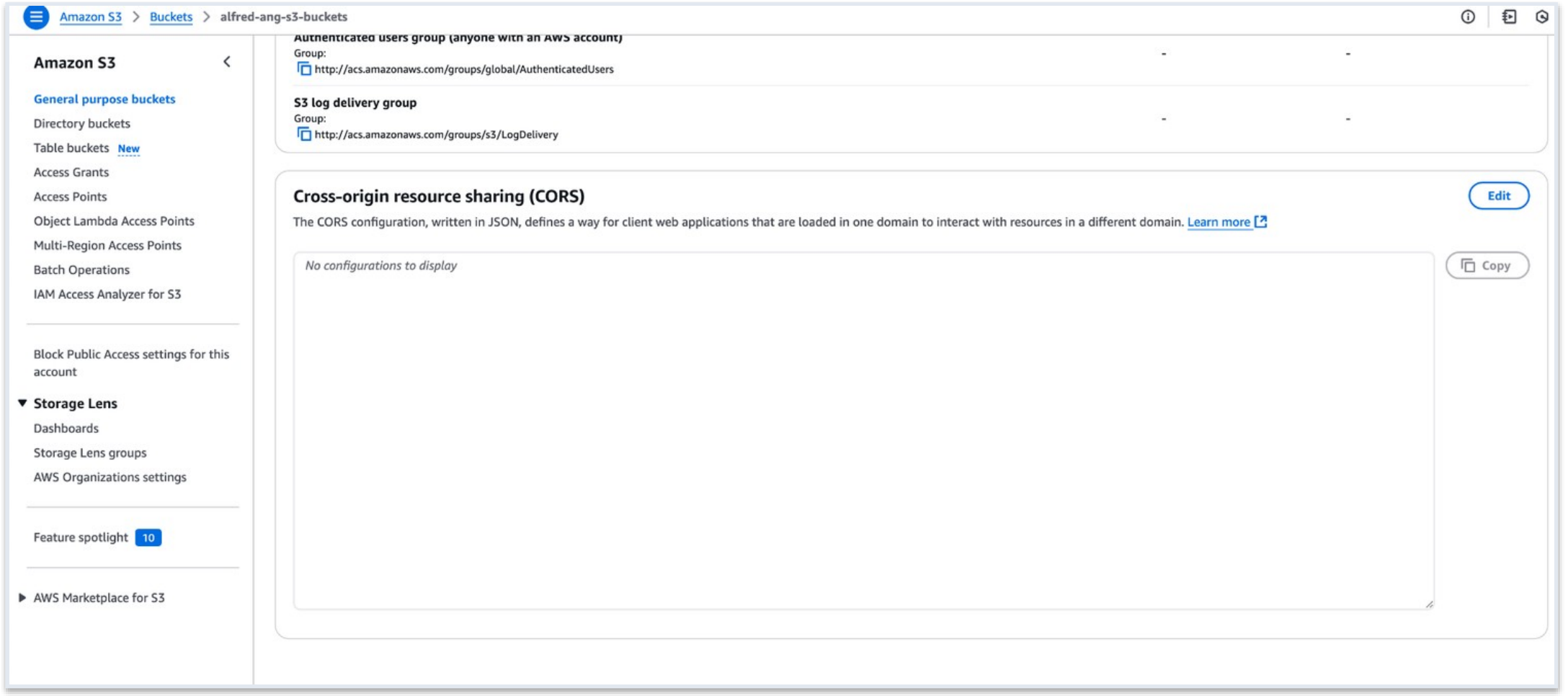
Activity: CORS



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Amazon S3 - MFA Delete

- MFA (Multi-Factor Authentication) - force users to generate a code on a device (usually a mobile phone or hardware) before doing important operations on S3
- MFA will be required to:
- Permanently delete an object version
- Suspend Versioning on the bucket
- MFA won’t be required to:
- Enable Versioning
- List deleted versions
- To use MFA Delete, Versioning must be enabled on the bucket
- Only the bucket owner (root account) can enable/disable MFA Delete
- Google Authenticator
- MFA Hardware Device

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S3 Access Logs

- For audit purpose, you may want to log all access to S3 buckets
- Any request made to S3, from any account, authorized or denied will be logged into another S3 bucket
- That data can be analyzed using data analysis tools…
- The target logging bucket must be in the same AWS region
- The log format is at: https://docs.aws.amazon.com/AmazonS3/latest/dev/LogFormat.html
- My-bucket
- requests
- Log all requests
- Logging Bucket

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S3 Access Logs: Warning

1. Do not set your logging bucket to be the monitored bucket
2. It will create a logging loop, and your bucket will grow exponentially
3. App Bucket & Logging Bucket
4. Logging loop
5. PutObject
6. Do not try this at home ©

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Amazon S3 - Pre-Signed URLs

- Generate pre-signed URLs using the S3 Console, AWS CLI or SDK
- URL Expiration
- S3 Console - 1 min up to 720 mins (12 hours)
- AWS CLI - configure expiration with --expires-in parameter in seconds (default 3600 secs, max. 604800 secs ~ 168 hours)
- Users given a pre-signed URL inherit the permissions of the user that generated the URL for GET / PUT
- Examples:
- Allow only logged-in users to download a premium video from your S3 bucket
- Allow an ever-changing list of users to download files by generating URLs dynamically
- Allow temporarily a user to upload a file to a precise location in your S3 bucket
- S3 Bucket
- (Private)

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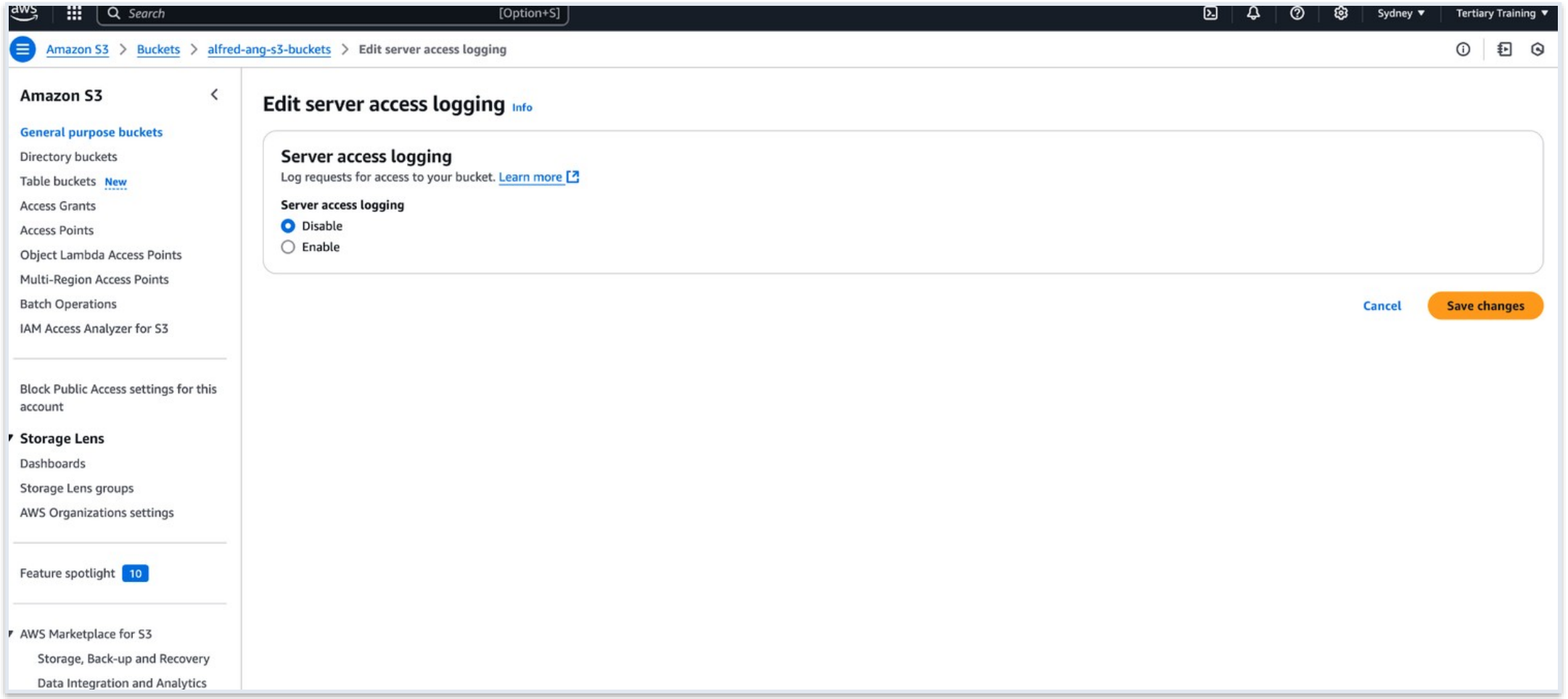
Activity: Access Log



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S3 Glacier Vault Lock

- Adopt a WORM (Write Once Read Many) model
- Create a Vault Lock Policy
- Lock the policy for future edits
- (can no longer be changed or deleted)
- Helpful for compliance and data retention
- Object
- Vault Lock Policy
- Object can’t be deleted

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S3 Object Lock (versioning must be enabled)

- Adopt a WORM (Write Once Read Many) model
- Block an object version deletion for a specified amount of time
- Retention mode - Compliance:
- Object versions can't be overwritten or deleted by any user, including the root user
- Objects retention modes can't be changed, and retention periods can't be shortened
- Retention mode - Governance:
- Most users can't overwrite or delete an object version or alter its lock settings
- Some users have special permissions to change the retention or delete the object
- Retention Period: protect the object for a fixed period, it can be extended
- Legal Hold:
- protect the object indefinitely, independent from retention period
- can be freely placed and removed using the s3:PutObjectLegalHold IAM permission

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S3 - Access Points

- S3 Bucket
- /finance/…
- /sales/…
- Simple Bucket Policy
- Finance Access Point
- Sales Access Point
- Analytics Access Point
- Policy
- Grant R/W to
- /finance prefix
- Policy
- Grant R/W to
- /sales prefix

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S3 - Access Points - VPC Origin

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Principal": "*",
            "Action": [
                "s3:GetObject"
            ],
            "Effect": "Allow",
            "Resource": [
                "arn:aws:s3:::awsexamplebucket1/*",
                "arn:aws:s3:us-west-2:123456789012:accesspoint/example-vpc-ap/object/*"
            ]
        }
    ]
}
```

We can define the access point to be accessible only from within the VPC

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S3 - Multi-Region Access Points

- Provide a global endpoint that span S3 buckets in multiple AWS regions
- Dynamically route requests to the nearest S3 bucket (lowest latency)
- Bi-directional S3 bucket replication rules are created to keep data in sync across regions
- Failover Controls - allows you to shift requests across S3 buckets in different AWS regions within minutes (Active-Active or Active-Passive)
- S3 Multi-Region Access Point
- Bi-directional replication

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S3 - Multi-Region Access Points

- S3 Multi-Region Access Point
- us-east-1
- S3 Bucket
- eu-west-1
- S3 Bucket
- ap-southeast-1
- S3 Bucket
- Bi-directional Cross-Region Replication
- Application
- request S3 object
- request routed to AWS region with the lowest latency
- Bi-directional Cross-Region Replication
- Bi-directional Cross-Region Replication

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Multi-Region Access Points - Failover Controls

- S3 Multi-Region Access Point
- us-east-1
- Selected Active S3 Bucket
- eu-west-1
- Selected Passive S3 Bucket
- Application
- request S3 object
- Failover initiated
- Active/Passive Failover Controls
- Regional traffic disruption
- Works with active/active or active/passive setups
- Bi-directional Cross-Region Replication

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VPC Endpoint Gateway for S3

- VPC Endpoint Gateway
- Private Instance
- Public Instance
- S3 Bucket
- Bucket policy by AWS:SourceIP (public IP)
- Public www
- VPC
- S3 Bucket Bucket policy by
- AWS:SourceVpce
- (one or few endpoints)
- OR
- AWS:SourceVpc
- (encompass all possible VPC endpoints)

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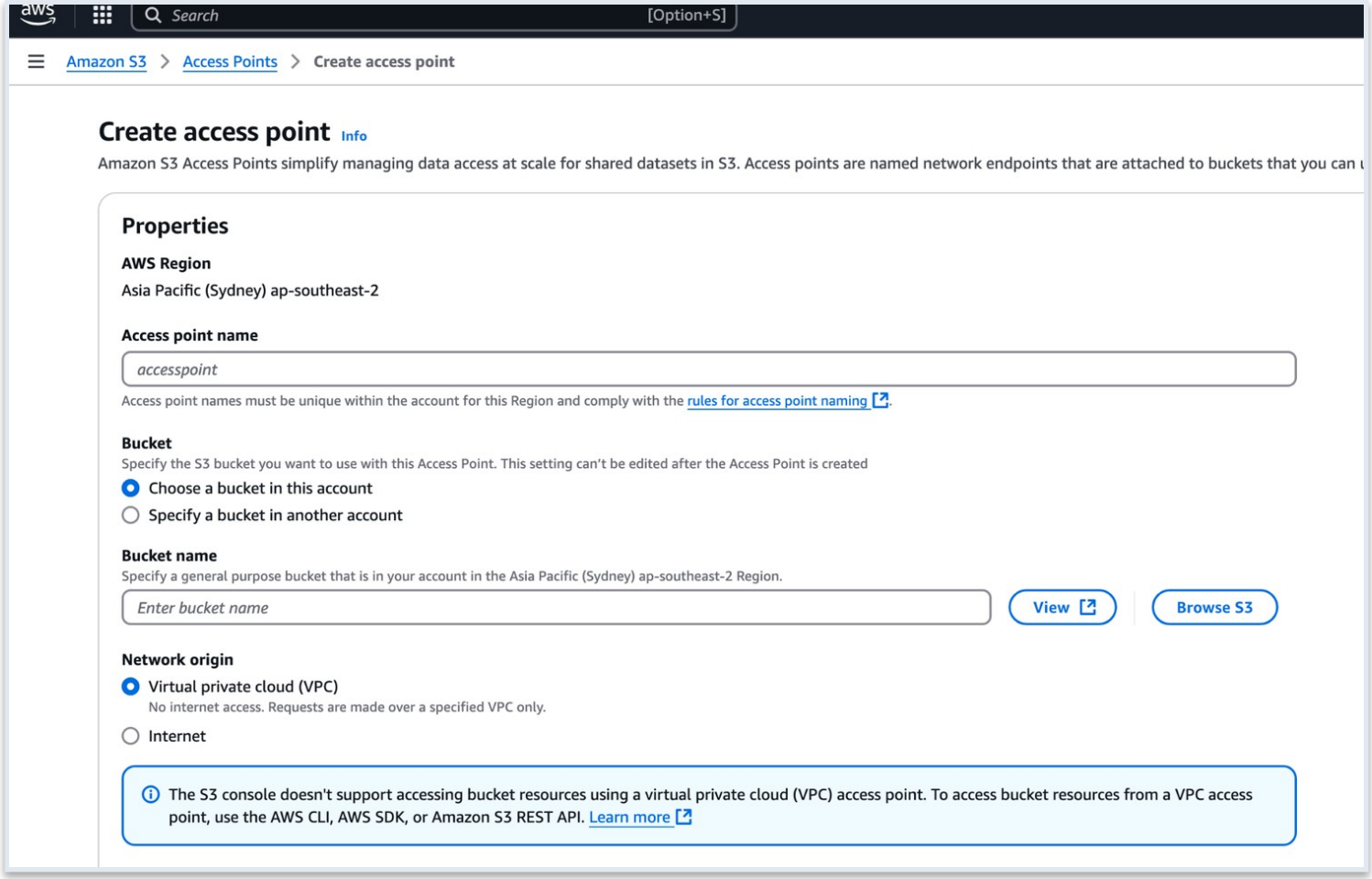
Activity: Access Point



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TOPIC 10

Advanced Storage Solutions

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TOPIC 10

AWS Snow Family

- Highly-secure, portable devices to collect and process data at the edge and migrate data into and out of AWS
- Snowball Edge
- Snowcone

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TOPIC 10

AWS Snow Family

Storage

AWS Snow Family

Process data at the edge and migrate data into and out of AWS

Ready to get started?

Order an AWS Snow family device

Create your large data migration plan for migration greater than 500TB

Pricing

AWS Snowball

Overview

The AWS Snow Family helps customers that need to run operations in austere, non-data center environments, and in locations where there's a lack of consistent network connectivity. The Snow Family, comprised of AWS Snowcone, AWS Snowball, and AWS

Getting started

Snowball Edge data migration guide

Migrate HDFS files to an Amazon S3 data lake with AWS Snowball Edge

Building a Linux edge-computing solution with AWS Snowball Edge and Amazon EC2

Deploying a LAMP-based multi-tier web application on AWS Snowball Edge

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Data Migrations with AWS Snow Family

- Challenges:
- Limited connectivity
- Limited bandwidth
- High network cost
- Shared bandwidth (can’t maximize the line)
- Connection stability
- AWS Snow Family: offline devices to perform data migrations
- If it takes more than a week to transfer over the network, use Snowball devices!

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Diagrams

- Direct upload to S3:
- With Snow Family:
- client
- Amazon S3 bucket
- import/Amazon S3
- export bucket
- AWS
- Snowball
- client
- AWS
- Snowball
- www: 10Gbit/s
- ship

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Snow Family - Usage Process

1. Request Snowball devices from the AWS console for delivery
2. Install the snowball client / AWS OpsHub on your servers
3. Connect the snowball to your servers and copy files using the client
4. Ship back the device when you're done (goes to the right AWS facility)
5. Data will be loaded into an S3 bucket
6. Snowball is completely wiped

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What is Edge Computing?

- Process data while it’s being created on an edge location
- A truck on the road, a ship on the sea, a mining station underground...
- These locations may have limited internet and no access to computing power
- We setup a Snowball Edge / Snowcone device to do edge computing
- Snowcone: 2 CPUs, 4 GB of memory, wired or wireless access
- Snowball Edge Compute Optimized (dedicated for that use case) & Storage Optimized
- Run EC2 Instances or Lambda functions at the edge
- Use cases: preprocess data, machine learning, transcoding media

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Amazon FSx - Overview

- Launch 3rd party high-performance file systems on AWS
- Fully managed service
- FSx for Lustre
- FSx for Windows File Server
- FSx for
- NetApp ONTAP
- FSx for OpenZFS

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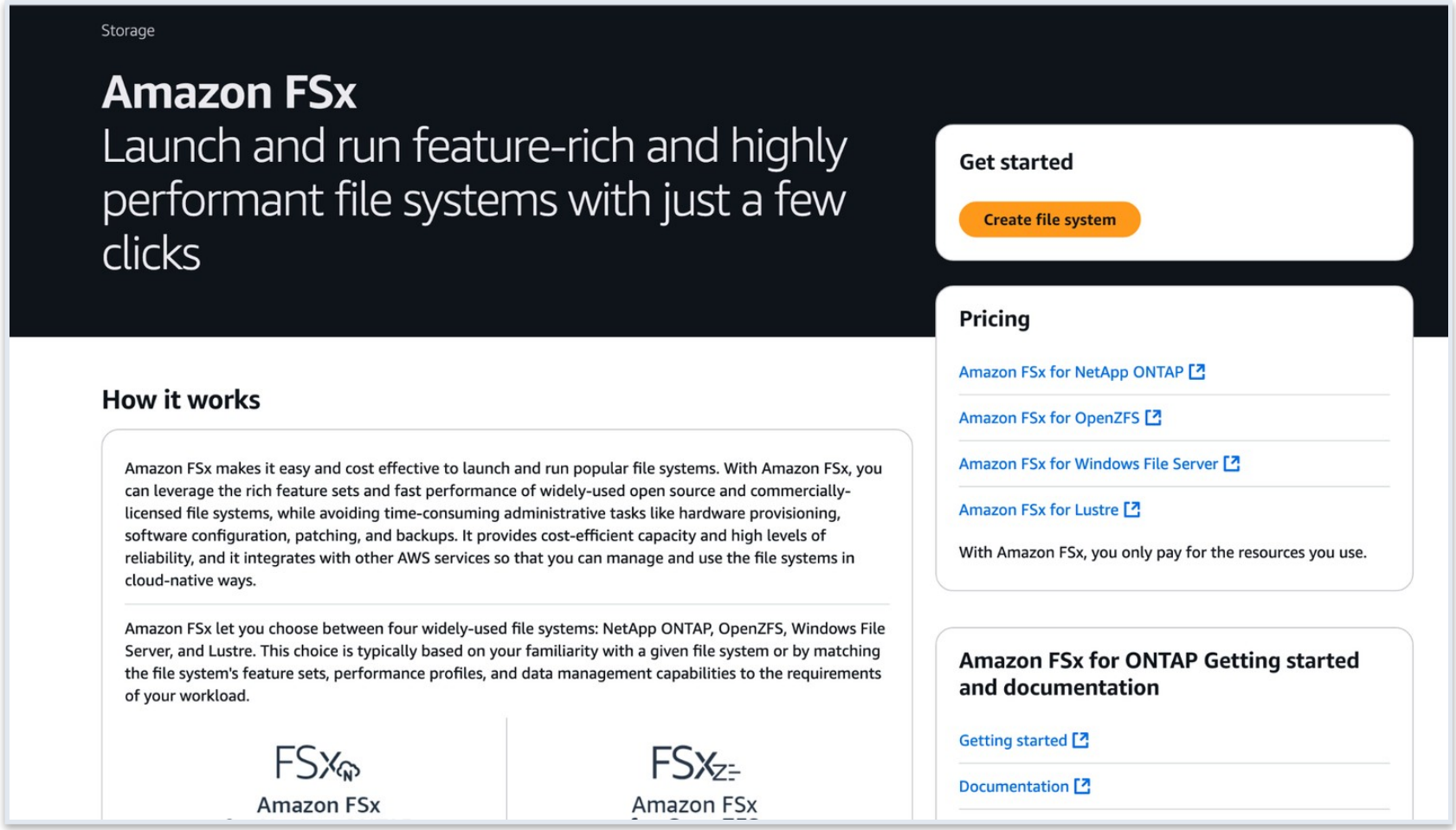
AWS FSx



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Amazon FSx for Windows (File Server)

- FSx for Windows is a fully managed Windows file system share drive
- Supports SMB protocol & Windows NTFS
- Microsoft Active Directory integration, ACLs, user quotas
- Can be mounted on Linux EC2 instances
- Supports Microsoft's Distrubuted File System (DFS) Namespaces (group files across multiple FS)
- Scale up to 10s of GB/s, millions of IOPS, 100s PB of data
- Storage Options:
- SSD - latency sensitive workloads (databases, media processing, data analytics, …)
- HDD - broad spectrum of workloads (home directory, CMS, …)
- Can be accessed from your on-premises infrastructure (VPN or Direct Connect)
- Can be configured to be Multi-AZ (high availability)
- Data is backed-up daily to S3

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Amazon FSx for Lustre

- Lustre is a type of parallel distributed file system, for large-scale computing
- The name Lustre is derived from “Linux” and “cluster
- Machine Learning, High Performance Computing (HPC)
- Video Processing, Financial Modeling, Electronic Design Automation
- Scales up to 100s GB/s, millions of IOPS, sub-ms latencies
- Storage Options:
- SSD - low-latency, IOPS intensive workloads, small & random file operations
- HDD - throughput-intensive workloads, large & sequential file operations
- Seamless integration with S3
- Can “read S3” as a file system (through FSx)
- Can write the output of the computations back to S3 (through FSx)
- Can be used from on-premises servers (VPN or Direct Connect)

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FSx Lustre - File System Deployment Options

- Scratch File System
- Temporary storage
- Data is not replicated (doesn’t persist if file server fails)
- High burst (6x faster, 200MBps per TiB)
- Usage: short-term processing, optimize costs
- Persistent File System
- Long-term storage
- Data is replicated within same AZ
- Replace failed files within minutes
- Usage: long-term processing, sensitive data
- Region
- Availability Zone 1
- Compute

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- Amazon FSx for NetApp ONTAP
- Managed NetApp ONTAP on AWS
- File System compatible with NFS, SMB, iSCSI protocol
- Move workloads running on ONTAP or NAS to AWS
- Works with:
- Linux
- Windows
- MacOS
- VMware Cloud on AWS
- Amazon Workspaces & AppStream 2.0
- Amazon EC2, ECS and EKS
- Storage shrinks or grows automatically
- Snapshots, replication, low-cost, compression and data de-duplication

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Amazon FSx for OpenZFS

- Managed OpenZFS file system on AWS
- File System compatible with NFS (v3, v4, v4.1, v4.2)
- Move workloads running on ZFS to AWS
- Works with:
- Linux
- Windows
- MacOS
- VMware Cloud on AWS
- Amazon Workspaces & AppStream 2.0
- Amazon EC2, ECS and EKS
- Up to 1,000,000 IOPS with < 0.5ms latency
- Snapshots, compression and low-cost
- Point-in-time instantaneous cloning (helpful for testing new workloads)

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FSx for SysOps

- FSx for Windows - Single-AZ
- Automatically replicates data within an AZ
- Two generations: Single-AZ 1 (SSD), Single-AZ 2 (SSD & HDD)
- us-east-1
- us-east-1
- us-east-1a
- File System
- FSx for Windows - Multi-AZ
- Automatically replicates data across AZs (synchronous)
- Standby file server in a different AZ (automatic failover)
- us-east-1a
- File System
- File System

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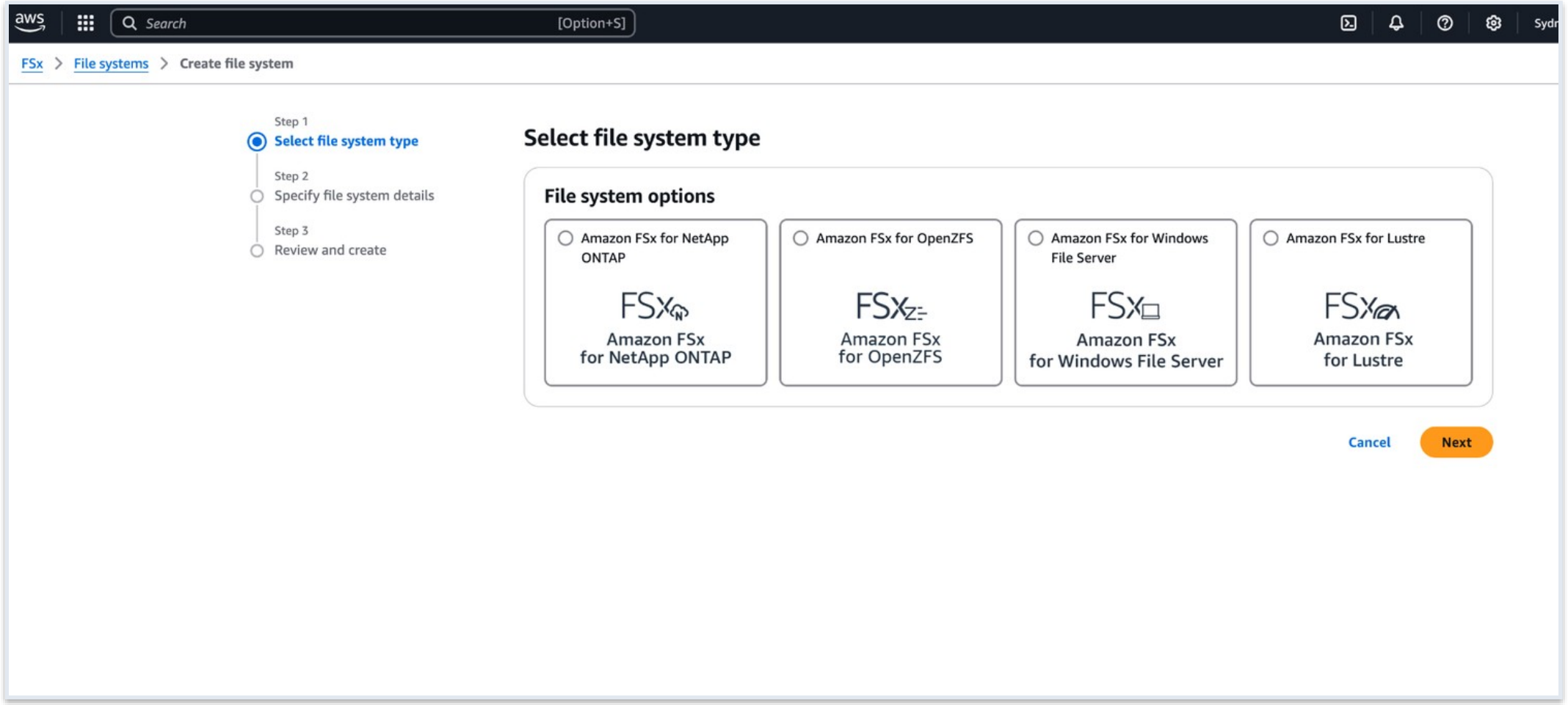
Activity: FSx



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Hybrid Cloud for Storage

- AWS is pushing for ”hybrid cloud”
- Part of your infrastructure is on the cloud
- Part of your infrastructure is on-premises
- This can be due to
- Long cloud migrations
- Security requirements
- Compliance requirements
- IT strategy
- S3 is a proprietary storage technology (unlike EFS / NFS), so how do you expose the S3 data on-premises?
- AWS Storage Gateway!

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AWS Storage Cloud Native Options

- Block
- File
- Object
- Amazon EBS
- EC2 Instance
- Store
- Amazon EFS
- Amazon FSx
- Amazon S3
- Amazon Glacier

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AWS Storage Gateway

- Bridge between on-premises data and cloud data
- Use cases:
- disaster recovery
- backup & restore
- tiered storage
- on-premises cache & low-latency files access
- Types of Storage Gateway:
- S3 File Gateway
- FSx File Gateway
- Volume Gateway
- Tape gateway
- Storage Gateway

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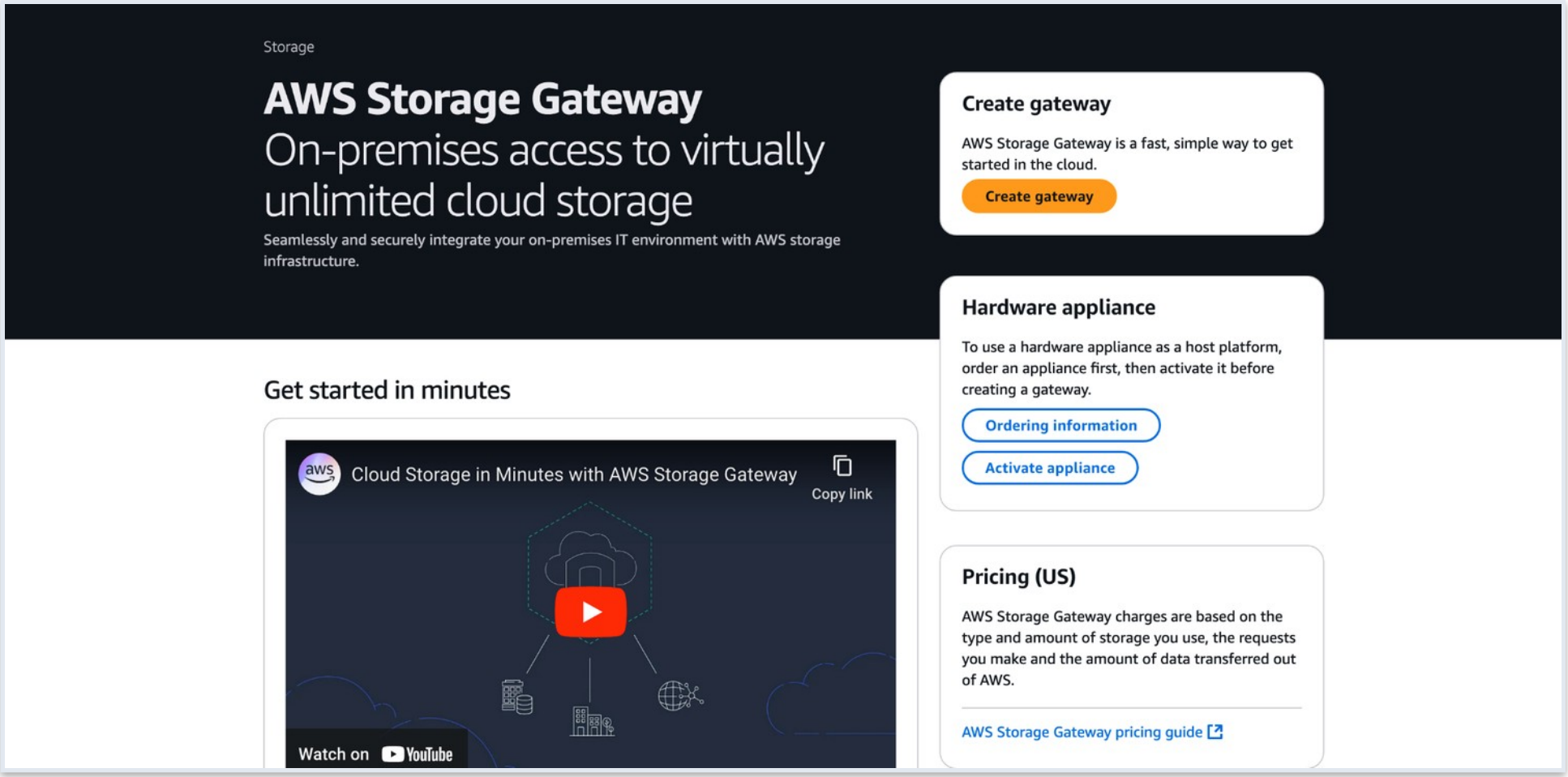
TOPIC 10

AWS Storage Gateway



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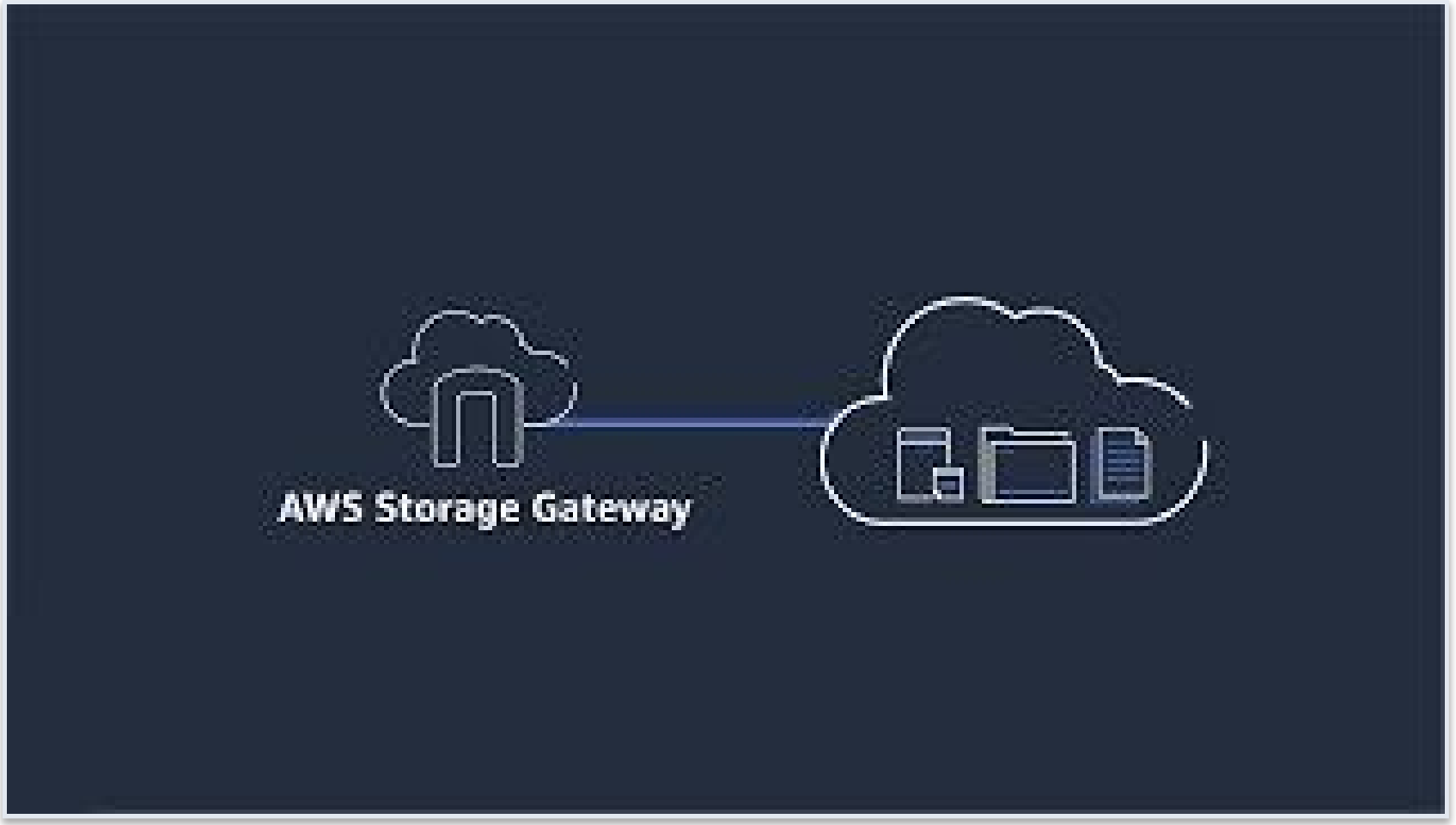
Reference



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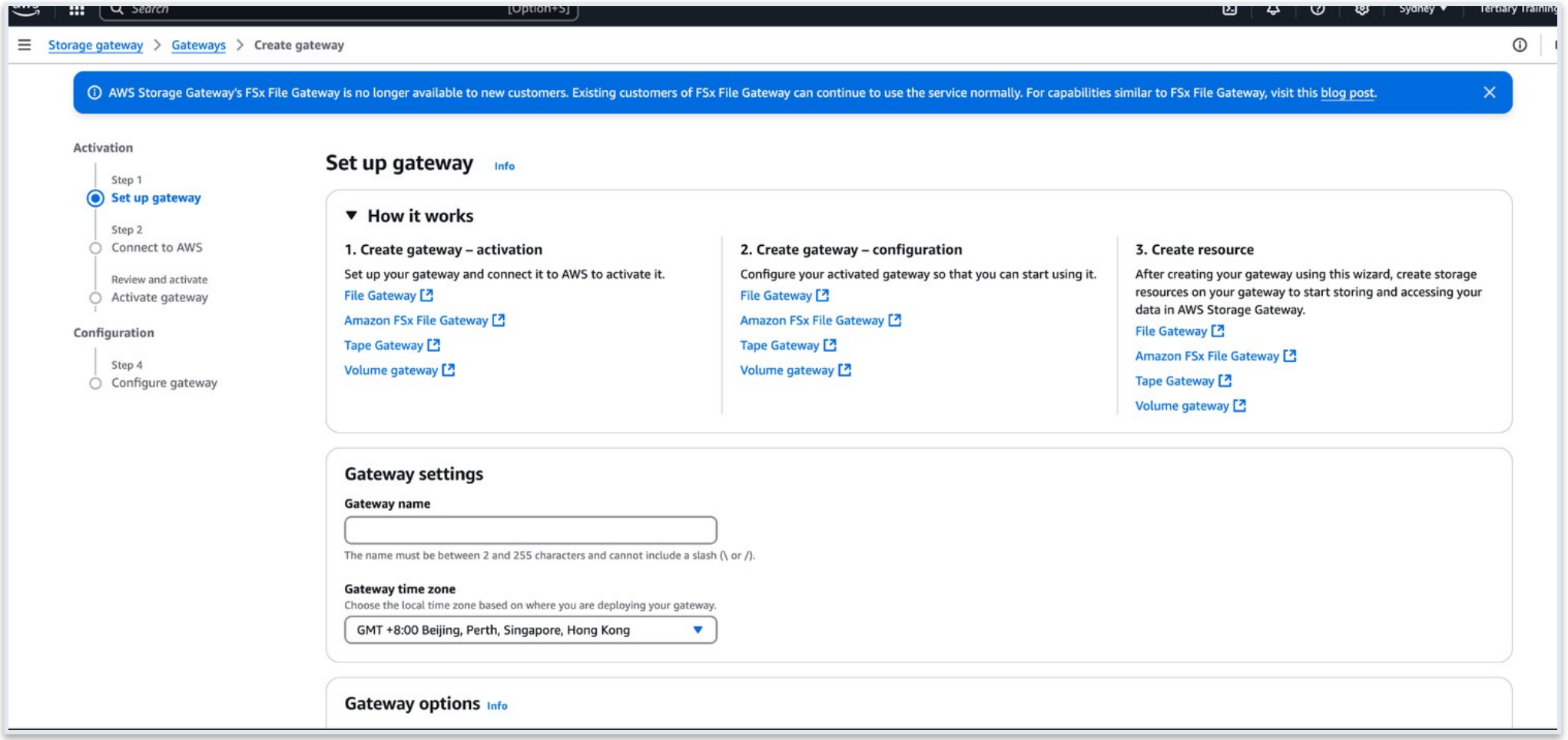
Activity: Storage Gateway



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TOPIC 10

Amazon S3 File Gateway

- Configured S3 buckets are accessible using the NFS and SMB protocol
- Most recently used data is cached in the file gateway
- Supports S3 Standard, S3 Standard IA, S3 One Zone A, S3 Intelligent Tiering
- Transition to S3 Glacier using a lifecycle Policy
- Bucket access using IAM roles for each File Gateway
- SMB Protocol has integration with Active Directory (AD) for user authentication
- Corporate Data Center
- .
- HTTPS
- Application Server
- S3 File Gateway
- S3 Standard S3 Standard-IA
- S3 One Zone-IA

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TOPIC 10

Amazon FSx File Gateway

- Native access to Amazon FSx for Windows File Server
- Local cache for frequently accessed data
- Windows native compatibility (SMB, NTFS, Active Directory...)
- Useful for group file shares and home directories
- Corporate Data Center
- AWS Cloud
- SMB Clients
- Amazon FSx File Gateway
- File systems
- Amazon FSx
- for Windows File Server

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TOPIC 10

Volume Gateway

- Block storage using iSCSI protocol backed by S3
- Backed by EBS snapshots which can help restore on-premises volumes!
- Cached volumes: low latency access to most recent data
- Stored volumes: entire dataset is on premise, scheduled backups to S3
- Corporate Data Center
- AWS Cloud
- Region
- S3 Bucket
- Amazon EBS Snapshots
- Application Server
- iSCSI
- Volume Gateway
- HTTPS

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TOPIC 10

Tape Gateway

- Some companies have backup processes using physical tapes (!)
- With Tape Gateway, companies use the same processes but, in the cloud
- Virtual Tape Library (VTL) backed by Amazon S3 and Glacier
- Back up data using existing tape-based processes (and iSCSI interface)
- Works with leading backup software vendors
- Corporate Data Center
- AWS Cloud
- Region
- Virtual Tapes stored in Amazon S3
- Archived Tapes stored in Amazon Glacier
- iSCSI
- Backup Server
- HTTPS

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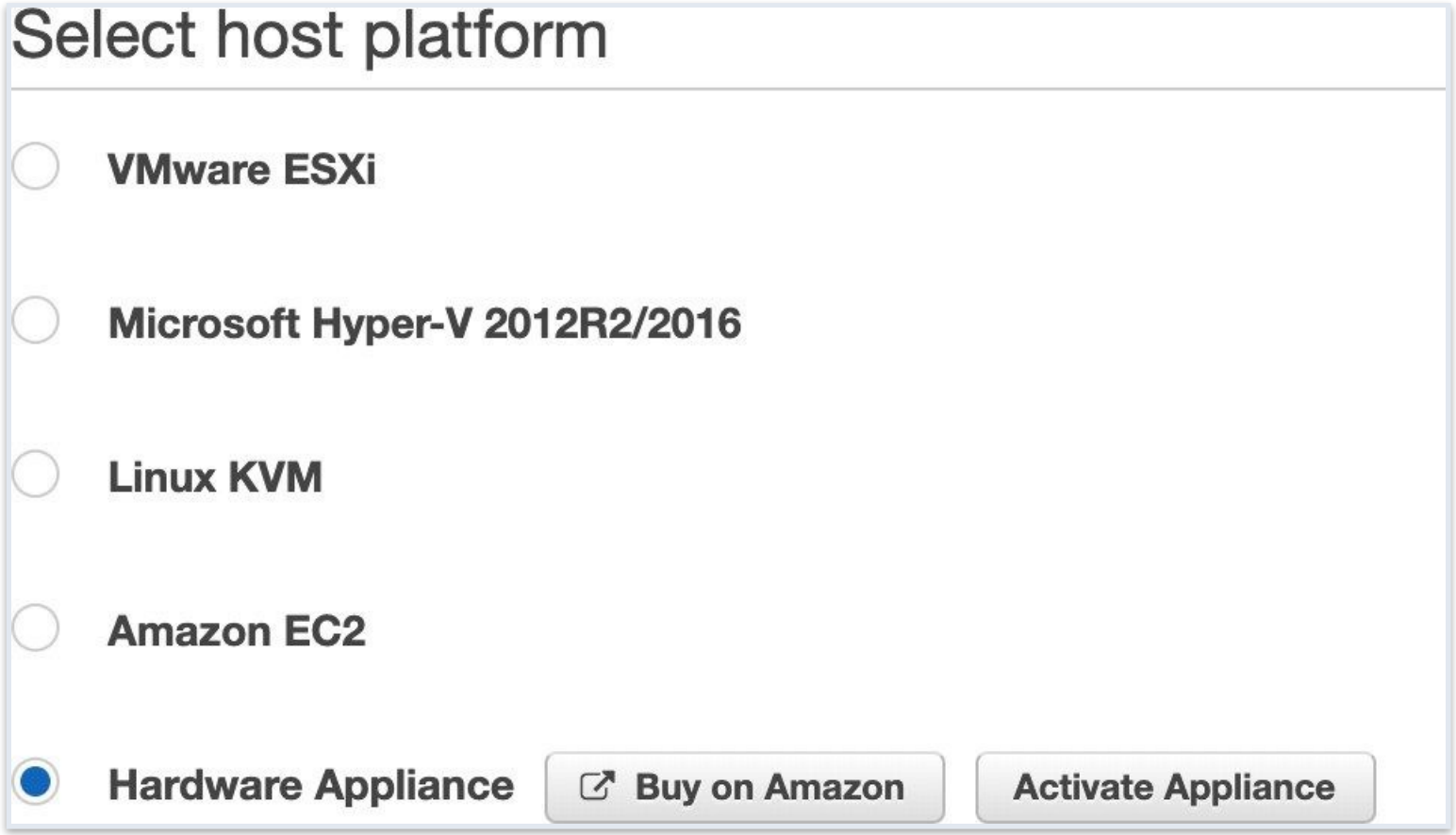
Storage Gateway - Hardware appliance



Using Storage Gateway means you need on-premises virtualization

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TOPIC 10

AWS Storage Gateway

- On-Premises
- File Gateway
- local cache
- Volume Gateway
- local cache
- Tape Gateway
- local cache
- Application Server
- Backup Application
- User/group file shares
- AWS Cloud
- NFS/SMB
- iSCSI

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Amazon FSx File Gateway

- Native access to Amazon FSx for Windows File Server
- Local cache for frequently accessed data
- Windows native compatibility (SMB, NFTS, Active Directory...)
- Useful for group file shares and home directories
- Corporate Data Center
- AWS Cloud
- SMB Clients
- Amazon FSx File Gateway
- File systems
- Amazon FSx
- for Windows File Server

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Storage Gateway - SysOps

- File Gateway is POSIX compliant (Linux file system)
- POSIX metadata ownership, permissions, and timestamps stored in the object's metadata in S3
- Reboot Storage Gateway VM: (e.g., maintenance)
- File Gateway: simply restart the Storage Gateway VM
- Volume and Tape Gateway:
- Stop Storage Gateway Service (AWS Console,VM local Console, Storage Gateway API)
- Reboot the Storage Gateway VM
- Start Storage Gateway Service (AWS Console,VM local Console, Storage Gateway API)

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Storage Gateway - Activations

- Two ways to get Activation Key:
- Using the Gateway VM CLI
- Make a web request to the Gateway VM (Port 80)
- Troubleshooting Activation Failures:
- Make sure the Gateway VM has port 80 opened
- Check that the Gateway VM has the correct time and synchronizing its time automatically to a Network Time Protocol (NTP) server
- File Gateway
- curl ‘vm_ip_address/?activationRegion= activation_region&no_redirect’
- F9KRH-….
- AWS Console
- F9KRH-….
- Corporate Data Center

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Storage Gateway - Volume Gateway Cache

- Cached mode: only the most recent data is stored
- Looking at cache efficiency
- Look at the CacheHitPercent metric (you want it to be high)
- Look at the CachePercentUsed (you don’t want it to be too high)
- Create a larger cache disk
- Use the cached volume to clone a new volume of a larger size
- Select the new disk as the cached volume
- AWS Cloud
- S3 Bucket
- Cached
- Volume
- Gateway
- Client

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TOPIC 11

Amazon CloudFront

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TOPIC 11

Amazon CloudFront

- Content Delivery Network (CDN)
- Improves read performance, content is cached at the edge
- Improves users experience
- 216 Point of Presence globally (edge locations)
- DDoS protection (because worldwide), integration with Shield, AWS Web Application Firewall
- Source: https://aws.amazon.com/cloudfront/features/?nc=sn&loc=2

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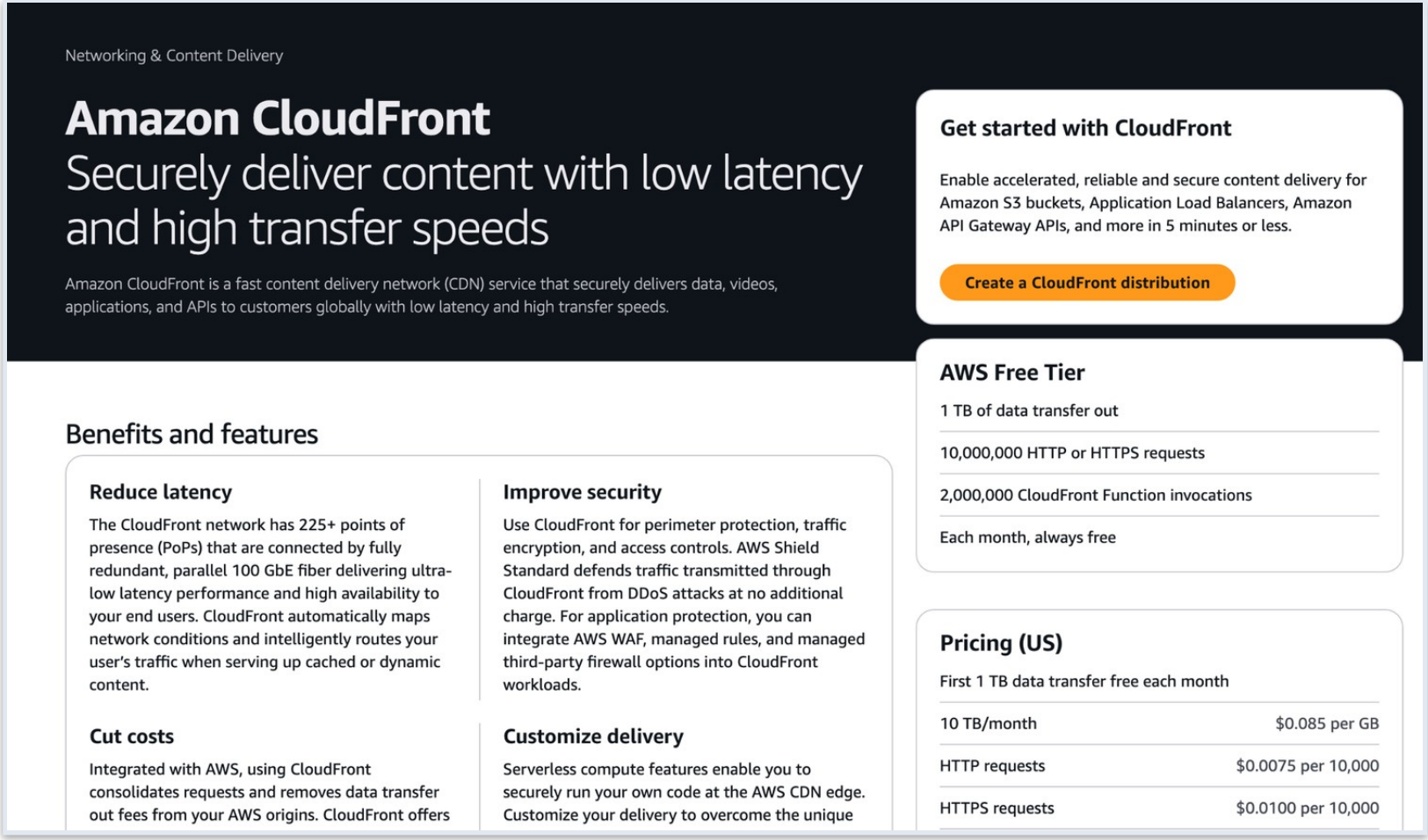
Amazon CloudFront



Networking & Content Delivery

Amazon CloudFront

Securely deliver content with low latency and high transfer speeds

Amazon CloudFront is a fast content delivery network (CDN) service that securely delivers data, videos, applications, and APIs to customers globally with low latency and high transfer speeds.

Get started with CloudFront

Enable accelerated, reliable and secure content delivery for Amazon S3 buckets, Application Load Balancers, Amazon API Gateway APIs, and more in 5 minutes or less.

Create a CloudFront distribution

AWS Free Tier

1 TB of data transfer out

10,000,000 HTTP or HTTPS requests

2,000,000 CloudFront Function invocations

Each month, always free

Pricing (US)

First 1 TB data transfer free each month

10 TB/month	$0.085 per GB
HTTP requests	$0.0075 per 10,000
HTTPS requests	$0.0100 per 10,000

Benefits and features

Reduce latency

The CloudFront network has 225+ points of presence (PoPs) that are connected by fully redundant, parallel 100 GbE fiber delivering ultra-low latency performance and high availability to your end users. CloudFront automatically maps network conditions and intelligently routes your user's traffic when serving up cached or dynamic content.

Improve security

Use CloudFront for perimeter protection, traffic encryption, and access controls. AWS Shield Standard defends traffic transmitted through CloudFront from DDoS attacks at no additional charge. For application protection, you can integrate AWS WAF, managed rules, and managed third-party firewall options into CloudFront workloads.

Cut costs

Integrated with AWS, using CloudFront consolidates requests and removes data transfer out fees from your AWS origins. CloudFront offers

Customize delivery

Serverless compute features enable you to securely run your own code at the AWS CDN edge. Customize your delivery to overcome the unique

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CloudFront - Origins

- S3 Bucket
- For distributing files and caching them at the edge
- Enhanced security with CloudFront Origin Access Control (OAC)
- OAC is replacing Origin Access Identity (OAI)
- CloudFront can be used as an ingress (to upload files to S3)
- Custom Origin (HTTP)
- Application Load Balancer
- EC2 instance
- S3 website (must first enable the bucket as a static S3 website)
- Any HTTP backend you want

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CloudFront at a high level

- CloudFront Edge Location
- Forward Request to your Origin
- Local Cache
- Origin
- S3
- or
- HTTP
- Client
- GET /beach.jpg?size=300x300 HTTP/1.1
- User-Agent: Mozilla/4.0 (compatible; MSIE5.01; Windows NT) Host: www.example.com
- Accept-Encoding: gzip, deflate

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CloudFront - S3 as an Origin

- Edge
- Los Angeles
- Edge Mumbai
- Edge
- Melbourne
- Public www
- Public www
- Edge
- São Paulo
- Origin Access Control
- + S3 bucket policy
- Origin (S3 bucket)
- OAC

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CloudFront vs S3 Cross Region Replication

- CloudFront:
- Global Edge network
- Files are cached for a TTL (maybe a day)
- Great for static content that must be available everything
- S3 Cross Region Replication:
- Must be setup for each region you want replication to happen
- Files are updated in near real-time
- Read only
- Great for dynamic content that needs to be available at low-latency in few regions

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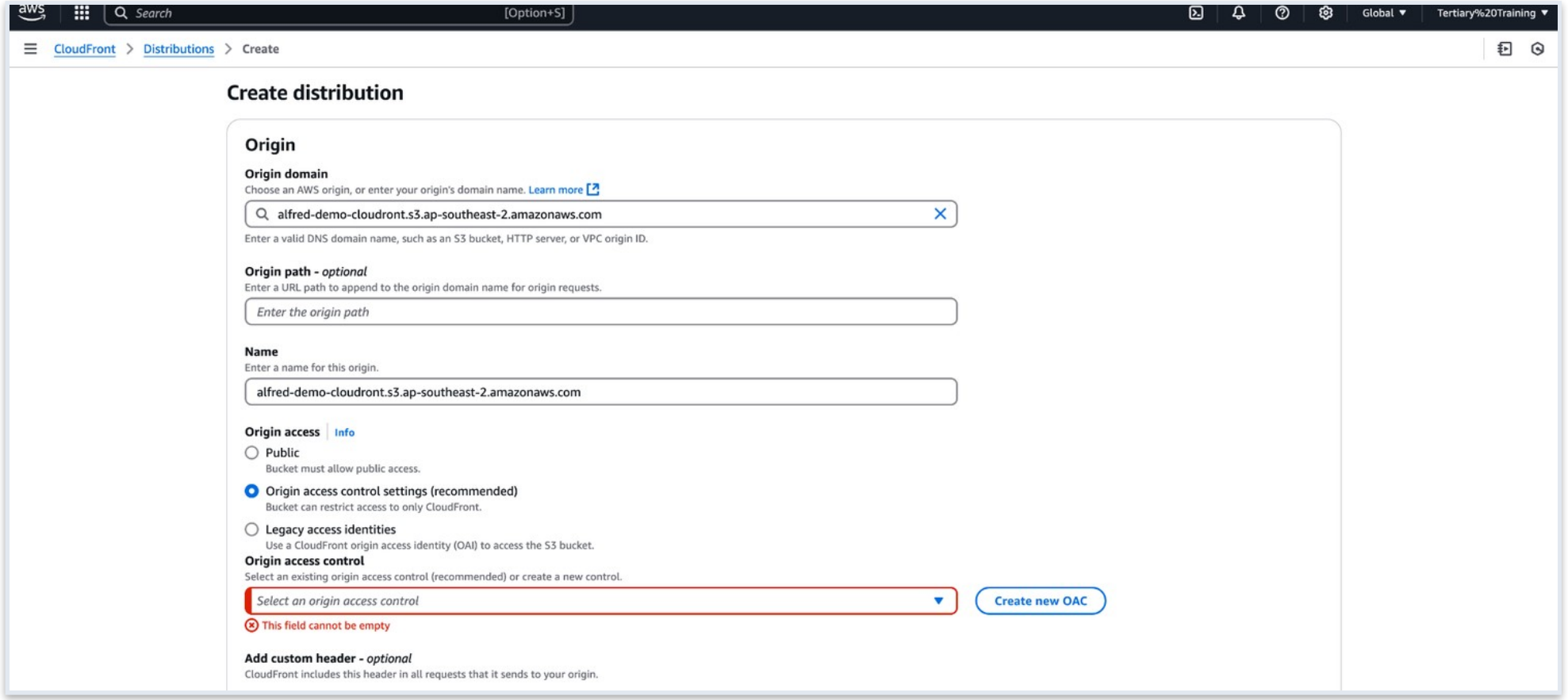
Activity: CloudFront



Upload some images and index.html to a new S3 bucket

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CloudFront - ALB or EC2 as an origin

- http://d7uri8nf7uskq.cloudfront.net/tools/list-cloudfront-ips
- Edge Location
- Public IPs
- Allow Security Group of Load Balancer
- Allow Public IP of Edge Locations
- Security group
- EC2 Instances
- Can be Private
- Security group
- Application Load Balancer
- Must be Public
- Edge Location
- Allow Public IP of Edge Locations

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CloudFront Geo Restriction

- You can restrict who can access your distribution
- Allowlist: Allow your users to access your content only if they're in one of the countries on a list of approved countries.
- Blocklist: Prevent your users from accessing your content if they're in one of the countries on a list of banned countries.
- The “country” is determined using a 3rd party Geo-IP database
- Use case: Copyright Laws to control access to content

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CloudFront Access Logs

- Logs every request made to CloudFront into a logging S3 bucket
- Edge Location
- Edge Location
- Edge Location
- Edge Location
- Website A
- http://d111111abcdef8.cloudfront.net
- Website B
- http://d222222abcdef8.cloudfront.net
- Amazon S3
- Origin Bucket
- Logs Bucket
- Users

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CloudFront Reports

- It’s possible to generate reports on:
- Cache Statistics Report
- Popular Objects Report
- Top Referrers Report
- Usage Reports
- Viewers Report
- These reports are based on the data from the Access Logs.

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CloudFront troubleshooting

- CloudFront caches HTTP 4xx and 5xx status codes returned by S3 (or the origin server)
- 4xx error code indicates that user doesn’t have access to the underlying bucket (403) or the object user is requesting is not found (404)
- 5xx error codes indicates Gateway issues

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CloudFront Caching

- Cache based on
- Headers
- Session Cookies
- Query String Parameters
- The cache lives at each CloudFront Edge Location
- You want to maximize the cache hit rate to minimize requests on the origin
- Control the TTL (0 seconds to 1 year), can
- be set by the origin using the Cache- Control header, Expires header…
- You can invalidate part of the cache using the CreateInvalidation API
- Request
- Origin
- forwards
- Expire based on TTL

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CloudFront Cache Behaviour for Headers

- Forward all headers to your origin
- => no caching, every request to origin
- => TTL must be set to 0
- Forward a whitelist of headers
- caching based on values in all the specified headers
- (None) == Forward only the default headers
- no caching based on request headers
- Best caching performance
- Request
- Client
- GET /image/cat.jpg HTTP/1.1
- Host: pics.mywebsite.com
- User-Agent: Mozilla/5.0 (Mac OS X 10_15_2,…) Date: Tue, 28 Jan 2020 17:01:57 GMT

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TOPIC 11

CloudFront Caching – Whitelist Headers

- Request
- Origin
- forward
- Edge Location
- Client
- GET /image/cat.jpg HTTP/1.1
- Host: pics.mywebsite.com
- User-Agent: Mozilla/5.0 (Mac OS X 10_15_2….) Date: Tue, 28 Jan 2020 17:01:57 GMT
- Authorization: SAPISIDHASH fdd00ecee39fe….
- Keep-Alive: 300 Accept-Ranges: bytes
- GET /image/cat.jpg HTTP/1.1
- Host: pics.mywebsite.com
- Authorization: SAPISIDHASH fdd00ecee39fe….

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TOPIC 11

CloudFront Origin Headers vs Cache Behavior

1. Origin Custom Headers:
2. Origin-level setting
3. Set a constant header / header value for all requests to origin
4. Behavior setting:
5. Cache-related settings
6. Contains the whitelist of headers to forward

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CloudFront Caching TTL

- “Cache-Control: max-age” is preferred to “Expires” header
- If the origin always sends back the header Cache-Control then you can set the TTL to be controlled only by that header
- In case you want to set min/max boundaries, you choose “customize” for the Object Caching setting
- In case the Cache-Control header is missing, it will default to “default value”
- Min. TTL
- Max. TTL
- Default TTL
- Cache-Control (optional)

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CloudFront Cache Behaviour for Cookies

- Cookies is a specific request header
- Default: do not process the cookies
- Caching is not based on cookies
- Cookies are not forwarded
- Forward a whitelist of cookies
- caching based on values in all the specified cookies
- Forward all cookies
- Worst caching performance
- Client
- GET /image/cat.jpg HTTP/1.1
- …
- Cookie: username=John Doe; location=uk; lang = eng; user_id = 12342
- …

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TOPIC 11

CloudFront Caching - Cookies

- Request
- Origin
- forward
- Edge Location
- Client
- GET /image/cat.jpg HTTP/1.1
- …
- Cookie: username=John Doe; location=uk; lang = eng; user_id = 12342
- …
- GET /image/cat.jpg HTTP/1.1
- …
- Cookie: user_id = 12342
- …

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CloudFront Cache Behaviour for Query Strings

- Query Strings Parameters are in the URL
- Default: do not process the query strings
- Caching is not based on query strings
- Parameters are not forwarded
- Forward a whitelist of query strings
- Caching based on the parameter whitelist
- Forward all query strings
- Caching based on all parameters
- Client
- GET /image/cat.jpg?border=red&size=large HTTP/1.1
- …
- Request

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TOPIC 11

CloudFront Caching - Query String Parameters

- Request
- Origin
- forward
- Edge Location
- Client
- GET /image/cat.jpg?border=red&size=large HTTP/1.1
- …
- GET /image/cat.jpg?size=large HTTP/1.1
- …
- Whitelisting on size
- Cache based on size
- Better caching because less query string values

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TOPIC 11

CloudFront - Maximize cache hits by separating static and dynamic distributions

- Dynamic Content (REST, HTTP server):
- ALB + EC2
- API Gateway + Lambda
- CDN Layer
- CloudFront
- Dynamic
- Static requests
- Static content
- Cache based on correct headers and cookie
- No headers / session caching rules Required for maximizing cache hits

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CloudFront - Increasing Cache Ratio

1. Monitor the CloudWatch metric CacheHitRate
2. Specify how long to cache your objects: Cache-Control max-age header
3. Specify none or the minimally required headers
4. Specify none or the minimally required cookies
5. Specify none or the minimally required query string parameters
6. Separate static and dynamic distributions (two origins)

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CloudFront with ALB sticky sessions

- Must forward / whitelist the cookie that controls the session affinity to the origin to allow the session affinity to work
- Set a TTL to a value lesser than when the authentication cookie expires
- requests
- forward cookies
- Edge Location
- GET /orders/all HTTP/1.1
- …
- Cookie: AWSALB=…
- Whitelisting on
- AWSALB cookie
- CloudFront
- Target Group

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TOPIC 12

Database For SysOps

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TOPIC 12

Amazon RDS Overview

- RDS stands for Relational Database Service
- It’s a managed DB service for DB use SQL as a query language.
- It allows you to create databases in the cloud that are managed by AWS
- Postgres
- MySQL
- MariaDB
- Oracle
- Microsoft SQL Server
- IBM DB2
- Aurora (AWS Proprietary database)

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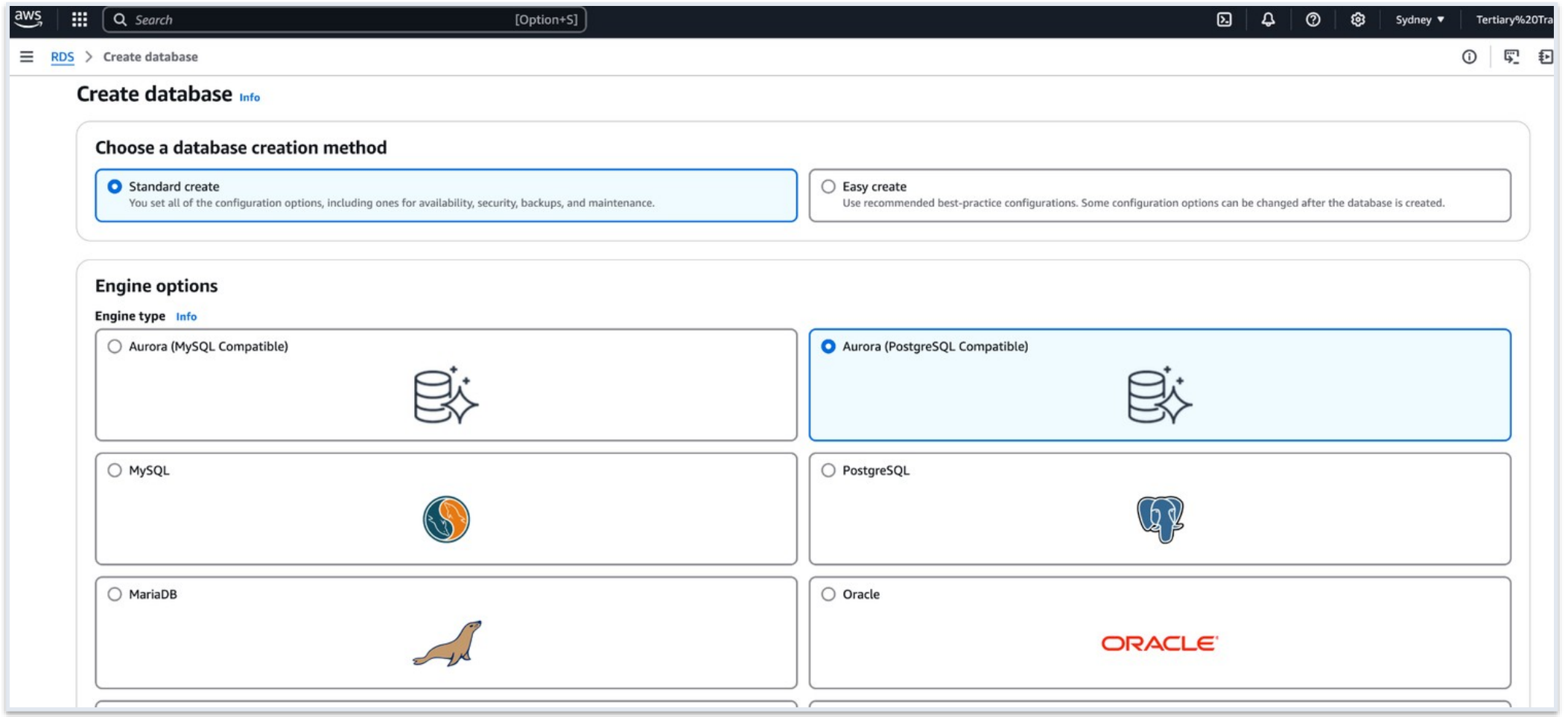
Amazon RDS Overview



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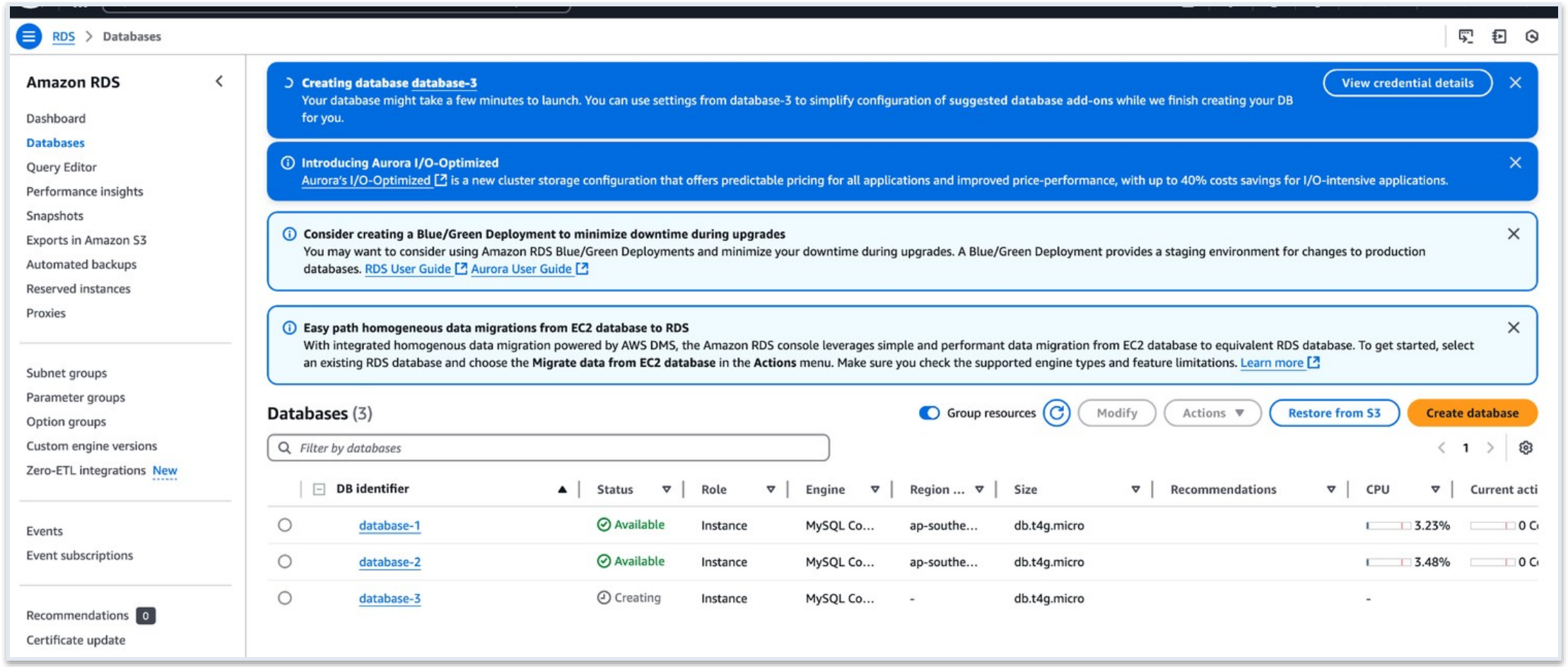
Activity: Create RDS



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Advantage over using RDS versus deploying DB on EC2

- RDS is a managed service:
- Automated provisioning, OS patching
- Continuous backups and restore to specific timestamp (Point in Time Restore)!
- Monitoring dashboards
- Read replicas for improved read performance
- Multi AZ setup for DR (Disaster Recovery)
- Maintenance windows for upgrades
- Scaling capability (vertical and horizontal)
- Storage backed by EBS
- BUT you can’t SSH into your instances

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RDS - Storage Auto Scaling

- Helps you increase storage on your RDS DB instance dynamically
- When RDS detects you are running out of free database storage, it scales automatically
- Avoid manually scaling your database storage
- You have to set Maximum Storage Threshold (maximum limit for DB storage)
- Automatically modify storage if:
- Free storage is less than 10% of allocated storage
- Low-storage lasts at least 5 minutes
- 6 hours have passed since last modification
- Useful for applications with unpredictable workloads
- Supports all RDS database engines
- Read/Write
- Application
- Storage

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TOPIC 12

RDS Read Replicas for read scalability

- RDS DB
- instance
- RDS DB
- instance read replica
- RDS DB
- instance read replica
- Application
- writes
- reads
- reads
- reads
- ASYNC
- replication

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RDS Read Replicas - Use Cases

- You have a production database that is taking on normal load
- You want to run a reporting application to run some analytics
- You create a Read Replica to run the new workload there
- The production application is unaffected
- Read replicas are used for SELECT (=read) only kind of statements (not INSERT, UPDATE, DELETE)
- RDS DB
- instance
- Production
- Application
- RDS DB
- instance read replica
- reads

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RDS Read Replicas - Network Cost

- ASYNC
- Replication
- Cross-Region
- $$$
- Region/AZ
- us-east-1a
- RDS DB
- instance
- Region/AZ
- eu-west-1b
- RDS DB
- instance read replica

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- In AWS there’s a network cost when data goes from one AZ to another

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RDS Multi AZ (Disaster Recovery)

- RDS Master DB instance (AZ A)
- Application
- writes
- reads
- SYNC
- replication
- SYNC replication
- One DNS name - automatic app failover to standby
- Increase availability
- Failover in case of loss of AZ, loss of network, instance or storage failure
- No manual intervention in apps
- Not used for scaling
- Note:The Read Replicas be setup as Multi AZ for Disaster Recovery (DR)

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RDS - From Single-AZ to Multi-AZ

- Zero downtime operation (no need to stop the DB)
- Just click on “modify” for the database
- The following happens internally:
- A snapshot is taken
- A new DB is restored from the snapshot in a new AZ
- Synchronization is established between the two databases
- RDS DB
- instance
- SYNC
- Replication
- Standby DB
- snapshot
- restore

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RDS Multi AZ - Failover Conditions

- The primary DB instance
- Failed
- OS is undergoing software patches
- Unreachable due to loss of network connectivity
- Modified (e.g., DB instance type changed)
- Busy and unresponsive
- Underlying storage failure
- An Availability Zone outage
- A manual failover of the DB instance was initiated using Reboot with failover

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Lambda by default

- AWS Cloud
- Private RDS
- DynamoDB
- VPC & Private Subnet
- Not working
- Default Lambda Deployment
- Public www works
- By default, your Lambda function is launched outside your own VPC (in an AWS-owned VPC)
- Therefore, it cannot access resources in your VPC (RDS, ElastiCache, internal ELB…)

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Private subnet

- Lambda in VPC
- You must define the VPC ID, the Subnets and the Security Groups
- Lambda will create an ENI (Elastic Network Interface) in your subnets
- AWSLambdaVPCAccessExecutionRole
- Lambda Function
- Lambda Security group
- RDS Security group
- Amazon RDS In VPC
- Elastic Network Interface (ENI)

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Public subnet

- VPC
- Private subnet
- RDS Proxy for AWS Lambda
- When using Lambda functions with RDS, it opens and maintains a database connection
- This can result in a "TooManyConnections" exception
- With RDS Proxy, you no longer need code
- that handles cleaning up idle connections and managing connection pools
- Supports IAM authentication or DB authentication, auto-scaling
- The Lambda function must have connectivity to the Proxy (public proxy => public Lambda, private proxy => Lambda in VPC)
- …
- Lambda functions
- RDS Proxy

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DB Parameter Groups

- You can configure the DB engine using Parameter Groups
- Dynamic parameters are applied immediately
- Static parameters are applied after instance reboot
- You can modify parameter group associated with a DB (must reboot)
- See documentation for list of parameters for a DB technology
- Must-know parameter
- PostgreSQL / SQL Server: rds.force_ssl=1 => force SSL connections
- MySQL / MariaDB: require_secure_transport=1 => force SSL connections

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RDS Backup vs Snapshots

- Backups are “continuous” and allow point in time recovery
- Backups happen during maintenance windows
- When you delete a DB instance, you can retain automated backups
- Backups have a retention period you set between 0 and 35 days
- To disable backups, set retention period to 0
- Snapshots takes IO operations and can stop the database from seconds to minutes
- Snapshots taken on Multi AZ DB don’t impact the master – just the standby
- Snapshots are incremental after the first snapshot (which is full)
- You can copy & share DB Snapshots
- Manual Snapshots don’t expire
- You can take a “final snapshot” when you delete your DB
- Backups
- Snapshots

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RDS Snapshots Sharing

- Manual snapshots: can be shared with AWS accounts
- Automated snapshots: can’t be shared, copy first
- You can only share unencrypted snapshots and snapshots encrypted with a customer managed key
- If you share an encrypted snapshots, you must also share any customer managed keys used to encrypt them
- Account B
- Share with Account B
- DB Instance
- (Snapshot: mydbsnapshot)
- RDS DB
- Snapshot
- Account A
- Snapshot: mydbsnapshot

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RDS Snapshots Sharing with KMS Encryption

- Account A
- us-east-1
- CMK - A
- Source
- Account B
- Target
- IAM Permissions
- kms:DescribeKey kms:CreateGrant kms:Encrypt kms:Decrypt kms:GenerateDataKey kms:ReEncrypt
- share
- us-east-1
- DB Instance
- (Snapshot: mydbsnapshot)

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RDS Events & Event Subscriptions

- RDS keeps record of events related to:
- DB instances
- Snapshots
- Parameter groups, security groups …
- Example: DB state changed from pending to running
- RDS Event Subscription
- Subscribe to events to be notified when an event occurs using SNS
- Specify the Event Source (instances, SGs, …) and the Event Category (creation, failover, …)
- RDS delivers events to EventBridge
- DB Instance
- (backup event)
- SNS
- event

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RDS Database Log Files

- DB Instance
- CW Logs
- Store Logs:
- General
- Audit
- Error
- Slow Query
- CW Alarm
- Alert
- Amazon SNS
- DB Admin
- notification
- CloudWatch

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TOPIC 12

RDS with CloudWatch

- CloudWatch metrics associated with RDS (gathered from the hypervisor):
- DatabaseConnections
- SwapUsage
- ReadIOPS / WriteIOPS
- ReadLatency / WriteLatency
- ReadThroughPut / WriteThroughPut
- DiskQueueDepth
- FreeStorageSpace
- Enhanced Monitoring (gathered from an agent on the DB instance)
- Useful when you need to see how different processes or threads use the CPU
- Access to over 50 new CPU, memory, file system, and disk I/O metrics

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RDS Performance Insights

- Visualize your database performance and analyze any issues that affect it
- With the Performance Insights dashboard, you can visualize the database load and filter the load:
- By Waits => find the resource that is the bottleneck (CPU, IO, lock, etc…)
- By SQL statements => find the SQL statement that is the problem
- By Hosts => find the server that is using the most our DB
- By Users => find the user that is using the most our DB
- DBLoad = the number of active sessions for the DB engine
- You can view the SQL queries that are putting load on your database

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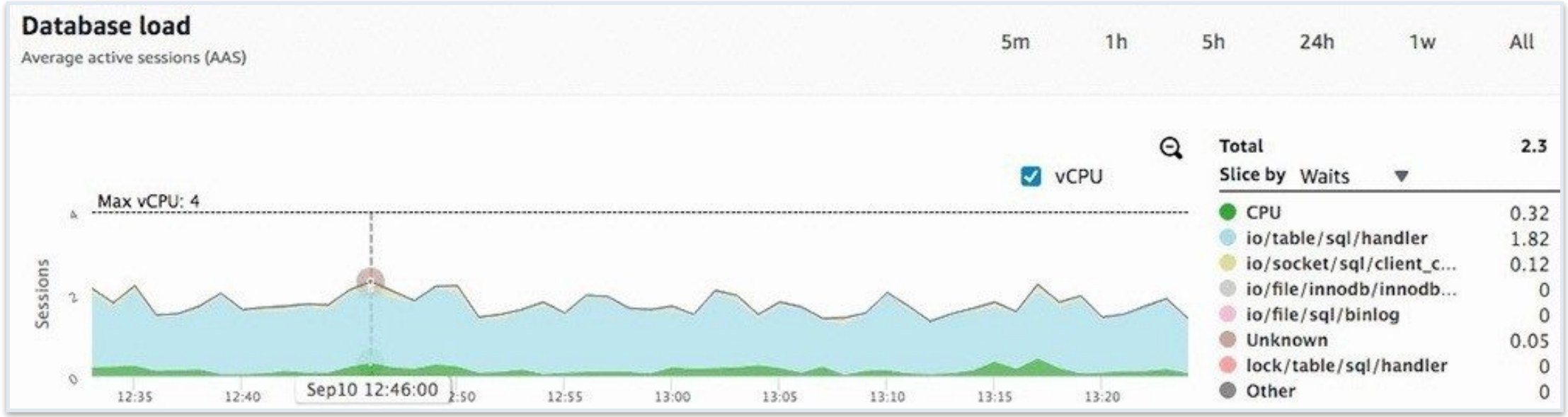
Performance Insights Screenshots DB Waits



From: https://aws.amazon.com/blogs/database/tuning-amazon-rds-for-mysql-with-performance-insights/

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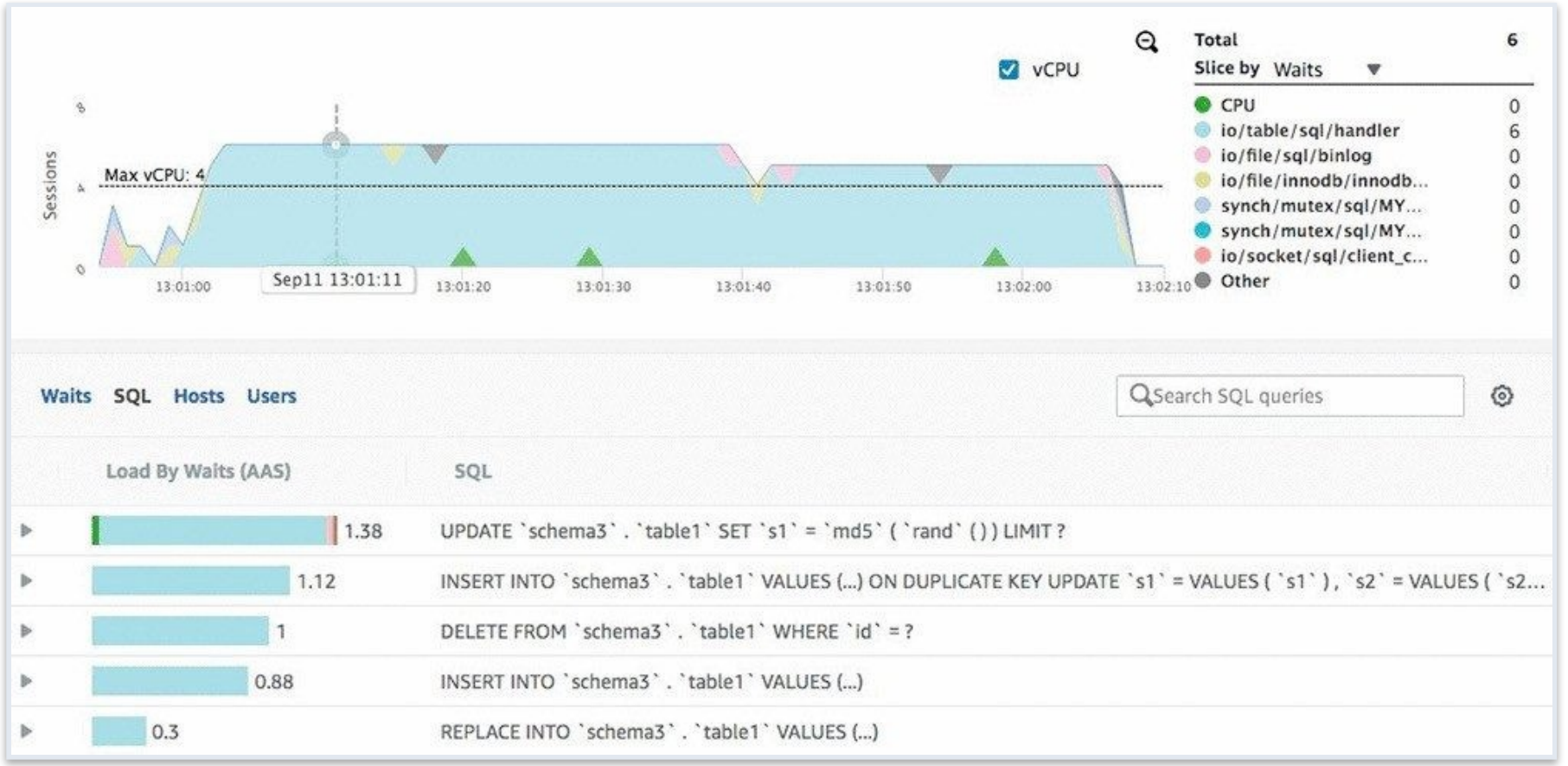
Performance Insights Screenshots SQL



From: https://aws.amazon.com/blogs/database/tuning-amazon-rds-for-mysql-with-performance-insights/

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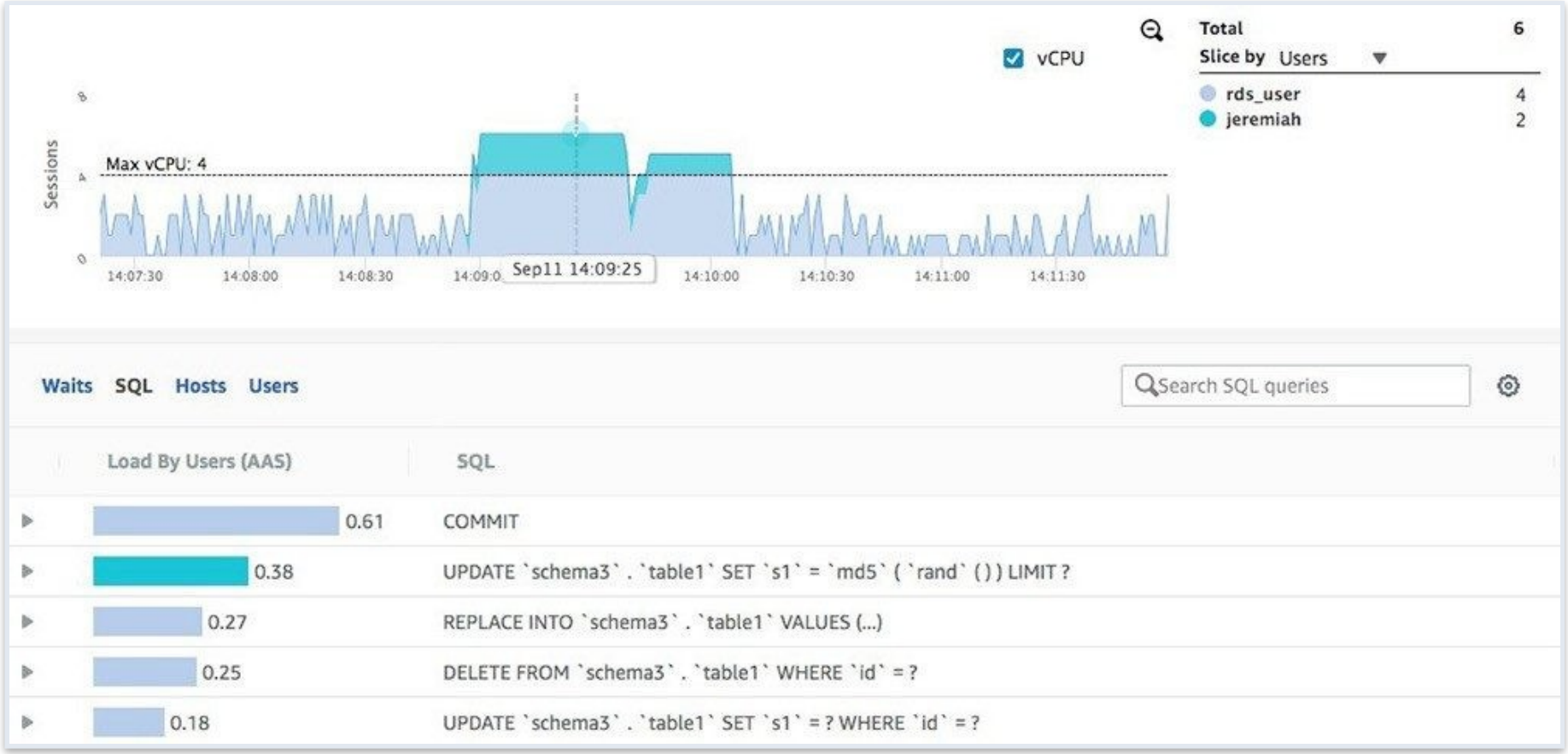
Performance Insights Screenshots Users



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TOPIC 13

Aurora

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Amazon Aurora

- Aurora is a proprietary technology from AWS (not open sourced)
- Postgres and MySQL are both supported as Aurora DB (that means your drivers will work as if Aurora was a Postgres or MySQL database)
- Aurora is “AWS cloud optimized” and claims 5x performance improvement over MySQL on RDS, over 3x the performance of Postgres on RDS
- Aurora storage automatically grows in increments of 10GB, up to 128TB.
- Aurora can have up to 15 replicas and the replication process is faster than MySQL (sub 10 ms replica lag)
- Failover in Aurora is instantaneous. It’s HA (High Availability) native.
- Aurora costs more than RDS (20% more) – but is more efficient

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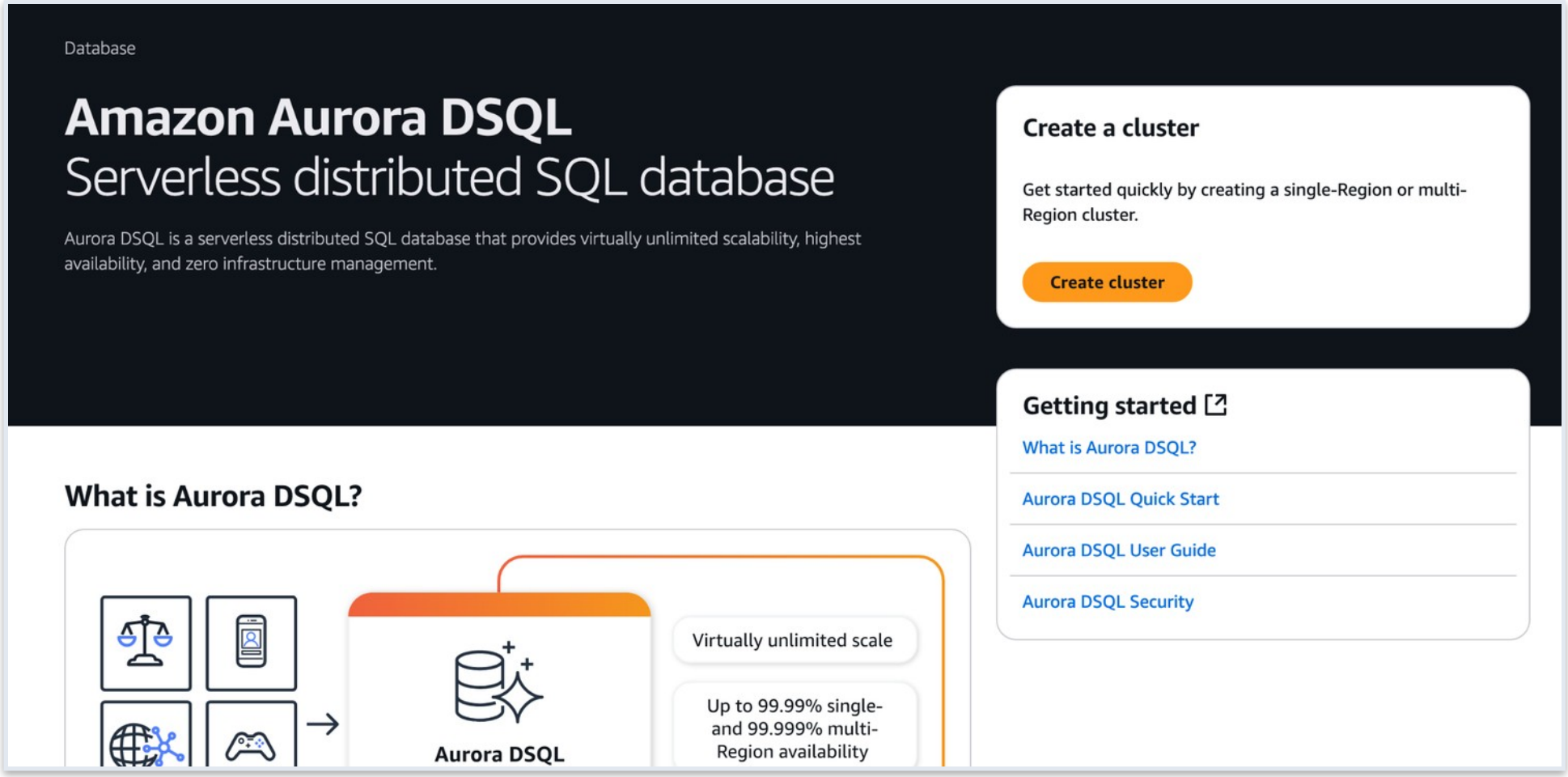
Amazon Aurora



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Aurora High Availability and Read Scaling

- 6 copies of your data across 3 AZ:
- 4 copies out of 6 needed for writes
- 3 copies out of 6 need for reads
- Self healing with peer-to-peer replication
- Storage is striped across 100s of volumes
- One Aurora Instance takes writes (master)
- Automated failover for master in less than 30 seconds
- Master + up to 15 Aurora Read Replicas serve reads
- Support for Cross Region Replication

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Auto Scaling

- Aurora DB Cluster
- W
- R
- R R R
- R
- Reader Endpoint
- Connection Load Balancing
- Writer Endpoint
- Pointing to the master
- client
- Shared storage Volume
- Auto Expanding from 10G to 128 TB

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Features of Aurora

- Automatic fail-over
- Backup and Recovery
- Isolation and security
- Industry compliance
- Push-button scaling
- Automated Patching with Zero Downtime
- Advanced Monitoring
- Routine Maintenance
- Backtrack: restore data at any point of time without using backups

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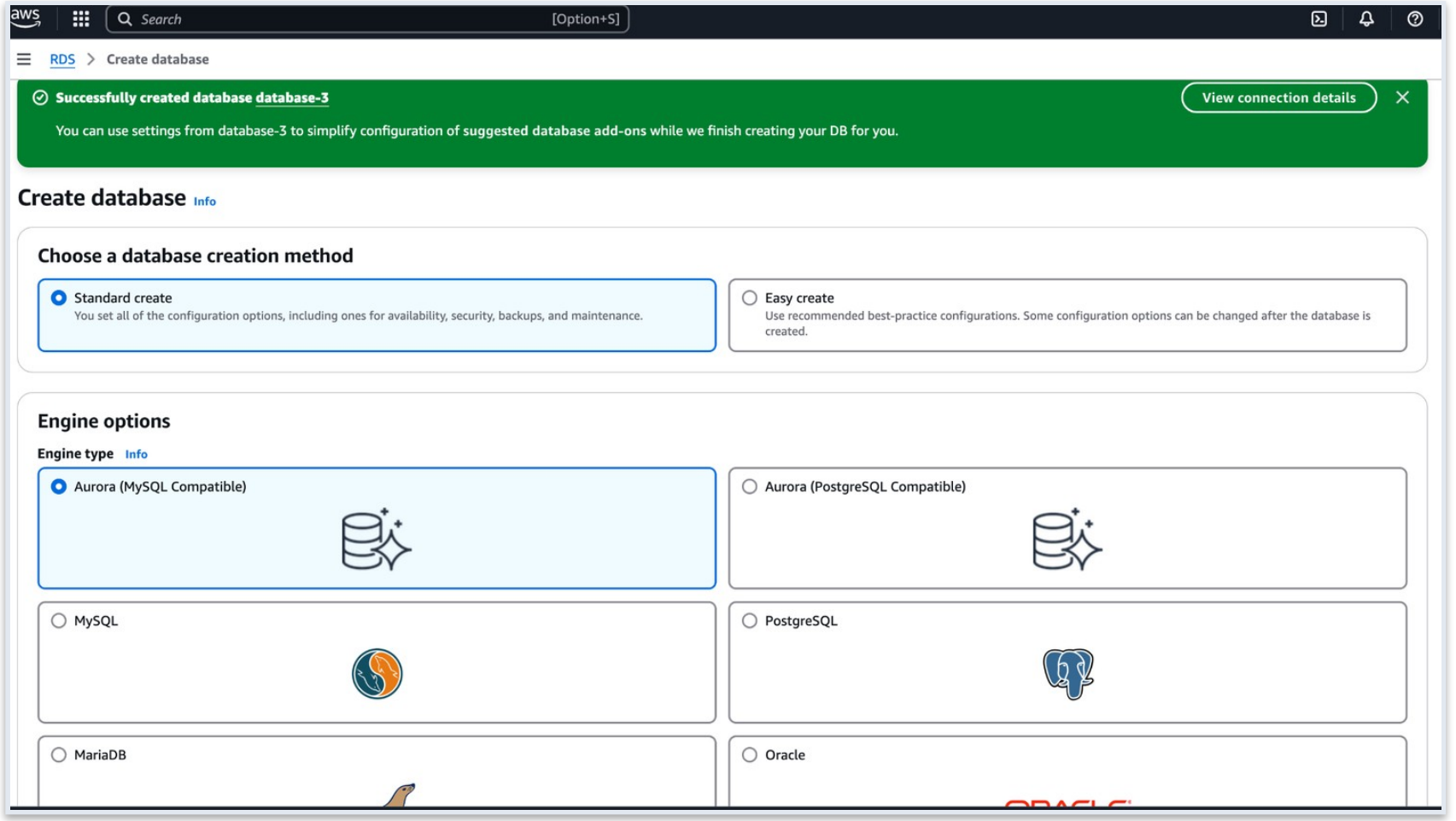
Activity: Create Aurora



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Backups, Backtracking & Restores in Aurora

- Automatic Backups
- Retention period 1-35 days (can’t be disabled)
- PITR, restore your DB cluster within 5 minutes of the current time
- Restore to a new DB cluster
- Aurora Backtracking
- Rewind the DB cluster back and forth in time (up to 72 hours)
- Doesn’t create a new DB cluster (in-place restore)
- Supports Aurora MySQL only
- Aurora Database Cloning
- Creates a new DB cluster that uses the same DB cluster volume as the original cluster
- Uses copy-on-write protocol (use the original/single copy of the data and allocate storage only when changes made to the data)
- Example: create a test environment using your production data

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Aurora for SysOps

- You can associate a priority tier (0-15) on each Read Replica
- Controls the failover priority
- RDS will promote the Read Replica with the highest priority (lowest tier)
- If replicas have the same priority, RDS promotes the largest in size
- If replicas have the same priority and size, RDS promotes arbitrary replica
- You can migrate an RDS MySQL snapshot to Aurora MySQL Cluster

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Aurora: CloudWatch metrics

- AuroraReplicaLag: amount of lag when replicating updates from the primary instance
- AuroraReplicaLagMaximum: max. amount of lag across all DB instances in the cluster
- AuroraReplicaLagMinimum: min. amount of lag across all DB instances in the cluster
- If replica lag is high, that means the users will
- have a different experience based on which replica they get the data from (eventual consistency)
- DatabaseConnections: current number of connections to a DB instance
- InsertLatency: average duration of insert operations
- AZ 1
- AZ 2
- AZ 3
- W
- R
- R R

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RDS & Aurora Security

- At-rest encryption:
- Database master & replicas encryption using AWS KMS – must be defined as launch time
- If the master is not encrypted, the read replicas cannot be encrypted
- To encrypt an un-encrypted database, go through a DB snapshot & restore as encrypted
- In-flight encryption: TLS-ready by default, use the AWS TLS root certificates client-side
- IAM Authenication: IAM roles to connect to your database (instead of username/pw)
- Security Groups: control Network access to your RDS / Aurora DB
- No SSH available except on RDS Custom
- Audit Logs can be enabled and sent to CloudWatch Logs for longer retention

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Amazon ElastiCache Overview

- The same way RDS is to get managed Relational Databases…
- ElastiCache is to get managed Redis or Memcached
- Caches are in-memory databases with really high performance, low latency
- Helps reduce load off of databases for read intensive workloads
- Helps make your application stateless
- AWS takes care of OS maintenance / patching, optimizations, setup, configuration, monitoring, failure recovery and backups
- Using ElastiCache involves heavy application code changes

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ElastiCache Solution Architecture - DB Cache

- Applications queries ElastiCache, if not available, get from RDS and store in ElastiCache.
- Helps relieve load in RDS
- Cache must have an invalidation strategy to make sure only the most current data is used in there.
- application
- Cache hit
- Amazon RDS
- Amazon ElastiCache
- Cache miss
- Read from DB
- Write to cache

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ElastiCache Solution Architecture - User Session Store

- User logs into any of the application
- The application writes the session data into ElastiCache
- The user hits another instance of our application
- The instance retrieves the data and the user is already logged in
- application
- Retrieve session
- Amazon ElastiCache
- application
- application
- Write session
- User

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ElastiCache - Redis vs Memcached

- REDIS
- Multi AZ with Auto-Failover
- Read Replicas to scale reads and have high availability
- Data Durability using AOF persistence
- Backup and restore features
- Supports Sets and Sorted Sets
- MEMCACHED
- Multi-node for partitioning of data (sharding)
- No high availability (replication)
- Non persistent
- Backup and restore (Serverless)
- Multi-threaded architecture
- +

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One primary node, up to 5 replicas

- Asynchronous Replication
- The primary node is used for read/write
- The other nodes are read-only
- One shard, all nodes have all the data
- Guard against data loss if node failure
- Multi-AZ enabled by default for failover
- Helpful to scale read performance
- ElastiCache Replication: Cluster Mode Disabled
- Shard
- Primary Node
- …
- 0-5 Replica Nodes
- Redis (cluster mode disabled) Cluster with Replication

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Redis Scaling - Cluster Mode Disabled

- Primary Node
- …
- 0-5 Replica Nodes
- Horizontal:
- Scale out/in by adding/removing read replicas (max. 5 replicas)
- Vertical:
- Scale up/down to larger/smaller node type
- ElastiCache will internally create a new node group, then data replication and DNS update
- Old Node Group New Node Group
- Primary Node
- …
- 0-5 Replica Nodes
- Replication

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ElastiCache Replication: Cluster Mode Enabled

- Data is partitioned across shards (helpful to scale writes)
- Each shard has a primary and up to 5 replica nodes (same concept as before)
- Multi-AZ capability
- Up to 500 nodes per cluster:
- 500 shards with single master
- 250 shards with 1 master and 1 replica
- …
- 83 shards with one master and 5 replicas
- Availability Zone
- Redis (cluster mode enabled)
- Cluster with Replication
- Shard 1
- Primary Node

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ElastiCache Redis (Cluster Mode Enabled)

- ElastiCache for Redis - Auto Scaling
- Shard 1
- Primary
- Replicas
- Shard 2
- Primary
- Replicas
- CloudWatch Metrics
- CloudWatch Alarm
- ElastiCachePrimaryEngineCPUUtilization
- CloudWatch Metric
- Target Tracking Policy ElastiCachePrimaryEngineCPUUtilization = 60%
- trigger

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ElastiCache - Redis Connection Endpoints

- Standalone Node
- One endpoint for read and write operations
- Cluster Mode Disabled Cluster
- Primary Endpoint - for all write operations
- Reader Endpoint - evenly split read operations across all read replicas
- Node Endpoint - for read operations
- Cluster Mode EnabledCluster
- Configuration Endpoint - for all read/write operations that support Cluster Mode Enabled commands
- Node Endpoint - for read operations
- Primary
- Read Replicas
- Primary Endpoint
- Client

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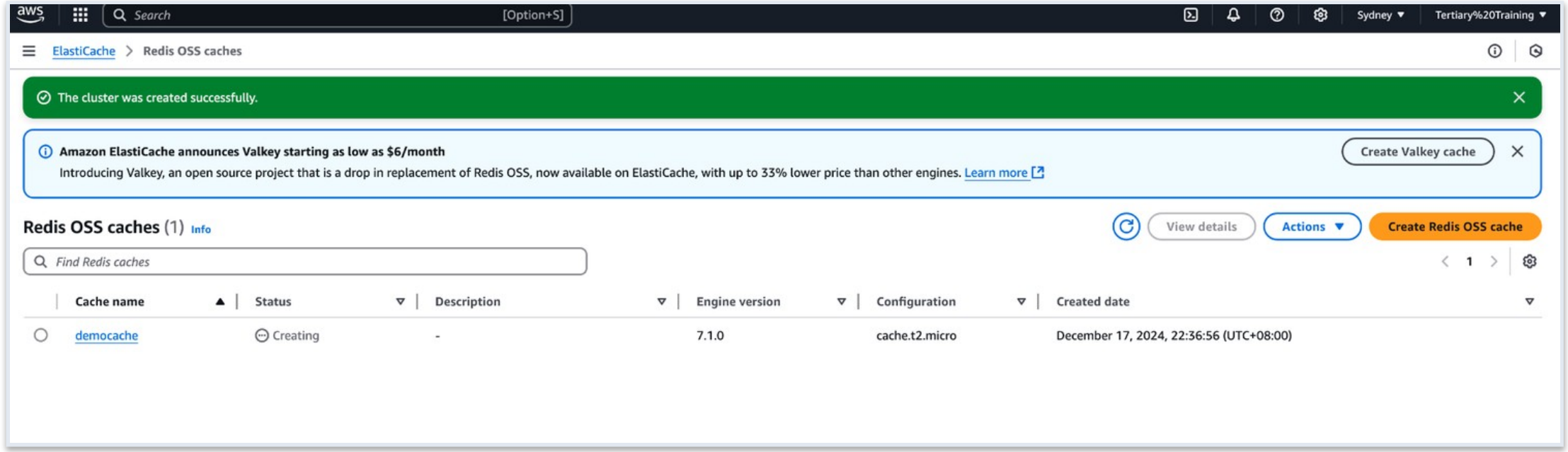
Activity: Create Elastic Cache



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Redis Scaling - Cluster Mode Disabled

- Horizontal:
- Scale out/in by adding/removing read replicas (max. 5 replicas)
- Vertical:
- Scale up/down to larger/smaller node type
- ElastiCache will internally create a new node group, then data replication and DNS update
- Primary Node
- …
- 0-5 Replica Nodes
- Primary Node
- …
- 0-5 Replica Nodes
- Replication
- DNS update

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Redis Scaling - Cluster Mode Enabled

- Two Modes:
- Online Scaling: continue serving requests during the scaling process (no downtime, some degradation in performance)
- Offline Scaling: unable to serve requests during the scaling process (backup and
- restore). Additional configurations supported (change node type, upgrade engine version, …)
- Horizontal: (Resharding and Shard Rebalancing)
- Resharding: scale out/in by adding/removing shards
- Shard Rebalancing: equally distribute the keyspaces among the shards as possible
- Supports Online and Offline Scaling
- Vertical: (change read/write capacity)
- Scale up/down to larger/smaller node type
- Supports Online Scaling

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Redis Metrics to Monitor

- Evictions: the number of non-expired items the cache evicted to allow space for new writes (memory is overfilled). Solution:
- Choose an eviction policy to evict expired items (e.g., evict least recently used (LRU) items)
- Scale up to larger node type (more memory) or scale out by adding more nodes
- CPUUtilization: monitor CPU utilization for the entire host
- Solution: scale up to larger node type (more memory) or scale out by adding more nodes
- SwapUsage: should not exceed 50 MB
- Solution: verify that you have configured enough reserved memory

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Redis Metrics to Monitor

1. CurrConnections: the number of concurrent and active connections
2. Solution: investigate application behavior to address the issue
3. DatabaseMemoryUsagePercentage: the percentage of memory utilization
4. NetworkBytesIn/Out & NetworkPacketsIn/Out
5. ReplicationBytes: the volume of data being replicated
6. ReplicationLag: how far behind the replica is from the primary node

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Memcached Scaling

- Memcached Clusters can have 1-40 nodes (soft limit)
- Horizontal:
- Add/remove nodes from the cluster
- Auto-discovery allow your app to find nodes
- Vertical:
- Scale up/down to larger/smaller node type
- Scale up process:
- Create a new cluster with the new node type
- Update your application to use the new cluster’s endpoints
- Delete the old cluster
- Memcached clusters/nodes start out empty
- Horizontal Scaling
- Memcached Cluster

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Memcached Auto Discovery

- Typically you need to manually connect to individual cache nodes in the cluster using its DNS endpoints
- Auto Discovery automatically identifies all of the nodes
- All the cache nodes in the cluster maintain a list of metadata about all other nodes
- This is seamless from a client perspective
- Memcached Cluster
- Cache Node 1
- Cache Node 2
- Cache Node 3
- Cache Node 4
- Client
- connect to configuration endpoint
- Cache Node 1 (IP address)

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Memcached Metrics to Monitor

- Evictions: the number of non-expired items the cache evicted to allow space for new writes (memory is overfilled). Solution:
- Choose an eviction policy to evict expired items (e.g., LRU items)
- Scale up to larger node type (more memory) or scale out by adding more nodes
- CPUUtilization
- Solution: scale up to larger node type or scale out by adding more nodes
- SwapUsage: should not exceed 50 MB
- CurrConnections: the number of concurrent and active connections
- Solution: investigate application behavior to address the issue
- Freeable Memory: amount of free memory on the host

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Example for RDS

- AWS responsibility:
- Manage the underlying EC2 instance, disable SSH access
- Automated DB patching
- Automated OS patching
- Audit the underlying instance and disks & guarantee it functions
- Your responsibility:
- Check the ports / IP / security group inbound rules in DB's SG
- In-database user creation and permissions
- Creating a database with or without public access
- Ensure parameter groups or DB is configured to only allow SSL connections
- Database encryption setting

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Example for S3

- AWS responsibility:
- Guarantee you get unlimited storage
- Guarantee you get encryption
- Ensure separation of the data between different customers
- Ensure AWS employees can’t access your data
- Your responsibility:
- Bucket configuration
- Bucket policy / public setting
- IAM user and roles
- Enabling encryption

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TOPIC 14

AWS Monitoring, Audit & Performance

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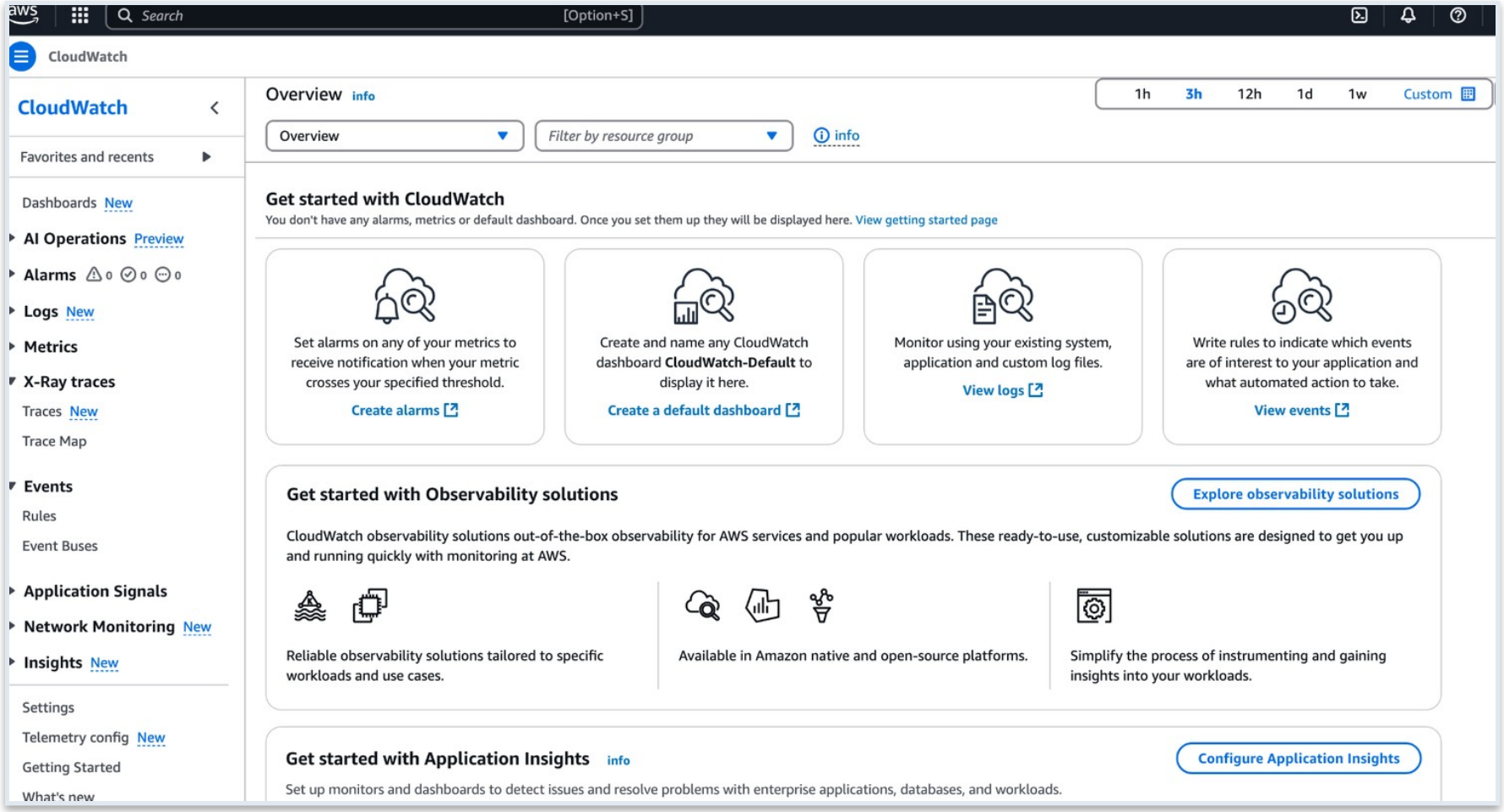
AWS CloudWatch



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Reference

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AWS CloudWatch Metrics

- CloudWatch provides metrics for every services in AWS
- Metric is a variable to monitor (CPUUtilization, NetworkIn…)
- Metrics belong to namespaces
- Dimension is an attribute of a metric (instance id, environment, etc…).
- Up to 30 dimensions per metric
- Metrics have timestamps
- Can create CloudWatch dashboards of metrics

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EC2 Detailed monitoring

- EC2 instance metrics have metrics “every 5 minutes”
- With detailed monitoring (for a cost), you get data “every 1 minute”
- Use detailed monitoring if you want to scale faster for your ASG!
- The AWS Free Tier allows us to have 10 detailed monitoring metrics
- Note: EC2 Memory usage is by default not pushed (must be pushed from inside the instance as a custom metric)

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CloudWatch Custom Metrics

- Possibility to define and send your own custom metrics to CloudWatch
- Example: memory (RAM) usage, disk space, number of logged in users …
- Use API call PutMetricData
- Ability to use dimensions (attributes) to segment metrics
- Instance.id
- Environment.name
- Metric resolution (StorageResolution API parameter - two possible value):
- Standard: 1 minute (60 seconds)
- High Resolution: 1/5/10/30 second(s) - Higher cost
- Important: Accepts metric data points two weeks in the past and two hours in the future (make sure to configure your EC2 instance time correctly)

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CloudWatch Dashboards

- Great way to setup custom dashboards for quick access to key metrics and alarms
- Dashboards are global
- Dashboards can include graphs from different AWS accounts and regions
- You can change the time zone & time range of the dashboards
- You can setup automatic refresh (10s, 1m, 2m, 5m, 15m)
- Dashboards can be shared with people who don’t have an AWS account (public, email address, 3rd party SSO provider through Amazon Cognito)
- Pricing:
- 3 dashboards (up to 50 metrics) for free
- $3/dashboard/month afterwards

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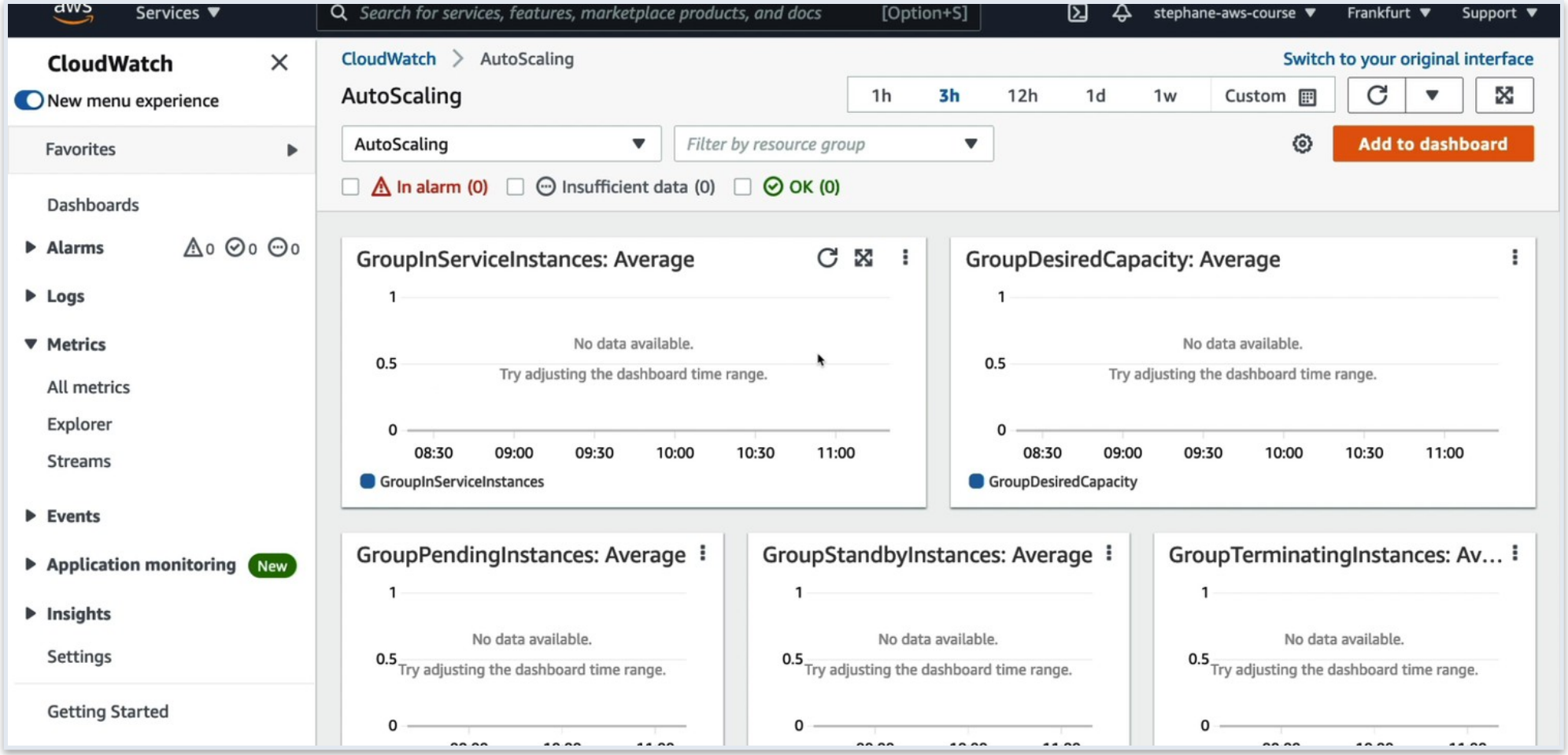
Activity: CloudWatch Dashboards



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CloudWatch Logs

- Logs group: arbitrary name, usually representing an application
- Log stream: instances within application / log files / containers
- Can define log expiration policies (never expire, 1 day to 10 years…)
- CloudWatch Logs can send logs to:
- Amazon S3 (exports)
- Kinesis Data Streams
- Kinesis Data Firehose
- AWS Lambda
- OpenSearch
- Logs are encrypted by default
- Can setup KMS-based encryption with your own keys

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CloudWatch Logs - Sources

- SDK, CloudWatch Logs Agent, CloudWatch Unified Agent
- Elastic Beanstalk: collection of logs from application
- ECS: collection from containers
- AWS Lambda: collection from function logs
- VPC Flow Logs:VPC specific logs
- API Gateway
- CloudTrail based on filter
- Route53: Log DNS queries

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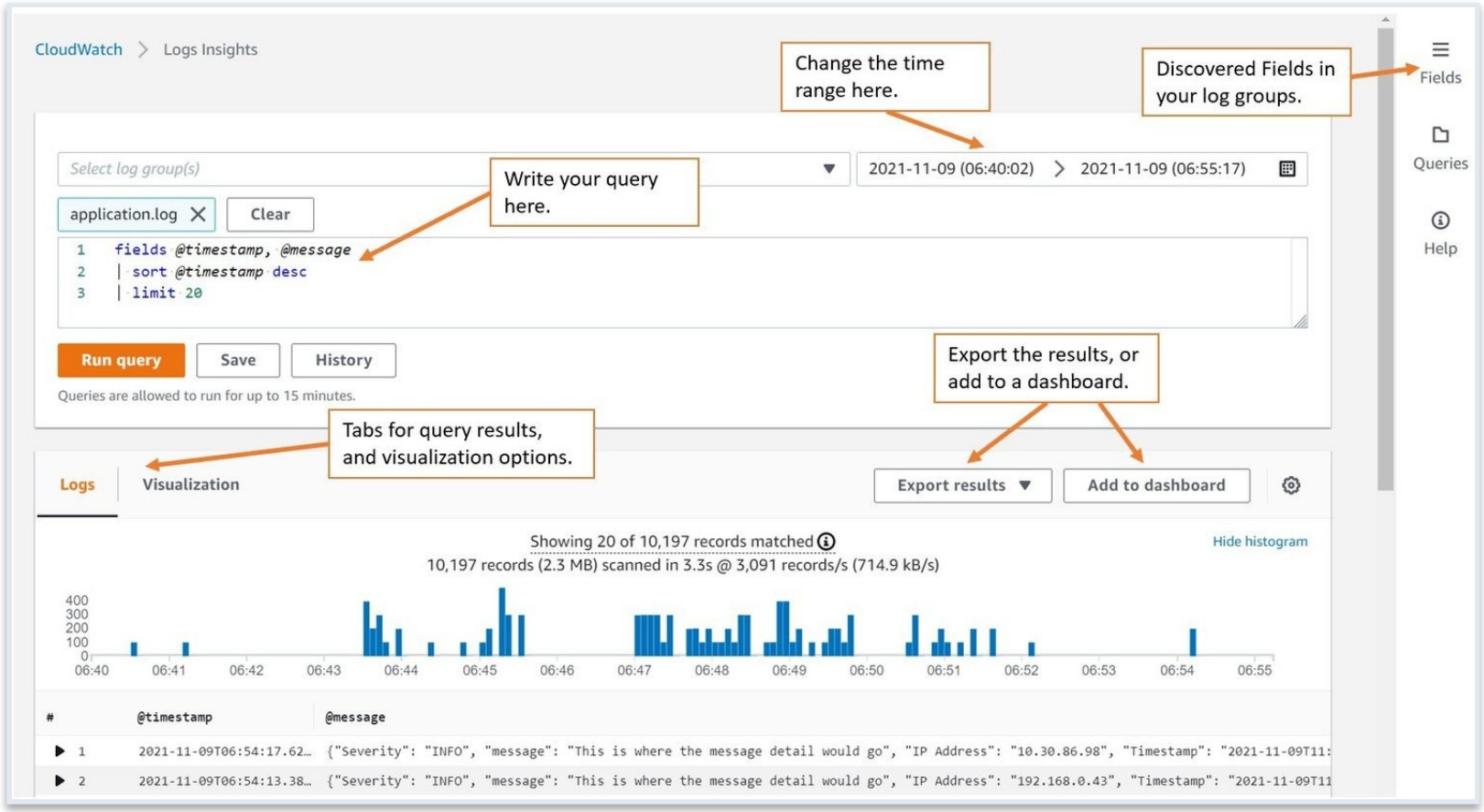
CloudWatch Logs Insights



https://mng.workshop.aws/operations-2022/detect/cwlogs.html

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CloudWatch Logs Insights

- Search and analyze log data stored in CloudWatch Logs
- Example: find a specific IP inside a log, count occurrences of “ERROR” in your logs…
- Provides a purpose-built query language
- Automatically discovers fields from AWS services and JSON log events
- Fetch desired event fields, filter based on conditions, calculate aggregate statistics, sort events, limit number of events…
- Can save queries and add them to CloudWatch Dashboards
- Can query multiple Log Groups in different AWS accounts
- It’s a query engine, not a real-time engine

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CloudWatch Logs - S3 Export

- Log data can take up to 12 hours to become available for export
- The API call is CreateExportTaskNot near-real time or real-time… use Logs Subscriptions instead
- CloudWatch Logs
- Amazon S3

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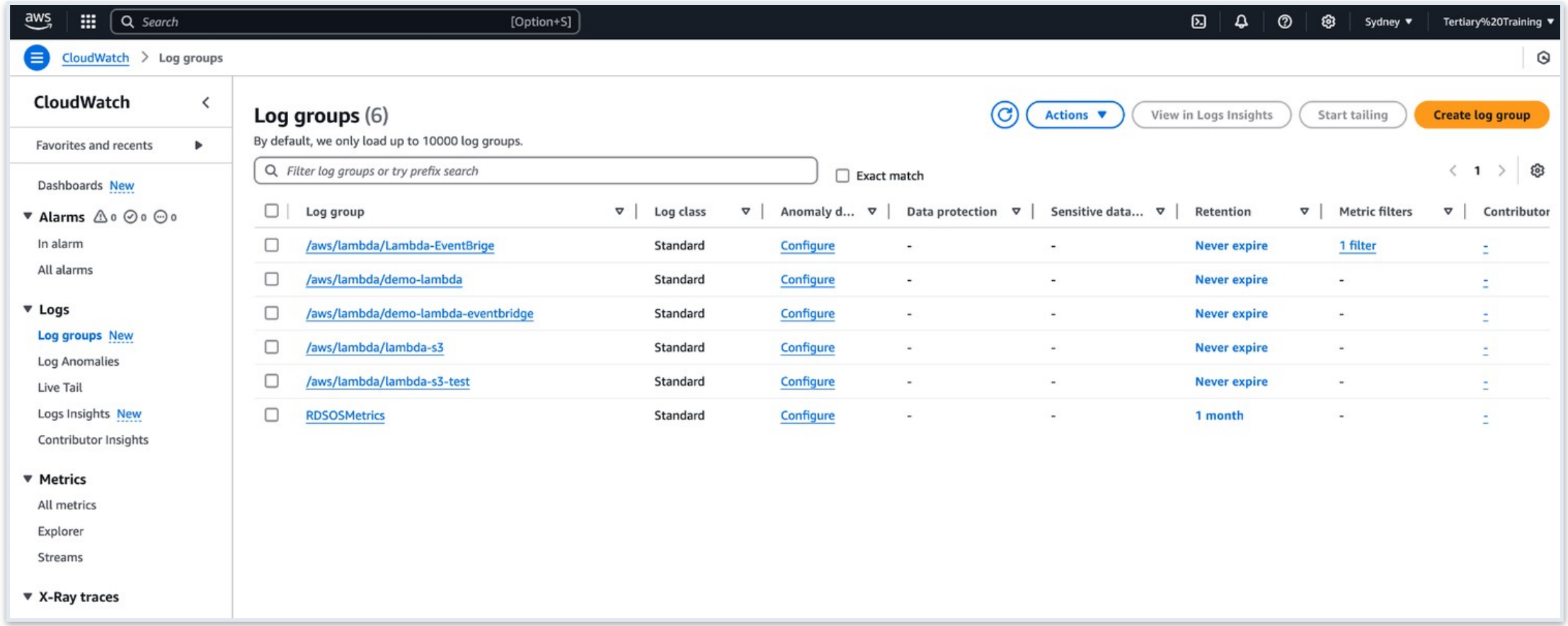
Activity: CloudWatch Logs



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CloudWatch Logs Subscriptions

- Get a real-time log events from CloudWatch Logs for processing and analysis
- Send to Kinesis Data Streams, Kinesis Data Firehose, or Lambda
- Subscription filter - filter which logs are events delivered to your destination
- CloudWatch Logs
- Subscription Filter
- Lambda
- Kinesis Data Firehose
- Kinesis Data Streams
- OpenSearch Service
- S3
- logs
- real-time
- near real-time

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CloudWatch Logs Aggregation Multi-Account & Multi Region

- Kinesis Data Streams
- Kinesis Data Firehose
- Near Real Time
- Amazon S3
- ACCOUNT A REGION 1
- CloudWatch Logs Subscription Filter
- ACCOUNT B REGION 2
- CloudWatch Logs Subscription Filter
- ACCOUNT B REGION 3
- CloudWatch Logs Subscription Filter

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CloudWatch Logs Subscriptions

- Cross-Account Subscription - send log events to resources in a different AWS account (KDS, KDF)
- Account - Sender
- (111111111111)
- CloudWatch Logs
- Subscription
- Filter
- logs
- Account - Recipient
- (999999999999)
- Kinesis Data Streams
- (RecipientStream)
- Subscription Destination
- Destination Access Policy

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CloudWatch Alarms

- Alarms are used to trigger notifications for any metric
- Various options (sampling, %, max, min, etc…)
- Alarm States:
- OK
- INSUFFICIENT_DATA
- ALARM
- Period:
- Length of time in seconds to evaluate the metric
- High resolution custom metrics: 10 sec, 30 sec or multiples of 60 sec

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CloudWatch Alarm Targets

1. Stop,Terminate, Reboot, or Recover an EC2 Instance
2. Trigger Auto Scaling Action
3. Send notification to SNS (from which you can do pretty much anything)
4. Amazon EC2
5. EC2 Auto Scaling
6. Amazon SNS

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CloudWatch Alarms - Composite Alarms

- CloudWatch Alarms are on a single metric
- Composite Alarms are monitoring the states of multiple other alarms
- AND and OR conditions
- Helpful to reduce “alarm noise” by creating complex composite alarms
- EC2 Instance
- monitor CPU
- monitor IOPS
- Amazon SNS
- Composite Alarm
- ALARM
- CW Alarm - A
- ALARM
- CW Alarm - B

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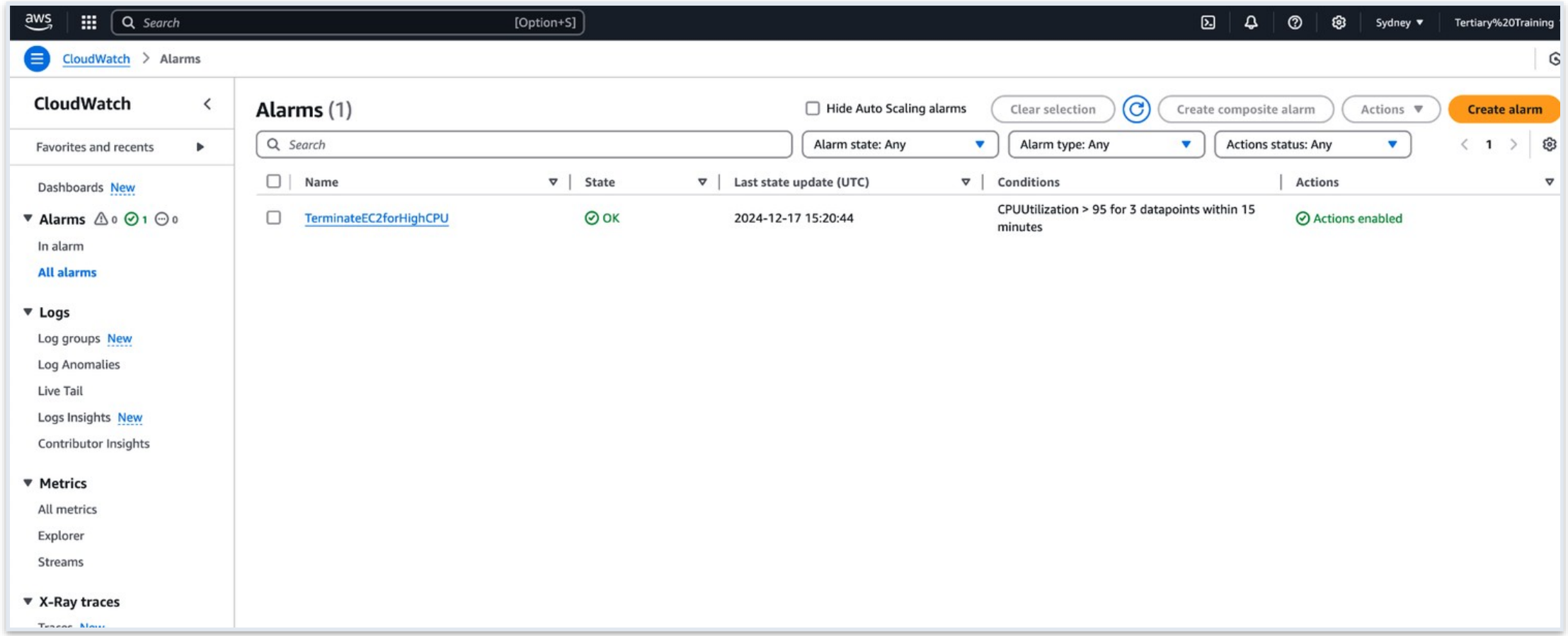
Activity: CloudWatch Alarms



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EC2 Instance Recovery

- CloudWatch Alarm
- StatusCheckFailed_System
- Status Check:
- Instance status = check the EC2 VM
- System status = check the underlying hardware
- Attached EBS status = check attached EBS volumes
- EC2 Instance
- monitor
- EC2 Instance Recovery
- Recovery: same Private, Public, Elastic IP, metadata, placement group
- SNS Topic
- alert

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CloudWatch Alarm: good to know

- Alarms can be created based on CloudWatch Logs Metrics Filters
- CW Logs
- CW Alarm
- Alert
- Amazon SNS
- To test alarms and notifications, set the alarm state to Alarm using CLI
- aws cloudwatch set-alarm-state --alarm-name "myalarm" --state-value ALARM --state-reason "testing purposes"
- CloudWatch
- Metric Filter

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CloudWatch Synthetics Canary

- Configurable script that monitor your APIs, URLs, Websites, …
- Reproduce what your customers do programmatically to find issues before customers are impacted
- Checks the availability and latency of your
- endpoints and can store load time data and screenshots of the UI
- Integration with CloudWatch Alarms
- Scripts written in Node.js or Python
- Programmatic access to a headless Google Chrome browser
- Can run once or on a regular schedule
- CloudWatch Synthetics Canary
- us-east-1
- Route 53
- us-west-2
- monitor

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CloudWatch Synthetics Canary Blueprints

- Heartbeat monitor - load URL, store screenshot and an HTTP archive file
- API canary - test basic read and write functions of REST APIs
- Broken link checker - check all links inside the URL that you are testing
- Visual monitoring - compare a screenshot taken during a canary run with a baseline screenshot
- Canary recorder - used with CloudWatch Synthetics Recorder (record your actions on a website and automatically generates a script for that)
- GUI flow builder - verifies that actions can be taken on your webpage (e.g., test a webpage with a login form)

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Amazon EventBridge (formerly CloudWatch Events)

- Schedule: Cron jobs (scheduled scripts)
- Event Pattern: Event rules to react to a service doing something
- Trigger Lambda functions, send SQS/SNS messages…
- IAM Root User Sign in Event
- SNS Topic with Email Notification
- Schedule Every hour
- Trigger script on Lambda function

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Amazon EventBridge Rules

- Example Source
- EC2 Instance
- (ex: Start Instance)
- CodeBuild
- (ex: failed build)
- S3 Event
- (ex: upload object)
- Trusted Advisor
- (ex: new Finding)
- CloudTrail
- (any API call)
- Schedule or Cron
- (ex: every 4 hours)

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Amazon EventBridge

- Event buses can be accessed by other AWS accounts using Resource-based Policies
- You can archive events (all/filter) sent to an event bus (indefinitely or set period)
- Ability to replay archived events
- AWS Services
- Default Event Bus
- Partner Event Bus
- Custom Event Bus
- AWS SaaS
- Partners
- Custom Apps

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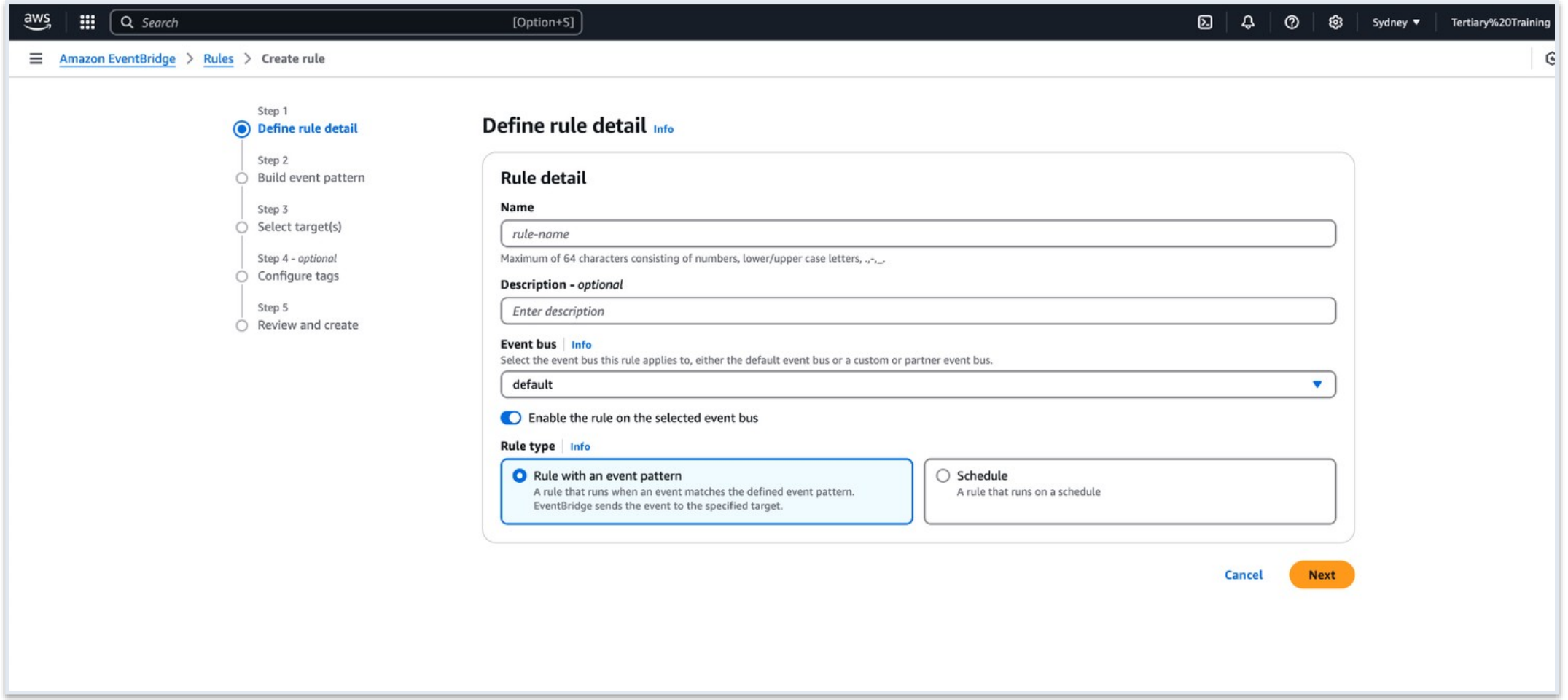
Activity: Amazon EventBridge



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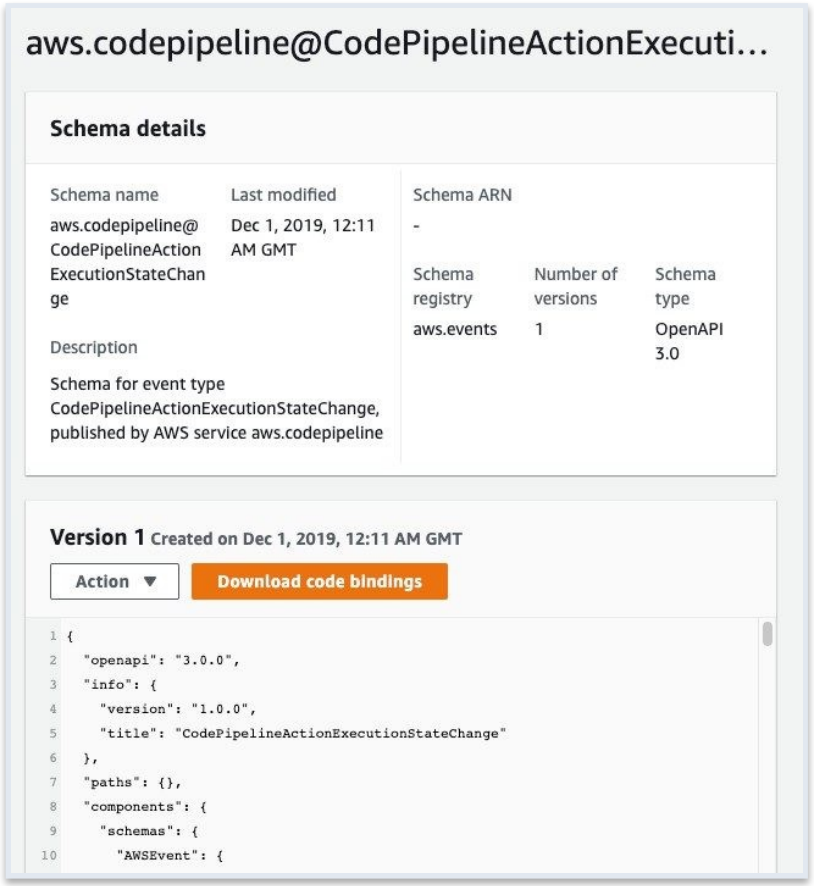
Amazon EventBridge - Schema Registry



EventBridge can analyze the events in your bus and infer the schema

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Amazon EventBridge - Resource-based Policy

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": "events:PutEvents",
            "Principal": { "AWS": "111122223333" },
            "Resource": "arn:aws:events:us-east-1:123456789012:
event-bus/central-event-bus"
        }
    ]
}
```

Manage permissions for a specific Event Bus

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Service Quotas CloudWatch Alarms

- AWS Service Quotas
- Notify you when you’re close to a service quota value threshold
- Create CloudWatch Alarms on the Service Quotas console
- Example: Lambda concurrent executions
- Helps you know if you need to request a quota increase or shutdown resources before limit is reached
- AWS Lambda Quota
- monitor
- CW Alarm
- trigger

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Alternative:Trusted Advisor + CW Alarms

- Limited number of Service Limits checks in Trusted Advisor (~50)
- Trusted Advisor publishes its check results to CloudWatch
- You can create CloudWatch Alarms on service quota usage (Service Limits)
- AWS Trusted Advisor
- CloudWatch
- CW Alarm
- Service Limits
- Lambda Concurrent Executions: 1000
- trigger

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AWS CloudTrail

- Provides governance, compliance and audit for your AWS Account
- CloudTrail is enabled by default!
- Get a history of events / API calls made within your AWS Account by:
- Console
- SDK
- CLI
- AWS Services
- Can put logs from CloudTrail into CloudWatch Logs or S3
- A trail can be applied to All Regions (default) or a single Region
- If a resource is deleted in AWS, investigate CloudTrail first!

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CloudTrail Diagram

- SDK
- CLI
- Console
- CloudTrail Console
- Inspect & Audit
- CloudWatch Logs
- S3 Bucket
- IAM Users & IAM Roles

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CloudTrail Events

- Management Events:
- Operations that are performed on resources in your AWS account
- Examples:
- Configuring security (IAM AttachRolePolicy)
- Configuring rules for routing data (Amazon EC2 CreateSubnet)
- Setting up logging (AWS CloudTrail CreateTrail)
- By default, trails are configured to log management events.
- Can separate Read Events (that don’t modify resources) from Write Events (that may modify resources)
- Data Events:
- By default, data events are not logged (because high volume operations)
- Amazon S3 object-level activity (ex: GetObject, DeleteObject, PutObject): can separate Read and Write Events

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CloudTrail Insights

- Enable CloudTrail Insights to detect unusual activity in your account:
- inaccurate resource provisioning
- hitting service limits
- Bursts of AWS IAM actions
- Gaps in periodic maintenance activity
- CloudTrail Insights analyzes normal management events to create a baseline
- And then continuously analyzes write events to detect unusual patterns
- Anomalies appear in the CloudTrail console
- Event is sent to Amazon S3
- An EventBridge event is generated (for automation needs)
- Management Events
- CloudTrail Insights
- Continuous analysis

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CloudTrail Events Retention

- Events are stored for 90 days in CloudTrail
- To keep events beyond this period, log them to S3 and use Athena
- CloudTrail
- S3 Bucket Long-term retention
- Data Events
- Management Events
- Insights Events
- log
- 90 days retention
- analyze
- Athena

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Amazon EventBridge - Intercept API Calls

- CloudTrail
- (any API call)
- Amazon EventBridge
- SNS
- event
- DynamoDB
- Log API call
- alert
- User
- DeleteTable API Call

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Amazon EventBridge + CloudTrail

- CloudTrail
- EventBridge
- IAM
- IAM Role
- User
- AssumeRole
- API Call logs
- event
- SNS
- CloudTrail
- EC2
- User
- AuthorizeSecurityGroupIngress

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CloudTrail - Log File Integrity Validation

- Digest files:
- References the log files for the last hour and contains a hash of each
- Stored in the same S3 bucket as log files (different folder)
- Helps you determine whether a log file was modified/deleted after CloudTrail delivered it
- Hashing using SHA-256, Digital Signing using SHA- 256 with RSA
- Protect the S3 bucket using bucket policy, versioning, MFA Delete protection, encryption, object lock
- Protect CloudTrail using IAM
- Log Files (1 hour)
- …
- reference
- contains a hash of each log file
- Digest File
- Amazon EC2

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CloudTrail - Integration with EventBridge

- AWS CloudTrail
- Amazon EventBridge
- trigger
- Lambda
- SNS SQS
- …
- Used to react to any API call being made in your account
- CloudTrail is not “real-time”:
- Delivers an event within 15 minutes of an API call
- Delivers log files to an S3 bucket every 5 minutes

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CloudTrail - Organizations Trails

- A trail that will log all events for all AWS accounts in an AWS Organization
- Log events for management and member accounts
- Trail with the same name will be created in every AWS account (IAM permissions)
- Member accounts can’t remove or modify the organization trail (view only)
- AWS Organization (o-exampleorgid)
- Management Account (111111111111)
- Member Account (222222222222)
- Member Account (444444444444)
- Member Account (333333333333)
- create
- AWS CloudTrail (MyOrganizationTrail)
- S3 Bucket
- (my-organization-bucket)

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AWS Config

- Helps with auditing and recording compliance of your AWS resources
- Helps record configurations and changes over time
- Questions that can be solved by AWS Config:
- Is there unrestricted SSH access to my security groups?
- Do my buckets have any public access?
- How has my ALB configuration changed over time?
- You can receive alerts (SNS notifications) for any changes
- AWS Config is a per-region service
- Can be aggregated across regions and accounts
- Possibility of storing the configuration data into S3 (analyzed by Athena)

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Config Rules

- Can use AWS managed config rules (over 75)
- Can make custom config rules (must be defined in AWS Lambda)
- Ex: evaluate if each EBS disk is of type gp2
- Ex: evaluate if each EC2 instance is t2.micro
- Rules can be evaluated / triggered:
- For each config change
- And / or: at regular time intervals
- AWS Config Rules does not prevent actions from happening (no deny)
- Pricing: no free tier, $0.003 per configuration item recorded per region,
- $0.001 per config rule evaluation per region

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AWS Config Resource

1. View compliance of a resource over time
2. View configuration of a resource over time
3. View CloudTrail API calls of a resource over time

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Config Rules - Remediations

- Automate remediation of non-compliant resources using SSM Automation Documents
- Use AWS-Managed Automation Documents or create custom Automation Documents
- Tip: you can create custom Automation Documents that invokes Lambda function
- You can set Remediation Retries if the resource is still non-compliant after auto-remediation
- IAM Access Key (NON_COMPLIANT)
- expired
- AWS Config
- monitor
- Auto-Remediation Action
- (SSM Document: AWSConfigRemediation- RevokeUnusedIAMUserCredentials)
- trigger
- Retries: 5
- deactivate

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Config Rules - Notifications

- Use EventBridge to trigger notifications when AWS resources are non- compliant
- Ability to send configuration changes and compliance state notifications to SNS (all events – use SNS Filtering or filter at client-side)
- Security group
- AWS Resources
- …
- monitor
- AWS Config
- EventBridge
- NON_COMPLIANT
- trigger
- Lambda
- SNS

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AWS Config - Aggregators

- The aggregator is created in one central aggregator account
- Aggregates rules, resources, etc… across multiple accounts & regions
- If using AWS Organizations, no need for individual Authorization
- Rules are created in each individual source AWS account
- Can deploy rules to multiple target accounts using CloudFormation StackSets
- AWS Account (Aggregator)
- Aggregator
- AWS Account – A (Source)
- Authorization
- AWS Account – B (Source)
- Authorization
- authorize Aggregator
- to collect data

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CloudWatch vs CloudTrail vs Config

- CloudWatch
- Performance monitoring (metrics, CPU, network, etc…) & dashboards
- Events & Alerting
- Log Aggregation & Analysis
- CloudTrail
- Record API calls made within your Account by everyone
- Can define trails for specific resources
- Global Service
- Config
- Record configuration changes
- Evaluate resources against compliance rules
- Get timeline of changes and compliance

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For an Elastic Load Balancer

- CloudWatch:
- Monitoring Incoming connections metric
- Visualize error codes as a % over time
- Make a dashboard to get an idea of your load balancer performance
- Config:
- Track security group rules for the Load Balancer
- Track configuration changes for the Load Balancer
- Ensure an SSL certificate is always assigned to the Load Balancer (compliance)
- CloudTrail:
- Track who made any changes to the Load Balancer with API calls

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TOPIC 15

AWS Account Management

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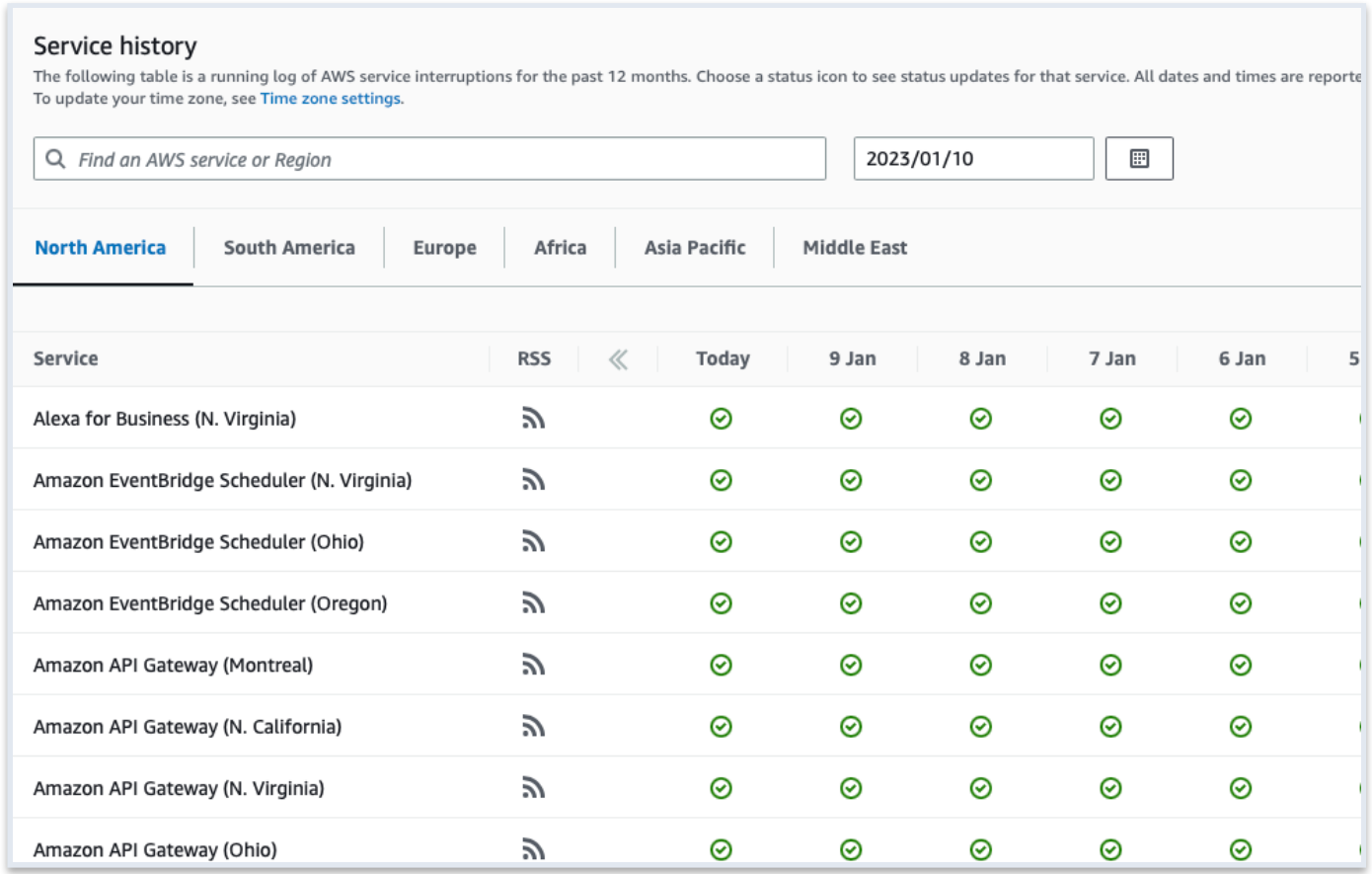
TOPIC 15

AWS Health Dashboard - Service History



Shows all regions, all services health

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AWS Health Dashboard - Your Account

- Previously called AWS Personal Health Dashboard (PHD)
- AWS Account Health Dashboard provides alerts and remediation guidance when AWS is experiencing events that may impact you.
- While the Service Health Dashboard displays the general status of AWS
- services, Account Health Dashboard gives you a personalized view into.
- the performance and availability of the AWS services underlying your AWS resources.
- The dashboard displays relevant and timely information to help you
- manage events in progress and provides proactive notification to help you plan for scheduled activities.
- Can aggregate data from an entire AWS Organization

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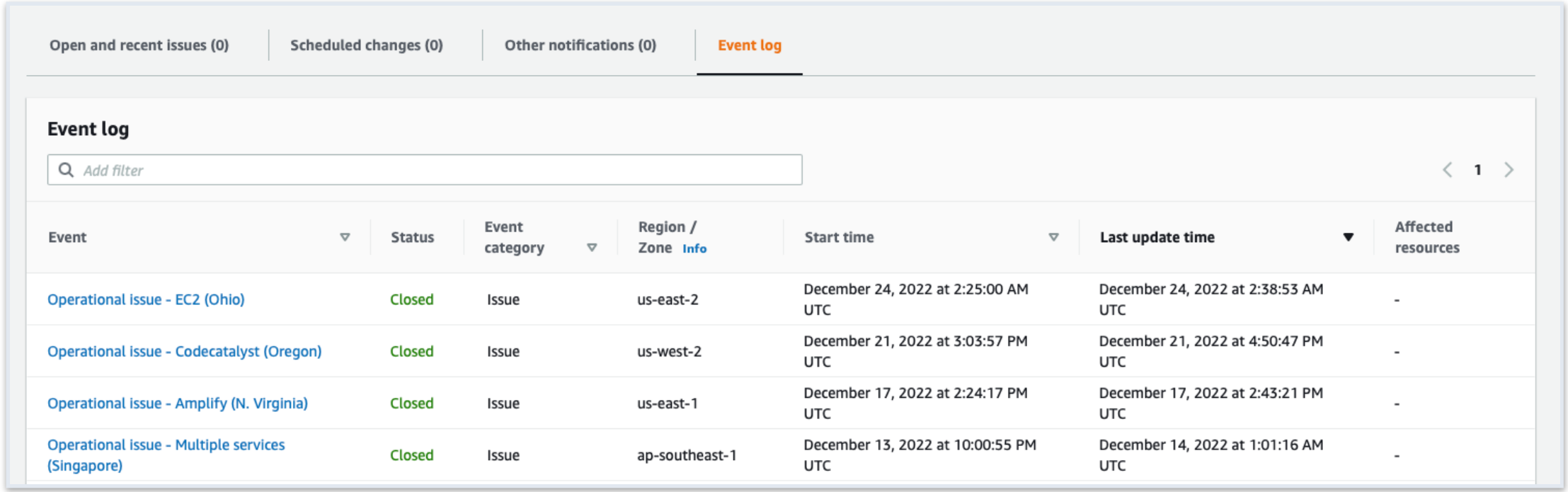
AWS Health Dashboard - Your Account



Global service

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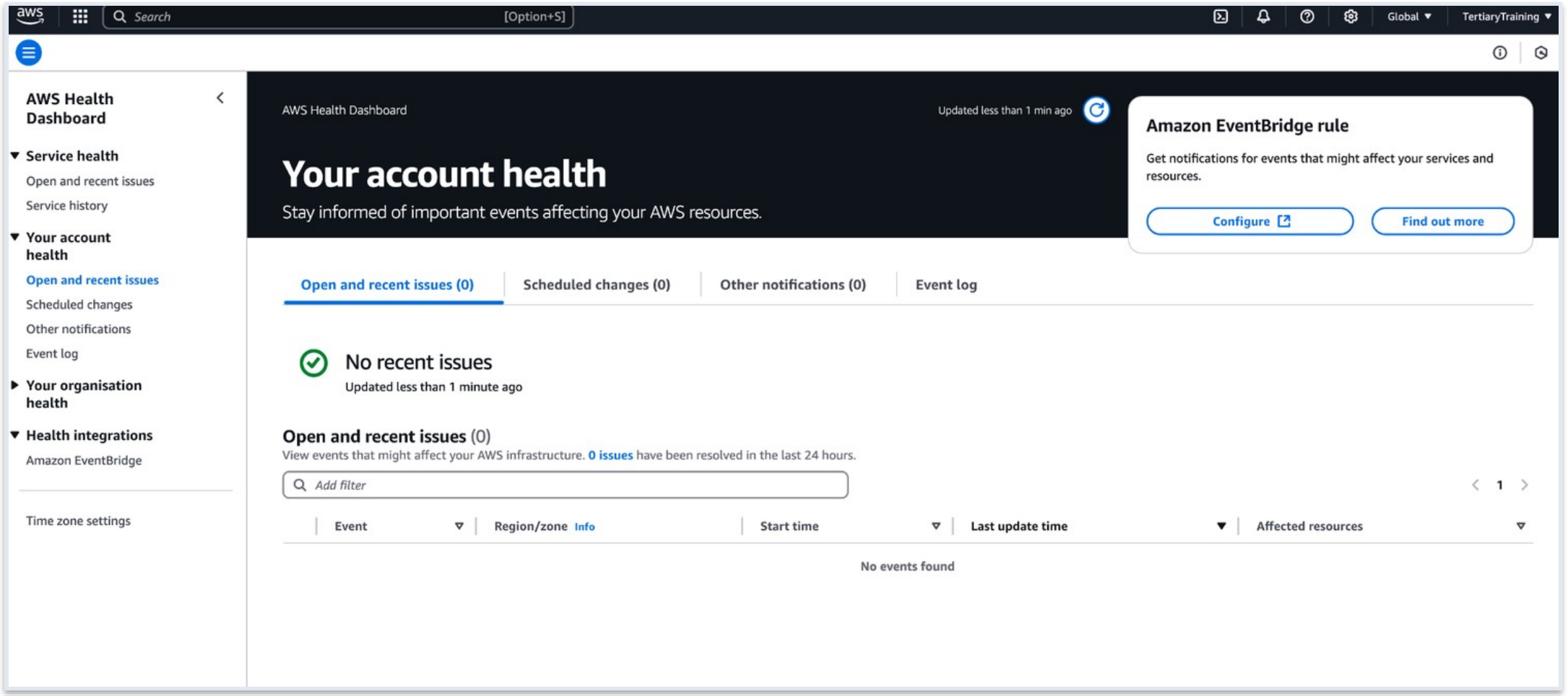
Activity: Your Health Dashboard



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Health Event Notifications

- Use EventBridge to react to changes for AWS Health events in your AWS account
- Example: receive email notifications when EC2 instances in your AWS account are scheduled for updates
- This is possible for Account events (resources that are affected in your account) and Public Events (Regional availability of a service)
- Use cases: send notifications, capture event information, take corrective action...
- AWS Health Dashboard
- EventBridge
- Lambda SNS
- SQS
- Kinesis Data Streams
- trigger
- invoke

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AWS Health Dashboard - Examples

- AWS Health Dashboard
- EventBridge
- Lambda
- IAM
- event
- Exposed Access Keys
- invoke
- Delete Access Keys
- Remediating Exposed IAM Access Keys
- AWS Health Dashboard
- EventBridge
- event
- Instance Scheduled for Retirement

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AWS Organizations

- Global service
- Allows to manage multiple AWS accounts
- The main account is the management account
- Other accounts are member accounts
- Member accounts can only be part of one organization
- Consolidated Billing across all accounts - single payment method
- Pricing benefits from aggregated usage (volume discount for EC2, S3…)
- Shared reserved instances and Savings Plans discounts across accounts
- API is available to automate AWS account creation

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AWS Organizations

- Root Organizational Unit (OU)
- Management Account
- OU (Dev)
- Member Accounts
- OU (Prod)
- OU (HR)
- OU (Finance)

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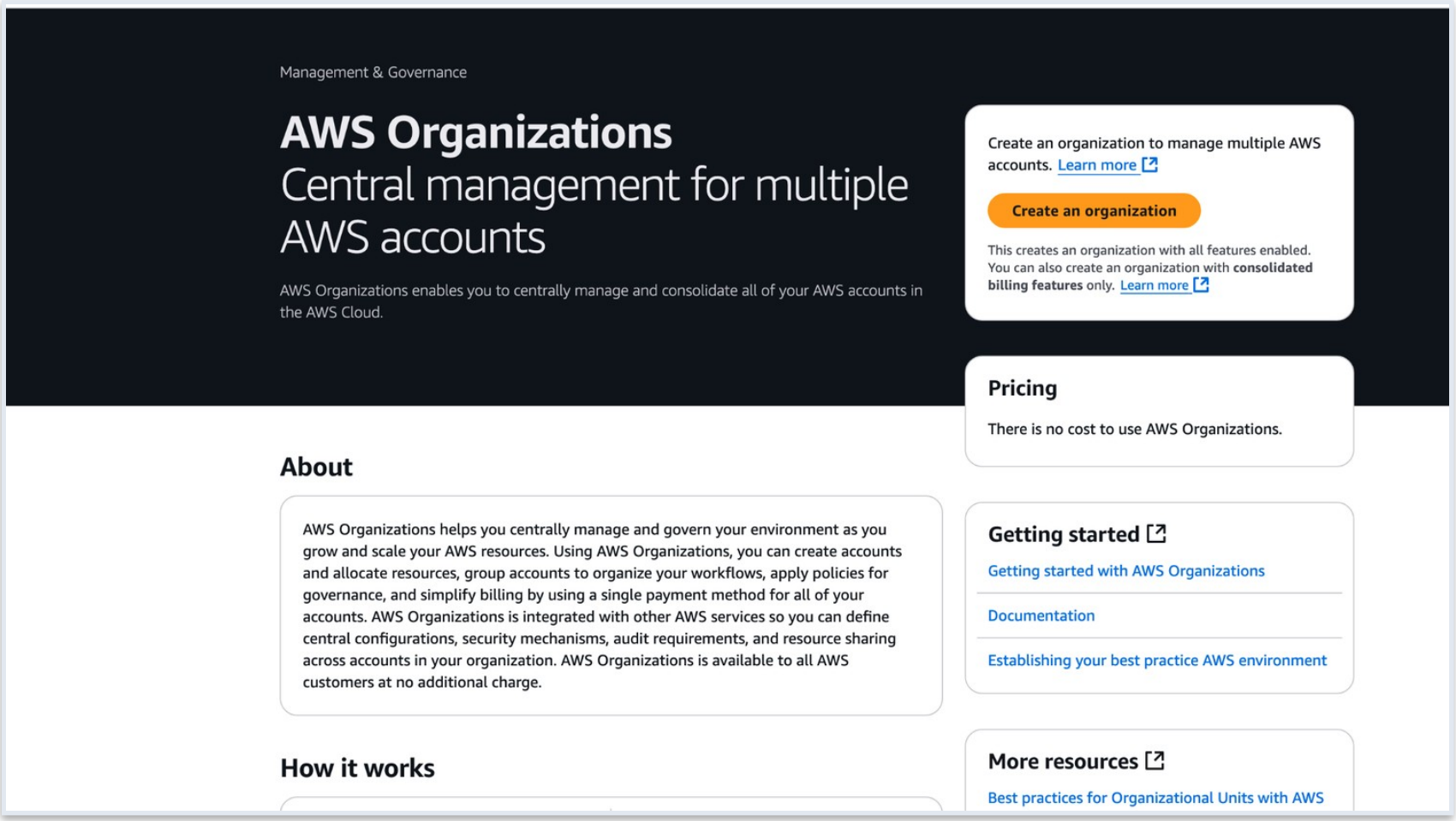
AWS Organizations



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Organizational Units (OU) - Examples

- Business Unit Environmental Lifecycle
- Management Account
- Sales OU
- Retail OU
- Finance OU
- Sales Account 1
- Sales Account 2
- Retail Account 1
- Retail Account 2
- Finance Account 1
- Finance Account 2
- Management Account
- Prod OU

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AWS Organizations

- Advantages
- Multi Account vs One Account Multi VPC
- Use tagging standards for billing purposes
- Enable CloudTrail on all accounts, send logs to central S3 account
- Send CloudWatch Logs to central logging account
- Establish Cross Account Roles for Admin purposes
- Security: Service Control Policies (SCP)
- IAM policies applied to OU or Accounts to restrict Users and Roles
- They do not apply to the management account (full admin power)
- Must have an explicit allow from the root through each OU in the direct path to the target account (does not allow anything by default – like IAM)

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SCP Hierarchy

- OU (Root)
- Management Account
- OU (Sandbox)
- Account A
- OU (Test)
- Account D
- FullAWSAccess
- Management Account
- Can do anything (no SCP apply)
- Account A
- Can do anything
- EXCEPT S3 (explicit Deny from Sandbox OU)
- EXCEPT EC2 (explicit Deny)

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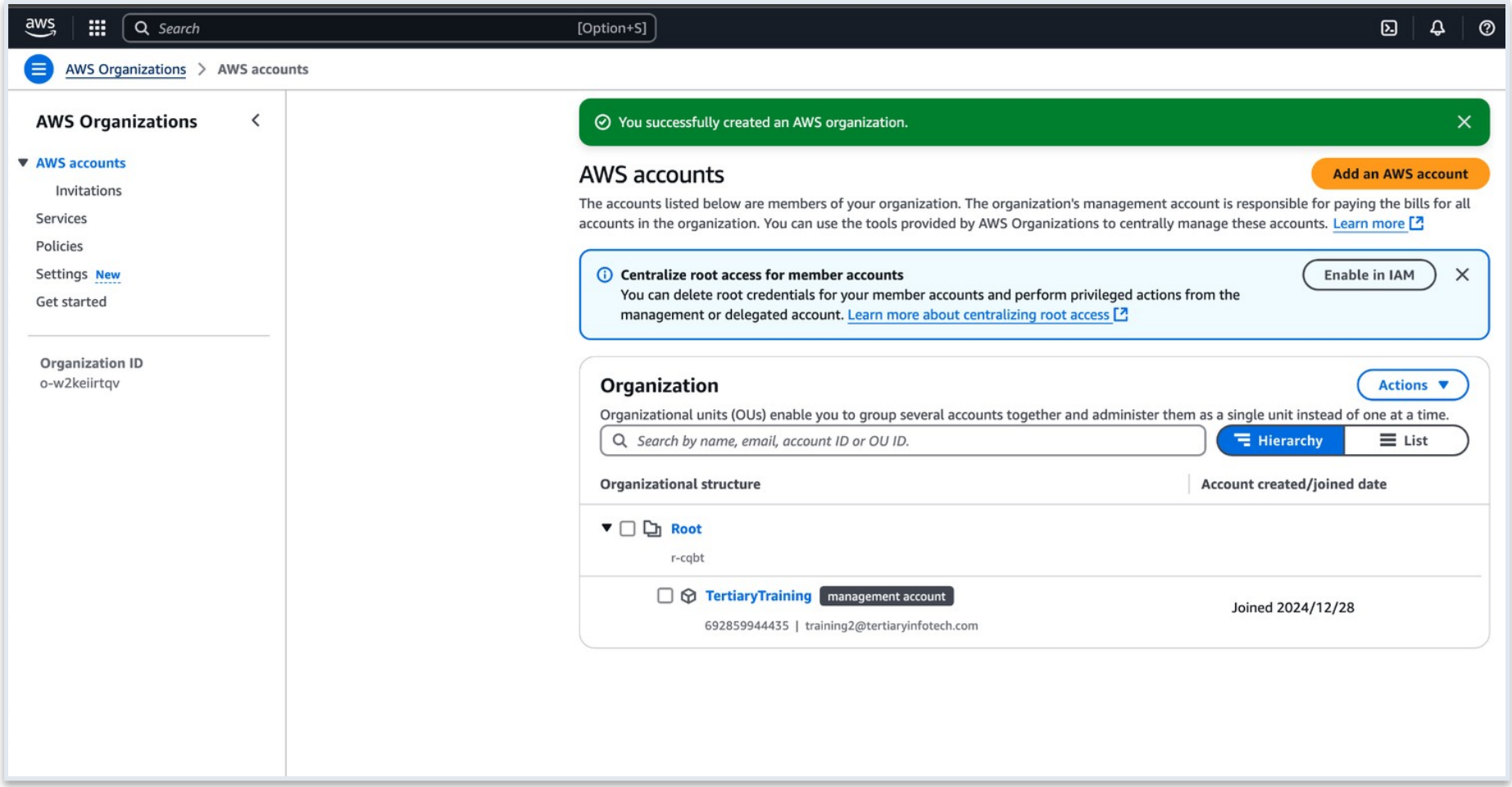
Activity: Organization



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SCP Examples

Blocklist and Allowlist strategies

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AWS Organizations - Reserved Instances

- For billing purposes, the consolidated billing feature of AWS Organizations treats all the accounts in the organization as one account.
- This means that all accounts in the organization can receive the hourly cost benefit of Reserved Instances that are purchased by any other account.
- The payer account (master account) of an organization can turn off
- Reserved Instance (RI) discount and Savings Plans discount sharing for any accounts in that organization, including the payer account
- This means that RIs and Savings Plans discounts aren't shared between any accounts that have sharing turned off.
- To share an RI or Savings Plans discount with an account, both accounts must have sharing turned on.

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AWS Organizations - IAM Policies

- Use AWS:PrincipalOrgID condition key in your resource-based policies to restrict access to IAM principals from accounts in an AWS Organization
- AWS Organization (o-exampleorgid)
- Management Account (111111111111)
- Member Account (123456789012)
- IAM User (stephane)
- Member Account (222222222222)
- IAM User (Olivia)
- S3 Bucket
- (financial-data-2021)
- Use the aws:PrincipalOrgID condition

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AWS Organizations - Tag Policies

```
{
    "tags": {
        "costcenter": {
            "tag_key": {
                "@@assign": "CostCenter"
            },
            "tag_value": {
                "@@assign": ["100", "200"]
            },
            "enforced_for": {
                "@@assign": ["secretsmanager:*"]
            }
        }
    }
}
```

Helps you standardize tags across resources in an AWS Organization

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AWS Control Tower

- Easy way to set up and govern a secure and compliant mutli-account AWS environment based on best practices
- Benefits:
- Automate the set up of your environment in a few clicks
- Automate ongoing policy management using guardrails
- Detect policy violations and remediate them
- Monitor compliance through an interactive dashboard
- AWS Control Tower runs on top of AWS Organizations:
- It automatically sets up AWS Organizations to organize accounts and implement SCPs (Service Control Policies)

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AWS Service Catalog

- Users that are new to AWS have too many options, and may create stacks that are not compliant / in line with the rest of the organization
- Some users just want a quick self-service portal to launch a set of
- authorized products pre-defined by admins
- Includes: virtual machines, databases, storage options, etc…
- Enter AWS Service Catalog!

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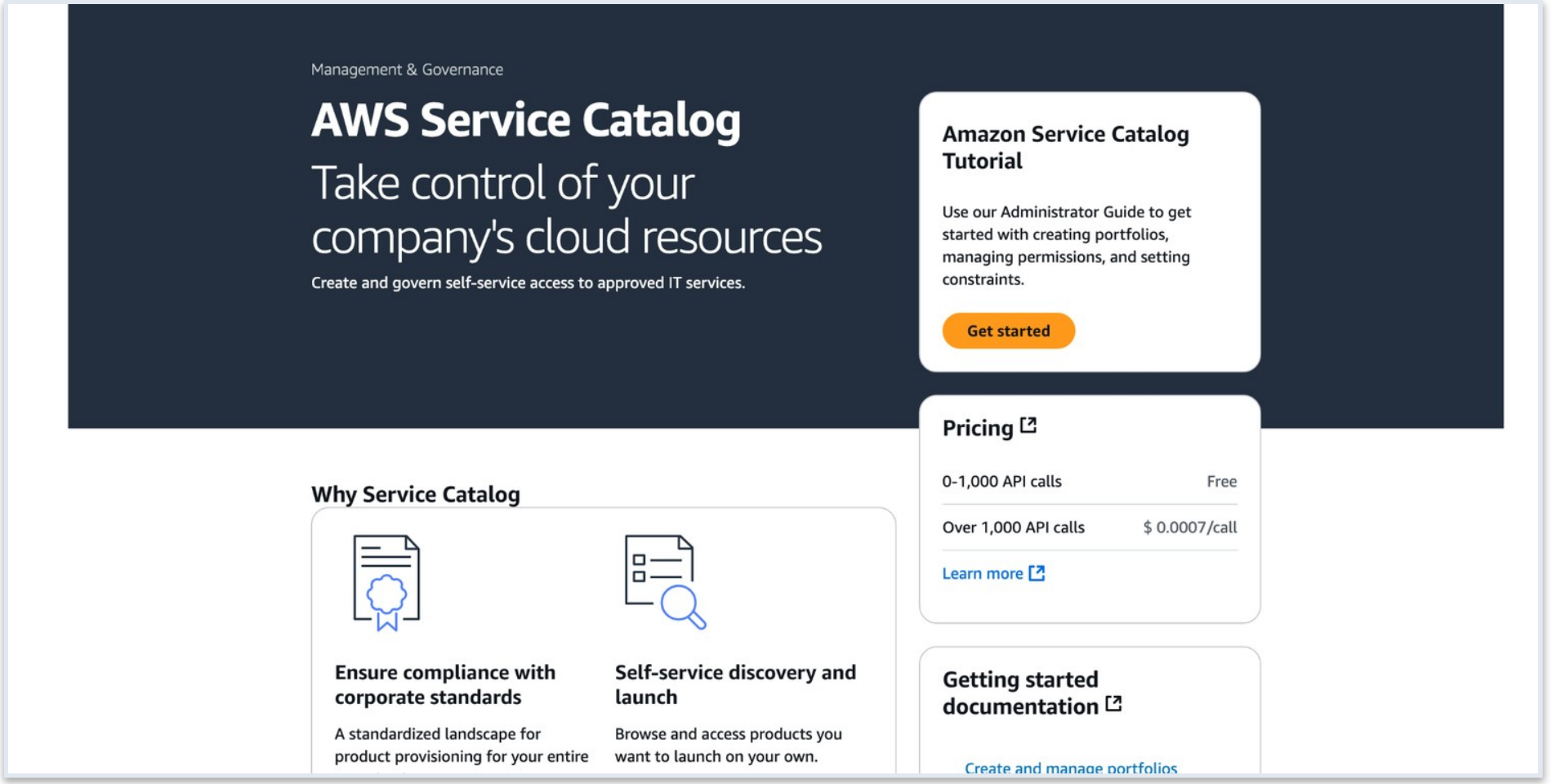
AWS Service Catalog



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Service Catalog diagram

- Product Portfolio
- CloudFormation Templates
- Collection of Products
- ADMIN TASKS
- IAM Permissions to Access Portfolios
- Control
- Product List
- USER TASKS
- Authorized by IAM
- launch
- Provisioned Products
- Ready to use Properly Configured Properly Tagged

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AWS Service Catalog - Sharing Catalogs

- Share portfolios with individual AWS accounts or AWS Organizations
- Sharing options:
- Share a reference of the portfolio, then import the shared portfolio in the recipient account (stays in-sync with the original portfolio)
- Deploy a copy of the portfolio into the recipient account (must re-deploy any updates)
- Ability to add products from the
- imported portfolio to the local portfolio
- Account A
- Portfolio A (port-xxxxxxxxxx)
- …
- Product 1 Product 2 Product 3
- Account B
- share (in-sync)

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AWS Service Catalog - TagOptions Library

- Easily manage tags on provisioned products
- TagOption:
- Key-value pair managed in AWS Service Catalog
- Used to create an AWS Tag
- Can be associated with Portfolios and Products
- Use cases: proper resources tagging, defined allowed tags, …
- Can be shared with other AWS accounts and AWS Organizations
- Portfolio A (port-xxxxxxxxxx)
- Product 1 Product 2 Product 3
- TagOption
- …
- Stack 1
- launch

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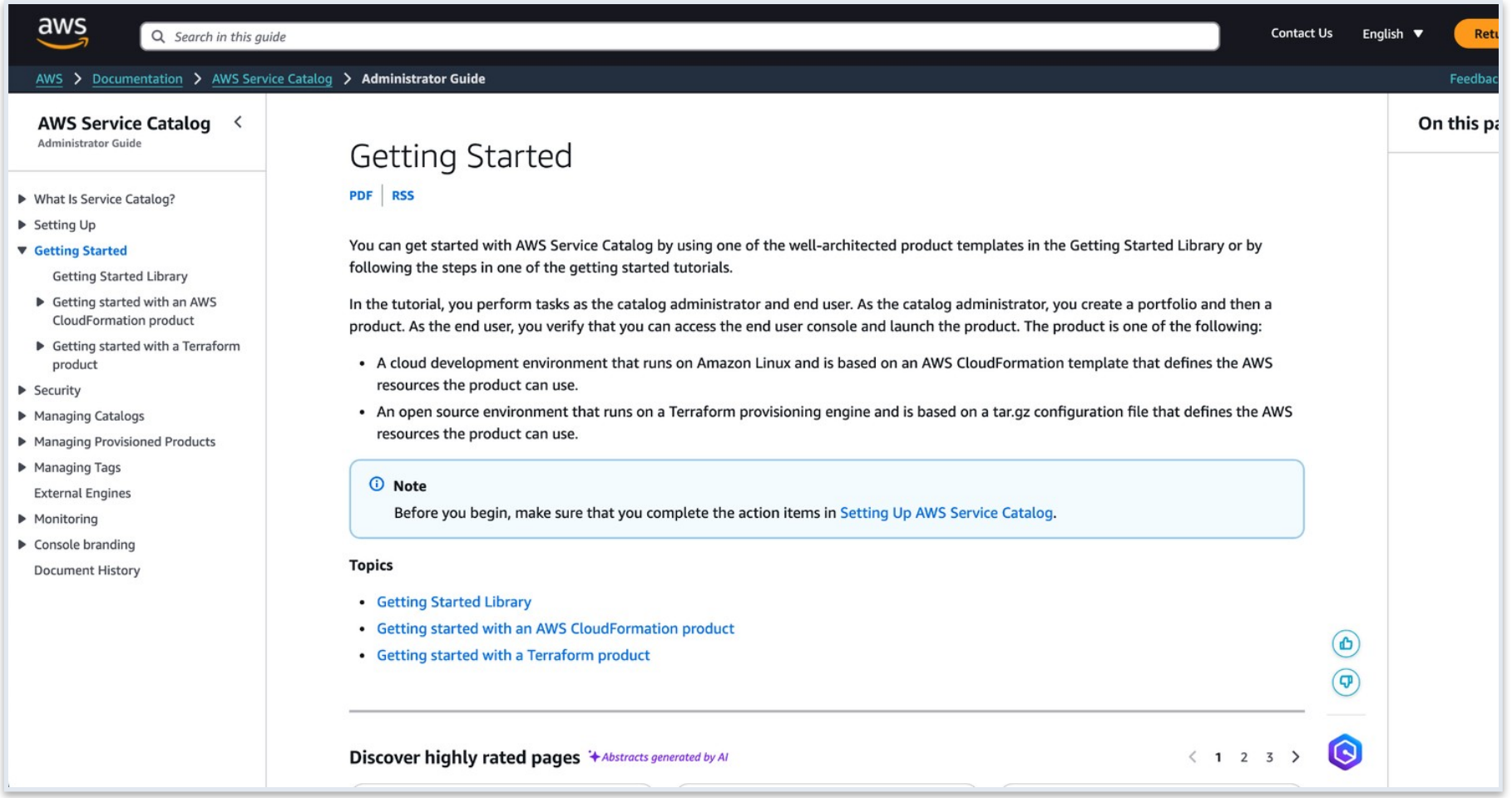
Activity: AWS Service Catalog



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AWS Billing Alarms

1. Billing data metric is stored in CloudWatch us-east-1
2. Billing data are for overall worldwide
3. AWS costs
4. It’s for actual cost, project costs
5. Let’s create a billing alarm together!

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Cost Explorer

- Visualize, understand, and manage your AWS costs and usage over time
- Create custom reports that analyze cost and usage data.
- Analyze your data at a high level: total costs and usage across all accounts
- Or Monthly, hourly, resource level granularity
- Choose an optimal Savings Plan(to lower prices on your bill)
- Forecast usage up to 12 months based on previous usage

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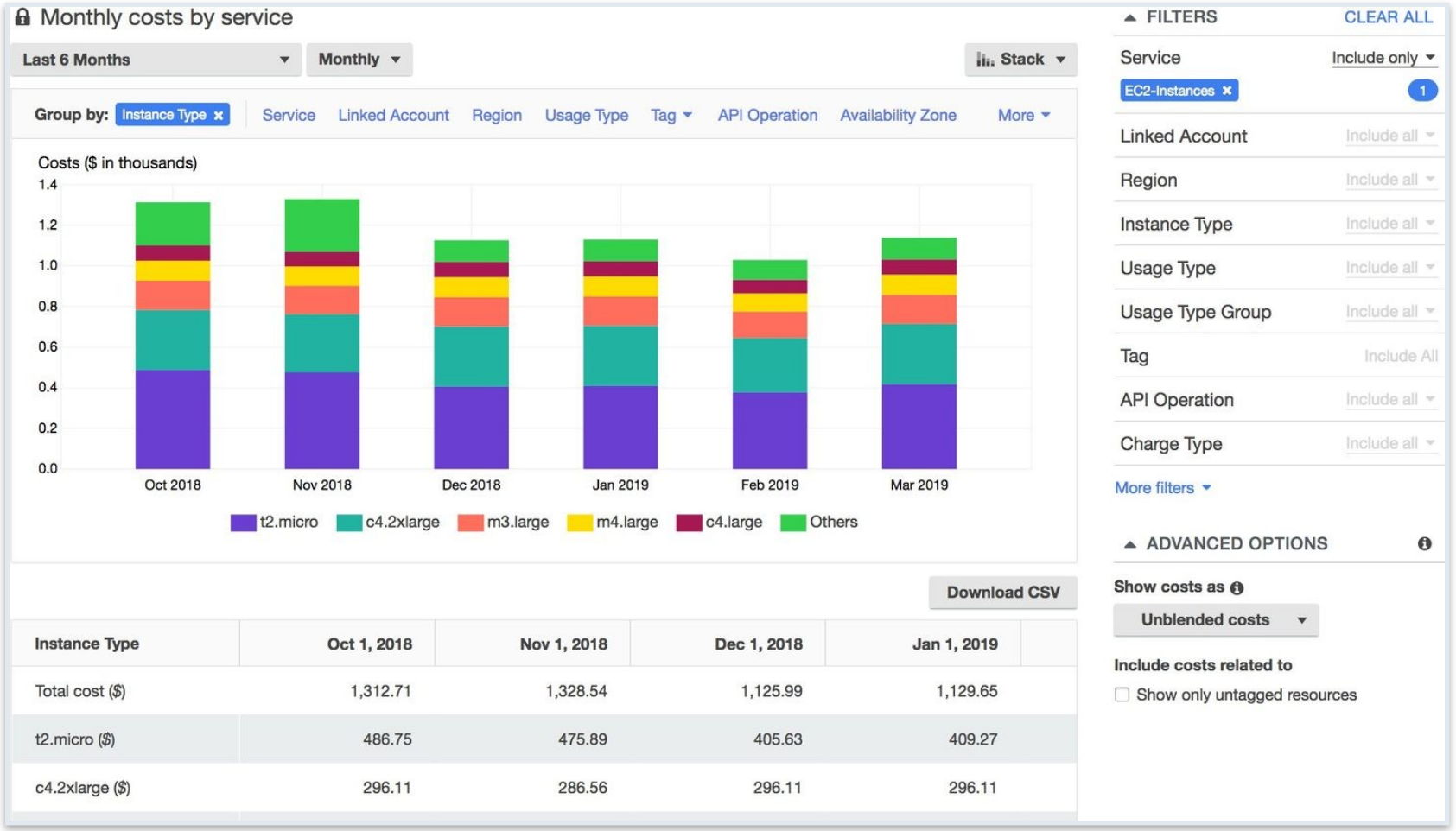
Cost Explorer - Monthly Cost by AWS Service



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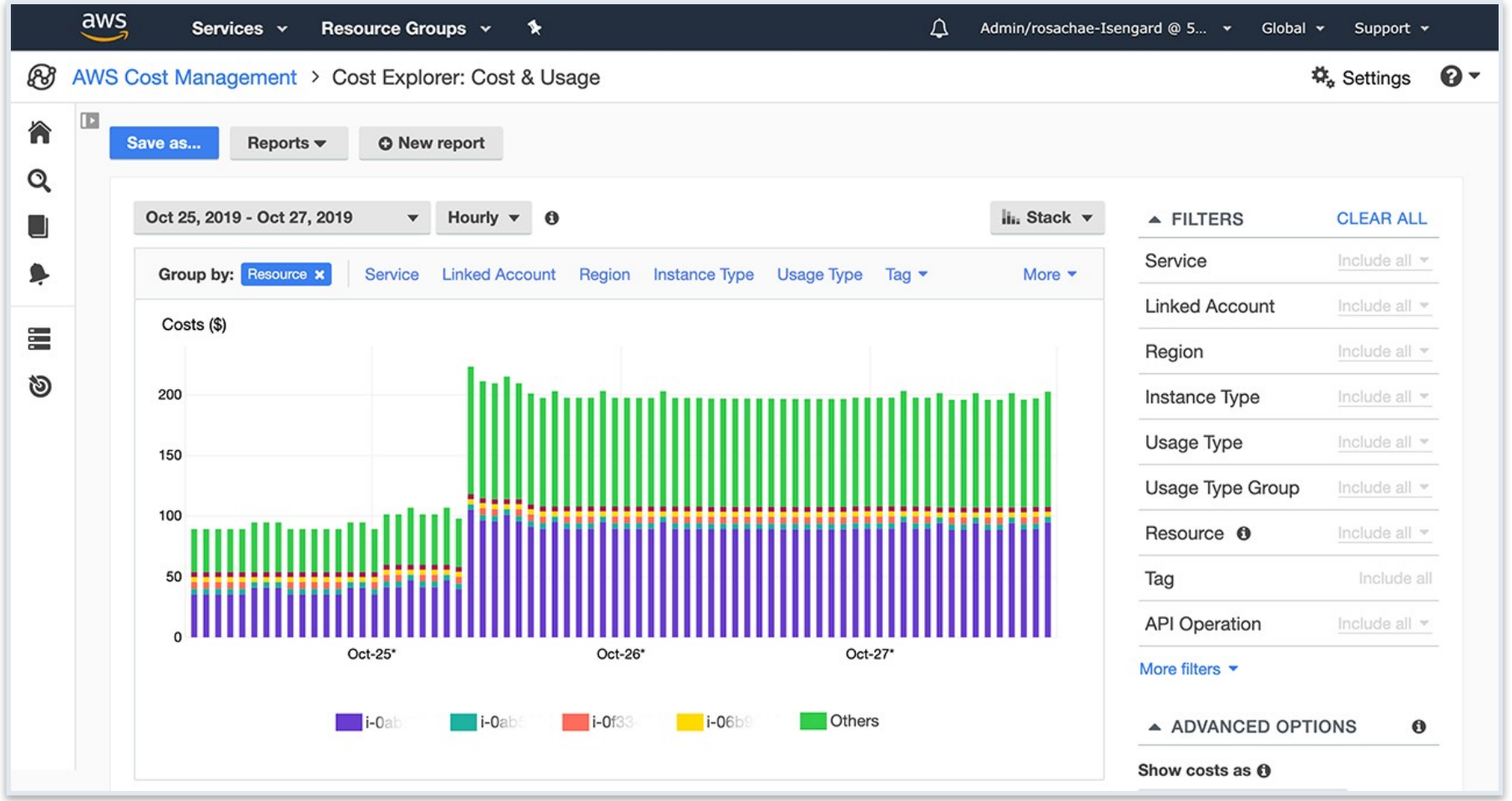
Cost Explorer - Hourly & Resource Level



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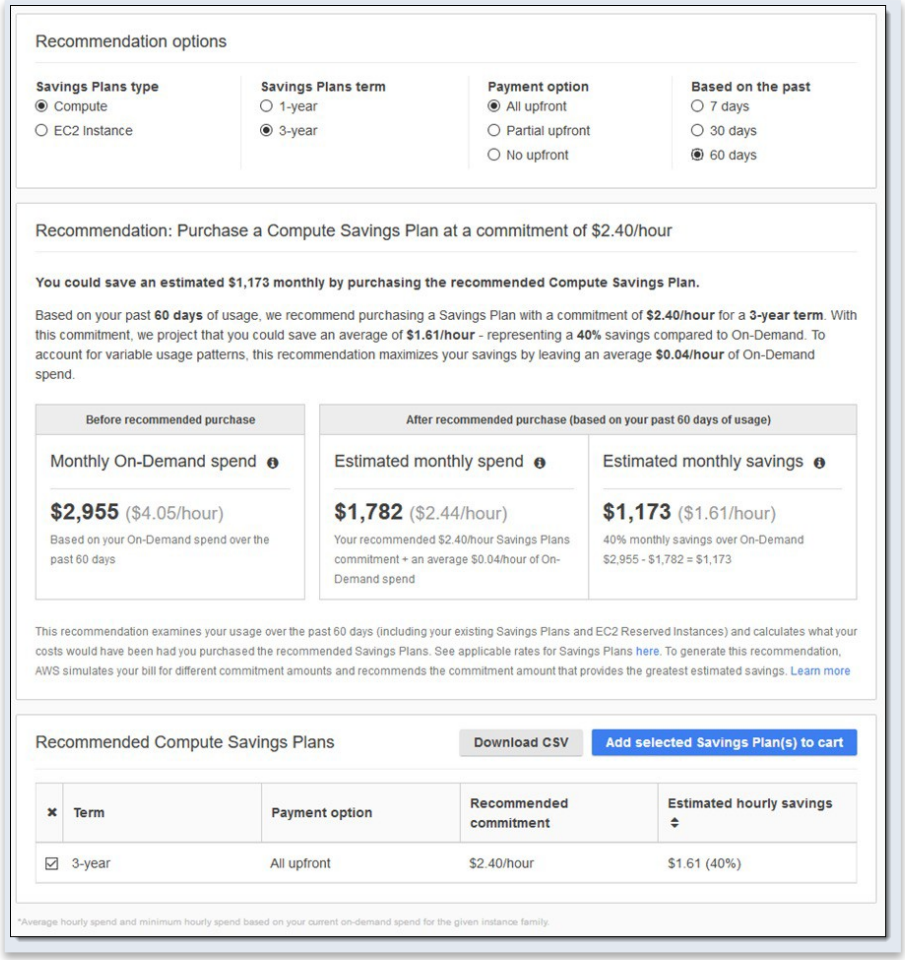
Cost Explorer - Savings Plan Alternative to Reserved Instances



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Cost Explorer - Forecast Usage

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AWS Budgets

- Create budget and send alarms when costs exceeds the budget
- 4 types of budgets: Usage, Cost, Reservation, Savings Plans
- For Reserved Instances (RI)
- Track utilization
- Supports EC2, ElastiCache, RDS, Redshift
- Up to 5 SNS notifications per budget
- Can filter by: Service, Linked Account,Tag, Purchase Option, Instance Type, Region, Availability Zone, API Operation, etc…
- Same options as AWS Cost Explorer!
- 2 budgets are free, then $0.02/day/budget

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Cost Allocation Tags

Total Cost	user:Owner	user:Stack	user:Cost Center	user:Application
0.95	DbAdmin	Test	80432	Widget2
0.01	DbAdmin	Test	80432	Widget2
3.84	DbAdmin	Prod	80432	Widget2
6.00	DbAdmin	Test	78925	Widget1
234.63	SysEng	Prod	78925	Widget1
0.73	DbAdmin	Test	78925	Widget1
0.00	DbAdmin	Prod	80432	Portal
2.47	DbAdmin	Prod	78925	Portal

Use cost allocation tags to track your AWS costs on a detailed level

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Cost and Usage Reports

- Dive deeper into your AWS costs and usage
- The AWS Cost & Usage Report contains the most comprehensive set of AWS cost and usage data available
- Includes additional metadata about AWS services, pricing, and reservations (e.g. Amazon EC2 Reserved Instances (RIs))
- The AWS Cost & Usage Report lists AWS usage for each:
- service category used by an account
- in hourly or daily line items
- any tags that you have activated for cost allocation purposes
- Can be configured for daily exports to S3
- Can be integrated with Athena, Redshift or QuickSight
- AWS Cost & Usage Report
- S3 Bucket

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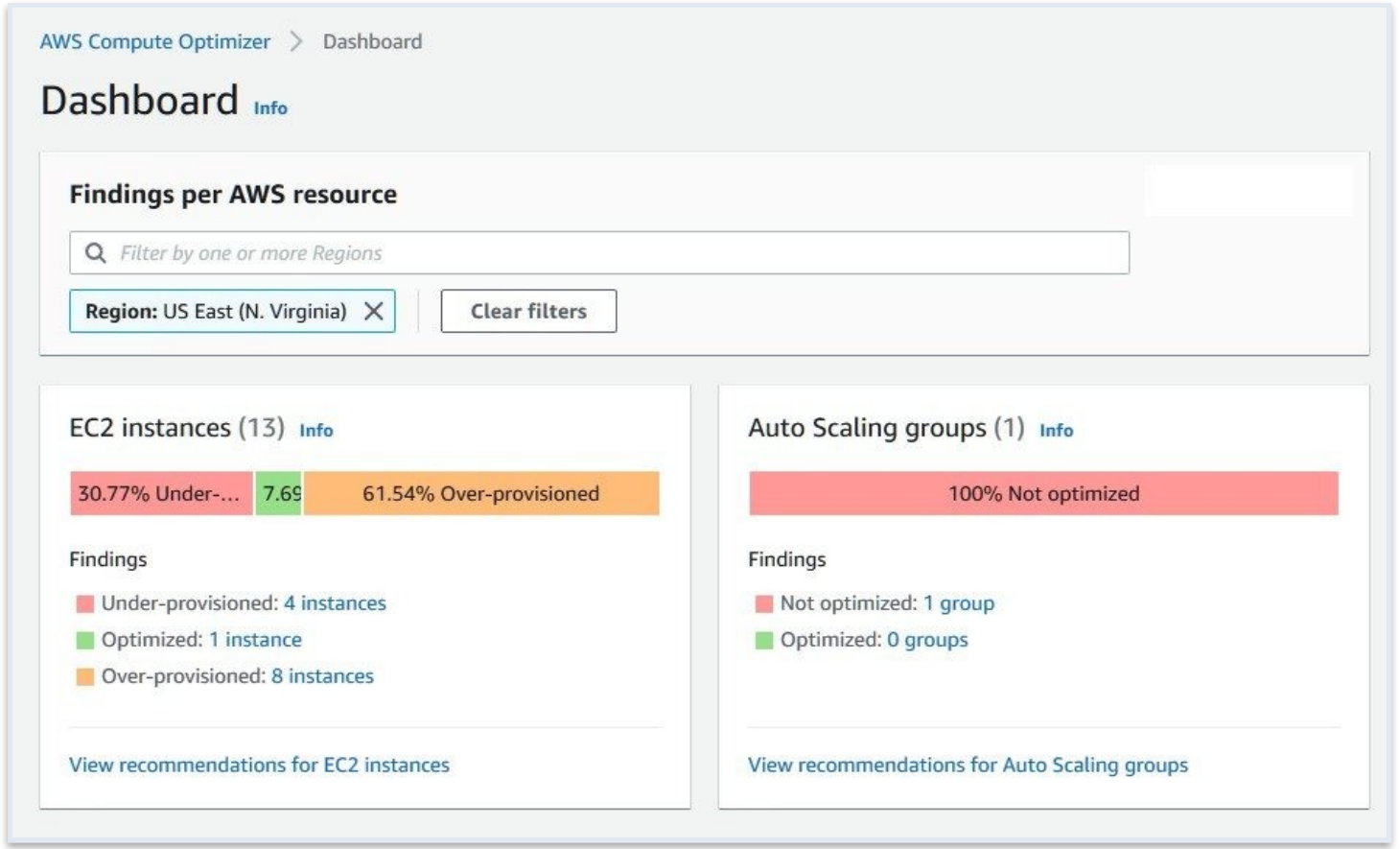
AWS Compute Optimizer



Reduce costs and improve performance by

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Cost and Usage Reports

	M	N	O	P	R	S	T
1	lineItem/ProductCode	lineItem/UsageType	lineItem/Operation	lineItem/AvailabilityZone	lineItem/UsageAmount	lineItem/CurrencyCode	lineItem/LineItemDescription
2	AmazonEC2	CW:AlarmMonitorUsage	Unknown		0.00134409	USD	$0.00 per alarm-month - first 10 alarms
3	AmazonS3	Requests-Tier1	ListAllMyBuckets		2	USD	$0.00 per request - PUT, COPY, POST, or LIST requests under the monthly global free tier
4	AmazonEC2	CW:AlarmMonitorUsage	Unknown		0.00134409	USD	$0.00 per alarm-month - first 10 alarms
5	AmazonEC2	APS2-EBS:VolumeUsage.gp2	CreateVolume-Gp2		0.01344086	USD	$0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier
6	AmazonEC2	APS2-EBS:VolumeUsage.gp2	CreateVolume-Gp2		0.01344086	USD	$0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier
7	AmazonEC2	USW2-BoxUsage:t2.micro	RunInstances:0002	us-west-2a	1	USD	$0.00 per Windows t2.micro instance-hour (or partial hour) under monthly free tier
8	AmazonEC2	USW2-USE1-AWS-Out-Bytes	PublicIP-Out		0.00000174	USD	$0.000 per GB - data transfer out under the monthly global free tier
9	AmazonEC2	USW2-USE1-AWS-In-Bytes	PublicIP-In		0.00000138	USD	$0.00 per GB - US West (Oregon) data transfer from US East (Northern Virginia)
10	AmazonEC2	USW2-USW1-AWS-In-Bytes	PublicIP-In		0.00000149	USD	$0.00 per GB - US West (Oregon) data transfer from US West (Northern California)
11	AmazonS3	Requests-Tier1	ListAllMyBuckets		2	USD	$0.00 per request - PUT, COPY, POST, or LIST requests under the monthly global free tier
12	AmazonEC2	USW2-DataTransfer-Out-Bytes	RunInstances		0.00038144	USD	$0.000 per GB - data transfer out under the monthly global free tier
13	AmazonEC2	USW2-USW1-AWS-Out-Bytes	PublicIP-Out		0.00000174	USD	$0.000 per GB - data transfer out under the monthly global free tier
14	AmazonEC2	USW2-DataTransfer-In-Bytes	RunInstances		0.00030951	USD	$0.000 per GB - data transfer in per month
15	AmazonEC2	USW2-BoxUsage:t2.micro	RunInstances:0002	us-west-2a	1	USD	$0.00 per Windows t2.micro instance-hour (or partial hour) under monthly free tier
16	AmazonEC2	USW2-USW1-AWS-Out-Bytes	PublicIP-Out		0.00000349	USD	$0.000 per GB - data transfer out under the monthly global free tier
17	AmazonEC2	USW2-USW1-AWS-In-Bytes	PublicIP-In		0.00000276	USD	$0.00 per GB - US West (Oregon) data transfer from US West (Northern California)
18	AmazonEC2	APS2-EBS:VolumeUsage.gp2	CreateVolume-Gp2		0.01344086	USD	$0.00 per GB-month of General Purpose (SSD) provisioned storage under monthly free tier
19	AmazonEC2	CW:AlarmMonitorUsage	Unknown		0.00134409	USD	$0.00 per alarm-month - first 10 alarms
20	AmazonEC2	USW2-BoxUsage:t2.micro	RunInstances:0002	us-west-2a	1	USD	$0.00 per Windows t2.micro instance-hour (or partial hour) under monthly free tier
21	AmazonEC2	USW2-DataTransfer-Regional-Bytes	PublicIP-Out		0.00000349	USD	$0.000 per GB - regional data transfer under the monthly global free tier
22	AmazonEC2	USW2-DataTransfer-In-Bytes	RunInstances		0.00032071	USD	$0.000 per GB - data transfer in per month
23	AmazonEC2	USW2-DataTransfer-Regional-Bytes	PublicIP-In		0.00000302	USD	$0.000 per GB - regional data transfer under the monthly global free tier
24	AmazonEC2	USW2-USE1-AWS-Out-Bytes	PublicIP-Out		0.00000174	USD	$0.000 per GB - data transfer out under the monthly global free tier
25	AmazonEC2	USW2-DataTransfer-Out-Bytes	RunInstances		0.00045736	USD	$0.000 per GB - data transfer out under the monthly global free tier
26	AmazonEC2	USW2-DataTransfer-In-Bytes	RunInstances		0.00036737	USD	$0.000 per GB - data transfer in per month
27	AmazonEC2	USW2-APN2-AWS-In-Bytes	PublicIP-In		0.00000005	USD	$0.00 per GB - US West (Oregon) data transfer from Asia Pacific (Seoul)
28	AmazonEC2	USW2-APN2-AWS-Out-Bytes	PublicIP-Out		0.00000018	USD	$0.000 per GB - data transfer out under the monthly global free tier
29	AmazonEC2	USW2-USE1-AWS-In-Bytes	PublicIP-In		0.00000153	USD	$0.00 per GB - US West (Oregon) data transfer from US East (Northern Virginia)
30	AmazonEC2	USW2-DataTransfer-Out-Bytes	RunInstances		0.00039945	USD	$0.000 per GB - data transfer out under the monthly global free tier
31	AmazonEC2	CW:AlarmMonitorUsage	Unknown		0.00134409	USD	$0.00 per alarm-month - first 10 alarms

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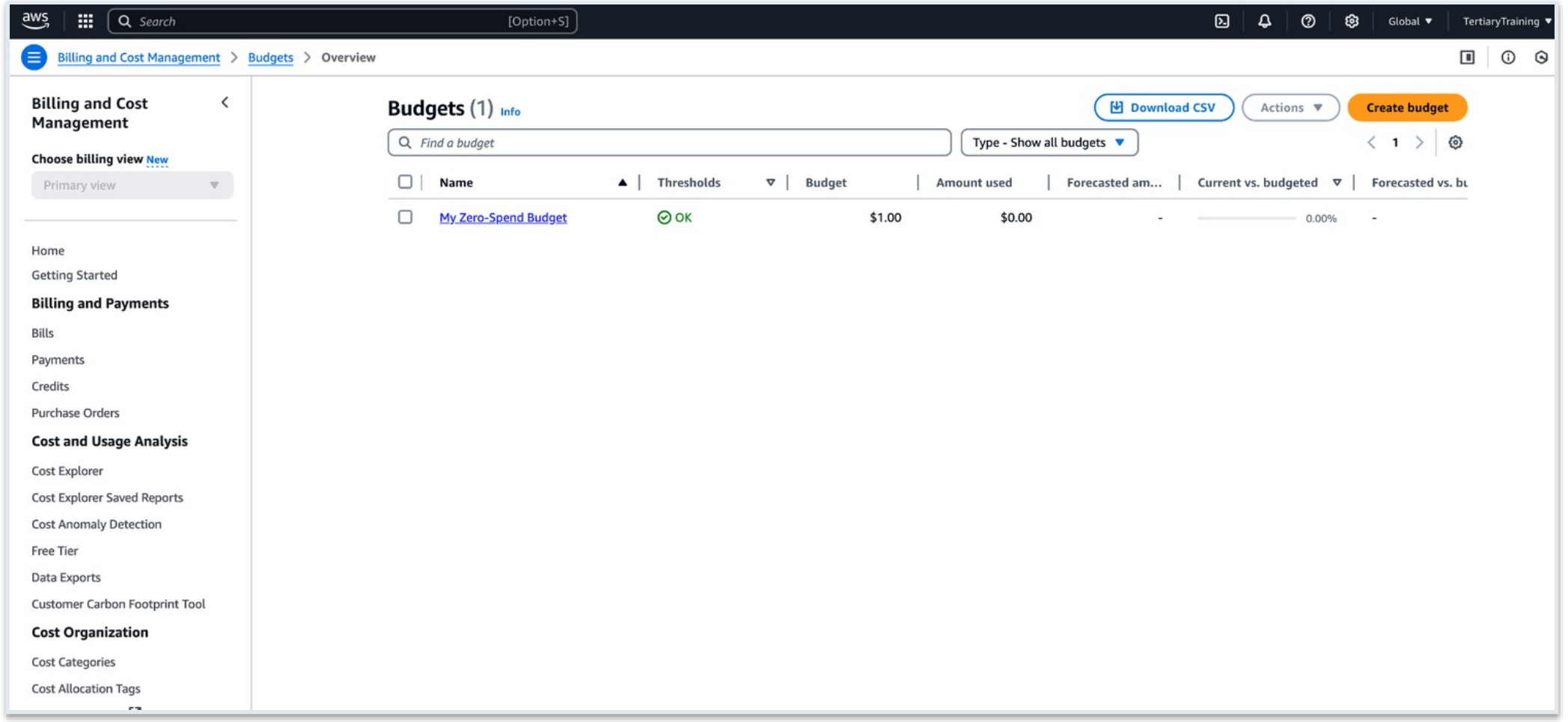
Activity: AWS Budgets



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Disaster Recovery

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AWS DataSync

- Move large amount of data to and from
- On-premises / other cloud to AWS (NFS, SMB, HDFS, S3 API…) - needs agent
- AWS to AWS (different storage services) – no agent needed
- Can synchronize to:
- Amazon S3 (any storage classes – including Glacier)
- Amazon EFS
- Amazon FSx (Windows, Lustre, NetApp, OpenZFS...)
- Replication tasks can be scheduled hourly, daily, weekly
- File permissions and metadata are preserved (NFS POSIX, SMB…)
- One agent task can use 10 Gbps, can setup a bandwidth limit

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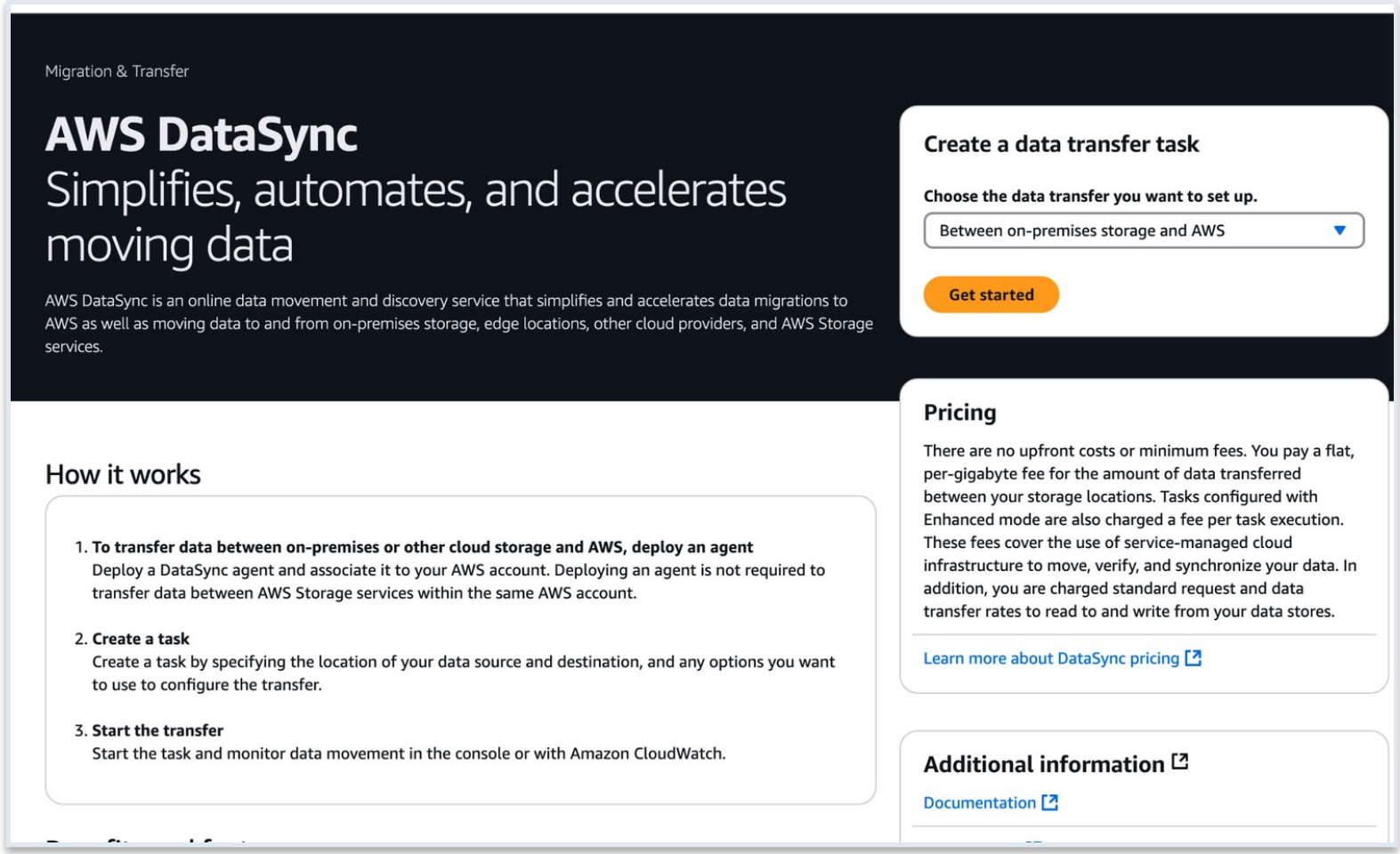
AWS DataSync



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AWS DataSync

- NFS / SMB to AWS (S3, EFS, FSx…)
- On-Premises
- NFS or SMB
- Server
- AWS DataSync Agent
- NFS or SMB
- Region
- AWS
- DataSync
- AWS Storage Resources
- S3 Standard
- S3 Intelligent- Tiering
- S3 Standard-IA

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AWS DataSync

- Transfer between AWS storage services
- AWS DataSync
- copy data and metadata between AWS Storage Services
- Amazon S3
- Amazon EFS
- Amazon FSx
- Amazon S3
- Amazon EFS
- Amazon FSx

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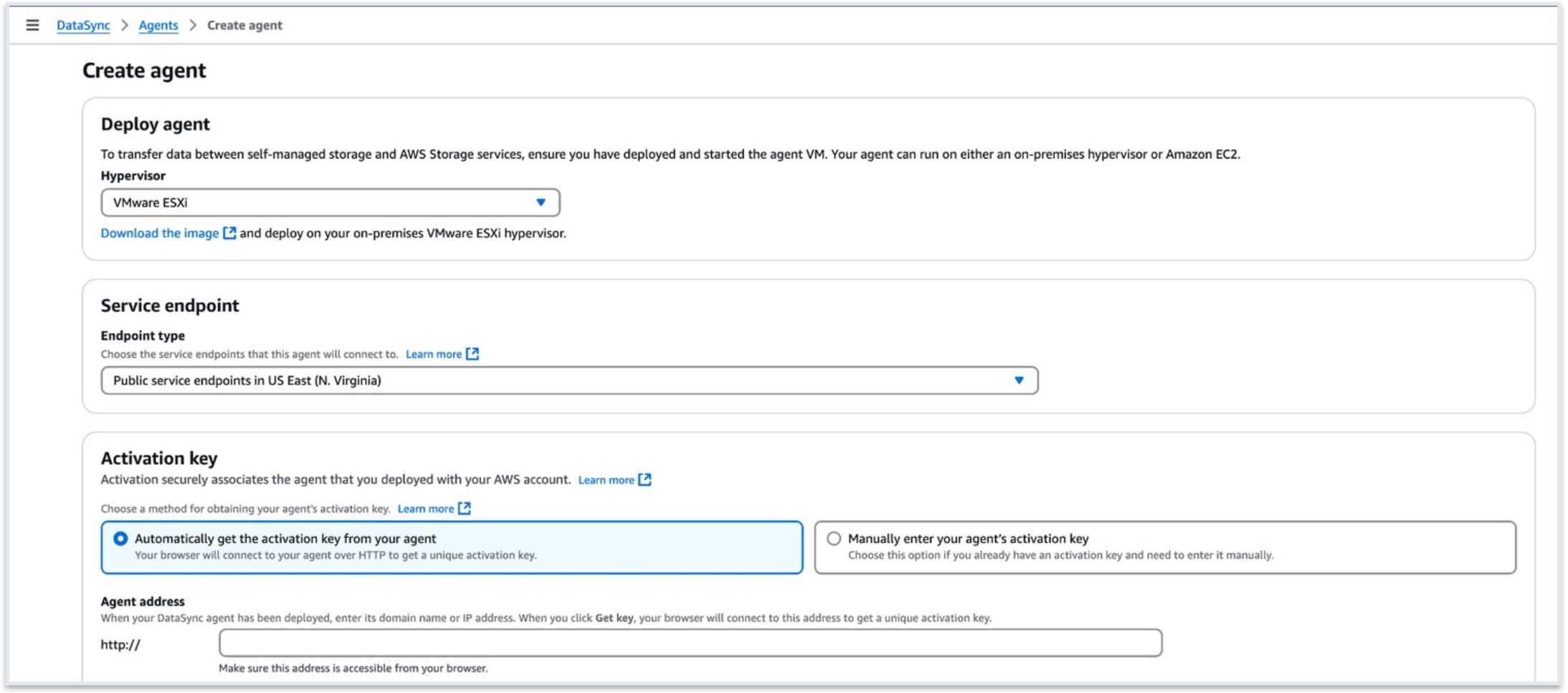
Activity: AWS DataSync



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AWS Backup

- Fully managed service
- Centrally manage and automate backups across AWS services
- No need to create custom scripts and manual processes
- Supported services:
- Amazon EC2 / Amazon EBS
- Amazon S3
- Amazon RDS (all DBs engines) / Amazon Aurora / Amazon DynamoDB
- Amazon DocumentDB / Amazon Neptune
- Amazon EFS / Amazon FSx (Lustre & Windows File Server)
- AWS Storage Gateway (Volume Gateway)
- Supports cross-region backups
- Supports cross-account backups

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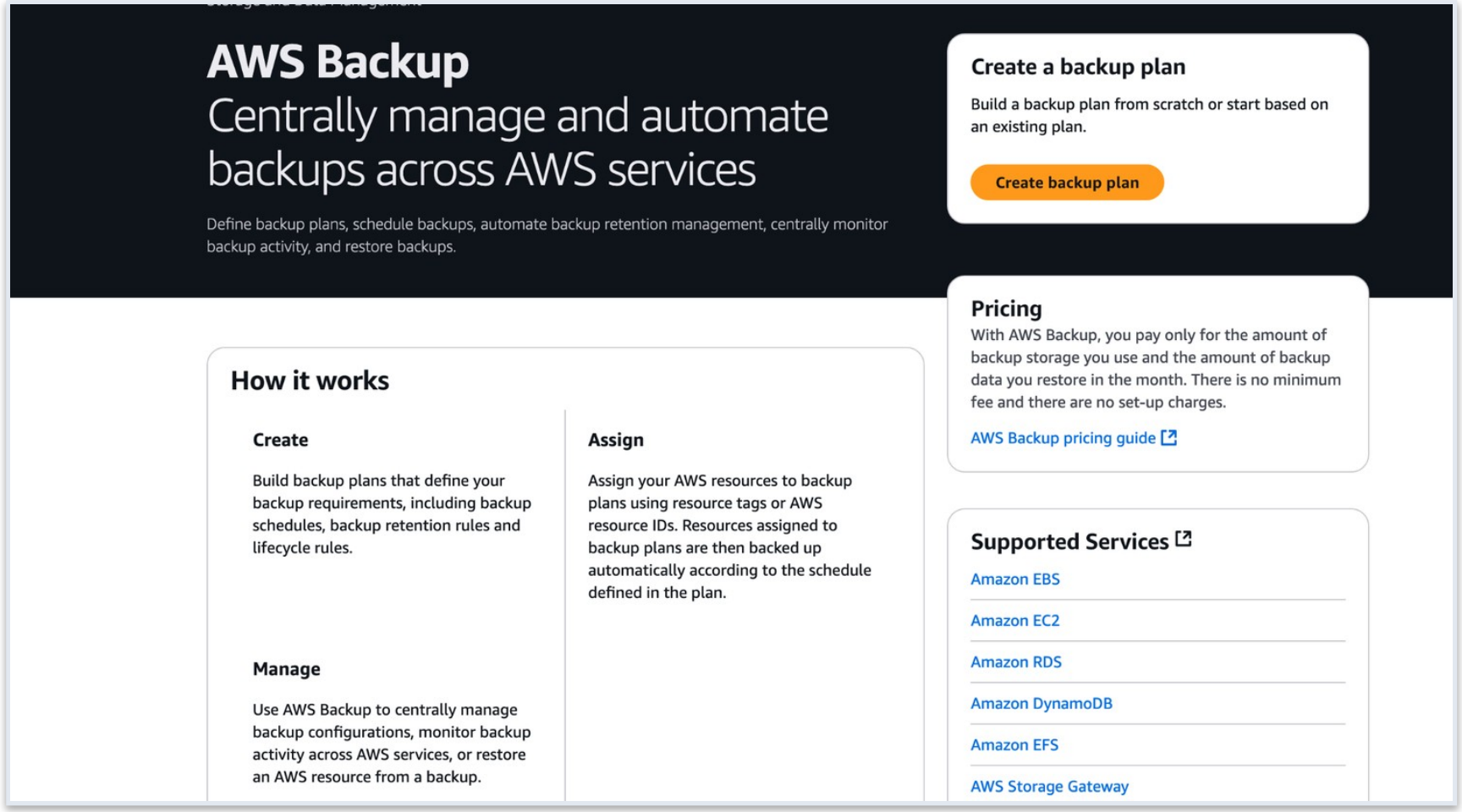
AWS Backup



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AWS Backup

- Supports PITR for supported services
- On-Demand and Scheduled backups
- Tag-based backup policies
- You create backup policies known as Backup Plans
- Backup frequency (every 12 hours, daily, weekly, monthly, cron expression)
- Backup window
- Transition to Cold Storage (Never, Days, Weeks, Months,Years)
- Retention Period (Always, Days, Weeks, Months,Years)

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AWS Backup

- AWS Backup
- Create Backup Plan
- (frequency, retention policy)
- Assign AWS Resources
- EC2
- EBS
- RDS
- EFS
- Aurora
- FSx
- Storage Gateway
- Amazon S3
- Automatically backed up to

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AWS Backup Vault Lock

- Enforce a WORM (Write Once Read Many) state for all the backups that you store in your AWS Backup Vault
- Additional layer of defense to protect your backups against:
- Inadvertent or malicious delete operations
- Updates that shorten or alter retention periods
- Even the root user cannot delete backups when enabled
- backup
- Backup Vault Lock Policy
- Backups can’t be deleted

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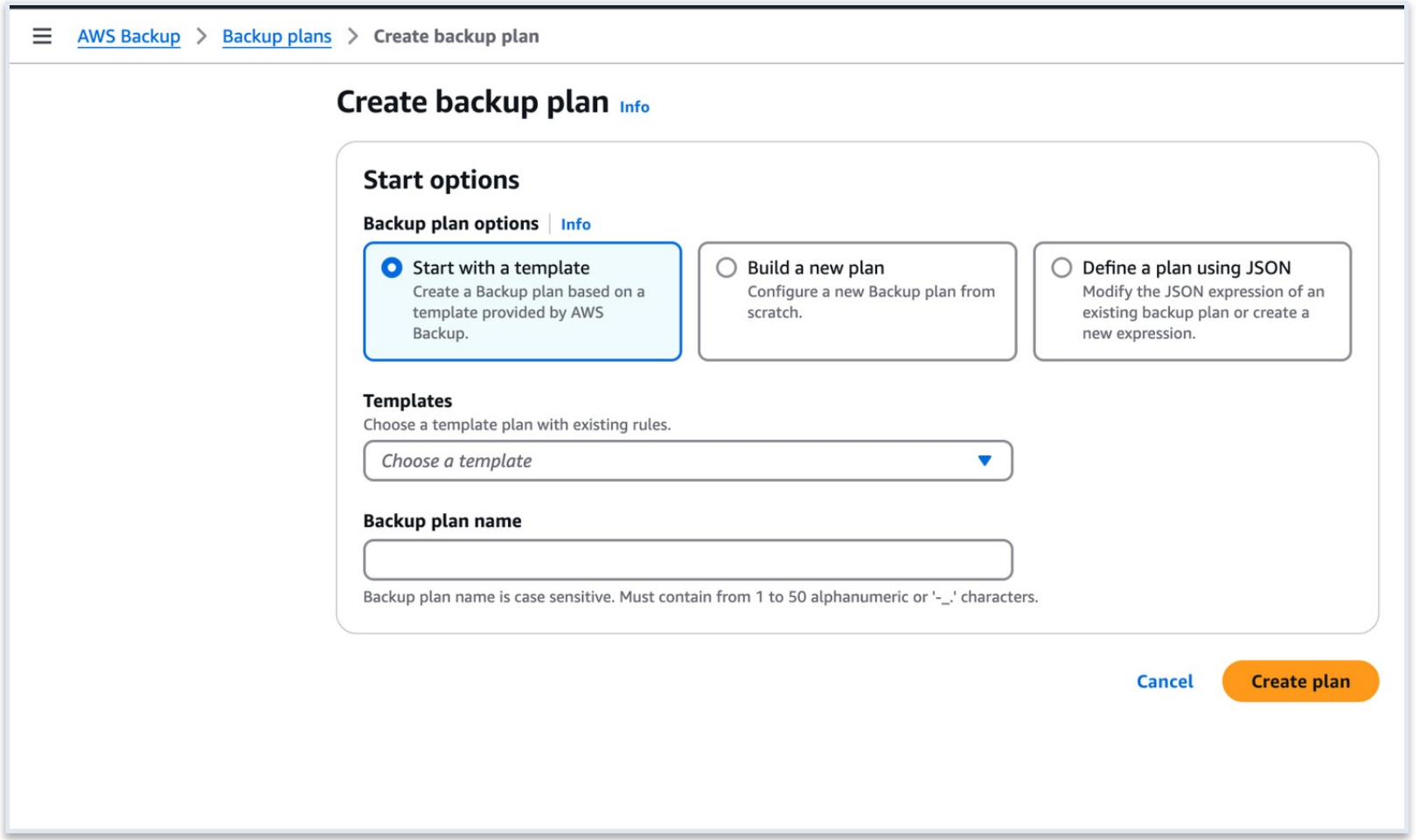
Activity: AWS Backup



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Security & Compliance

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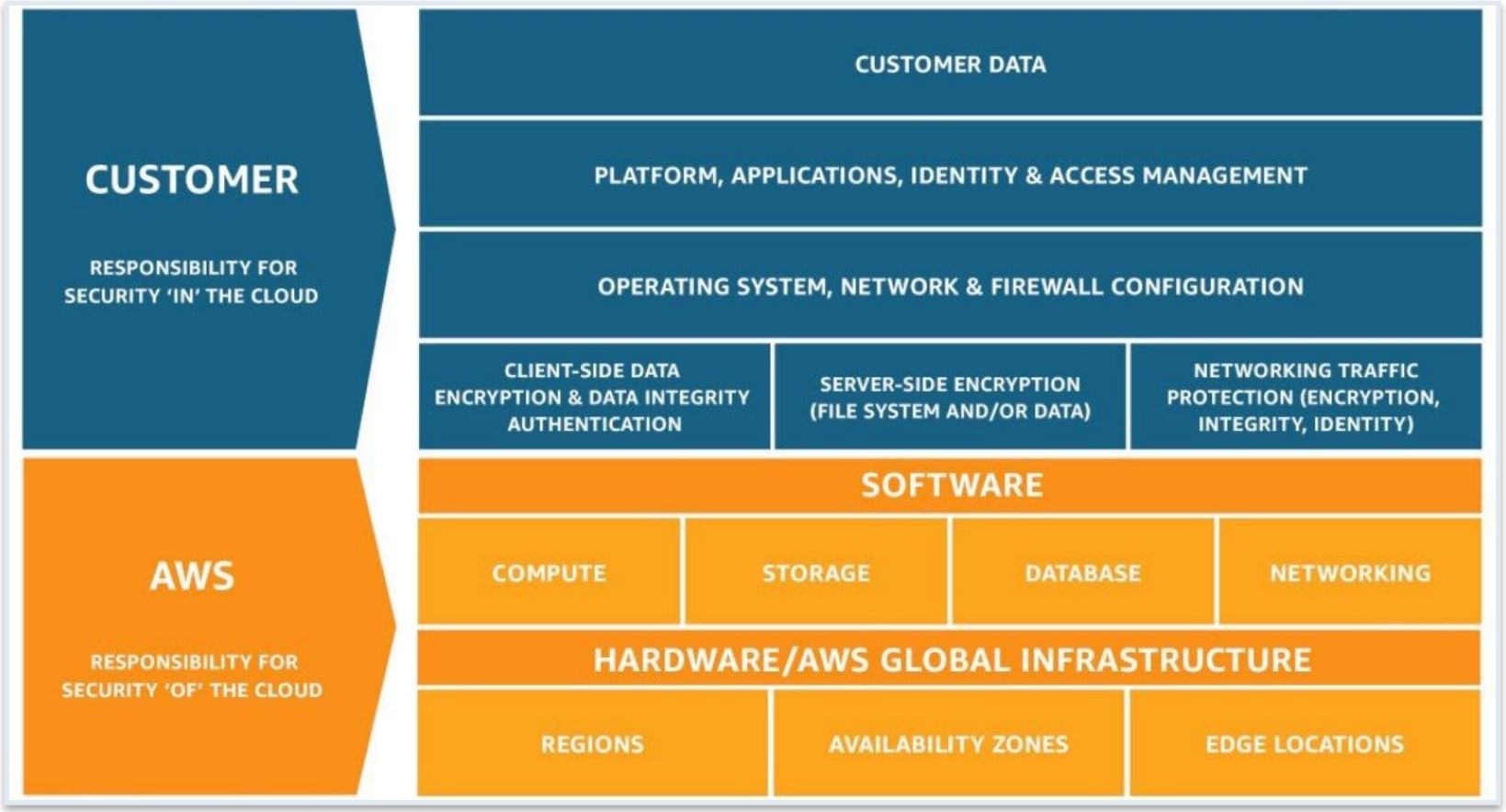
AWS Shared Responsibility Model



https://aws.amazon.com/compliance/shared-responsibility-model/

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AWS Shared Responsibility Model

- AWS responsibility - Security of the Cloud
- Protecting infrastructure (hardware, software, facilities, and networking) that runs all the AWS services
- Managed services like S3, DynamoDB, RDS, etc.
- Customer responsibility - Security in the Cloud
- For EC2 instance, customer is responsible for management of the guest OS (including security patches and updates), firewall & network configuration, IAM
- Encrypting application data
- Shared controls:
- Patch Management, Configuration Management, Awareness & Training

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attacker

- masters
- bots
- normal users
- Not accessible Not responsive
- application server
- What’s a DDOS* Attack?
- *Distributed Denial-of-Service

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DDoS Protection on AWS

- AWS Shield Standard: protects against DDoS attack for your website and applications, for all customers at no additional costs
- AWS Shield Advanced: 24/7 premium DDoS protection
- AWS WAF: filter specific requests based on rules
- CloudFront and Route 53:
- Availability protection using global edge network
- Combined with AWS Shield, provides attack mitigation at the edge
- Be ready to scale - leverage AWS Auto Scaling

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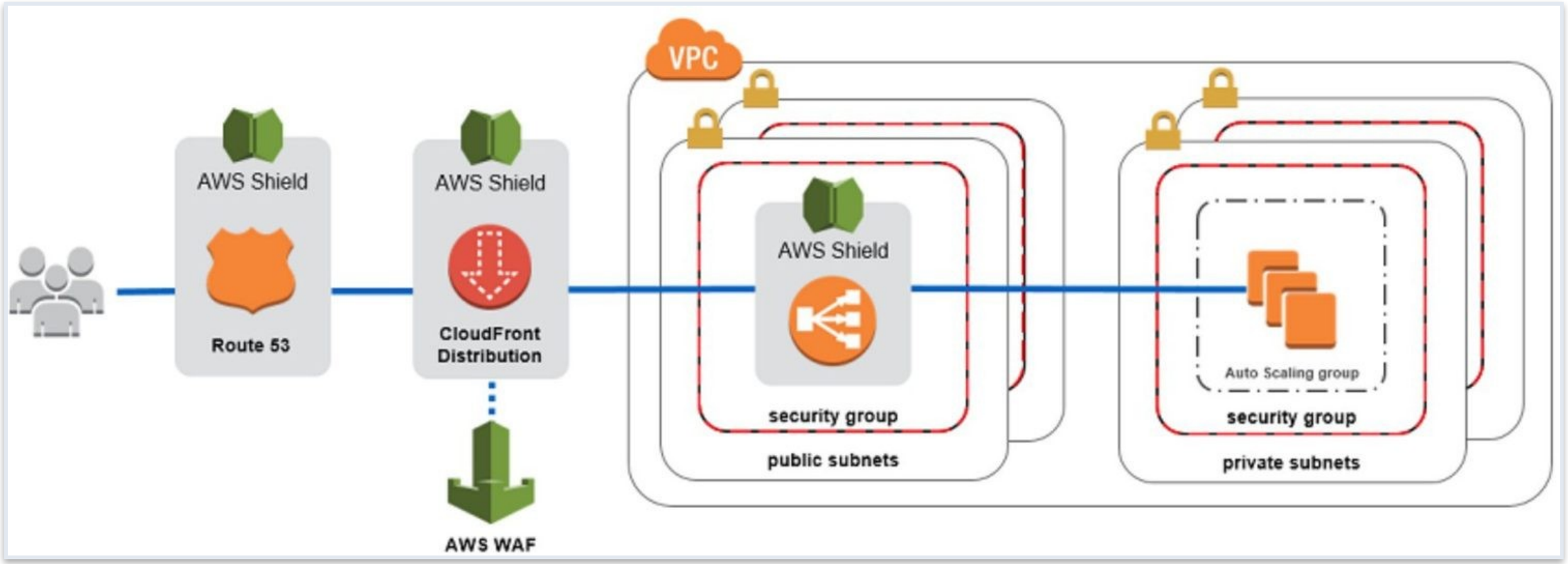
Sample Reference Architecture



https://aws.amazon.com/answers/networking/aws-ddos-attack-mitigation/

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AWS Shield

- AWS Shield Standard:
- Free service that is activated for every AWS customer
- Provides protection from attacks such as SYN/UDP Floods, Reflection attacks and other layer 3/layer 4 attacks
- AWS Shield Advanced:
- Optional DDoS mitigation service ($3,000 per month per organization)
- Protect against more sophisticated attack on Amazon EC2, Elastic Load Balancing (ELB), Amazon CloudFront, AWS Global Accelerator, and Route 53
- 24/7 access to AWS DDoS response team (DRP)
- Protect against higher fees during usage spikes due to DDoS

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AWS Shield

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AWS WAF - Web Application Firewall

- Protects your web applications from common web exploits (Layer 7)
- Layer 7 is HTTP (vs Layer 4 is TCP)
- Deploy on Application Load Balancer, API Gateway, CloudFront
- Define Web ACL (Web Access Control List):
- Rules can include: IP addresses, HTTP headers, HTTP body, or URI strings
- Protects from common attack - SQL injection and Cross-Site (XSS)
- Size constraints, geo-match (block countries)
- Rate-based rules (to count occurrences of events) - for DDOS protection

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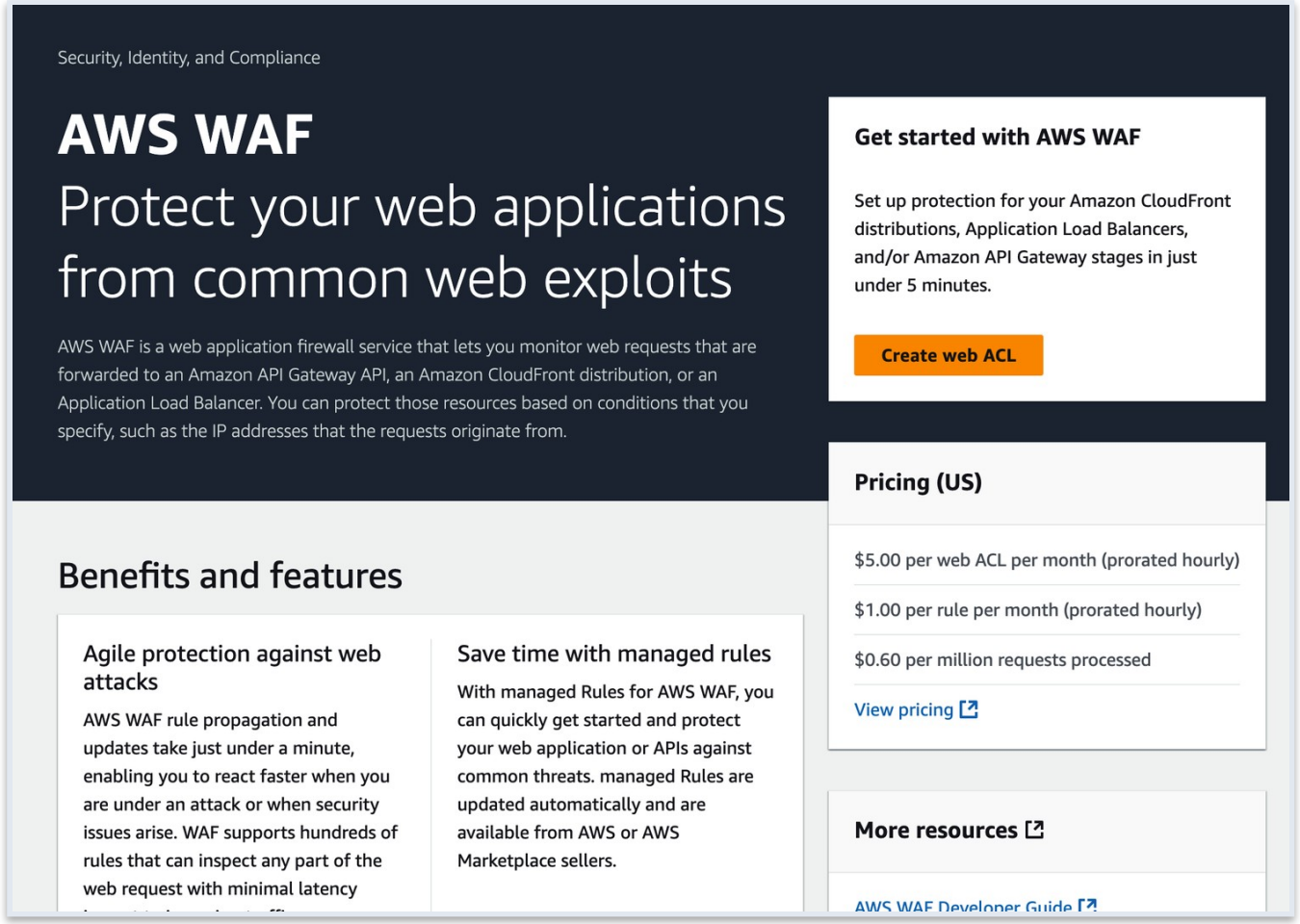
AWS WAF - Web Application Firewall



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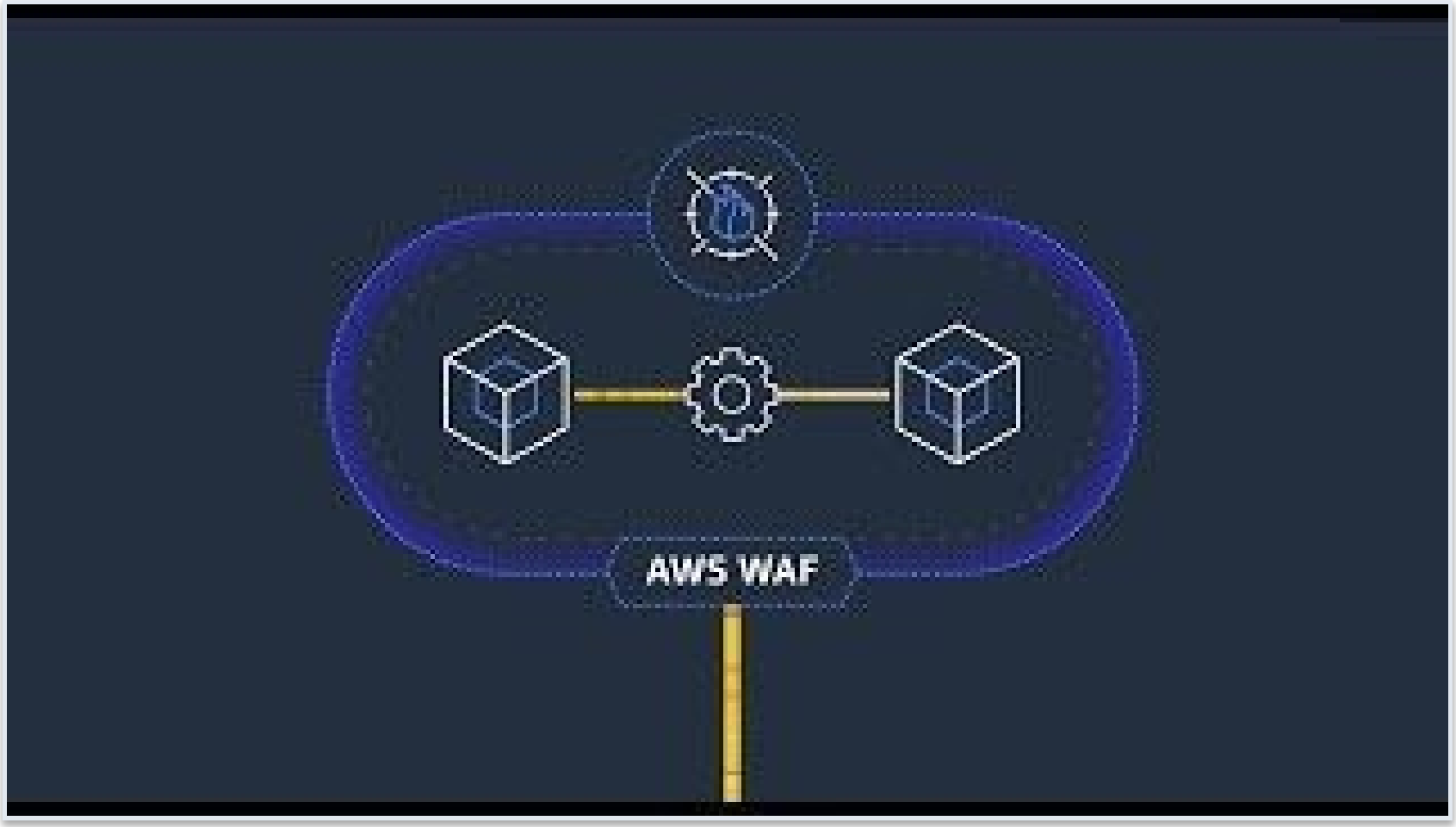

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Penetration Testing on AWS Cloud

- AWS customers are welcome to carry out security assessments or penetration tests against their AWS infrastructures without prior approval for 8 services:
- Amazon EC2 instances, NAT Gateways, and Elastic Load Balancers
- Amazon RDS
- Amazon CloudFront
- Amazon Aurora
- Amazon API Gateways
- AWS Lambda and Lambda Edge functions
- Amazon Lightsail resources
- Amazon Elastic Beanstalk environments
- List can increase over time (you won’t be tested on that at the exam)

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Penetration Testing on your AWS Cloud

- Prohibited Activities
- DNS zone walking via Amazon Route 53 Hosted Zones
- Denial of Service (DoS), Distributed Denial of Service (DDoS), Simulated DoS, Simulated DDoS
- Port flooding
- Protocol flooding
- Request flooding (login request flooding, API request flooding)
- For any other simulated events, contact aws-security-simulated-event@amazon.com
- Read more: https://aws.amazon.com/security/penetration-testing/

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Amazon Inspector

- Automated Security Assessments
- For EC2 instances
- Leveraging the AWS System Manager (SSM) agent
- Analyze against unintended network accessibility
- Analyze the running OS against known vulnerabilities
- For Container Images push to Amazon ECR
- Assessment of Container Images as they are pushed
- For Lambda Functions
- Identifies software vulnerabilities in function code and package dependencies
- Assessment of functions as they are deployed
- Reporting & integration with AWS Security Hub
- Send findings to Amazon Event Bridge
- SSM Agent

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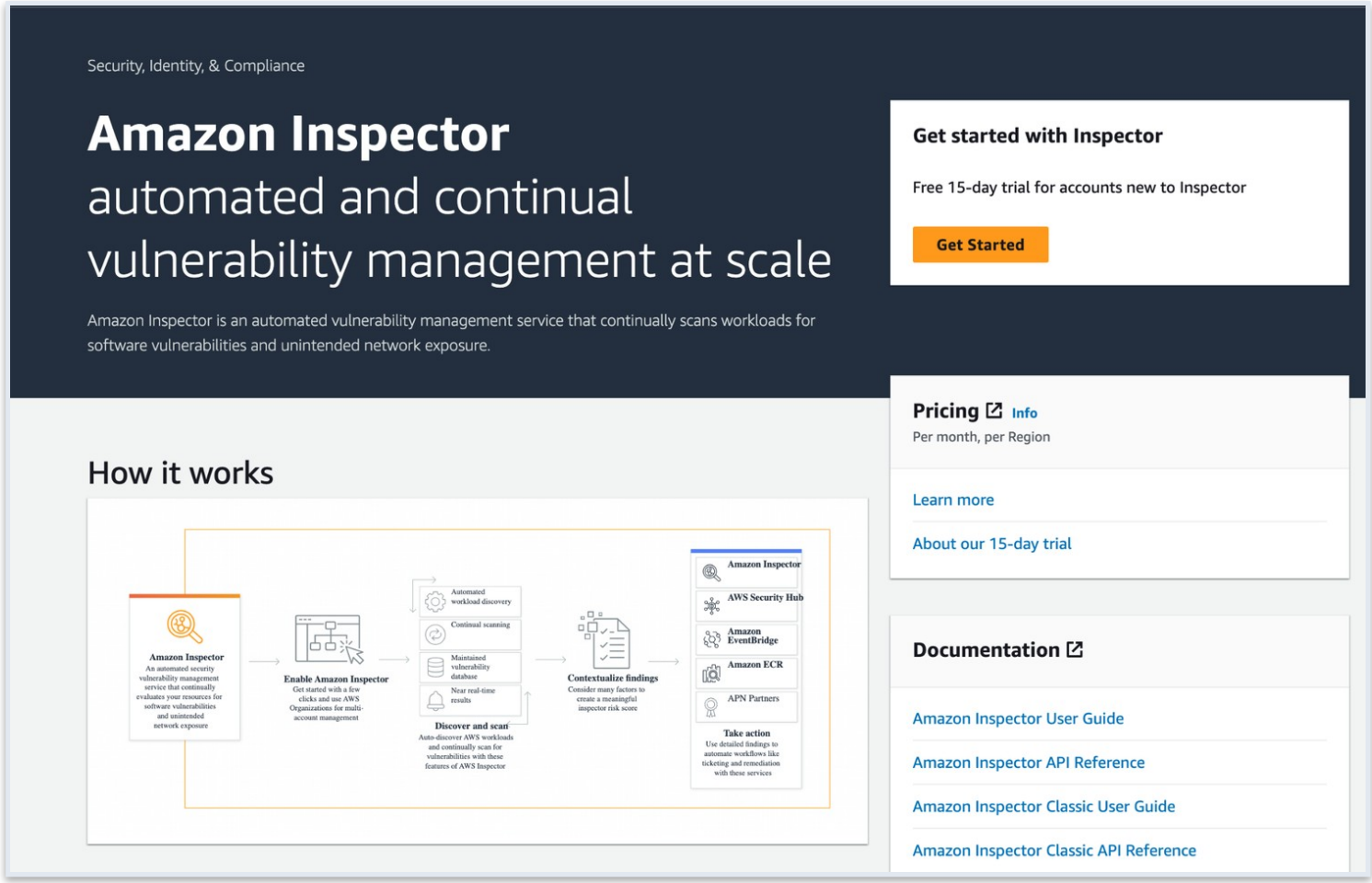
Amazon Inspector



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What does Amazon Inspector evaluate?

1. Remember: only for EC2 instances, Container Images & Lambda functions
2. Continuous scanning of the infrastructure, only when needed
3. Package vulnerabilities (EC2, ECR & Lambda) - database of CVE
4. Network reachability (EC2)
5. A risk score is associated with all vulnerabilities for prioritization

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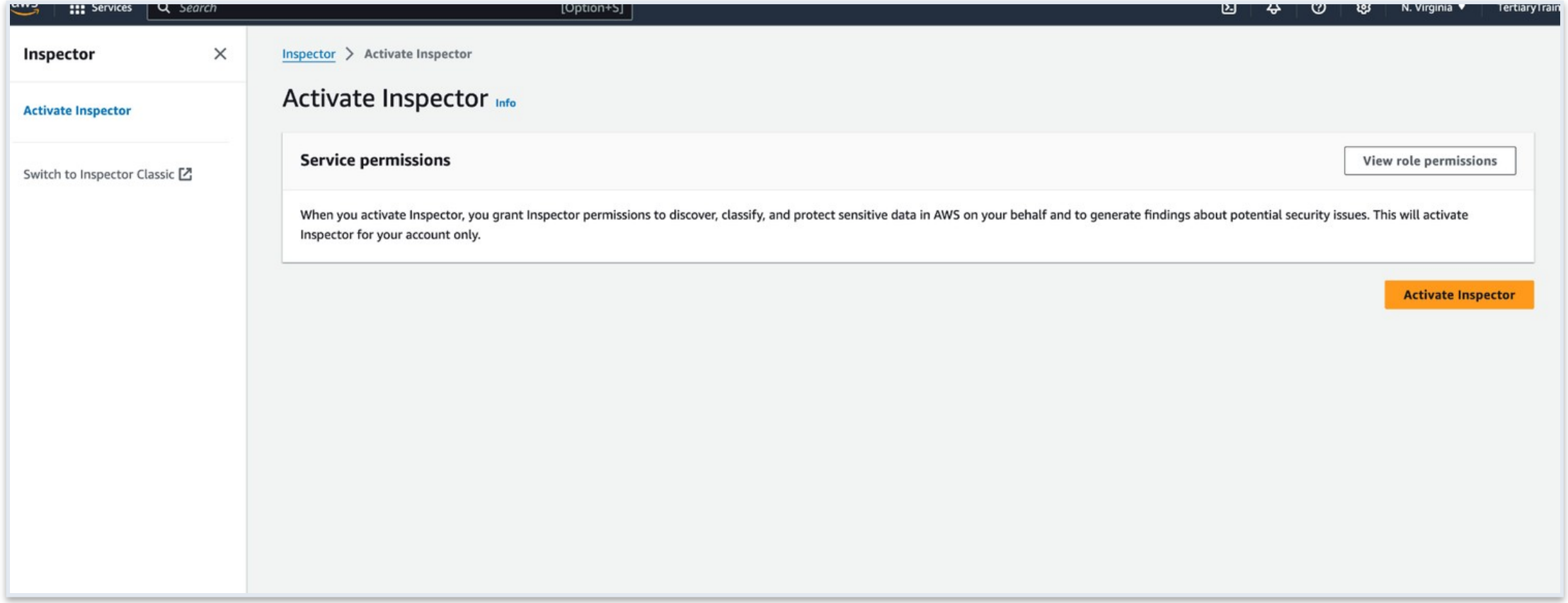
Activity: Amazon Inspector



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Logging in AWS for security and compliance

- To help compliance requirements, AWS provides many service-specific security and audit logs
- Service Logs include:
- CloudTrail trails - trace all API calls
- Config Rules - for config & compliance over time
- CloudWatch Logs - for full data retention
- VPC Flow Logs - IP traffic within your VPC
- ELB Access Logs - metadata of requests made to your load balancers
- CloudFront Logs - web distribution access logs
- WAF Logs - full logging of all requests analyzed by the service
- Logs can be analyzed using AWS Athena if they're stored in S3
- You should encrypt logs in S3, control access using IAM & Bucket Policies, MFA
- Move Logs to Glacier for cost savings
- Read whitepaper if interested at:

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Intelligent Threat discovery to protect your AWS Account

- Uses Machine Learning algorithms, anomaly detection, 3rd party data
- One click to enable (30 days trial), no need to install software
- Input data includes:
- CloudTrail Events Logs - unusual API calls, unauthorized deployments
- CloudTrail Management Events – create VPC subnet, create trail, …
- CloudTrail S3 Data Events – get object, list objects, delete object, …
- VPC Flow Logs - unusual internal traffic, unusual IP address
- DNS Logs - compromised EC2 instances sending encoded data within DNS queries
- Optional Feature - EKS Audit Logs, RDS & Aurora, EBS, Lambda, S3 Data Events…
- Can setup EventBridge rules to be notified in case of findings
- EventBridge rules can target AWS Lambda or SNS
- Can protect against CryptoCurrency attacks (has a dedicated "finding" for it)
- Amazon GuardDuty

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Amazon GuardDuty

- VPC Flow Logs
- CloudTrail Logs
- DNS Logs (AWS DNS)
- GuardDuty
- EventBridge
- SNS
- Lambda
- Optional Features
- S3 Logs
- EBS Volumes
- Lambda Network RDS & Aurora Activity Login Activity
- EKS Audit Logs & Runtime Monitoring

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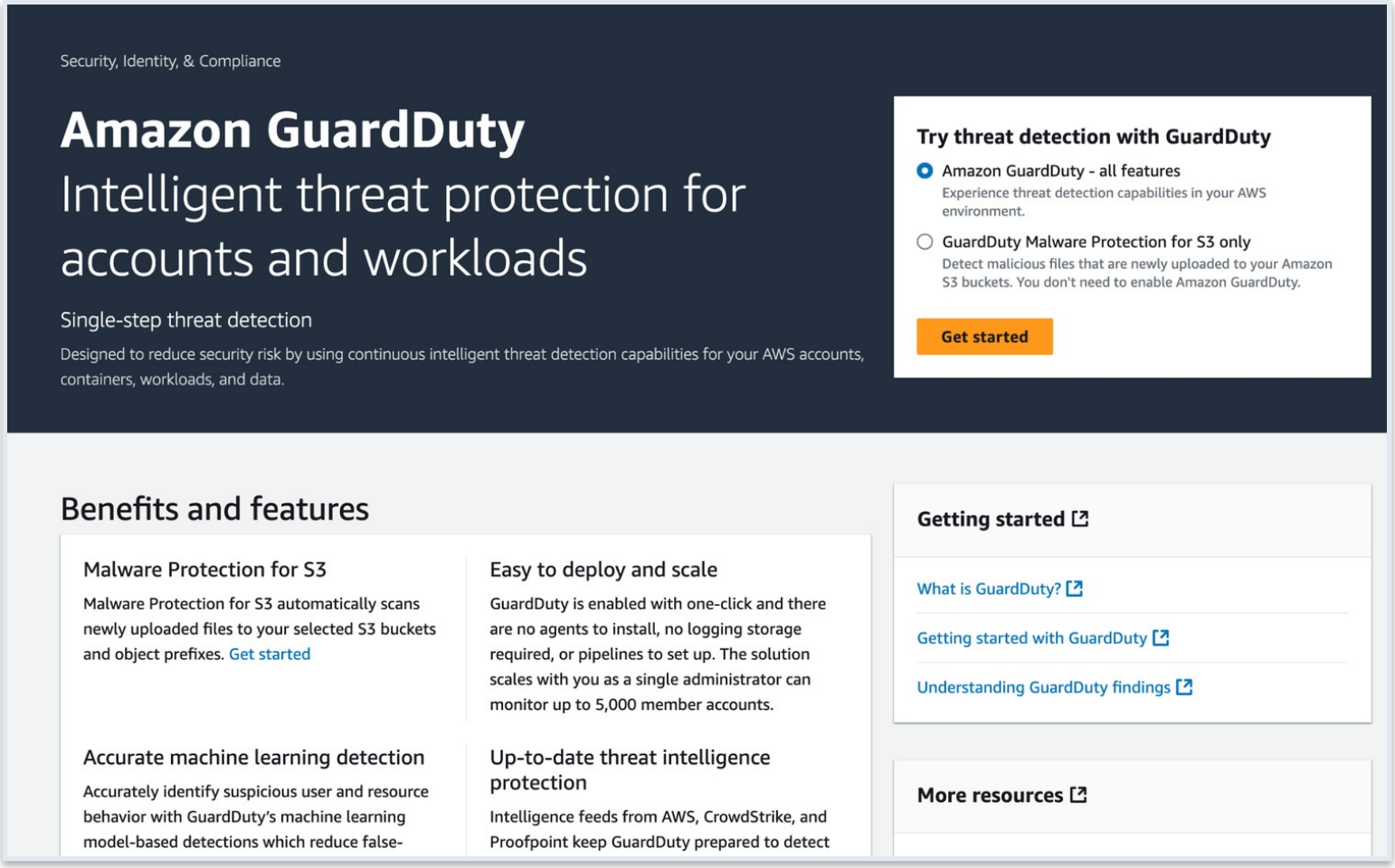
Amazon GuardDuty



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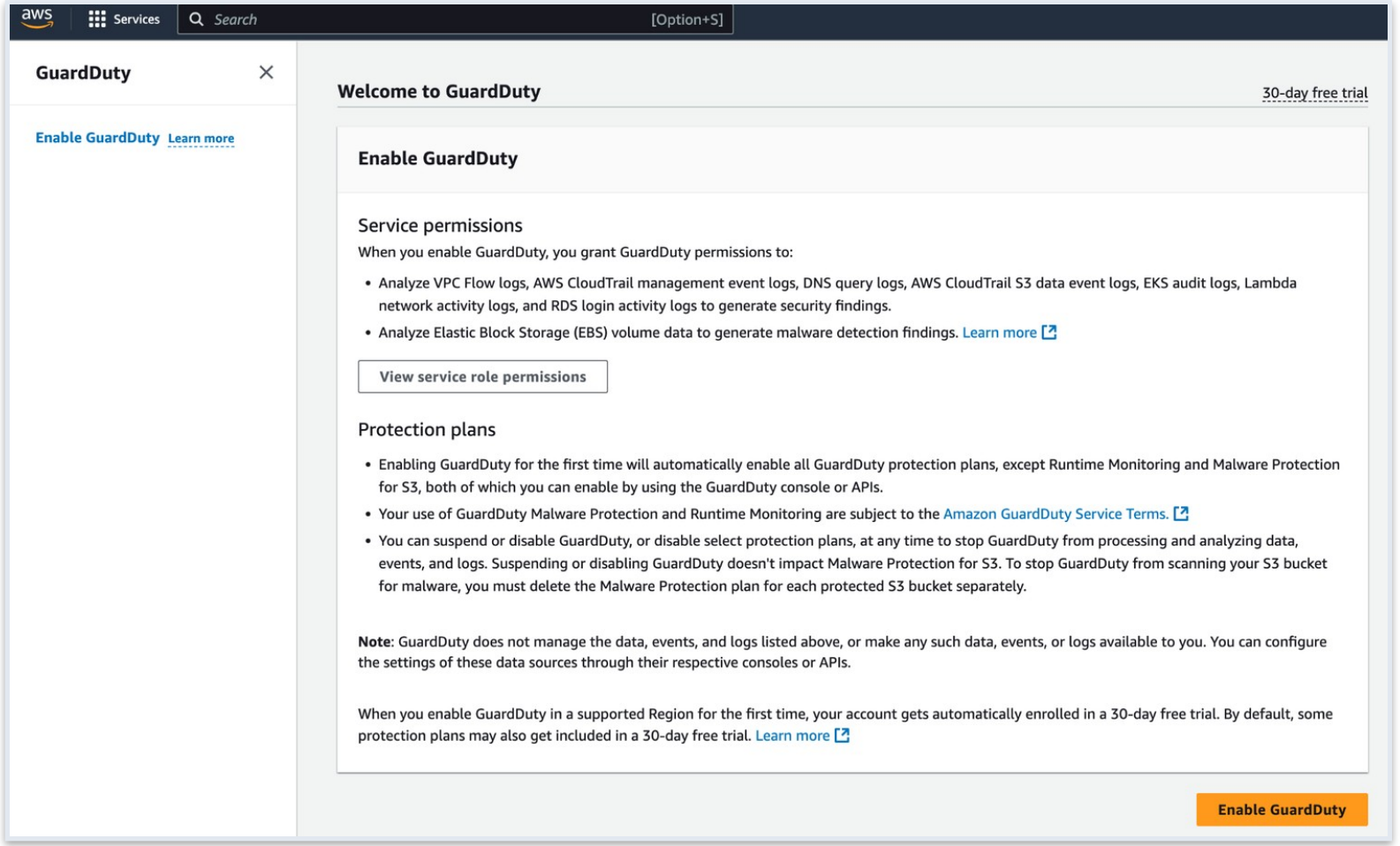
Activity: Amazon GuardDuty



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AWS Macie

- Amazon Macie is a fully managed data security and data privacy service that uses machine learning and pattern matching to discover and protect your sensitive data in AWS.
- Macie helps identify and alert you to sensitive data, such as personally identifiable information (PII)
- S3 Buckets
- Macie
- Discover Sensitive Data (PII)
- Amazon EventBridge
- analyze
- notify
- integrations

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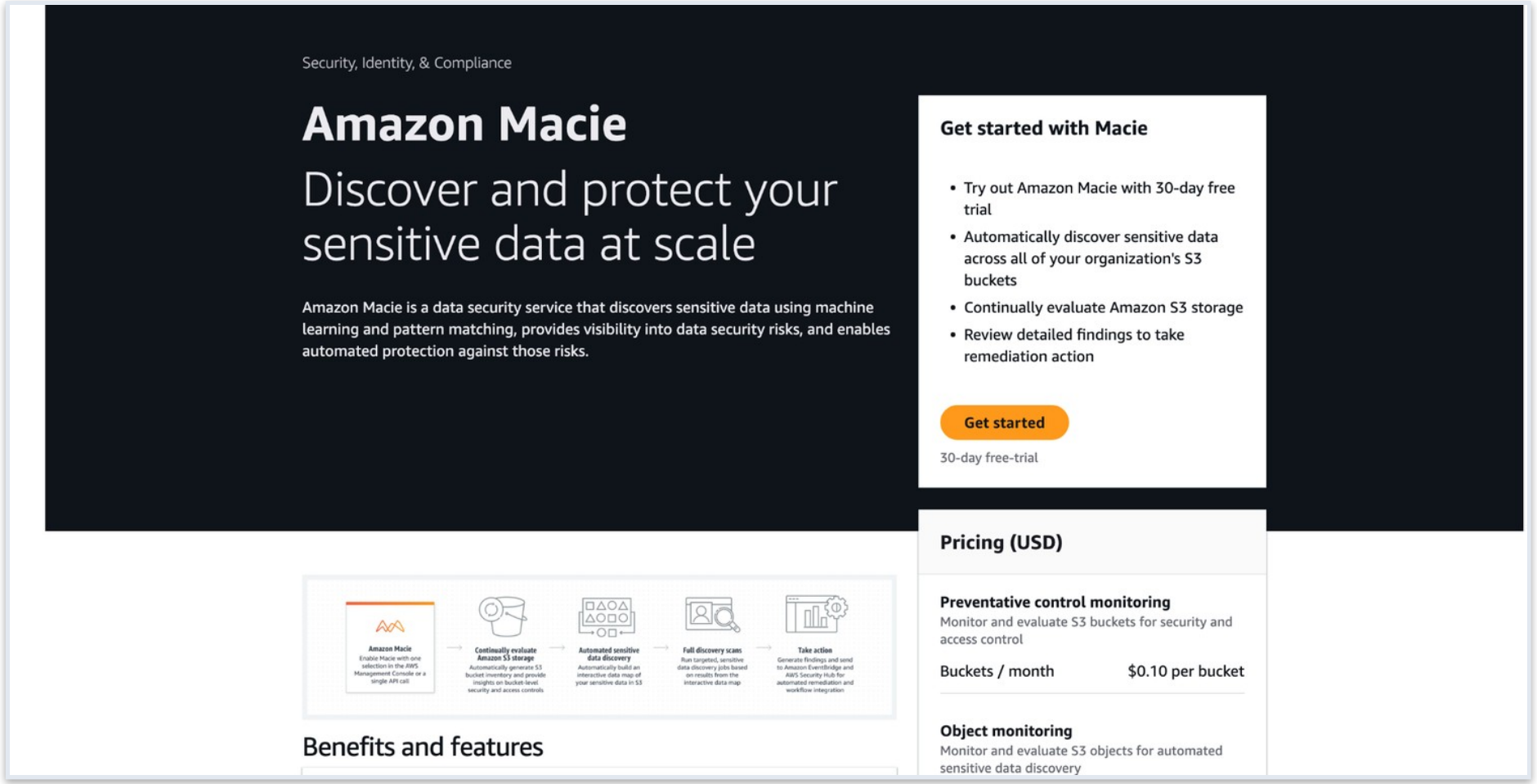
AWS Macie



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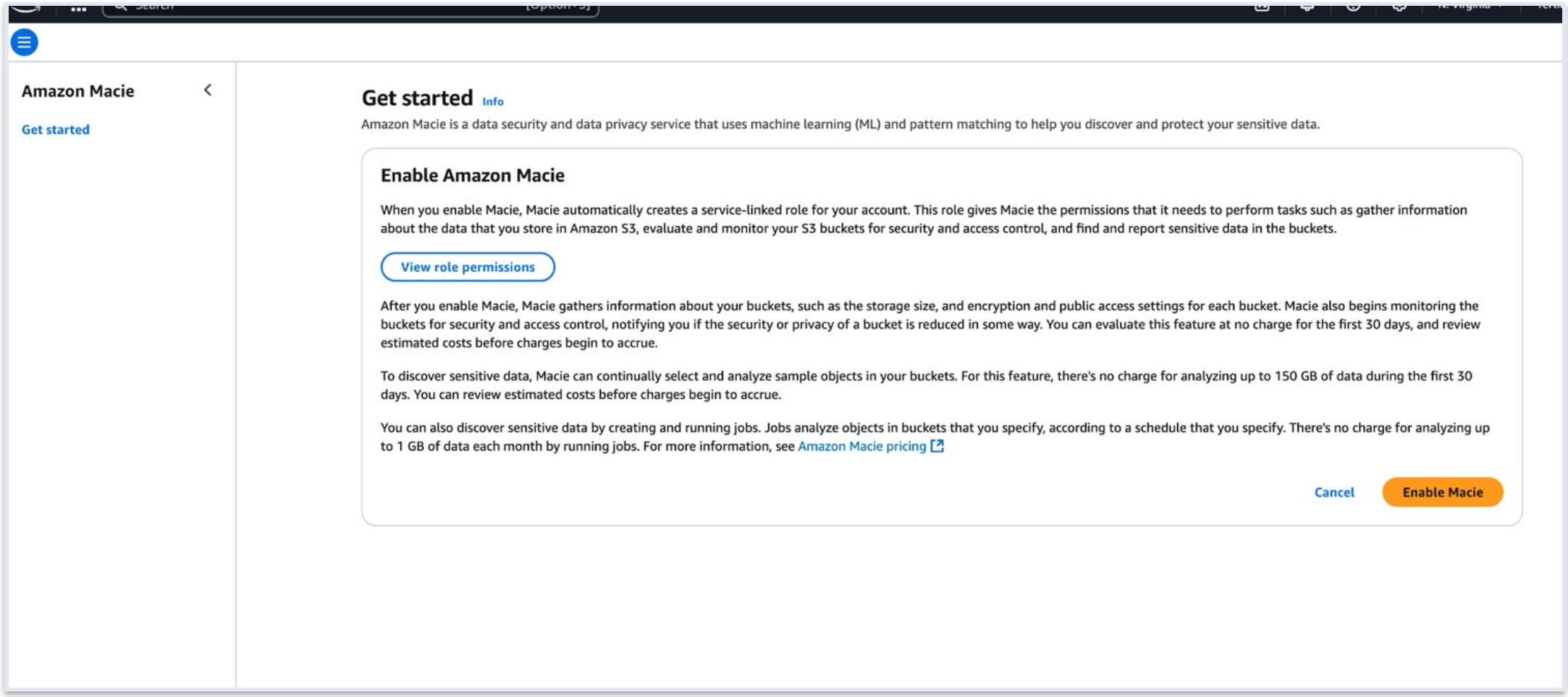
Activity: AWS Macie



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Trusted Advisor

- No need to install anything - high level AWS account assessment
- Analyze your AWS accounts and provides recommendation on 6 categories:
- Cost optimization
- Performance
- Security
- Fault tolerance
- Service limits
- Operational Excellence
- Business & Enterprise Support plan
- Full Set of Checks
- Programmatic Access using AWS Support API

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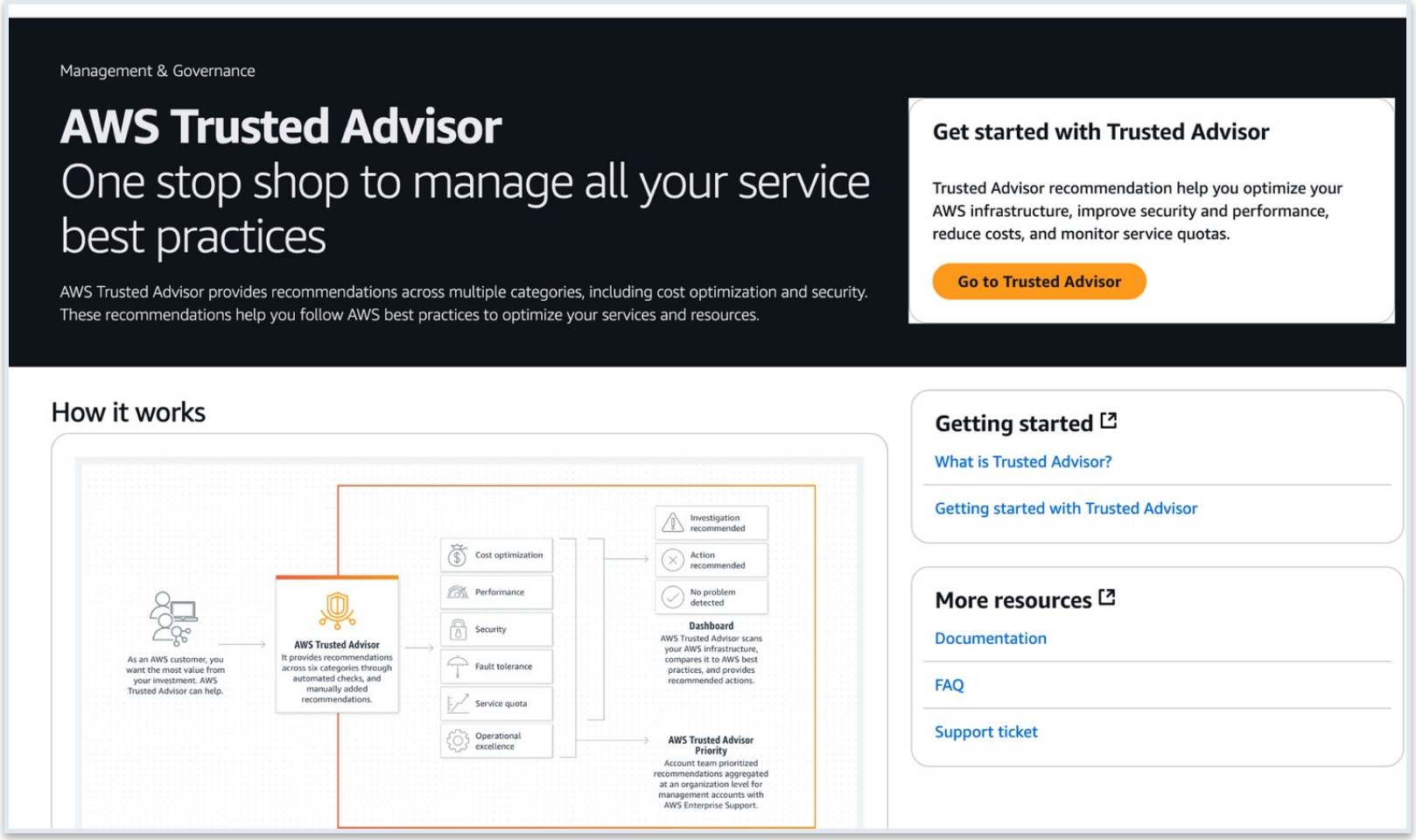
Trusted Advisor



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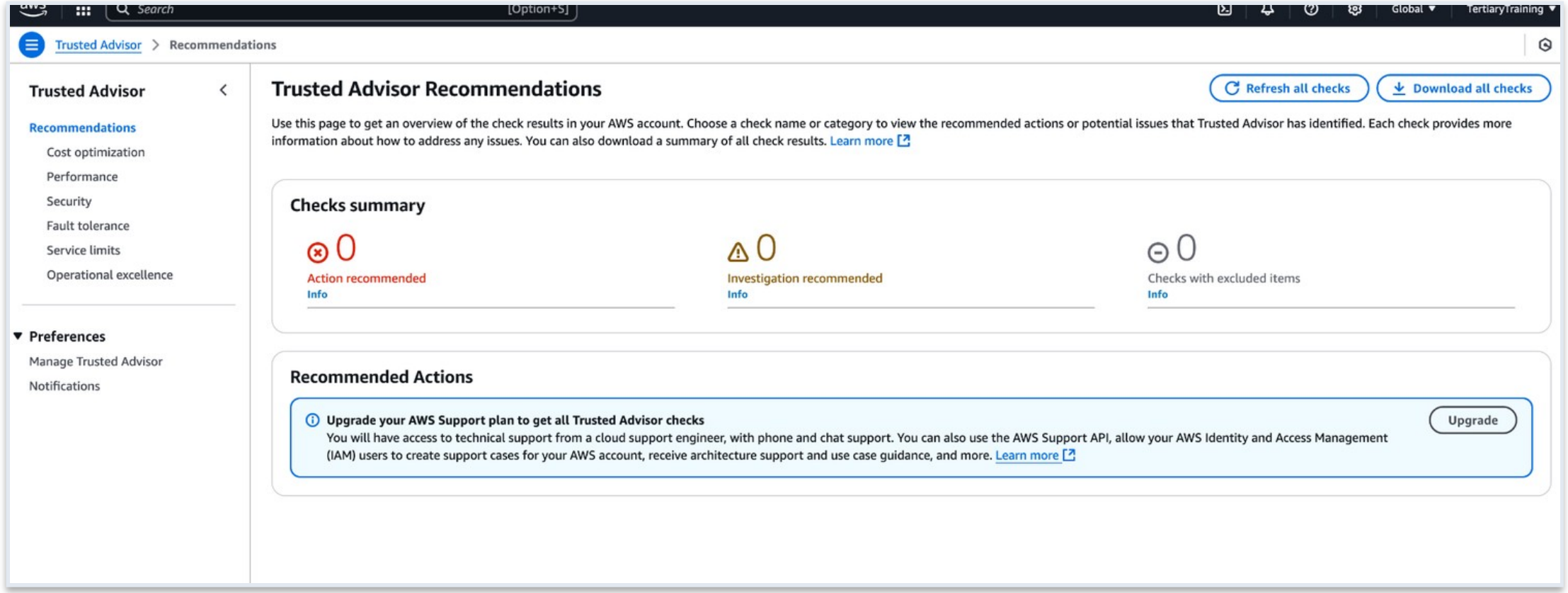
Activity: Trusted Advisor



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Why encryption?

- Encryption in flight (TLS / SSL)
- Data is encrypted before sending and decrypted after receiving
- TLS certificates help with encryption (HTTPS)
- Encryption in flight ensures no MITM (man in the middle attack) can happen
- HTTPS Website
- TLS Encryption
- TLS Decryption
- Username: admin Password: supersecret
- Username: admin Password: supersecret
- aGVsbG8gd29 ybGQgZWh…
- Client
- Server
- Client

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Why encryption?

- Server-side encryption at rest
- Data is encrypted after being received by the server
- Data is decrypted before being sent
- It is stored in an encrypted form thanks to a key (usually a data key)
- The encryption / decryption keys must be managed somewhere, and the server must have access to it
- +
- HTTP(S)
- HTTP(S)
- Object
- Data key
- Object
- +

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Why encryption?

- Client-side encryption
- Data is encrypted by the client and never decrypted by the server
- Data will be decrypted by a receiving client
- The server should not be able to decrypt the data
- Could leverage Envelope Encryption
- +
- Data key (client-side)
- Encryption
- Object
- store
- Encrypted object
- Any storage service (FTP, S3, …)
- +

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AWS KMS (Key ManagementService)

- Anytime you hear “encryption” for an AWS service, it’s most likely KMS
- AWS manages encryption keys for us
- Fully integrated with IAM for authorization
- Easy way to control access to your data
- Able to audit KMS Key usage using CloudTrail
- Seamlessly integrated into most AWS services (EBS, S3, RDS, SSM…)
- Never ever store your secrets in plaintext, especially in your code!
- KMS Key Encryption also available through API calls (SDK, CLI)
- Encrypted secrets can be stored in the code / environment variables

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KMS Keys Types

- KMs Keys is the new name of KMS Customer Master Key
- Symmetric (AES-256 keys)
- Single encryption key that is used to Encrypt and Decrypt
- AWS services that are integrated with KMS use Symmetric CMKs
- You never get access to the KMS Key unencrypted (must call KMS API to use)
- Asymmetric (RSA & ECC key pairs)
- Public (Encrypt) and Private Key (Decrypt) pair
- Used for Encrypt/Decrypt, or Sign/Verify operations
- The public key is downloadable, but you can’t access the Private Key unencrypted
- Use case: encryption outside of AWS by users who can’t call the KMS API

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AWS KMS (Key ManagementService)

- Types of KMS Keys:
- AWS Owned Keys (free): SSE-S3, SSE-SQS, SSE-DDB (default key)
- AWS Managed Key: free (aws/service-name, example: aws/rds or aws/ebs)
- Customer managed keys created in KMS: $1 / month
- Customer managed keys imported: $1 / month
- + pay for API call to KMS ($0.03 / 10000 calls)
- Automatic Key rotation:
- AWS-managed KMS Key: automatic every 1 year
- Customer-managed KMS Key: (must be enabled) automatic & on-demand
- Imported KMS Key: only manual rotation possible using alias

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Copying Snapshots across regions

- EBS Volume Encrypted With KMS
- EBS Snapshot Encrypted With KMS
- KMS Key A
- KMS Key A
- Region eu-west-2
- EBS Volume Encrypted With KMS
- EBS Snapshot Encrypted With KMS
- KMS Key B
- KMS Key B
- Region ap-southeast-2
- KMS ReEncrypt with KMS Key B

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KMS Key Policies

- Control access to KMS keys, “similar” to S3 bucket policies
- Difference: you cannot control access without them
- Default KMS Key Policy:
- Created if you don’t provide a specific KMS Key Policy
- Complete access to the key to the root user = entire AWS account
- Custom KMS Key Policy:
- Define users, roles that can access the KMS key
- Define who can administer the key
- Useful for cross-account access of your KMS key

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Copying Snapshots across accounts

- Create a Snapshot, encrypted with your own KMS Key (Customer Managed Key)
- Attach a KMS Key Policy to
- authorize cross-account access
- Share the encrypted snapshot
- (in target) Create a copy of the Snapshot, encrypt it with a CMK in your account
- Create a volume from the snapshot
- KMS Key Policy

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KMS Automatic Key Rotation

- 111122223333
- KMS Key ID = 1234abcd-12ab-34cd-56ef-1234567890ab
- 444455556666
- KMS Key ID = 1234abcd-12ab-34cd-56ef-1234567890ab
- Automatic Key rotation
- 111122223333
- AWS-managed KMS Keys: automatically rotated every 1 year
- For Customer-Managed Symmetric KMS Key:
- Automatic key rotation is optionally enabled
- Customize Rotation Period between 90 and 2560 days (default: 365 days)
- Previous key is kept active so you can decrypt old data
- New Key has the same KMS Key ID (only the backing key is changed)
- Backing Key Backing Key Saved Backing Key

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KMS On-Demand Key Rotation

- For Customer-Managed Symmetric KMS Key (not AWS managed CMK)
- Does NOT require Automatic Key Rotation to be enabled
- Does NOT change existing Automatic Rotation schedules
- Limit to how many times you can trigger an on-demand key rotation
- On Demand Key rotation
- 111122223333
- KMS Key ID = 1234abcd-12ab-34cd-56ef-1234567890ab
- 444455556666
- KMS Key ID = 1234abcd-12ab-34cd-56ef-1234567890ab
- 111122223333
- Backing Key Backing Key Saved Backing Key

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KMS Manual Key Rotation

- (Customer-Managed Symmetric KMS Key & Imports)
- 111122223333
- Alias = MyCustomKey
- KMS Key ID = 1234abcd-12ab-34cd-56ef-1234567890ab
- Client
- Client
- When you want to rotate key (example: every month)
- New Key has a different KMS Key ID
- Keep the previous key active so you can decrypt old data
- Better to use aliases in this case (to hide the change of key for the application)
- Good solution to rotate KMS Key that are not eligible for automatic rotation (like asymmetric KMS key)
- Rotate Key Manually

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KMS Alias Updating

- Key Alias = MyAppKey
- UpdateAlias API
- KMS Key ID = eba96e8f-7f66-4d5a-aeb6-f32ae8d65d85
- OLD KEY
- NEW KEY
- KMS Key ID = 881cd7f5-b3eb-43e2-82fc-d6f5bda81e00
- Better to use aliases in this case (to hide the change of key for the application)
- Application

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Changing The KMS Key For An Encrypted EBS Volume

- You can’t change the encryption keys used by an EBS volume
- Create an EBS snapshot and create a new EBS volume and specify the new KMS key
- EBS Volume
- (encrypted)
- CMK - A
- EBS Snapshot
- (encrypted)
- create snapshot
- EBS Volume
- (encrypted)
- CMK - B
- create volume
- CMK - A

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Sharing KMS Encrypted RDS DB Snapshots

- You can share RDS DB snapshots encrypted with KMS CMK with other accounts but must first share the KMS CMK with the target account using Key Policy
- Account A (111122223333)
- CMK - A
- Account B (444455556666)
- access through CMK Key Policy
- share
- us-east-1
- DB Instance
- (Snapshot: mydbsnapshot)
- RDS DB
- Snapshot
- (encrypted)

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KMS Key Deletion Considerations

- Schedule CMK for deletion with a waiting period of 7 to 30 days
- CMK’s status is "Pending deletion" during the waiting period
- During the CMK’s deletion waiting period:
- The CMK can't be used for cryptographic operations (e.g., can’t decrypt KMS- encrypted objects in S3 - SSE-KMS)
- The key is not rotated even if planned
- You can cancel the key deletion during the waiting period
- Consider disabling your key instead of deleting it if you’re not sure!

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KMS Key Deletion - CloudWatch Alarm

- Use CloudTrail, CloudWatch Logs, CloudWatch Alarms and SNS to be notified when someone tries to use a CMK that’s "Pending deletion" in a cryptographic operation (Encrypt, Decrypt, …)
- CW Logs
- CW Alarm
- Alert
- Amazon SNS
- CloudWatch
- Metric Filter:
- “* is pending deletion”
- AWS KMS
- CMK
- (Pending deletion)
- CloudTrail

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CloudHSM

- KMS => AWS manages the software for encryption
- CloudHSM => AWS provisions encryption hardware
- Dedicated Hardware (HSM = Hardware Security Module)
- You manage your own encryption keys entirely (not AWS)
- HSM device is tamper resistant, FIPS 140-2 Level 3 compliance
- Supports both symmetric and asymmetric encryption (SSL/TLS keys)
- No free tier available
- Must use the CloudHSM Client Software
- Redshift supports CloudHSM for database encryption and key management
- Good option to use with SSE-C encryption

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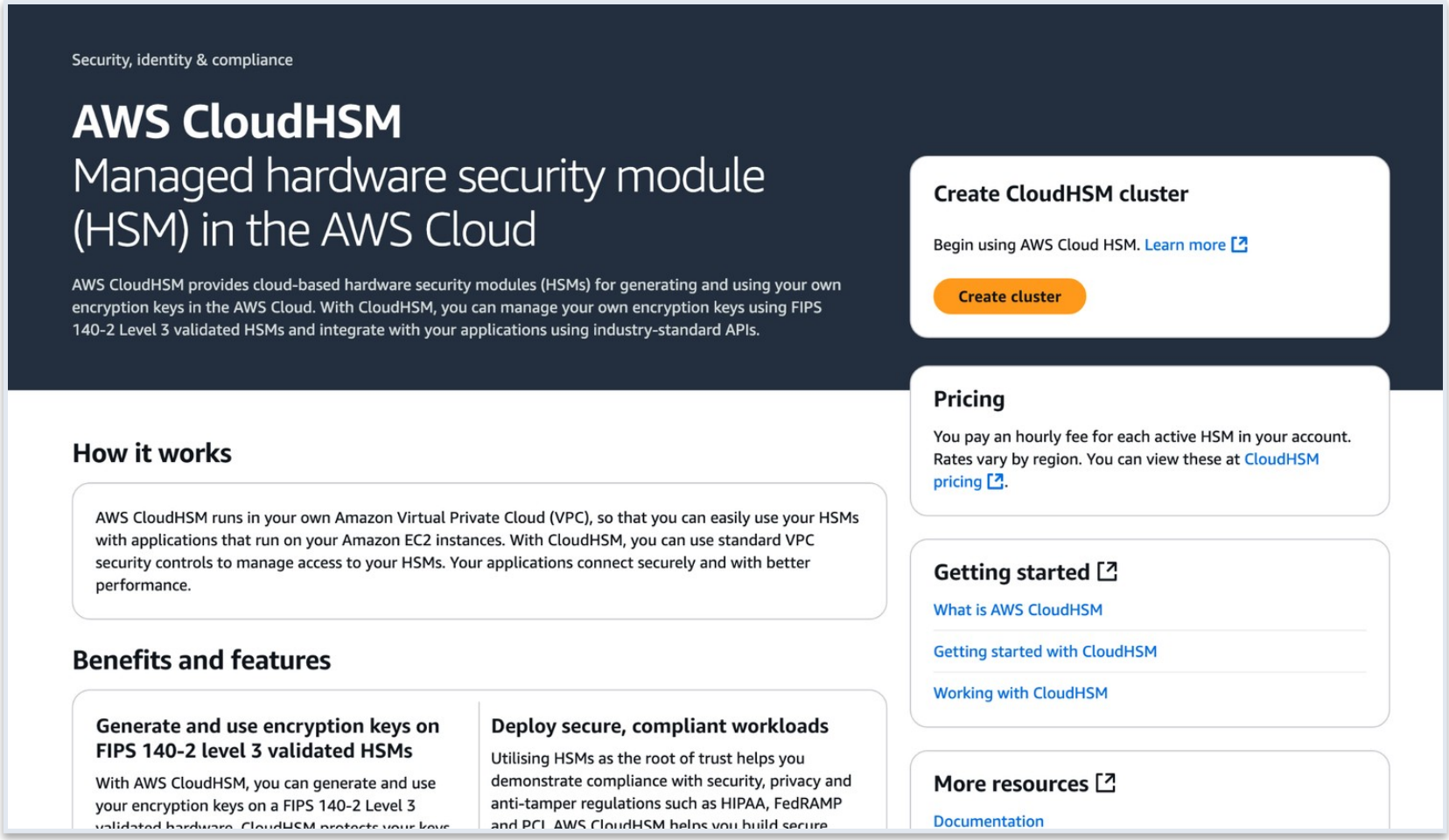
CloudHSM



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CloudHSM Diagram

- AWS CloudHSM
- CloudHSM Client
- AWS manages the Hardware
- SSL Connection
- User manages the Keys
- IAM permissions:
- CRUD an HSM Cluster
- CloudHSM Software:
- Manage the Keys
- Manage the Users

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CloudHSM - High Availability

- CloudHSM clusters are spread across Multi AZ (HA)
- Great for availability and durability
- CloudHSM 1
- CloudHSM Client
- Availability Zone 1
- CloudHSM 2
- Availability Zone 2

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CloudHSM - Integration with AWS Services

- Through integration with AWS KMS
- Configure KMS Custom Key Store with CloudHSM
- Example: EBS, S3, RDS …
- RDS DB
- Instance
- EBS Volume
- CloudHSM
- AWS KMS
- (Custom Key Store)
- KMS Encryption
- Connector
- CloudTrail
- keys usage logs

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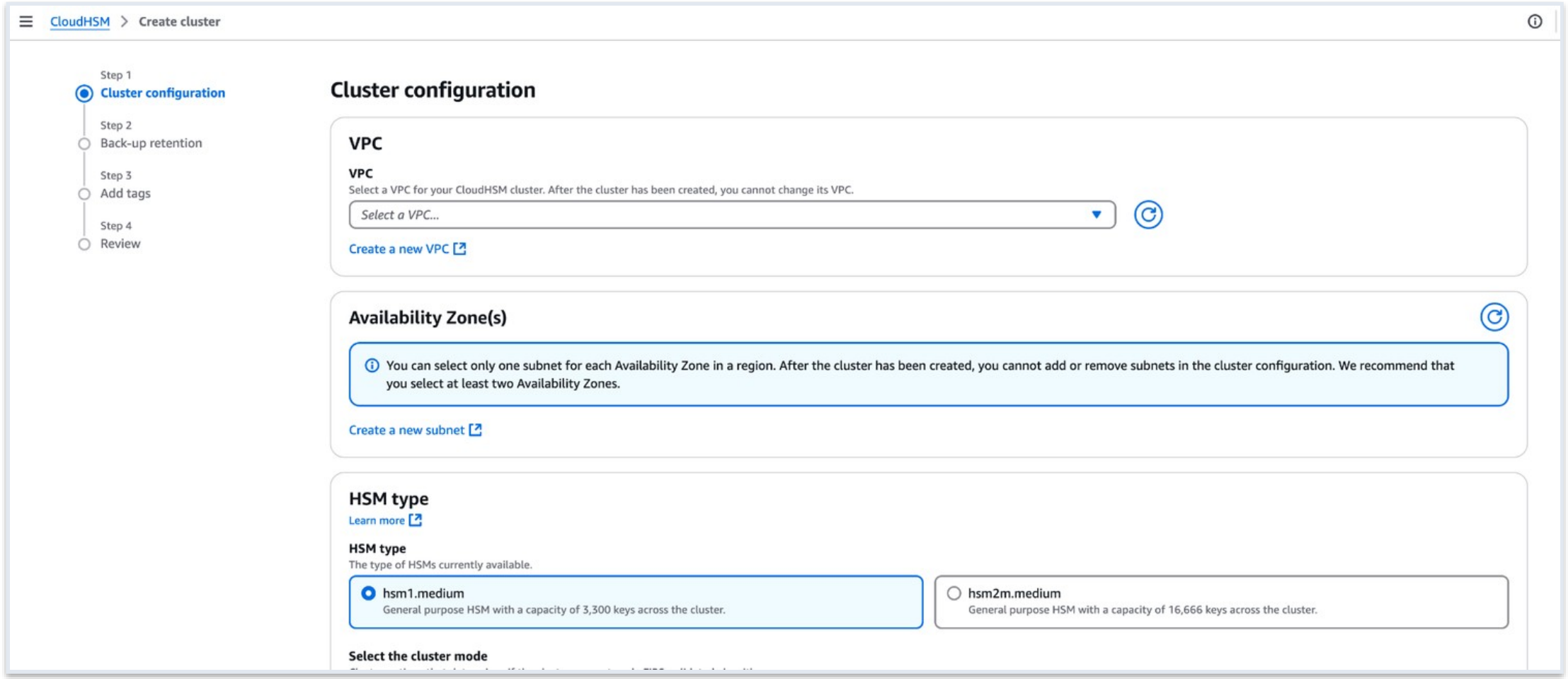
Activity: CloudHSM



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CloudHSM vs KMS

- CloudHSM vs KMS

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CloudHSM vs KMS

- CloudHSM vs KMS

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AWS Artifact (not really a service)

- Portal that provides customers with on-demand access to AWS compliance documentation and AWS agreements
- Artifact Reports - Allows you to download AWS security and compliance documents from third-party auditors, like AWS ISO certifications, Payment Card Industry (PCI), and System and Organization Control (SOC) reports
- Artifact Agreements - Allows you to review, accept, and track the status of AWS agreements such as the Business Associate Addendum (BAA) or the Health Insurance Portability and Accountability Act (HIPAA) for an individual account or in your organization
- Can be used to support internal audit or compliance

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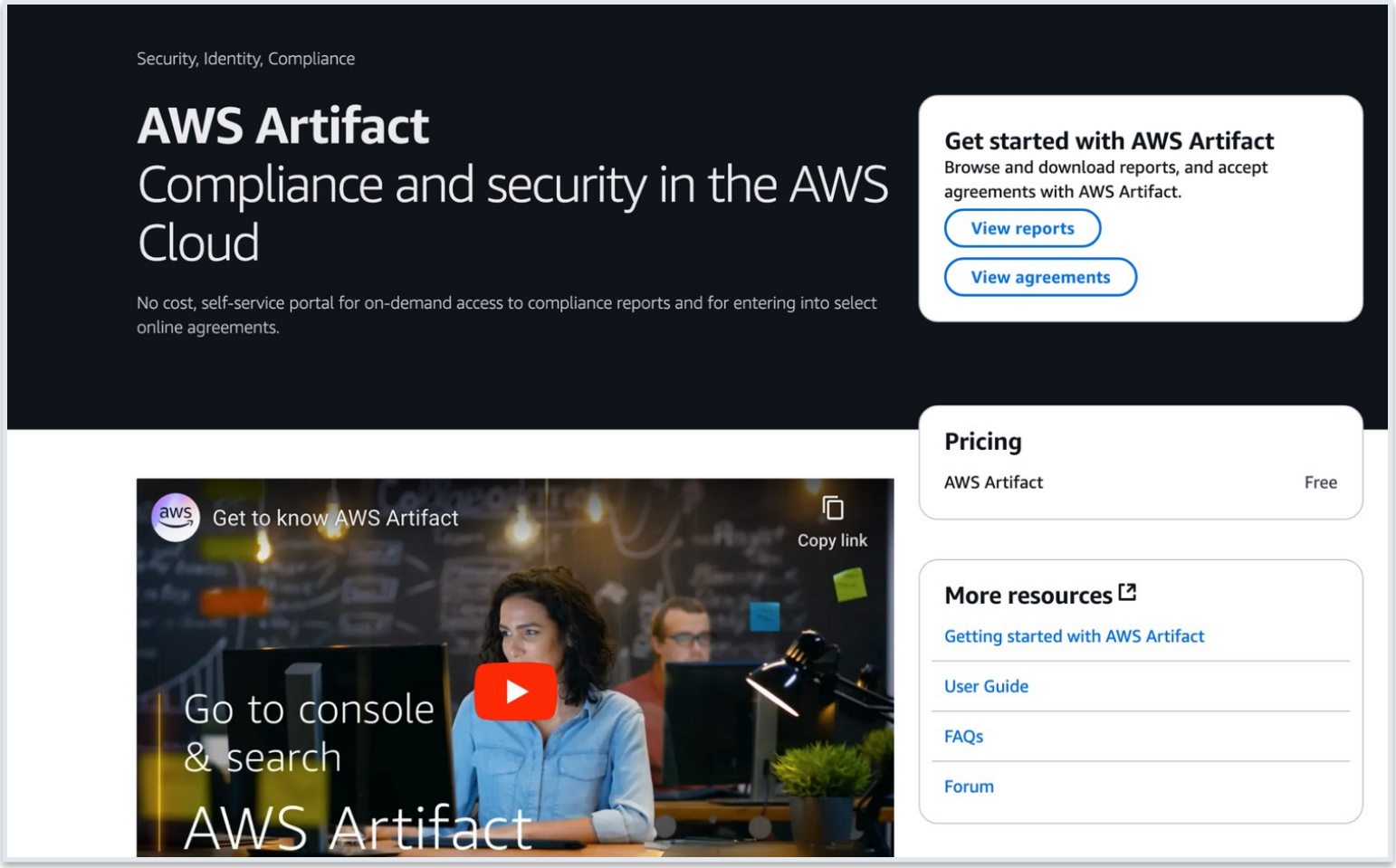
AWS Artifact (not really a service)



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AWS Certificate Manager (ACM)

- Easily provision, manage, and deploy TLS Certificates
- Provide in-flight encryption for websites (HTTPS)
- Supports both public and private TLS certificates
- Free of charge for public TLS certificates
- Automatic TLS certificate renewal
- Integrations with (load TLS certificates on)
- Elastic Load Balancers (CLB, ALB, NLB)
- CloudFront Distributions
- APIs on API Gateway
- Cannot use ACM with EC2 (can’t be extracted)
- Auto Scaling group
- EC2 Instance
- EC2 Instance

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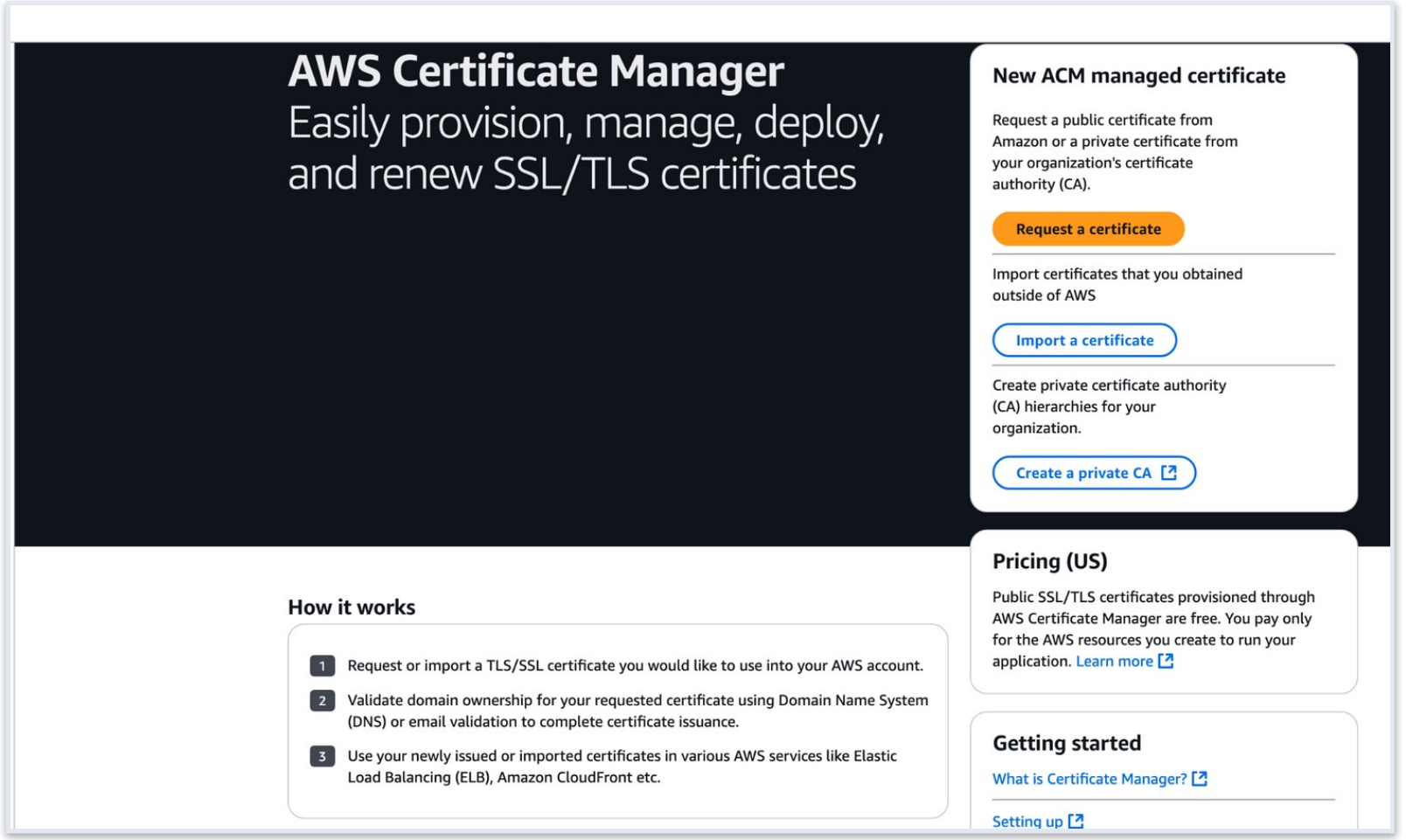
AWS Certificate Manager (ACM)



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ACM - Requesting Public Certificates

- List domain names to be included in the certificate
- Fully Qualified Domain Name (FQDN): corp.example.com
- Wildcard Domain: *.example.com
- Select Validation Method: DNS Validation or Email validation
- DNS Validation is preferred for automation purposes
- Email validation will send emails to contact addresses in the WHOIS database
- DNS Validation will leverage a CNAME record to DNS config (ex: Route 53)
- It will take a few hours to get verified
- The Public Certificate will be enrolled for automatic renewal
- ACM automatically renews ACM-generated certificates 60 days before expiry

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ACM - Importing Public Certificates

- Option to generate the certificate
- outside of ACM and then import it
- No automatic renewal, must import a new certificate before expiry
- ACM sends daily expiration events
- starting 45 days prior to expiration
- The # of days can be configured
- Events are appearing in EventBridge
- AWS Config was a managed rule named acm-certificate-expiration-check to check for expiring certificates (configurable number of days)
- AWS Config
- EventBridge
- Rule events: Non-compliance
- Lambda

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ACM - Integration with ALB

- Auto Scaling group
- EC2 Instance
- EC2 Instance
- AWS Certificate Manager
- Application Load Balancer
- With HTTP -> HTTPS redirect rule
- provision and maintain TLS certs
- HTTP
- Redirect to HTTPS HTTPS

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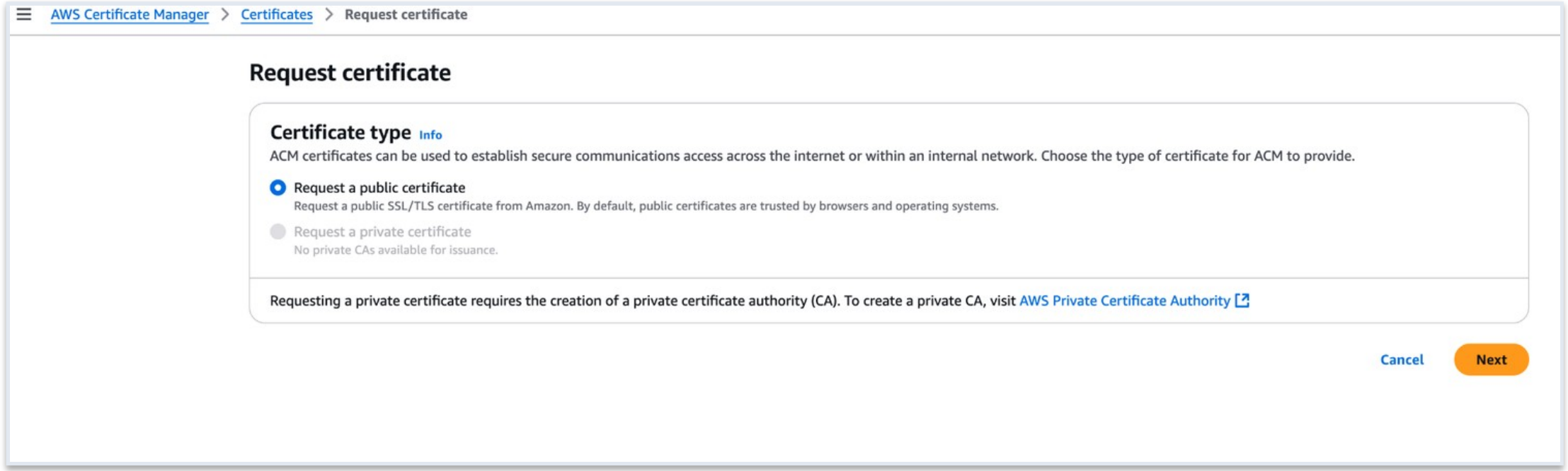
Activity: AWS Certificate Manager (ACM)



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ACM - Integration with API Gateway

- Create a Custom Domain Name in API Gateway
- Edge-Optimized (default): For global clients
- Requests are routed through the CloudFront Edge locations (improves latency)
- The API Gateway still lives in only one region
- The TLS Certificate must be in the same region as CloudFront, in us-east-1
- Then setup CNAME or (better) A-Alias record in Route 53
- Regional:
- For clients within the same region
- The TLS Certificate must be imported on API Gateway, in the same region as the API Stage
- Then setup CNAME or (better) A-Alias record in Route 53
- us-east-1
- CloudFront
- ACM

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API Gateway - Endpoint Types

- Edge-Optimized (default): For global clients
- Requests are routed through the CloudFront Edge locations (improves latency)
- The API Gateway still lives in only one region
- Regional:
- For clients within the same region
- Could manually combine with CloudFront (more control over the caching strategies and the distribution)
- Private:
- Can only be accessed from your VPC using an interface VPC endpoint (ENI)
- Use a resource policy to define access

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SSM Parameter Store

- Secure storage for configuration and secrets
- Optional Seamless Encryption using KMS
- Serverless, scalable, durable, easy SDK
- Version tracking of configurations / secrets
- Configuration management using path & IAM
- Notifications with CloudWatch Events
- Integration with CloudFormation
- SSM Parameter
- Store
- Applications
- Plaintext configuration
- Encrypted configuration
- Check IAM permissions

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SSM Parameter Store Hierarchy

- /my-department/
- my-app/
- dev/
- db-url
- db-password
- prod/
- db-url
- db-password
- other-app/
- /other-department/
- /aws/reference/secretsmanager/secret_ID_in_Secrets_Manager
- /aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2
- Dev Lambda Function

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Standard and advanced parameter tiers

- Standard and advanced parameter tiers

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Parameters Policies (for advanced parameters)

- Allow to assign a TTL to a parameter (expiration date) to force updating or deleting sensitive data such as passwords
- Can assign multiple policies at a time
- Expiration (to delete a parameter)
- ExpirationNotification (CW Events)
- NoChangeNotification (CW Events)

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AWS Secrets Manager

1. Newer service, meant for storing secrets
2. Capability to force rotation of secrets every X days
3. Automate generation of secrets on rotation (uses Lambda)
4. Integration with Amazon RDS (MySQL, PostgreSQL, Aurora)
5. Secrets are encrypted using KMS
6. Mostly meant for RDS integration

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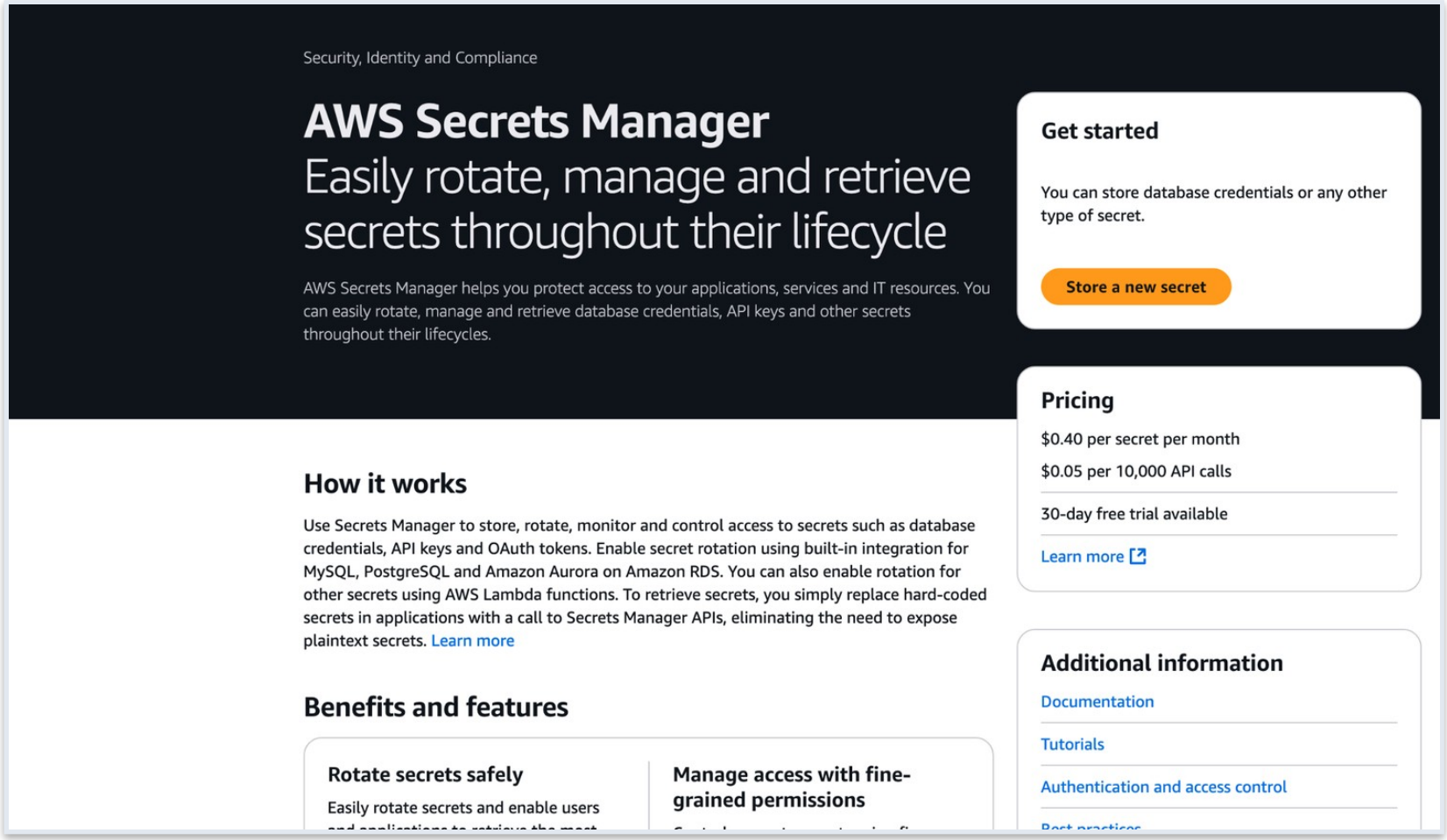
AWS Secrets Manager



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AWS Secrets Manager - Multi-Region Secrets

- Secrets Manager
- MySecret-A
- (primary)
- Replicate Secrets across multiple AWS Regions
- Secrets Manager keeps read replicas in sync with the primary Secret
- Ability to promote a read replica Secret to a standalone Secret
- Use cases: multi-region apps, disaster recovery strategies, multi-region DB…
- Secrets Manager
- MySecret-A
- (replica)
- replicate
- us-east-1 (Primary)

us-west-2 (Secondary)

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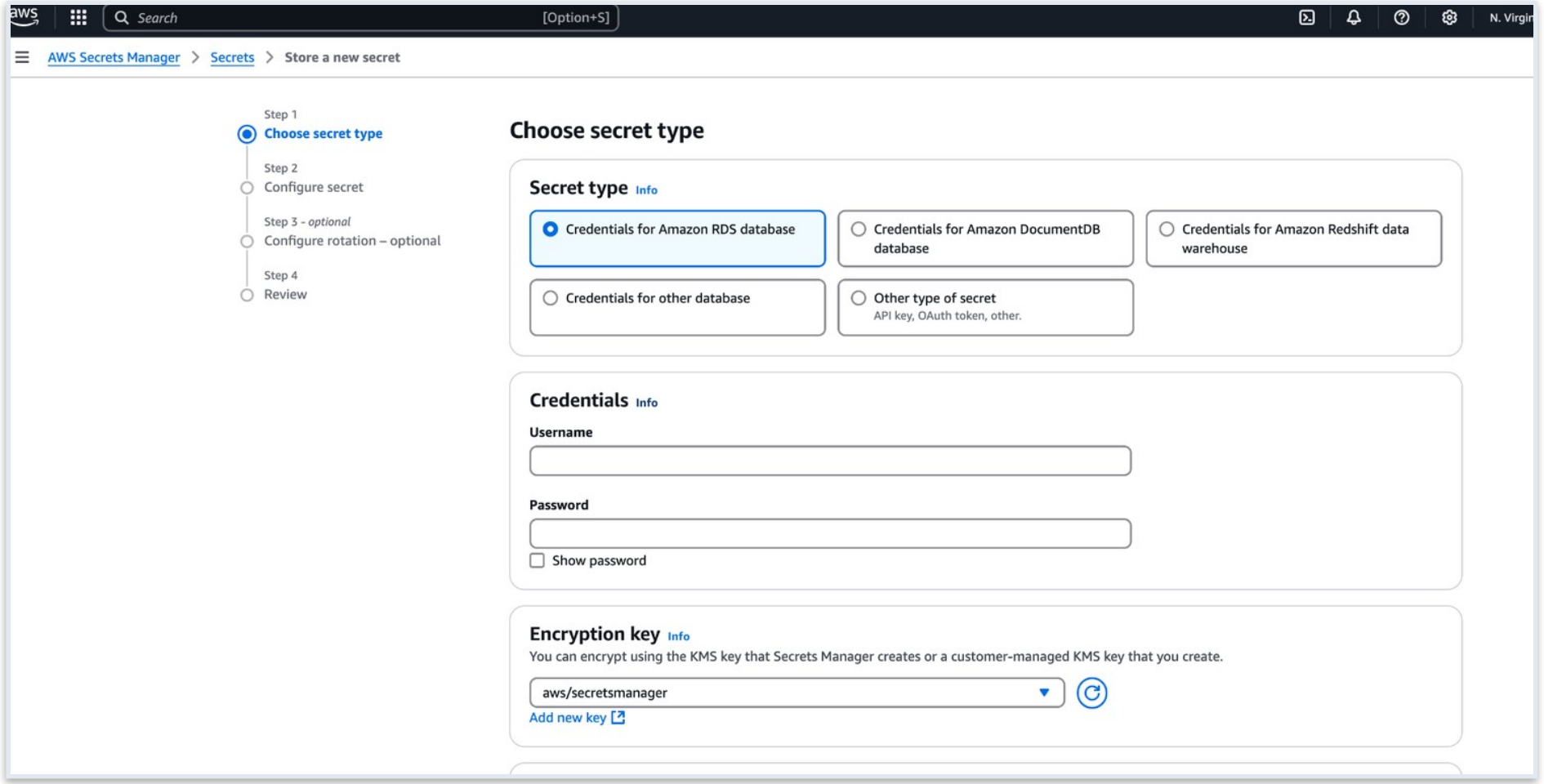
Activity: AWS Secrets Manager



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SSM Parameter Store vs Secrets Manager

- Secrets Manager ($$$):
- Automatic rotation of secrets with AWS Lambda
- Lambda function is provided for RDS, Redshift, DocumentDB
- KMS encryption is mandatory
- Can integration with CloudFormation
- SSM Paramter Store ($):
- Simple API
- No secret rotation (can enable rotation using Lambda triggered by EventBridge)
- KMS encryption is optional
- Can integration with CloudFormation
- Can pull a Secrets Manager secret using the SSM Parameter Store API

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SSM Parameter Store vs Secrets Manager Rotation

- AWS Secrets Manager
- AWS Secrets Manager
- invoke
- Amazon RDS
- every 30 days
- Lambda Function (can be provided)
- change password
- EventBridge
- Lambda Function
- invoke
- SSM Parameter Store
- change value
- every 30 days

SSM Parameter Store

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Secrets Manager - Monitoring

- CloudTrail captures API calls to the Secrets Manager API
- CloudTrail captures other related events that might have a security
- or compliance impact on your AWS account or might help you troubleshoot operational problems.
- CloudTrail records these events as non-API service events.
- RotationStarted event
- RotationSucceeded event
- RotationFailed event
- RotationAbandoned event – a manual change to a secret instead of automated rotation
- StartSecretVersionDelete event
- CancelSecretVersionDelete event
- EndSecretVersionDelete event
- Combine with CloudWatch Logs and CloudWatch alarms for automations
- CW Logs

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Secrets Manager - Troubleshooting Rotation

- AWS Secrets Manager
- invoke
- Amazon RDS
- Lambda Function
- change password
- CloudWatch Logs
- Error
- CloudTrail
- Administrator
- logs
- API calls
- & Non-API events
- debug issues

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TOPIC 18

Identity

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IAM Security Tools

- IAM Credentials Report (account-level)
- a report that lists all your account's users and the status of their various credentials
- IAM Access Advisor (user-level)
- Access advisor shows the service permissions granted to a user and when those services were last accessed.
- You can use this information to revise your policies.

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IAM Access Analyzer

- Find out which resources are shared externally
- S3 Buckets
- IAM Roles
- KMS Keys
- Lambda Functions and Layers
- SQS queues
- Secrets Manager Secrets
- Define Zone of Trust = AWS Account or AWS Organization
- Access outside zone of trusts => findings
- Role
- User
- Account
- S3 Bucket

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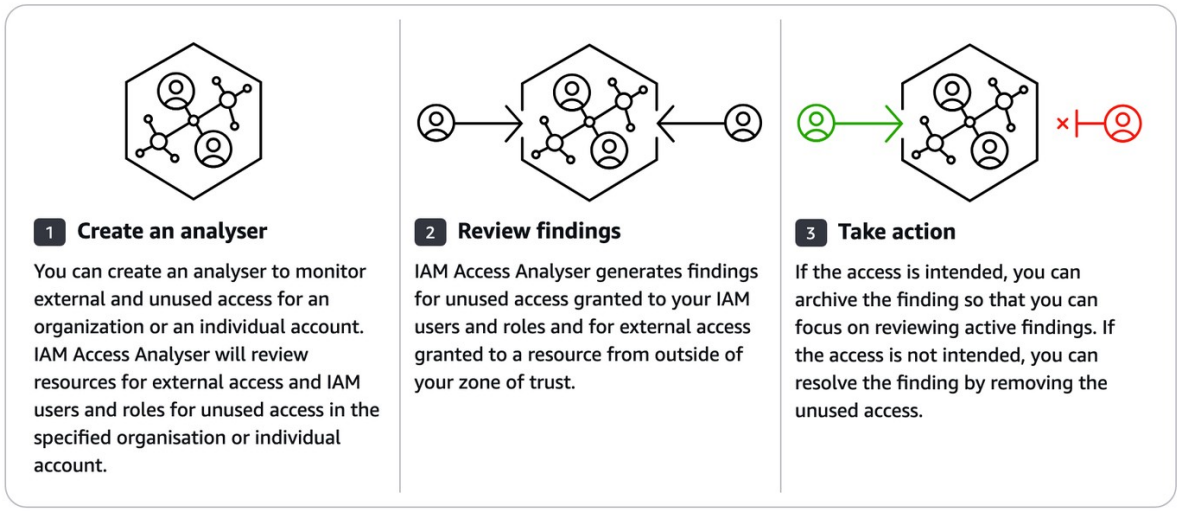
IAM Access Analyzer

Security, Identity & Compliance

IAM Access Analyser

Simplify your journey to least privilege

How it works



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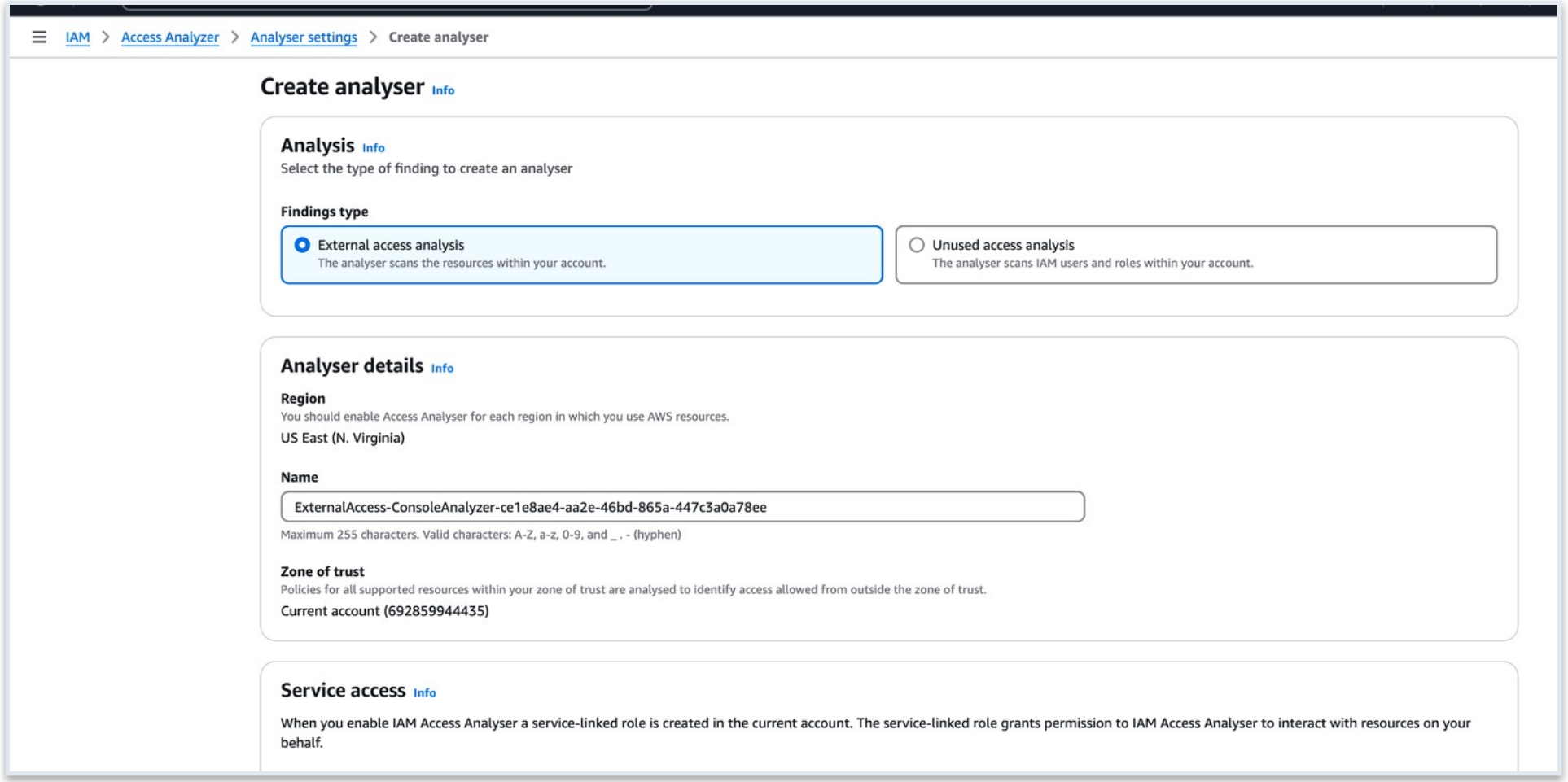
Activity: IAM Access Analyzer



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What’s Identity Federation?

- Federation lets users outside of AWS to assume temporary role for accessing AWS resources.
- These users assume identity provided access role.
- Federation assumes a form of 3rd party authentication
- LDAP
- Microsoft Active Directory (~= SAML)
- Single Sign On
- Open ID
- Cognito
- Using federation, you don't need to create IAM users (user management is outside of AWS)
- user
- 3rd party
- login
- Gives credentials

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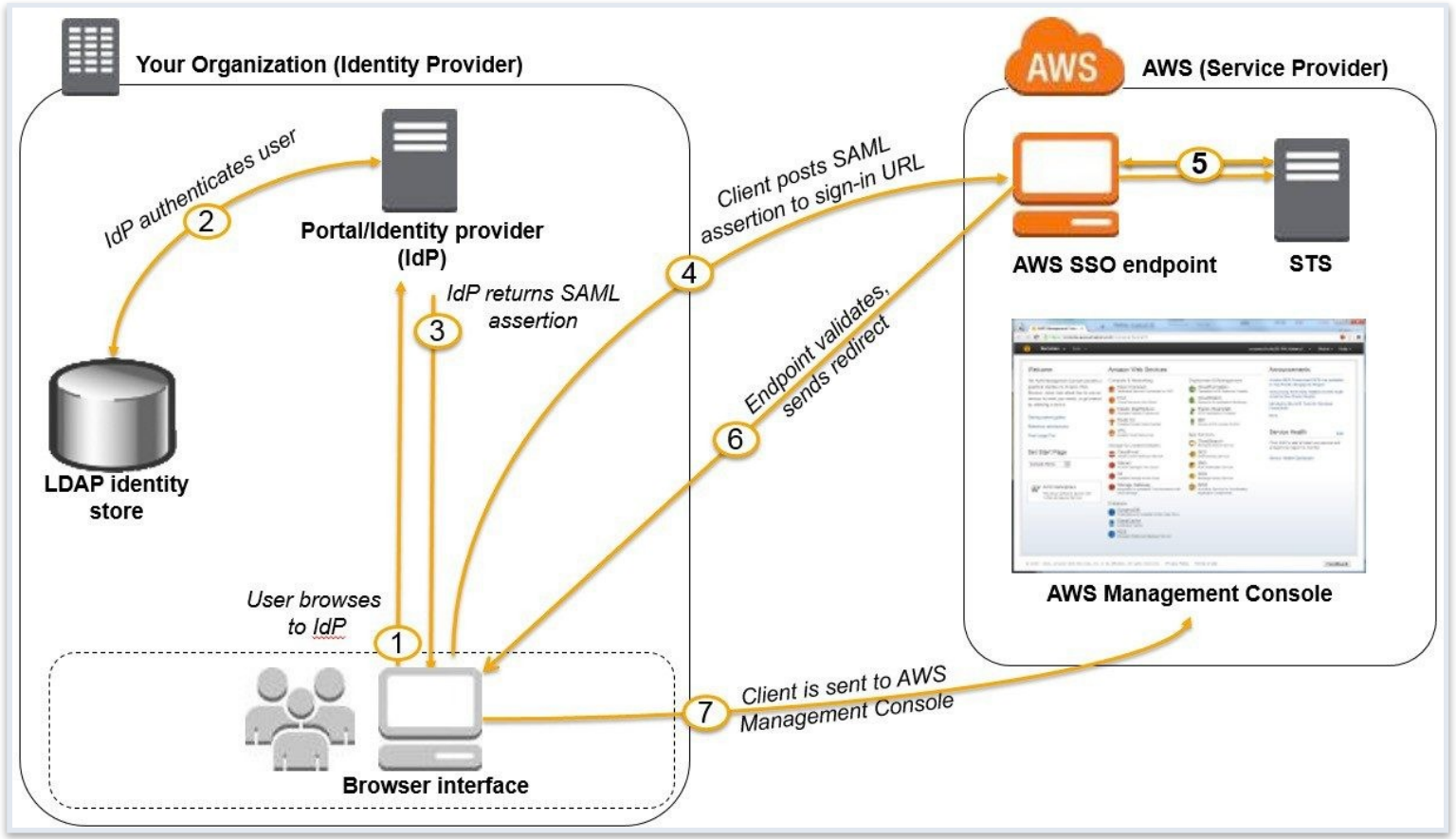
SAML Federation



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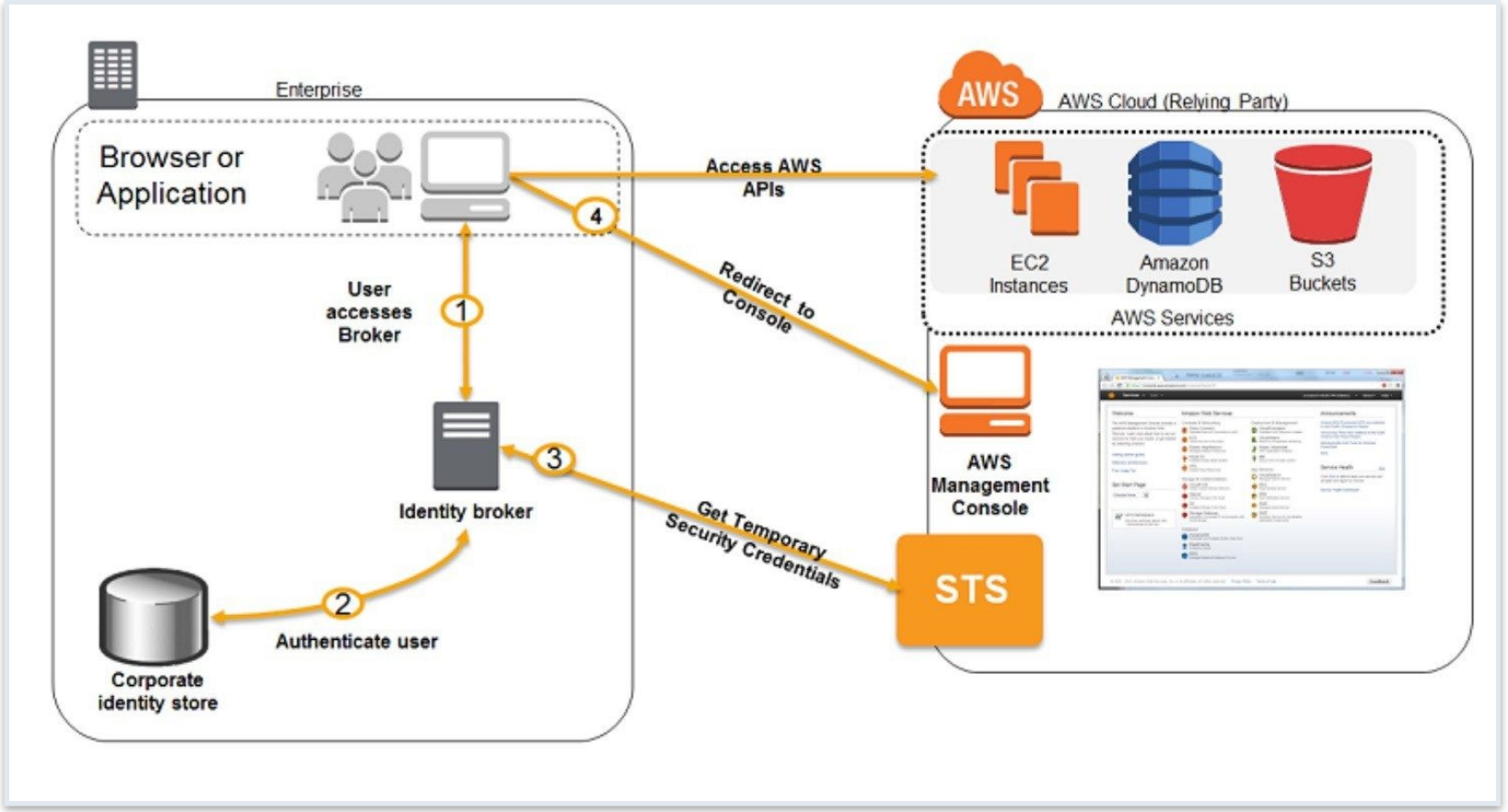
Custom Identity Broker Application



For Enterprises

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TOPIC 18

AWS Cognito - Federated Identity Pools

- For Public Applications
- Goal:
- Provide direct access to AWS Resources from the Client Side
- How:
- Log in to federated identity provider – or remain anonymous
- Get temporary AWS credentials back from the Federated Identity Pool
- These credentials come with a pre-defined IAM policy stating their permissions
- Example:
- provide (temporary) access to write to S3 bucket using Facebook Login
- Note:
- Web Identity Federation is an alternative to using Cognito but AWS recommends against it
- Federated Identity
- Authenticate to FIP

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AWS STS - Security Token Service

- Allows to grant limited and temporary access to AWS resources.
- Token is valid for up to one hour (must be refreshed)
- AssumeRole
- Within your own account: for enhanced security
- Cross Account Access: assume role in target account to perform actions there
- AssumeROleWithSAML
- return credentials for users logged with SAML
- AssumeRoleWithWebIdentity
- return creds for users logged with an IdP (Facebook Login, Google Login, OIDC compatible…)
- AWS recommends against using this, and using Cognito instead
- GetSessionToken
- for MFA, from a user or AWS account root user

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Using STS to Assume a Role

- Define an IAM Role within your account or cross-account
- Define which principals can access this IAM Role
- Use AWS STS (Security Token Service) to retrieve credentials and impersonate the IAM Role you have access to (AssumeRole API)
- Temporary credentials can be valid between 15 minutes to 1 hour
- AWS STS
- permissions
- Role (same or other account)
- temporary security credential
- IAM
- user
- AssumeRole API

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Cross - Account Access with STS

- IAM Role
- (UpdateApp)
- Production Account
- 1. Admin creates an IAM role that grants the Development account access to productionapp S3 bucket
- S3 Bucket
- (productionapp)
- IAM Group
- (Developers)
- IAM Group
- (Testers)
- Development Account
- 2. Admin grants members Developers

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Cognito User Pools (CUP) - User Features

- Create a serverless database of user for your web & mobile apps
- Simple login: Username (or email) / password combination
- Password reset
- Email & Phone Number Verification
- Multi-factor authentication (MFA)
- Federated Identities: users from Facebook, Google, SAML…
- Feature: block users if their credentials are compromised elsewhere
- Login sends back a JSON Web Token (JWT)

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Cognito User Pools (CUP) - Diagram

- Cognito User Pools
- Federation through
- Third Party Identity Provider (IdP)
- Database of Users
- Web Applications
- Mobile Applications
- Social Identity Provider
- Returns JSON Web Token
- JWT
- login

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Cognito User Pools (CUP) - Integrations

- CUP integrates with API Gateway and Application Load Balancer
- REST API +
- Pass Token
- Evaluate Cognito Token
- Authenticate Retrieve token
- Cognito User Pools
- Cognito User Pools
- Application Load Balancer
- + Listeners & Rules
- Authenticate
- Target Group
- Backend
- API Gateway

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Cognito Identity Pools (Federated Identities)

- Get identities for "users" so they obtain temporary AWS credentials
- Your identity pool (e.g identity source) can include:
- Public Providers (Login with Amazon, Facebook, Google, Apple)
- Users in an Amazon Cognito user pool
- OpenID Connect Providers & SAML Identity Providers
- Developer Authenticated Identities (custom login server)
- Cognito Identity Pools allow for unauthenticated (guest) access
- Users can then access AWS services directly or through API Gateway
- The IAM policies applied to the credentials are defined in Cognito
- They can be customized based on the user_id for fine grained control

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Cognito Identity Pools - Diagram

- Cognito Identity Pools
- Web & Mobile Applications
- Login and Get Token
- Exchange token for temporary AWS credentials
- validate
- Get temp creds
- Private S3 Bucket
- DynamoDB Table
- Direct access to AWS
- Social Identity Provider
- Cognito User Pools

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TOPIC 18

Cognito Identity Pools - Diagram with CUP

- Cognito Identity Pools
- Web & Mobile Applications
- Login and Get Token
- Exchange token for temporary AWS credentials
- validate
- Get temp creds
- Private S3 Bucket
- DynamoDB Table
- Direct access to AWS
- Cognito User Pools
- Internal DB of users

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Cognito Identity Pools - IAM Roles

1. Default IAM roles for authenticated and guest users
2. Define rules to choose the role for each user based on the user's ID
3. You can partition your users' access using policy variables
4. IAM credentials are obtained by Cognito Identity Pools through STS
5. The roles must have a "trust" policy of Cognito Identity Pools

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Cognito Identity Pools - Guest User example

- Cognito Identity Pools - Guest User example

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Cognito Identity Pools - Policy variable on S3

- Cognito Identity Pools - Policy variable on S3

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TOPIC 18

Cognito Identity Pools - DynamoDB

- Cognito Identity Pools - DynamoDB

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Cognito User Pools vs Identity Pools

- Cognito User Pools (for authentication = identity verification)
- Database of users for your web and mobile application
- Allows to federate logins through Public Social, OIDC, SAML…
- Can customize the hosted UI for authentication (including the logo)
- Has triggers with AWS Lambda during the authentication flow
- Adapt the sign-in experience to different risk levels (MFA, adaptive authentication, etc…)
- Cognito Identity Pools (for authorization = access control)
- Obtain AWS credentials for your users
- Users can login through Public Social, OIDC, SAML & Cognito User Pools
- Users can be unauthenticated (guests)
- Users are mapped to IAM roles & policies, can leverage policy variables
- CUP + CIP = authenication + authorization

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TOPIC 19

Amazon Route 53

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Whatis DNS?

- Domain Name System which translates the human friendly hostnames into the machine IP addresses
- www.google.com => 172.217.18.36
- DNS is the backbone of the Internet
- DNS uses hierarchical naming structure
- .com
- example.com www.example.com api.example.com

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DNS Terminologies

- TLD
- SLD
- Sub Domain
- FQDN (Fully Qualified Domain Name)
- Protocol
- Root
- Domain Registrar: Amazon Route 53, GoDaddy, …
- DNS Records: A, AAAA, CNAME, NS, …
- Zone File: contains DNS records
- Name Server: resolves DNS queries (Authoritative or Non-Authoritative)
- Top Level Domain (TLD): .com, .us, .in, .gov, .org, …
- Second Level Domain (STD): amazon.com, google.com, …
- URL

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How DNS Works

- Local DNS Server
- Assigned and Managed by your company or assigned by your ISP dynamically
- example.com?
- Root DNS Server
- example.com?
- TLD DNS Server
- (.com)
- .com NS 1.2.3.4
- example.com?
- example.com NS 5.6.7.8
- example.com IP 9.10.11.12
- Managed by ICANN
- Managed by Domain Registrar (e.g., Amazon Registrar, Inc.)

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Amazon Route 53

- A highly available, scalable, fully managed and Authoritative DNS
- Authoritative = the customer (you) can update the DNS records
- Route 53 is also a Domain Registrar
- Ability to check the health of your resources
- The only AWS service which provides 100% availability SLA
- Why Route 53? 53 is a reference to the traditional DNS port
- example.com?
- 54.22.33.44
- Client
- Amazon Route 53
- AWS Cloud
- Public IP
- 54.22.33.44

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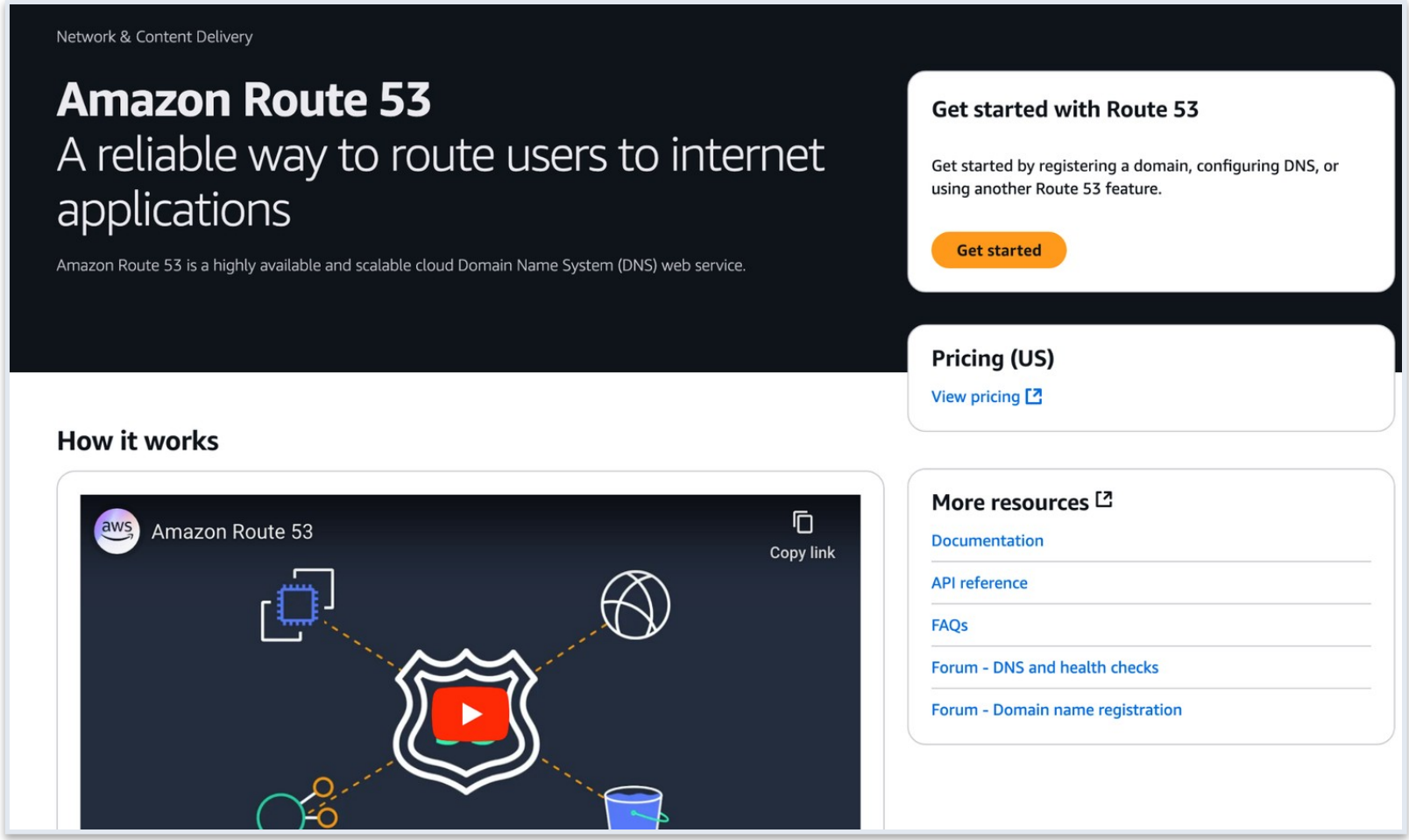
Amazon Route 53



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Reference

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Route 53 - Records

- How you want to route traffic for a domain
- Each record contains:
- Domain/SubDomain namer - e.g., example.com
- Record Type - e.g., A or AAAA
- Value - e.g., 12.34.56.78
- Routing Policy - how Route 53 responds to queries
- TTL - amount of time the record cached at DNS Resolvers
- Route 53 supports the following DNS record types:
- (must know) A / AAAA / CNAME / NS
- (advanced) CAA / DS / MX / NAPTR / PTR / SOA / TXT / SPF / SRV

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Route 53 - Record Types

- A - maps a hostname to IPv4
- AAAA - maps a hostname to IPv6
- CNAME - maps a hostname to another hostname
- The target is a domain name which must have an A or AAAA record
- Can’t create a CNAME record for the top node of a DNS namespace (Zone Apex)
- Example: you can’t create for example.com, but you can create for www.example.com
- NS - Name Servers for the Hosted Zone
- Control how traffic is routed for a domain

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Route 53 - Hosted Zones

- A container for records that define how to route traffic to a domain and its subdomains
- Puiblic Hosted Zones - contains records that specify how to route traffic on the Internet (public domain names) application1.mypublicdomain.com
- Private Hosted Zones - contain records that specify how you route traffic within one or more VPCs (private domain names) application1.company.internal
- You pay $0.50 per month per hosted zone

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Route 53 - Public vs Private Hosted Zones

- VPC
- EC2 Instance
- (Public IP)
- Application
- Load Balancer
- Public Hosted Zone
- Client
- S3 Bucket
- Amazon CloudFront
- 54.22.33.44
- VPC
- Private Hosted Zone
- EC2 Instance EC2 Instance

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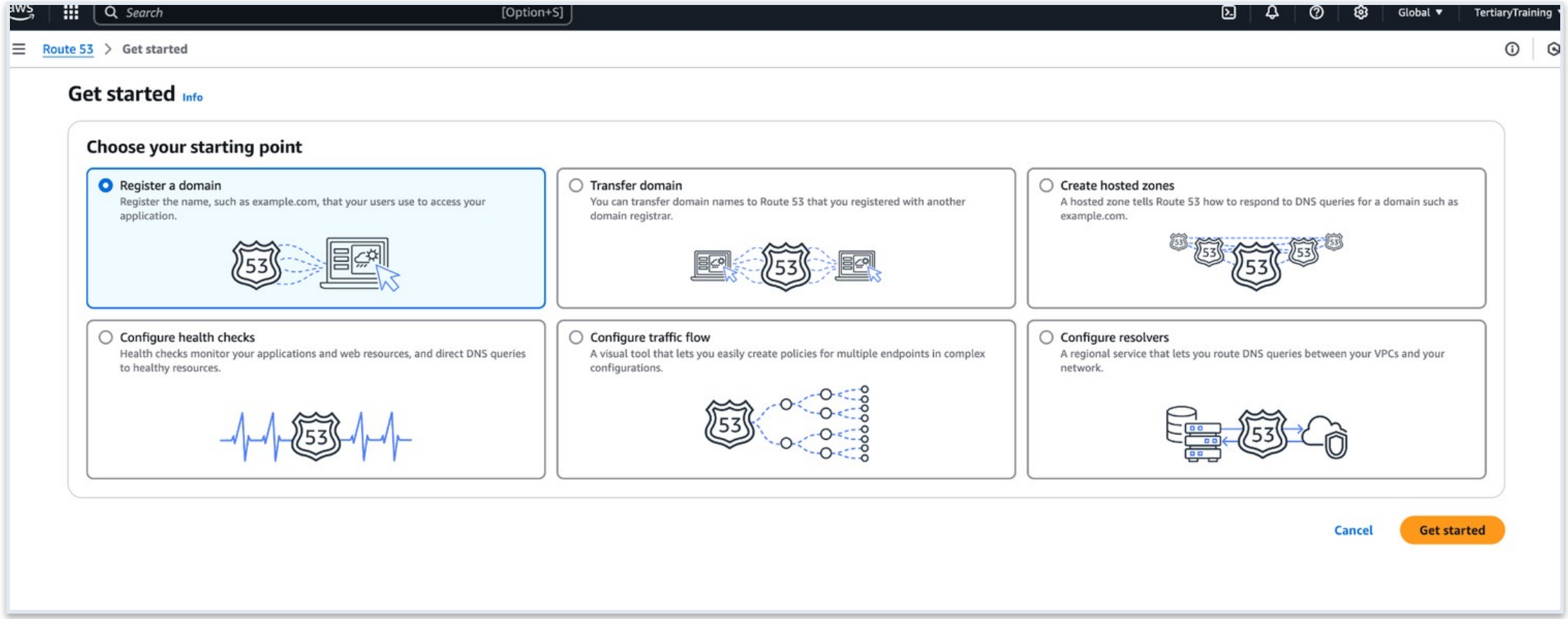
Activity: Route 53



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Route 53 - Records TTL (Time To Live)

- Web Server
- DNS Request
- myapp.example.com?
- TTL
- Amazon Route 53
- A 12.34.56.78
- (with TTL
- HTTP Request
- HTTP Response
- Client
- Will cache the result for The TTL of the record
- High TTL - e.g. 24 hr
- Less traffic on Route 53

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CNAME vs Alias

- AWS Resources (Load Balancer, CloudFront...) expose an AWS hostname:
- lb1-1234.us-east2.elb.amazonaws.com and you want myapp.mydomain.com
- CNAME:
- Points a hostname to any other hostname. (app.mydomain.com => blabla.anything.com)
- ONLY FOR NON ROOT DOMAIN (aka something.mydomain.com)
- Alias:
- Points a hostname to an AWS Resource (app.mydomain.com => blabla.amazonaws.com)
- Works for ROOT DOMAIN and NON ROOT DOMAIN (aka mydomain.com)
- Free of charge
- Native health check

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Route 53 - Alias Records

- Maps a hostname to an AWS resource
- An extension to DNS functionality
- Automatically recognizes changes in the resource’s IP addresses
- Unlike CNAME, it can be used for the top node of a DNS namespace (Zone Apex), e.g.: example.com
- Alias Record is always of type A/AAAA for
- AWS resources (IPv4 / IPv6)
- You can't set the TTL
- MyALB-123456789.us-east-1.elb.amazonaws.com
- Amazon Route 53
- Alias Record (Enabled)
- AWS-Managed
- (IP Addresses might change)

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Route 53 - Alias Records Targets

- Elastic Load Balancers
- CloudFront Distributions
- API Gateway
- Elastic Beanstalk environments
- S3 Websites
- VPC Interface Endpoints
- Global Accelerator accelerator
- Route 53 record in the same hosted zone
- You cannot set an ALIAS record for an EC2 DNS name
- Global Accelerator
- Elastic Load Balancer
- Amazon CloudFront
- Amazon API Gateway

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Route 53 - Routing Policies

- Define how Route 53 responds to DNS queries
- Don’t get confused by the word “Routing”
- It’s not the same as Load balancer routing which routes the traffic
- DNS does not route any traffic, it only responds to the DNS queries
- Route 53 Supports the following Routing Policies
- Simple
- Weighted
- Failover
- Latency based
- Geolocation
- Multi-Value Answer
- Geoproximity (using Route 53 Traffic Flow feature)

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Routing Policies - Simple

- Typically, route traffic to a single resource
- Can specify multiple values in the same record
- If mutliple values are returned a
- random one is chosen by the client
- When Alias enabled, specify only one AWS resource
- Can’t be associated with Health Checks
- foo.example.com
- A 11.22.33.44
- Client
- Amazon Route 53
- Single Value
- foo.example.com
- A 11.22.33.44

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Routing Policies - Weighted

- Control the % of the requests that go to each specific resource
- Assign each record a relative weight:
- $traffic$ (%) =
- !"#$%& ()* + ,-".#(#. *".)*/
- 012)(+33 &%" 4"#$%&, ()* +33 *".)*/,
- Weights don’t need to sum up to 100
- DNS records must have the same name and type
- Can be associated with Health Checks
- Use cases: load balancing between regions, testing new application versions…
- Assign a weight of 0 for a record to stop sending traffic to a resource
- If all records have a weight of 0 then all records will be returned equally
- Amazon Route 53
- Weight: 70

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Routing Policies - Latency-based

- Redirect to the resource that has the least latency close to us
- Super helpful when latency for users is a priority
- Latency is based on trafficbetween users and AWS Regions
- Germany users may be directed to the US (if that’s the lowest latency)
- Can be associated with Health
- Checks (has a failover capability)
- ALB
- (us-east-1)
- ALB
- (ap-southeast-1)

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Route 53 - Health Checks

- HTTP Health Checks are only for public resources
- Health Check => Automated DNS Failover:
- Health checks that monitor an endpoint (application, server, other AWS resource)
- Health checks that monitor other health checks (Calculated Health Checks)
- Health checks that monitor CloudWatch Alarms (full control !!) – e.g., throttles of DynamoDB, alarms on RDS, custom metrics,
- … (helpful for private resources)
- Health Checks are integrated with CW metrics
- Amazon Route 53
- DNS Record (latency, geoproximity, …)
- ALB
- Auto Scaling group
- EC2 Instance

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eu-west-1

- Health Checker (us-east-1)
- Health Checks - Monitor an Endpoint
- about 15 global health checkers will check the endpoint health
- Healthy/Unhealthy Threshold – 3 (default)
- Interval – 30 sec (can set to 10 sec – higher cost)
- Supported protocol: HTTP, HTTPS and TCP
- If> 18% of health checkers report the endpoint is healthy, Route 53 considers it Healthy. Otherwise, it's Unhealthy
- Ability to choose which locations you want Route 53 to use
- Health Checks pass only when the endpoint responds with the 2xx and 3xx status codes
- Health Checks can be setup to pass / fail based on the text in the first 5120 bytes of the response
- Configure your router/firewall to allow incoming requests from Route 53 Health Checkers
- HTTP Request

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Amazon Route 53

- Route 53 - Calculated Health Checks
- Combine the results of multiple Health Checks into a single Health Check
- You can use OR, AND or NOT
- Can monitor up to 256 Child Health Checks
- Specify how many of the health checks need to pass to make the parent pass
- Usage: perform maintenance to your website without causing all health checks to fail
- Health Check
- (Parent)
- Health Check
- (Child)
- Health Health
- Check Check
- (Child) (Child)

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Health Checks - Private Hosted Zones

- Route 53 health checkers are outside the VPC
- They can’t access private endpoints (private VPC or on-premises resource)
- You can create a CloudWatch Metric and associate a CloudWatch Alarm, then create a Health Check that checks the alarm itself
- Health Checker (us-east-1)
- VPC
- Private subnet
- monitor
- CloudWatch Alarm
- monitor

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Routing Policies - Failover (Active-Passive)

- EC2 Instance
- (Primary)
- Health Check
- (mandatory)
- Failover
- Client
- Amazon Route 53
- EC2 Instance
- (Secondary – Disaster Recovery)
- DNS Requests

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Routing Policies - Geolocation

- Different from Latency-based!
- This routing is based on user location
- Specify location by Continent, Country or by US State (if there’s overlapping, most precise location selected)
- Should create a "Default" record (in case there’s no match on location)
- Use cases: website localization, restrict content distribution, load balancing …
- Can be associated with Health Checks
- A 11.22.33.44
- A 55.66.77.88
- Default
- A 99.11.22.33

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Routing Policies - Geoproximity

- Route traffic to your resources based on the geographic location of users and resources
- Ability to shift more traffic to resources based on the defined bias
- To change the size of the geographic region, specify bias values:
- To expand (1 to 99) – more traffic to the resource
- To shrink (-1 to -99) – less traffic to the resource
- Resources can be:
- AWS resources (specify AWS region)
- Non-AWS resources (specify Latitude and Longitude)
- You must use Route 53 Traffic Flow to use this feature

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Routing Policies - Geoproximity

1 us-west-1 Bias: 0

2 us-east-1 Bias: 0

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TOPIC 19

Routing Policies - Geoproximity

1. us-west-1 Bias: 0
2. us-east-1 Bias: 50
3. Higher bias in us-east-1

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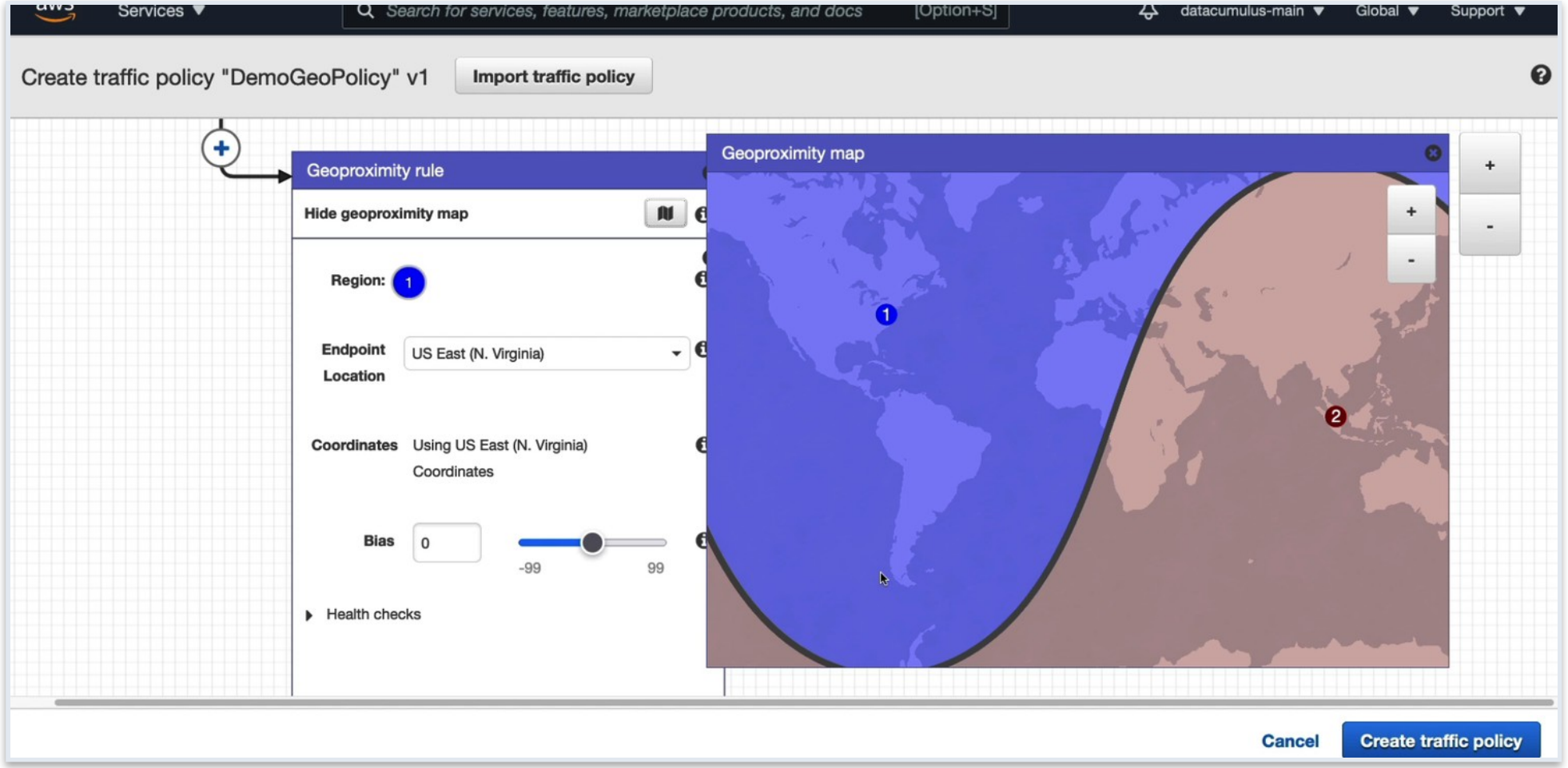
Activity: GeoPolicy



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Route 53 - Traffic flow

- Simplify the process of creating and maintaining records in large and complex configurations
- Visual editor to manage complex routing decision trees
- Configurations can be saved as
- Traffic Flow Policy
- Can be applied to different Route 53 Hosted Zones (different domain names)
- Supports versioning

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Routing Policies - IP-based Routing

- Routing is based on client's IP addresses
- You provide a list of CIDRs for your clients and the corresponding endpoints/locations (user-IP-to-endpoint mappings)
- Use cases: Optimize performance, reduce network costs…
- Example: route end users from a particular ISP to a specific endpoint
- CIDR Collection
- Route 53
- Records
- EC2 Instance
- (1.2.3.4)
- EC2 Instance
- (5.6.7.8)
- User A

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Routing Policies - Multi-Value

Name	Type	Value	TTL	Set ID	Health Check
www.example.com	A Record	192.0.2.2	60	Web1	A
www.example.com	A Record	198.51.100.2	60	Web2	B
www.example.com	A Record	203.0.113.2	60	Web3	C

Use when routing traffic to multiple resources

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Domain Registrar vs DNS Service

- You buy or register your domain name with a Domain Registrar typically by paying annual charges (e.g., GoDaddy, Amazon Registrar Inc., …)
- The Domain Registrar usually provides you with a DNS service to manage your DNS records
- But you can use another DNS service to manage your DNS records
- Example: purchase the domain from GoDaddy and use Route 53 to manage your DNS records
- Amazon Route 53
- purchase
- example.com
- User
- manage DNS records

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GoDaddy as Registrar & Route 53 as DNS Service

1. Amazon Route 53
2. Public Hosted Zone
3. stephanetheteacher.com

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3rd Party Registrar with Amazon Route 53

- If you buy your domain on a 3rd party registrar, you can still use route 53 as the DNS Service provider
- Create a Hosted Zone in Route 53
- Update NS Records on 3rd party website to use Route 53 Name Servers
- Domain Registrar != DNS Service
- But every Domain Registrar usually comes with some DNS features

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S3 Website with Route 53

- For acme.example.com:
- Create an S3 bucket with the same name as the target record (acme.example.com)
- Enable S3 website on the bucket (and enable S3 bucket public settings)
- Create a Route 53 Alias record to the S3 website endpoint or type A - IPv4 address
- This only works for HTTP traffic (for HTTPS, use CloudFront)
- Amazon Route 53
- Users
- S3 Bucket -
- Static Website Hosting Enabled
- (acme.example.com)
- http://acme.example.com

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Route 53 - Hybrid DNS

- By default, Route 53 Resolver automatically answers DNS queries for:
- Local domain names for EC2 instances
- Records in Private Hosted Zones
- Records in public Name Servers
- Hybrid DNS - resolving DNS queries between VPC (Route 53 Resolver) and your networks (other DNS Resolvers)
- Networks can be:
- VPC itself / Peered VPC
- On-premises Network (connected through Direct Connect or AWS VPN)
- Region
- EC2 Instance
- (ec2-192-0-2-44.compute-1.amazonaws.com)
- VPC

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Route 53 - Resolver Endpoints

- Inbound Endpoint
- DNS Resolvers on your network can forward DNS queries to Route 53 Resolver
- Allows your DNS Resolvers to resolve domain names for AWS resources (e.g., EC2 instances) and records in Route 53 Private Hosted Zones
- Outbound Endpoint
- Route 53 Resolver conditionally forwards DNS queries to your DNS Resolvers
- Use Resolver Rules to forward DNS queries to your DNS Resolvers
- Associated with one or more VPCs in the same AWS Region
- Create in two AZs for high availability
- Each Endpoint supports 10,000 queries per second per IP address

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TOPIC 19

Private Subnet 2

- Private Subnet 1
- Route 53 - Resolver Inbound Endpoints
- VPC
- us-east-1
- On-Premises Data Center
- DNS Resolvers
- (onpremise.private)
- Server
- (web.onpremise.private)
- Resolver
- Inbound Endpoint
- Route 53 Resolver
- (IP: x.x.x.2)

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Private Subnet 2

- Private Subnet 1
- Route 53 - Resolver Outbound Endpoints
- VPC
- us-east-1
- On-Premises Data Center
- DNS Resolvers
- (onpremise.private)
- Server
- (web.onpremise.private)
- Resolver
- Outbound Endpoint
- Route 53 Resolver
- (IP: x.x.x.2)

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Route 53 - Resolver Rules

- Control which DNS queries are forwarded to DNS Resolvers on your network
- Conditional Forwarding Rules (Forwarding Rules
- Forward DNS queries for a specified domain and all its subdomains to target IP addresses
- System Rules
- Selectively overriding the behavior defined in Forwarding Rules (e.g., don’t forward DNS queries for a subdomain acme.example.com)
- Auto-defined System Rules
- Defines how DNS queries for selected domains are resolved (e.g., AWS internal domain names, Privated Hosted Zones)
- If multiple rules matched, Route 53 Resolver chooses the most specific match
- Resolver Rules can be shared across accounts using AWS RAM
- Manage them centrally in one account
- Send DNS queries from multiple VPC to the target IP defined in the rule

TOPIC 20

Networking VPC

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TOPIC 20

Understanding CIDR - IPv4

- Classless Inter-Domain Routing - a method for allocating IP addresses
- Used in Security Groups rules and AWS networking in general
- They help to define an IP address range:
- We’ve seen WW.XX.YY.ZZ/32 => one IP
- We’ve seen 0.0.0.0/0 => all IPs
- But we can define:192.168.0.0/26 =>192.168.0.0 – 192.168.0.63 (64 IP addresses)

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Understanding CIDR - IPv4

- A CIDR consists of two components
- Base IP
- Represents an IP contained in the range (XX.XX.XX.XX)
- Example: 10.0.0.0, 192.168.0.0, …
- Subnet Mask
- Defines how many bits can change in the IP
- Example: /0, /24, /32
- Can take two forms:
- /8 ◆ 255.0.0.0
- /16 ◆ 255.255.0.0
- /24 ◆ 255.255.255.0
- /32 ◆ 255.255.255.255

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TOPIC 20

Understanding CIDR - Subnet Mask

- The Subnet Mask basically allows part of the underlying IP to get additional next values from the base IP
- 192
- 168
- 0
- 0
- 192
- 168
- 0
- 0
- 192
- 168
- 0

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TOPIC 20

Understanding CIDR - Little Exercise

- 192.168.0.0/24 = … ?
- 192.168.0.0 – 192.168.0.255 (256 IPs)
- 192.168.0.0/16 = … ?
- 192.168.0.0 – 192.168.255.255 (65,536 IPs)
- 134.56.78.123/32 = … ?
- Just 134.56.78.123
- 0.0.0.0/0
- All IPs!
- When in doubt, use this website https://www.ipaddressguide.com/cidr

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TOPIC 20

Public vs Private IP (IPv4)

- The Internet Assigned Numbers Authority (IANA) established certain blocks of IPv4 addresses for the use of private (LAN) and public (Internet) addresses
- Private IP can only allow certain values:
- 10.0.0.0 – 10.255.255.255 (10.0.0.0/8) +- in big networks
- 172.16.0.0 – 172.31.255.255 (172.16.0.0/12) +- AWS default VPC in that range
- 192.168.0.0 – 192.168.255.255 (192.168.0.0/16) +- e.g., home networks
- All the rest of the IP addresses on the Internet are Public

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TOPIC 20

Default VPC Walkthrough

- All new AWS accounts have a default VPC
- New EC2 instances are launched into the default VPC if no subnet is specified
- Default VPC has Internet connectivity and all EC2 instances inside it have public IPv4 addresses
- We also get a public and a private IPv4 DNS names

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TOPIC 20

VPC in AWS - IPv4

- VPC = Virtual Private Cloud
- You can have multiple VPCs in an AWS region (max. 5 per region – soft limit)
- Max. CIDR per VPC is 5, for each CIDR:
- Min. size is /28 (16 IP addresses)
- Max. size is /16 (65536 IP addresses)
- Because VPC is private, only the Private IPv4 ranges are allowed:
- 10.0.0.0 – 10.255.255.255 (10.0.0.0/8)
- 172.16.0.0 – 172.31.255.255 (172.16.0.0/12)
- 192.168.0.0 – 192.168.255.255 (192.168.0.0/16)
- Your VPC CIDR should NOT overlap with your other networks (e.g. corporate)

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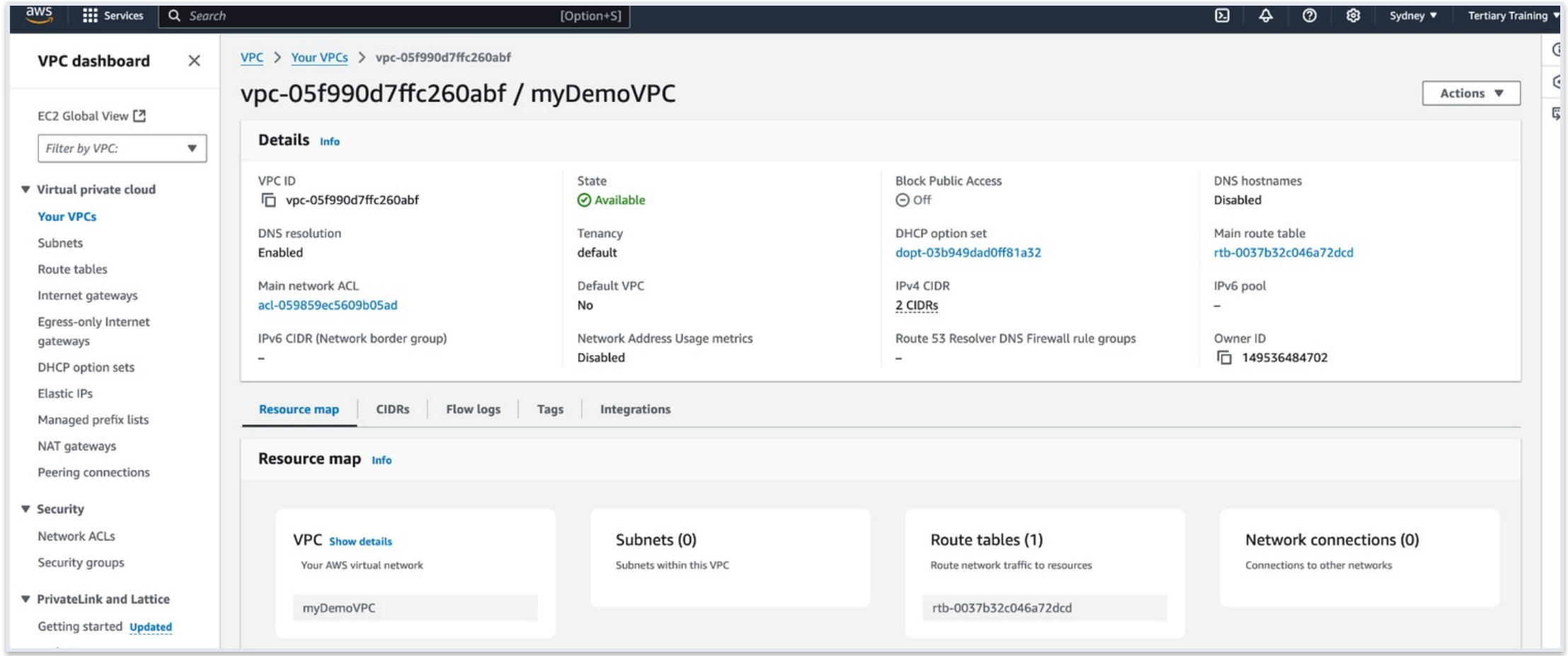
TOPIC 20

Activity: Create VPC



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TOPIC 20

Adding Subnets

1. Region
2. VPC
3. Availability Zone
4. Private Subnet
5. Public Subnet

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TOPIC 20

VPC - Subnet (IPv4)

- AWS reserves 5 IP addresses (first 4 & last 1) in each subnet
- These 5 IP addresses are not available for use and can’t be assigned to an EC2 instance
- Example: if CIDR block 10.0.0.0/24, then reserved IP addresses are:
- 10.0.0.0 - Network Address
- 10.0.0.1 - reserved by AWS for the VPC router
- 10.0.0.2 - reserved by AWS for mapping to Amazon-provided DNS
- 10.0.0.3 - reserved by AWS for future use
- 10.0.0.255 - Network Broadcast Address. AWS does not support broadcast in a VPC, therefore the address is reserved
- Exam tip if you need 29 IP addresses for EC2 instances:
- You can’t choose a subnet of size /27 (32 IP addresses, 32 – 5 = 27 < 29)
- You need to choose a subnet of size /26 (64 IP addresses, 64 – 5 = 59 > 29)

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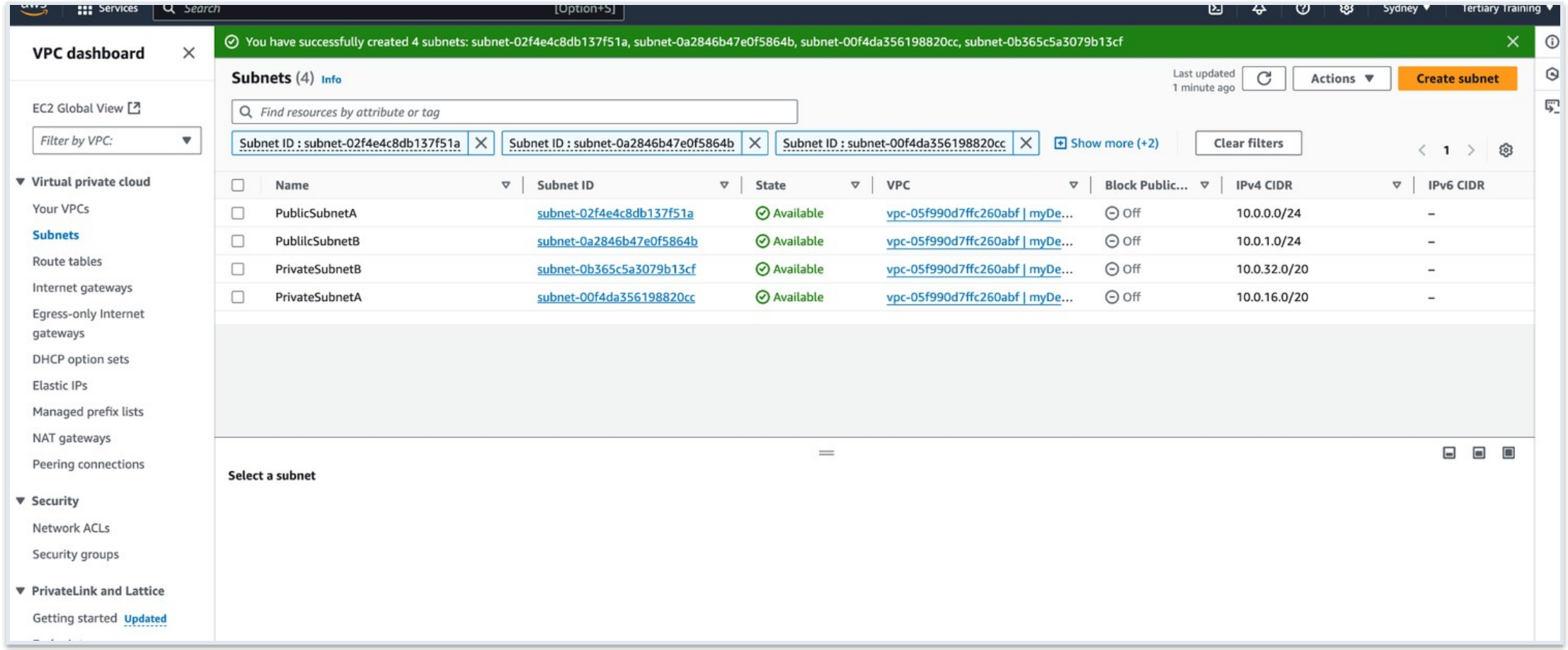
TOPIC 20

Activity: Create Subnets



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TOPIC 20

Internet Gateway (IGW)

- Allows resources (e.g., EC2 instances) in a VPC connect to the Internet
- It scales horizontally and is highly available and redundant
- Must be created separately from a VPC
- One VPC can only be attached to one IGW and vice versa
- Internet Gateways on their own do not allow Internet access…
- Route tables must also be edited!

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TOPIC 20

Adding Internet Gateway

1. Private Subnet
2. Public Subnet
3. Region
4. VPC
5. Internet Gateway
6. Availability Zone

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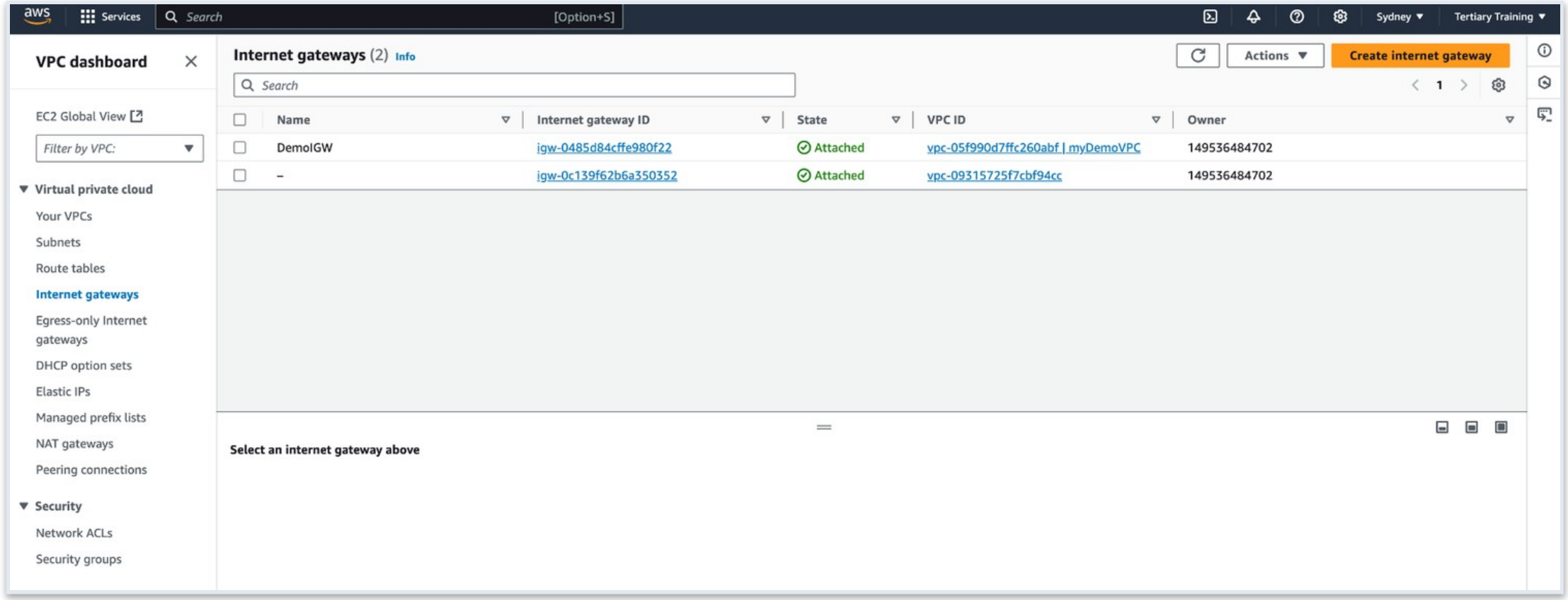
TOPIC 20

Activity: Create Internet Gateway



Create Internet Gateway

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TOPIC 20

Editing Route Tables

- Region
- VPC
- Availability Zone
- Private Subnet
- Public Subnet
- Internet Gateway
- Router
- Route
- Table
- Security Group
- Public EC2 Instance
- Internet
- www

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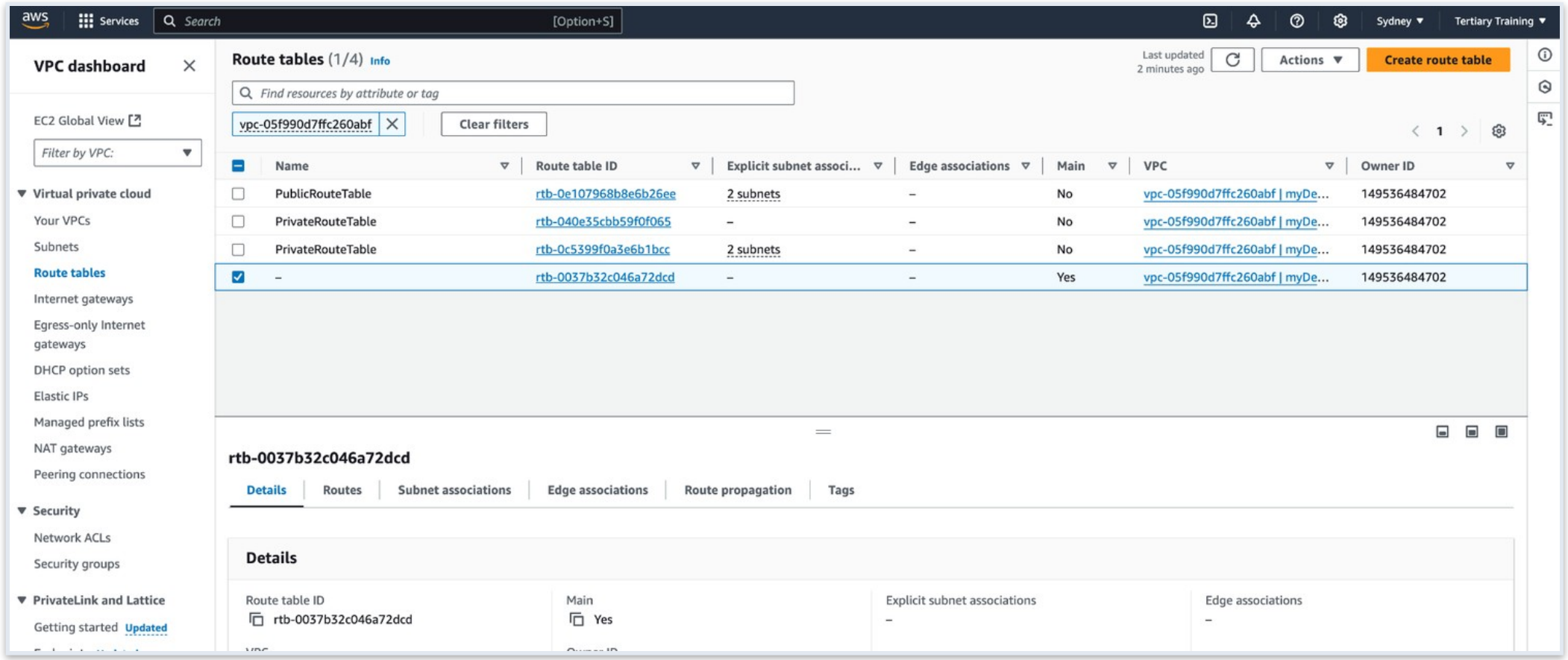
TOPIC 20

Activity: Create Route Tables



Create Route Tables

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TOPIC 20

Bastion Hosts

- We can use a Bastion Host to SSH into our private EC2 instances
- The bastion is in the public subnet which is then connected to all other private subnets
- Bastion host security group must allow inbound from the internet on port 22 from restricted CIDR, for example the public CIDR of your corporation
- Security group of the EC2 instance must allow the Security Group of the Bastion Host, or the private IP of the Bastion host
- VPC
- Private Subnet
- Public Subnet
- Security Group (BastionHost-SG)
- EC2 Instance
- (Bastion Host)
- Security Group (LinuxInstance-SG)

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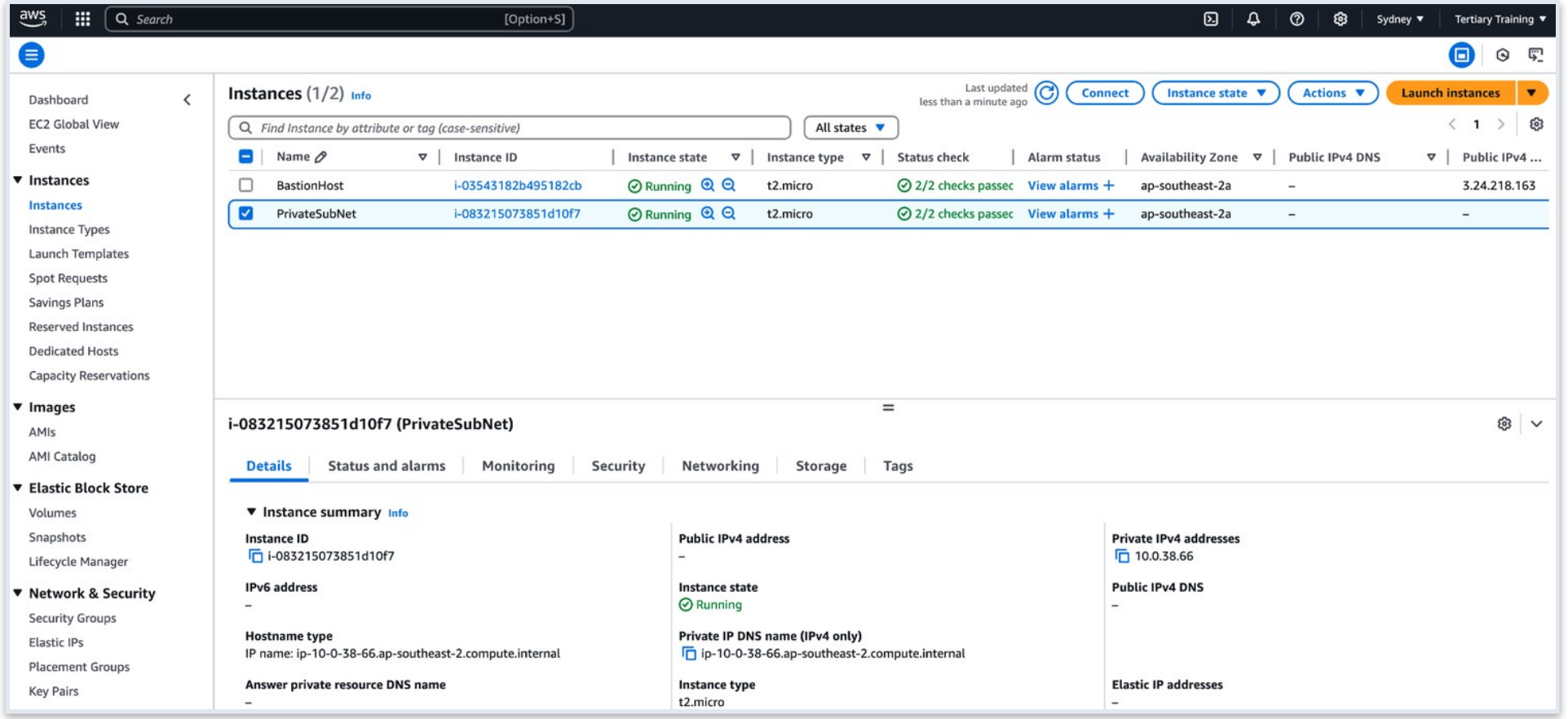
TOPIC 20

Activity: Bastion Host



Create a instance in the public subnet and a instance in the private subnet in the same VPC.

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TOPIC 20

NAT Instance (outdated but still at the exam)

- NAT = Network Address Translation
- Allows EC2 instances in private subnets to connect to the Internet
- Must be launched in a public subnet
- Must disable EC2 setting: Source / destination Check
- Must have Elastic IP attached to it
- Route Tables must be configured to route traffic from private subnets to the NAT Instance
- VPC
- Private Subnet
- Public Subnet
- NAT Instance
- Security Group (NATInstance-SG)
- (IP: 12.34.56.78)
- IP: 10.0.0.10

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TOPIC 20

NAT Instance

- Region
- VPC
- Private Subnet
- Public Subnet
- Internet Gateway
- Router
- Route
- Table
- Security Group
- Public EC2 Instance
- Availability Zone
- Internet
- www

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TOPIC 20

NAT Instance - Comments

- Pre-configured Amazon Linux AMI is available
- Reached the end of standard support on December 31, 2020
- Not highly available / resilient setup out of the box
- You need to create an ASG in multi-AZ + resilient user-data script
- Internet traffic bandwidth depends on EC2 instance type
- You must manage Security Groups & rules:
- Inbound:
- Allow HTTP / HTTPS traffic coming from Private Subnets
- Allow SSH from your home network (access is provided through Internet Gateway)
- Outbound:
- Allow HTTP / HTTPS traffic to the Internet

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TOPIC 20

NAT Gateway

- AWS-managed NAT, higher bandwidth, high availability, no administration
- Pay per hour for usage and bandwidth
- NATGW is created in a specific Availability Zone, uses an Elastic IP
- Can’t be used by EC2 instance in the same subnet (only from other subnets)
- Requires an IGW (Private Subnet => NATGW => IGW)
- 5 Gbps of bandwidth with automatic scaling up to 100 Gbps
- No Security Groups to manage / required

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TOPIC 20

NAT Gateway

- Region
- VPC
- Private Subnet
- Public Subnet
- Internet Gateway
- Router
- Route
- Table
- Security Group
- Public EC2 Instance
- Availability Zone
- Internet
- www

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TOPIC 20

NAT Gateway with High Availability

- NAT Gateway is resilient within
- a single Availability Zone
- Must create multiple NAT Gateways in multiple AZsfor fault-tolerance
- There is no cross-AZ failover needed because if an AZ goes down it doesn't need NAT
- Region
- VPC
- AZ - B
- AZ - A
- Public Subnet
- Private Subnet
- NAT Gateway
- EC2 Instance
- Router

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TOPIC 20

NAT Gateway vs NAT Instance

- More at: https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-comparison.html

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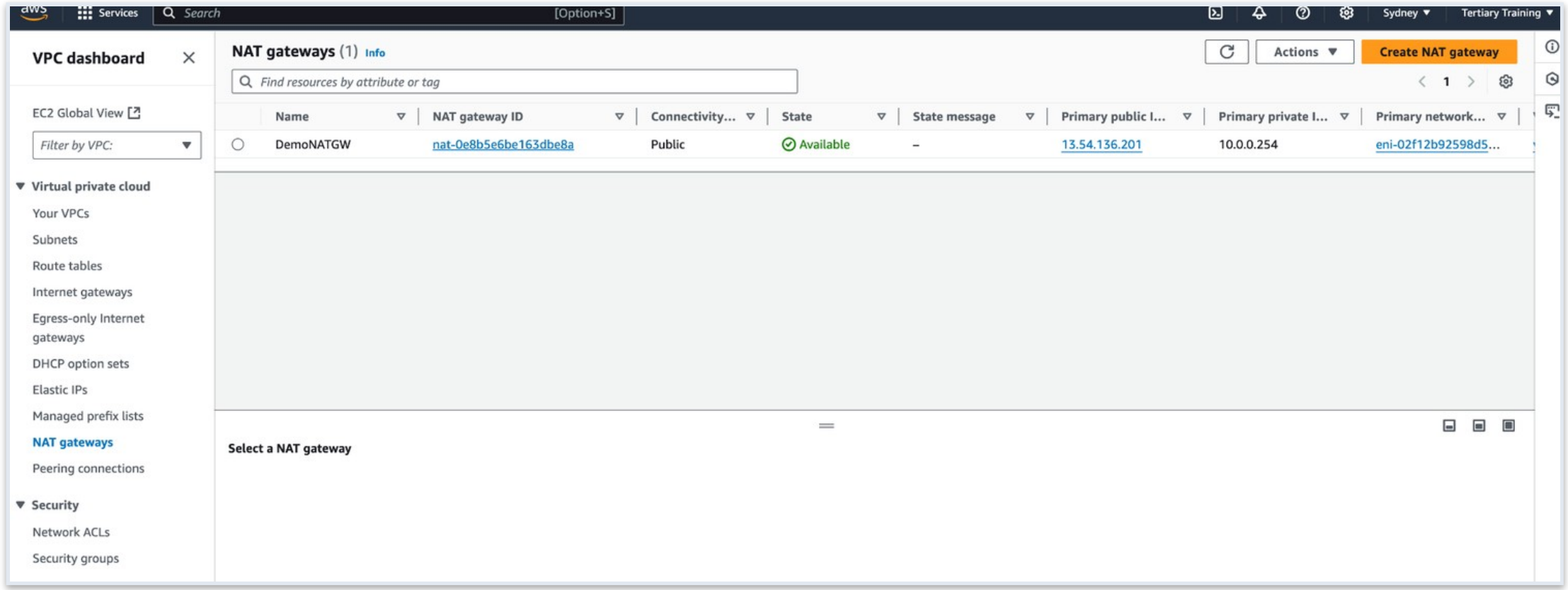
TOPIC 20

Activity: NAT Gateway



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TOPIC 20

DNS Resolution in VPC

- DNS Resolution (enableDnsSupport)
- Decides if DNS resolution from Route 53 Resolver server is supported for the VPC
- True (default): it queries the Amazon Provider DNS Server at 169.254.169.253 or the reserved IP address at the base of the VPC IPv4 network range plus two (.2)
- VPC
- DNS Hostnames
- DNS Support
- Public subnet
- EC2 Instance
- Custom DNS Server
- Internet
- www.google.com? A 142.251.37.36
- VPC

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TOPIC 20

DNS Resolution in VPC

- DNS Hostnames (enableDnsHostnames)
- By default,
- True => default VPC
- False => newly created VPCs
- Won’t do anything unless enableDnsSupport=true
- If True, assigns public hostname to EC2 instance if it has a public IPv4
- VPC
- DNS Hostnames
- DNS Support
- Public subnet
- EC2 Instance
- (Private DNS: ip-private-ipv4-address.ec2.internal)
- VPC

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TOPIC 20

Public subnet

- DNS Resolution in VPC
- If you use custom DNS domain names in a Private Hosted Zone in Route 53, you must set both these attributes (enableDnsSupport & enableDnsHostname) to true
- VPC
- DNS Hostnames
- DNS Support
- EC2 Instance
- IP: 169.254.169.253
- IP: xx.xx.xx.2
- Route 53 Resolver
- Web.mycompany.private ?
- 10.0.0.10
- Private subnet

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TOPIC 20

Subnet

- Security Groups & NACLs
- Security Group
- EC2 Instance
- NACL
- NACL Inbound Rules
- SG Inbound Rules
- Outbound Allowed
- (Stateful)
- NACL Outbound Rules (Stateless)
- Incoming Request
- Subnet
- Security Group
- EC2 Instance

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TOPIC 20

Network Access Control List (NACL)

- NACL are like a firewall which control traffic from and to Subnets to 1 NACL per subnet new subnets are assigned the Default NACL
- You define NACL rules:
- Rules have a number (1-32766), higher precedence with a lower number
- First rule match will drive the decision
- Example: if you define #100 ALLOW 10.0.0.10/32 and #200 DENY 10.0.0.10/32, the IP address will be allowed because 100 has a higher precedence over 200
- The last rule is an asterisk (*) and denies a request in case of no rule match
- AWS recommends adding rules by increment of 100
- Newly created NACLs will deny everything
- NACL are a great way of blocking a specific IP address at the subnet level

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TOPIC 20

NACLs

- Region
- VPC
- Private Subnet
- Public Subnet
- Internet Gateway
- Router
- Route
- Table
- Security Group
- Public EC2 Instance
- Availability Zone
- Internet
- www

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TOPIC 20

Default NACL

- Accepts everything inbound/outbound with the subnets it’s associated with
- Do NOT modify the Default NACL, instead create custom NACLs
- Default NACL for a VPC that supports IPv4
- Inbound Rules
- Outbound Rules

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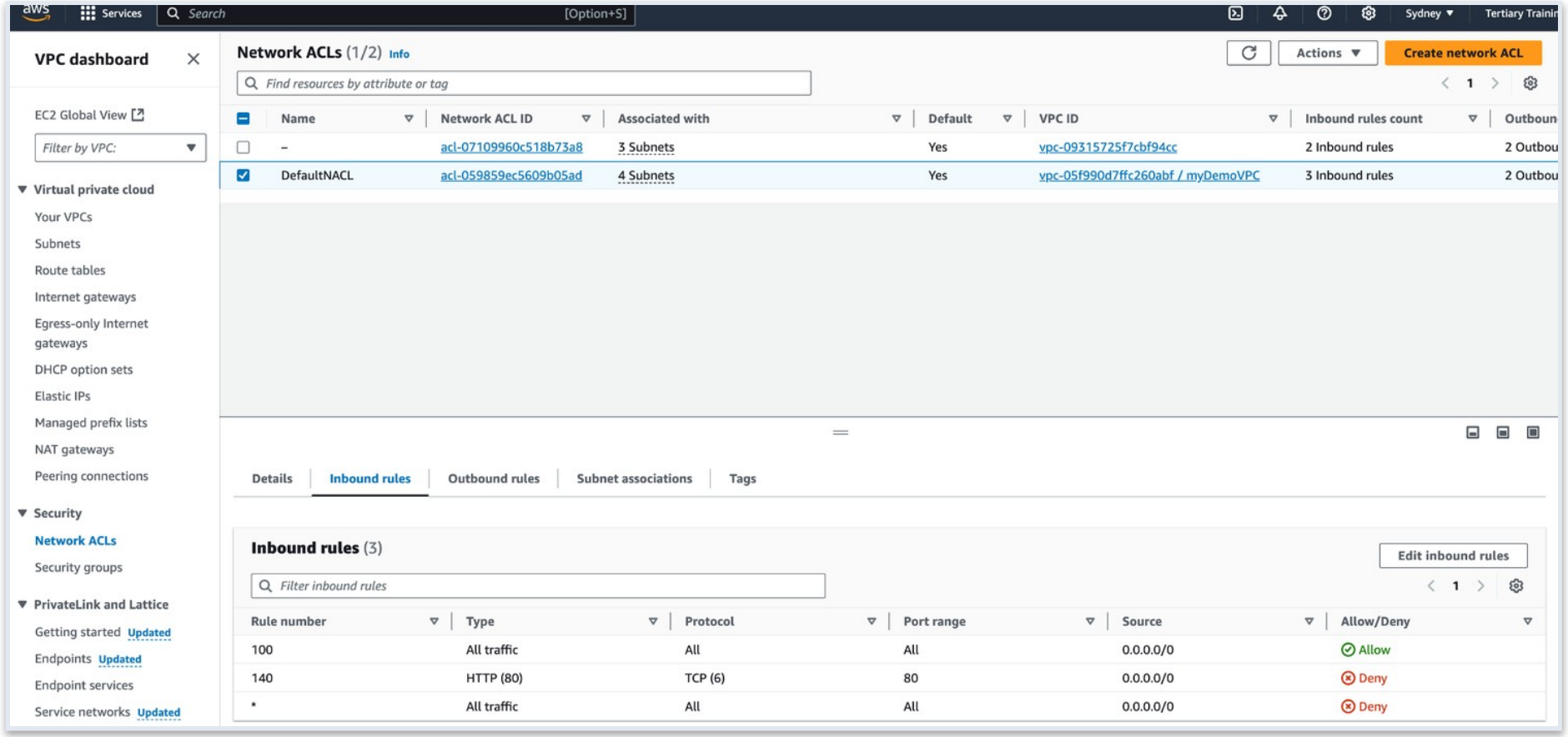
TOPIC 20

Activity: NACL and Security Group



Install HTTPD on Bastion Host

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TOPIC 20

Ephemeral Ports

- For any two endpoints to establish a connection, they must use ports
- Clients connect to a defined port, and expect a response on an ephemeral port
- Different Operating Systems use different port ranges, examples:
- IANA & MS Windows 10 ➔ 49152 – 65535
- Many Linux Kernels ➔ 32768 – 60999
- Web Server
- IP: 55.66.77.88
- Fixed Port: 443
- Dest. Port
- 443
- Src. IP
- 11.22.33.44
- Client

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TOPIC 20

NACL with Ephemeral Ports

- VPC
- Web Tier
- Web Subnet (Public)
- Database Tier
- DB Subnet (Private)
- DB Instance Port 3306
- Web-NACL
- DB-NACL
- Allow Outbound TCP
- On port 3306
- To DB Subnet CIDR
- Allow Inbound TCP
- On port 3306

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TOPIC 20

Create NACL rules for each target subnets CIDR

- Web Subnet - B (Public)
- DB Subnet – B (Private)
- VPC
- Web Tier
- Web Subnet - A (Public)
- Database Tier
- DB Subnet – A (Private)
- DB Instance
- DB Instance
- Web-NACL
- DB-NACL

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TOPIC 20

Security Group vs NACLs

- NACL Examples: https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html

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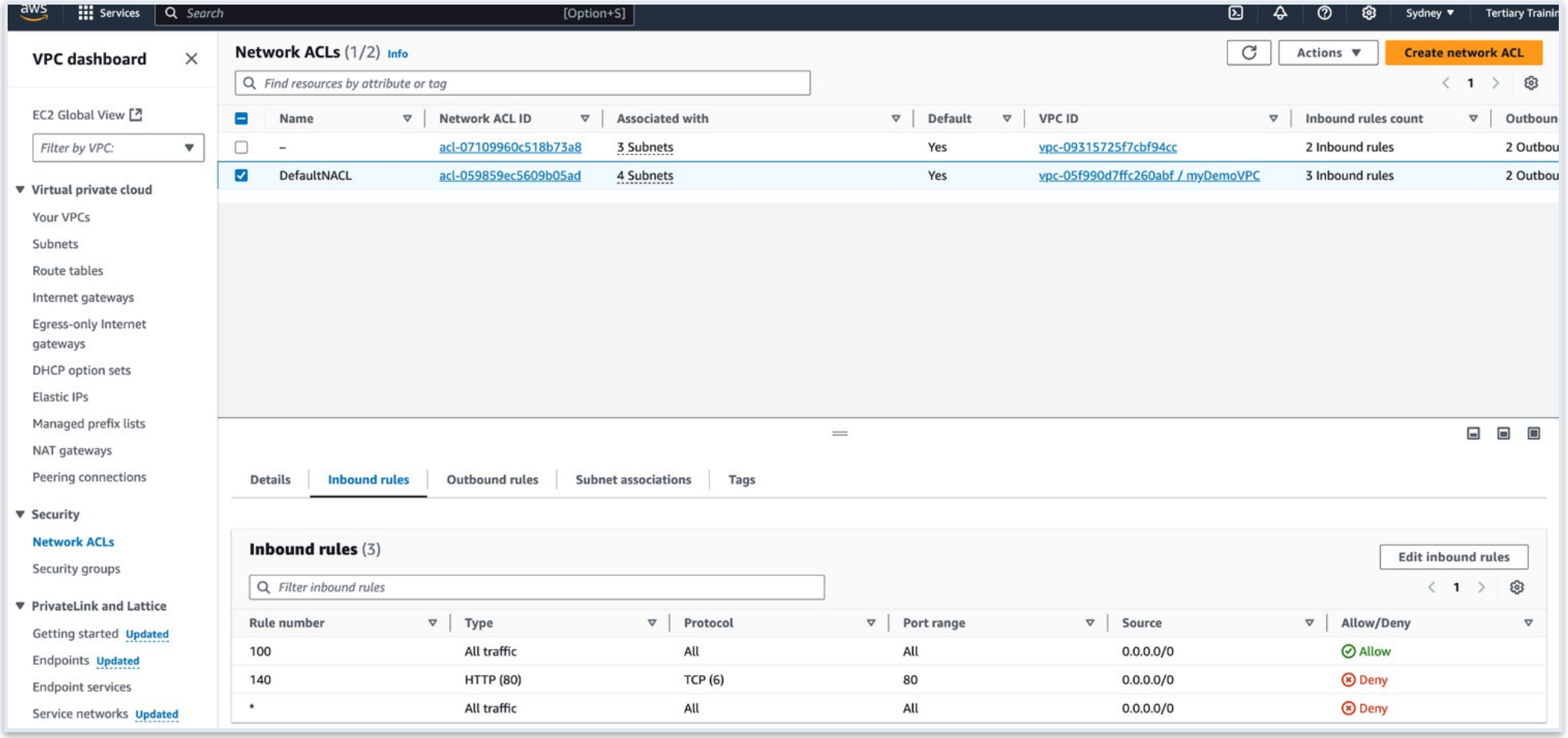
TOPIC 20

Activity: NACL and Security Group



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TOPIC 20

VPC - Reachability Analyzer

- A network diagnostics tool that troubleshoots
- network connectivity between two endpoints in your VPC(s)
- It builds a model of the network configuration,
- then checks the reachability based on these configurations (it doesn't send packets) When the destination is
- Reachable - it produces hop-by-hop details of the virtual network path
- Not Reachable - it identifies the blocking
- component(s) (e.g., configuration issues in SGs, NACLs, Route Tables, …)
- Use cases: troubleshoot connectivity issues, ensure network configuration is as intended, …
- Security Group - A
- EC2 Instance - A
- ENI - A
- Security Group - B

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TOPIC 20

VPC Peering

- Privately connect two VPCs using AWS' network
- Make them behave as if they were in the same network
- Must not have overlapping CIDRs
- VPC Peering connection is NOT transitive (must be established for each VPC that need to communicate with one another)
- You must update route tables in each VPC's subnets to ensure EC2 instances can communicate with each other
- VPC - A
- VPC - B
- VPC - C
- VPC Peering
- (A – B)
- VPC Peering

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TOPIC 20

VPC Peering - Good to know

- You can create VPC Peering connection between VPCs in different AWS accounts/regions
- You can reference a security group in a peered VPC (works cross accounts – same region)
- Account ID

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TOPIC 20

VPC Peering

- Region
- VPC
- Private Subnet
- Public Subnet
- Internet Gateway
- Router
- Route
- Table
- Security Group
- Public EC2 Instance
- Availability Zone
- Internet
- www

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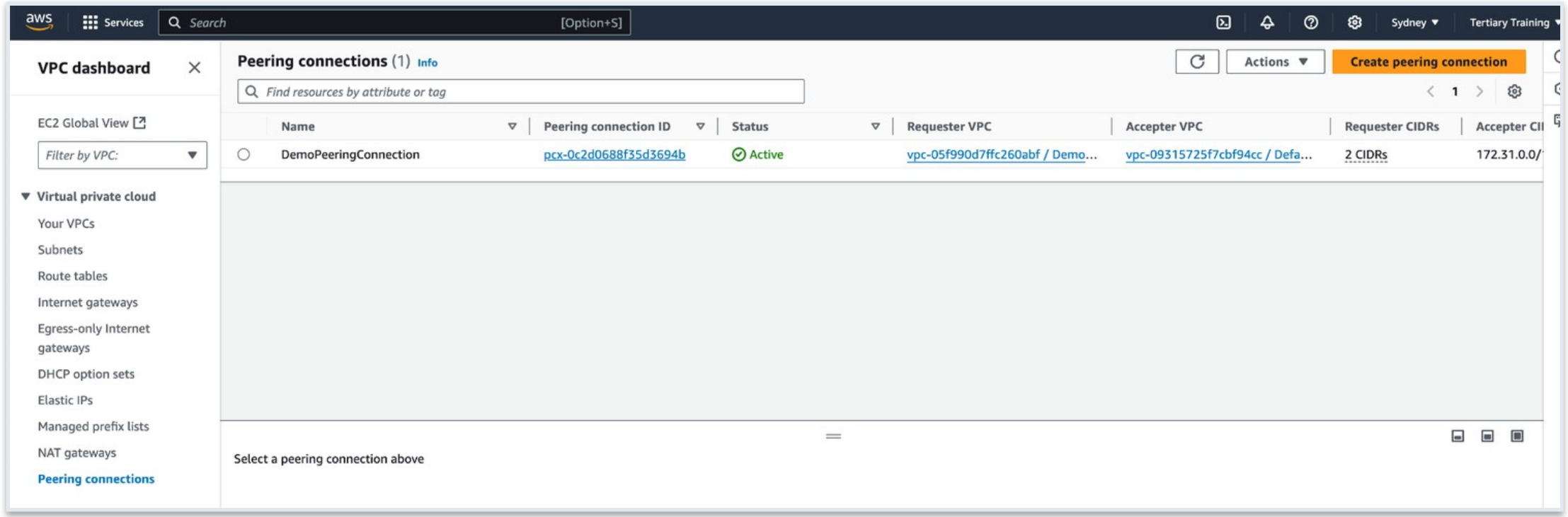
TOPIC 20

Activity: VPC Peering



Create a VPC Peering between two VPCs.

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TOPIC 20

VPC Endpoints

- Region
- VPC
- Private Subnet
- Public Subnet
- Internet Gateway
- Router
- Route
- Table
- Security Group
- Public EC2 Instance
- Availability Zone
- Internet
- www

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TOPIC 20

VPC Endpoints (AWS PrivateLink)

- Every AWS service is publicly exposed (public URL)
- VPC Endpoints (powered by AWS PrivateLink) allows you to connect to AWS services using a private network instead of using the public Internet
- They’re redundant and scale horizontally
- They remove the need of IGW, NATGW, … to access AWS Services
- In case of issues:
- Check DNS Setting Resolution in your VPC
- Check Route Tables
- VPC
- Private Subnet
- Public Subnet
- Internet Gateway
- Region

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TOPIC 20

Types of Endpoints

- Interface endpoints (powered by PrivateLink)
- Provisions an ENI (private IP address) as an entry point (must attach a Security Group)
- Supports most AWS services
- $ per hour + $ per GB of data processed
- Gateway Endpoints
- Provisions a gateway and must be used as a target in a route table (does not use security groups)
- Supports both S3 and DynamoDB
- Free
- Region
- VPC
- Private Subnet
- EC2 Instance
- VPC Endpoint

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TOPIC 20

Gateway or Interface Endpoint for S3?

- Cost: free for Gateway, $ for interface endpoint
- Interface Endpoint is preferred access is required from on- premises (Site to Site VPN or Direct Connect), a different VPC or a different region
- AWS Cloud
- Region
- VPC
- Direct Connect
- S2S VPN
- Amazon S3
- Interface Endpoint
- In-VPC
- Apps
- Gateway Endpoint

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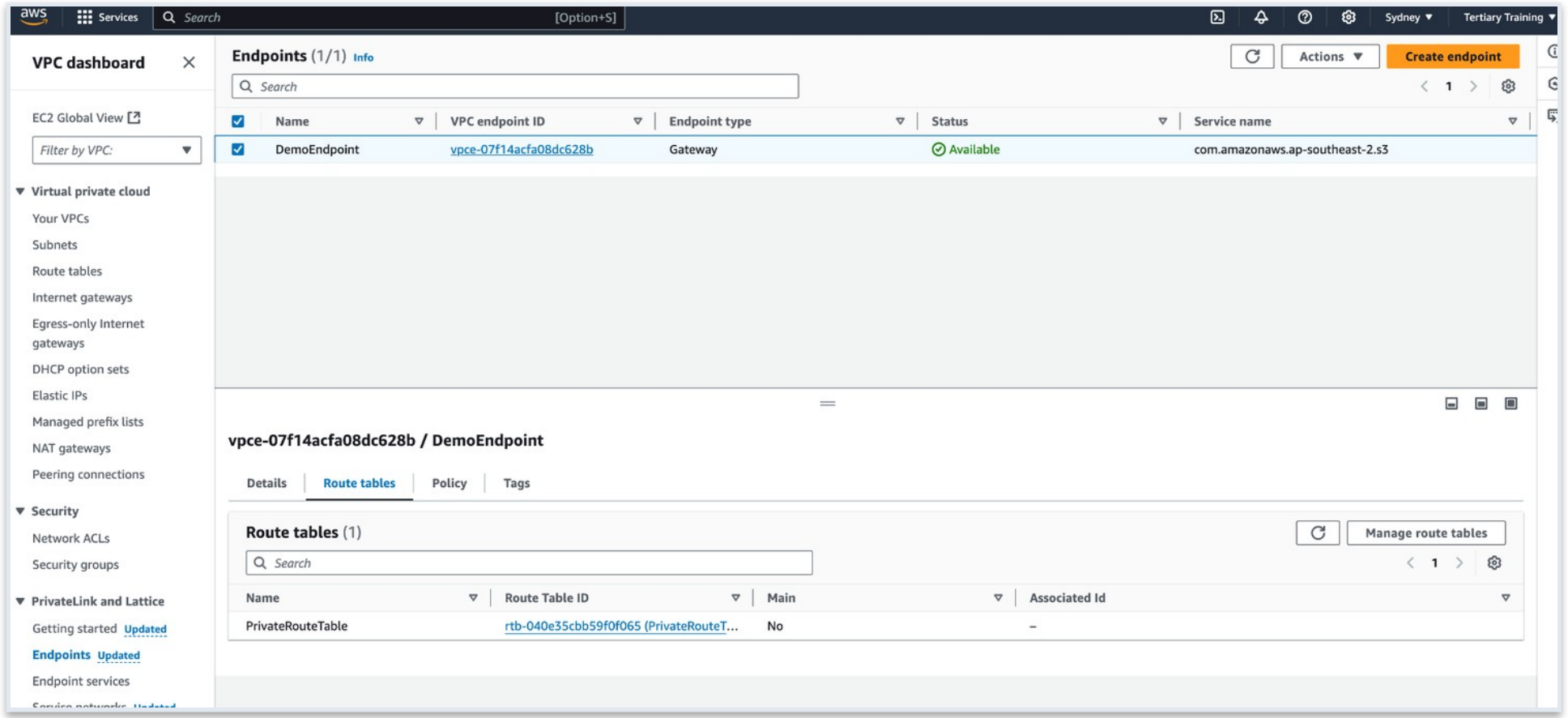
TOPIC 20

Activity: VPC Endpoint



Create a VPC Endpoint to access S3.

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TOPIC 20

Private subnet

- Lambda in VPC accessing DynamoDB
- AWS Cloud
- DynamoDB
- Public subnet
- NAT
- IGW
- VPC Gateway Endpoint For DynamoDB
- DynamoDB is a public service from AWS
- Option 1: Access from the public internet
- Because Lambda is in a VPC, it needs a NAT Gateway in a public subnet and an internet gateway
- Option 2 better and free access from the private VPC network
- Deploy a VPC Gateway endpoint for DynamoDB
- Change the Route Tables

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TOPIC 20

VPC Flow Logs

- Capture information about IP traffic going into your interfaces:
- VPC Flow Logs
- Subnet Flow Logs
- Elastic Network Interface (ENI) Flow Logs
- Helps to monitor & troubleshoot connectivity issues
- Flow logs data can go to S3, CloudWatch Logs, and Kinesis Data Firehose
- Captures network information from AWS managed interfaces too: ELB, RDS, ElastiCache, Redshift, WorkSpaces, NATGW,Transit Gateway…

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TOPIC 20

VPC Flow Logs

- Region
- VPC
- Private Subnet
- Public Subnet
- Internet Gateway
- Router
- Route
- Table
- Security Group
- Public EC2 Instance
- Availability Zone
- Internet
- www

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TOPIC 20

VPC Flow Logs Syntax

- srcaddr & dstaddr - help identify problematic IP
- srcport & dstport - help identity problematic ports
- action - success or failure of the request due to Security Group / NACL
- Can be used for analytics on usage patterns, or malicious behavior
- Query VPC flow logs using athena on S3 or CloudWatch Logs Insights
- Flow Logs examples: https://docs.aws.amazon.com/vpc/latest/userguide/flow-logs- records-examples.html
- version
- account-id
- interface-id
- srcaddr
- dstaddr
- srcport protocol bytes

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TOPIC 20

VPC Flow Logs - Troubleshoot SG & NACL issues

- Security Group
- EC2 Instance
- NACL
- NACL Inbound Rules
- Subnet
- SG Inbound Rules
- Outbound Allowed
- (Stateful)
- NACL Outbound Rules (Stateless)
- Security Group
- EC2 Instance
- NACL
- NACL Outbound Rules

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TOPIC 20

VPC Flow Logs - Architectures

- VPC Flow Logs
- CloudWatch Logs
- CloudWatch Contributor Insights
- Top-10 IP addresses
- CW Alarm
- Alert
- Amazon SNS
- VPC Flow Logs
- CloudWatch Logs
- Metric Filter
- SSH, RDP…
- VPC Flow Logs
- S3 Bucket

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TOPIC 20

AWS Site-to-Site VPN

- Region
- Availability Zone
- VPC
- VPC Peering Connections
- Public Subnet
- Internet Gateway
- Router
- Route
- Table
- NAT Gateway
- Security Group
- Public EC2 Instance
- NACL

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TOPIC 20

AWS Site-to-Site VPN

- Region
- VPC
- Private Subnet
- Public Subnet
- Internet Gateway
- Router
- Route
- Table
- Security Group
- Public EC2 Instance
- Availability Zone
- Internet
- www

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TOPIC 20

AWS Site-to-Site VPN

- Virtual Private Gateway (VGW)
- VPN concentrator on the AWS side of the VPN connection
- VGW is created and attached to the VPC from which you want to create the Site-to-Site VPN connection
- Possibility to customize the ASN (Autonomous System Number)
- Customer Gateway (CGW)
- Software application or physical device on customer side of the VPN connection
- https://docs.aws.amazon.com/vpn/latest/s2svpn/your-cgw.html#DevicesTested

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TOPIC 20

Private Subnet

- Site-to-Site VPN Connections
- Customer Gateway Device (On-premises)
- What IP address to use?
- Public Internet-routable IP address for your Customer Gateway device
- If it's behind a NAT device that's enabled for NAT traversal (NAT-T), use the public IP address of the NAT device
- Important step: enable Route Propagation for the Virtual Private Gateway in the route table that is associated with your subnets
- If you need to ping your EC2 instances from on-premises, make sure you add the ICMP protocol on the inbound of your security groups
- VPC
- Security Group
- Corporate Data Center
- Server

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TOPIC 20

AWS VPN CloudHub

- Provide secure communication between
- multiple sites, if you have multiple VPN connections
- Low-cost hub-and-spoke model for
- primary or secondary network connectivity between different locations (VPN only)
- It’s a VPN connection so it goes over the public Internet
- To set it up, connect multiple VPN connections on the same VGW, setup dynamic routing and configure route tables
- VPC
- Availability Zone
- Availability Zone
- Private Subnet 1
- EC2 Instances
- Private Subnet 2

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TOPIC 20

Direct Connect (DX)

- Provides a dedicated private connection from a remote network to your VPC
- Dedicated connection must be setup between your DC and AWS Direct Connect locations
- You need to setup a Virtual Private Gateway on your VPC
- Access public resources (S3) and private (EC2) on same connection
- Use Cases:
- Increase bandwidth throughput - working with large data sets - lower cost
- More consistent network experience - applications using real-time data feeds
- Hybrid Environments (on prem + cloud)
- Supports both IPv4 and IPv6

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TOPIC 20

Direct Connect Diagram

- Region (us-east-1)
- VPC
- Private Subnet
- EC2 Instances
- Virtual
- Private Gateway
- Amazon S3
- Amazon Glacier
- AWS Direct Connect Location
- AWS Cage
- Customer or partner cage
- AWS Direct Connect Endpoint
- Customer or partner router

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TOPIC 20

Direct Connect Gateway

- If you want to setup a Direct Connect to one or more VPC in many different regions (same account), you must use a Direct Connect Gateway
- Region (us-east-1)
- VPC
- Region (us-west-1)
- VPC
- 10.0.0.0/16
- 172.16.0.0/16
- Direct Connect Gateway
- Customer network
- AWS Direct Connect
- Private virtual interface
- Private virtual interface

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TOPIC 20

Direct Connect - Connection Types

- Dedicated Connections: 1Gbps,10 Gbps and 100 Gbps capacity
- Physical ethernet port dedicated to a customer
- Request made to AWS first, then completed by AWS Direct Connect Partners
- Hosted Connections: 50Mbps, 500 Mbps, to 10 Gbps
- Connection requests are made via AWS Direct Connect Partners
- Capacity can be added or removed on demand
- 1, 2, 5, 10 Gbps available at select AWS Direct Connect Partners
- Lead times are often longer than 1 month to establish a new connection

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TOPIC 20

Direct Connect - Encryption

- Data in transit is not encrypted but is private
- AWS Direct Connect + VPN provides an IPsec-encrypted private connection
- Good for an extra level of security, but slightly more complex to put in place
- Region (us-east-1)
- VPC
- Availability Zone (us-east-1a)
- Availability Zone (us-east-1b)
- Private Subnet 1
- EC2 Instances
- Private Subnet 2
- EC2 Instances
- AWS Direct Connect Location
- Customer Network

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TOPIC 20

Direct Connect - Resiliency

- High Resiliency for Critical Workloads
- One connection at multiple locations
- Maximum Resiliency for Critical Workloads
- Maximum resilience is achieved by separate connections terminating on separate devices in more than one location.
- Region
- AWS Direct Connect Location - 1
- Corporate data center
- AWS Direct Connect Location - 2
- Corporate data center
- Region
- AWS Direct Connect Location - 1
- Corporate data center

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TOPIC 20

Site-to-Site VPN connection as a backup

- In case Direct Connect fails, you can set up a backup Direct Connect connection (expensive), or a Site-to-Site VPN connection
- Direct Connect
- Primary Connection
- Site-to-Site VPN
- Backup Connection
- AWS Cloud
- VPC
- Corporate DC

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TOPIC 20

Exposing services in your VPC to other VPC

- Option 1: make it public
- Goes through the public www
- Tough to manage access
- Option 2:VPC peering
- Must create many peering relations
- Opens the whole network
- Peering connection
- Service VPC
- Customer VPC
- Application service
- Service VPC
- Customer VPC With IGW
- Application service

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TOPIC 20

AWS PrivateLink (VPC Endpoint Services)

- Most secure & scalable way to expose a service to 1000s of VPC (own or other accounts)
- Does not require VPC peering, internet gateway, NAT, route tables…
- Requires a network load balancer (Service VPC) and ENI (Customer VPC) or GWLB
- If the NLB is in multiple AZ, and the ENIs in multiple AZ, the solution is fault tolerant!
- Service VPC
- Customer VPC
- Application service
- Network Load Balancer
- Elastic Network Interface (ENI)
- AWS Private Link
- Consumer Application
- AWS
- private

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TOPIC 20

Private subnet

- AWS Private Link & ECS
- VPC
- Network Load Balancer
- Application Load Balancer
- AWS PrivateLink
- Direct Connect
- Customer Network
- Corporate data center
- Virtual Private Gateway
- Customer Gateway
- VPN
- Connection
- Amazon ECS service

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TOPIC 20

EC2-Classic & AWS ClassicLink (deprecated)

- EC2 Classic: instances run in a single network shared with other customers
- Amazon VPC: your instances run logically isolated to your AWS account
- ClassicLink: allows you to link EC2-Classic instances to a VPC in your account
- Must associate a security group
- Enables communication using private IPv4 addresses
- Removes the need to make use of public IPv4 addresses or Elastic IP addresses
- Likely to be distractors at the exam

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TOPIC 20

Network topologies can become complicated

- Customer Gateway
- Amazon VPC
- VPN Connection
- Amazon VPC
- VPC Peering Connection
- Amazon VPC
- VPC Peering Connection
- Amazon VPC
- VPC Peering Connection
- VPC Peering Connection
- VPC Peering Connection
- VPN Connection
- VPN Connection

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TOPIC 20

Transit Gateway

- For having transitive peering between thousands of VPC and on-premises, hub-and-spoke (star) connection
- Regional resource, can work cross-region
- Share cross-account using Resource Access Manager (RAM)
- You can peer Transit Gateways across regions
- Route Tables: limit which VPC can talk with other VPC
- Works with Direct Connect Gateway,VPN connections
- Supports IP Multicast (not supported by any other AWS service)
- AWS Direct Connect Gateway
- Amazon VPC
- Amazon VPC
- Amazon VPC
- Amazon VPC

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TOPIC 20

Transit Gateway: Site-to-Site VPN ECMP

- ECMP = Equal-cost multi-path routing
- Routing strategy to allow to forward a packet over multiple best path
- Use case: create multiple Site- to-Site VPN connections to
- increase the bandwidth of
- your connection to AWS
- VPC
- VPC
- VPC
- VPC
- AWS Transit Gateway
- VPC attachment
- VPC attachment
- VPC attachment

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TOPIC 20

Transit Gateway: throughput with ECMP

- VPN to virtual private gateway
- 1x
- =
- 1x
- VPC
- 1x
- =
- 1.25 Gbps
- VPN connection (2 tunnels)
- VPN to transit gateway
- 1x
- = 1x
- VPC

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TOPIC 20

Transit Gateway - Share Direct Connect between multiple accounts

- AWS Direct Connect Location
- Corporate data center
- Clients
- Clients
- Servers
- AWS Direct Connect endpoint
- Customer router/firewall
- Transit VIF
- VLAN
- Account 1
- Direct Connect Gateway
- AWS Cloud
- Region

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TOPIC 20

VPC - Traffic Mirroring

- Allows you to capture and inspect network traffic in your VPC
- Route the traffic to security appliances that you manage
- Capture the traffic
- From (Source) - ENIs
- To (Targets) - an ENI or a Network Load Balancer
- Capture all packets or capture the packets of your interest (optionally, truncate packets)
- Source and Target can be in the same VPC or different VPCs (VPC Peering)
- Use cases: content inspection, threat monitoring, troubleshooting, …
- Source B
- Source A
- Auto Scaling group
- Network Load Balancer
- Inbound & Outbound traffic

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TOPIC 20

What is IPv6?

- IPv4 designed to provide 4.3 Billion addresses (they'll be exhausted soon)
- IPv6 is the successor of IPv4
- IPv6 is designed to provide 3.4 × 10!" unique IP addresses
- Every IPv6 address in AWS is public and Internet-routable (no private range)
- Format -+ x.x.x.x.x.x.x.x (x is hexadecimal, range can be from 0000 to ffff)
- Examples:
- 2001:db8:3333:4444:5555:6666:7777:8888
- 2001:db8:3333:4444:cccc:dddd:eeee:ffff
- :: ➔ all 8 segments are zero
- 2001:db8:: ➔ the last 6 segments are zero
- ::1234:5678 ➔ the first 6 segments are zero
- 2001:db8::1234:5678 ➔ the middle 4 segments are zero

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TOPIC 20

IPv6 in VPC

- IPv4 cannot be disabled for your VPC and subnets
- You can enable IPv6 (they’re public IP addresses) to operate in dual-stack mode
- Your EC2 instances will get at least a private internal IPv4 and a public IPv6
- They can communicate using either IPv4 or IPv6 to the internet through an Internet Gateway
- EC2 Instance
- (Private IP: 10.0.0.5) (IPv6: 2001:db8::ff00:42:8329)
- VPC
- Internet
- Gateway IPv4 & IPv6
- Internet

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TOPIC 20

IPv6 Troubleshooting

- IPv4 cannot be disabled for your VPC and subnets
- So, if you cannot launch an EC2 instance in your subnet
- It’s not because it cannot acquire an IPv6 (the space is very large)
- It’s because there are no available IPv4 in your subnet
- Solution: create a new IPv4 CIDR in your subnet
- User
- create
- VPC
- (IPv4: 192.168.0.0/24) (IPv4: 10.0.0.0/24)
- (IPv6: 2001:db8:1234:5678::/56)
- …
- 192.168.0.10 192.168.0.15
- 10.0.0.35

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TOPIC 20

Egress-only Internet Gateway

- Used for IPv6 only
- (similar to a NAT Gateway but for IPv6)
- Allows instances in your VPC outbound connections over IPv6 while preventing the internet to initiate an IPv6 connection to your instances
- You must update the Route Tables
- VPC
- Internet Gateway
- Public Subnet
- IPv6: 2001:db8::b1c2
- Egress-only Internet Gateway
- initiate connections from both sides
- Internet
- can't initiate connections from Internet

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TOPIC 20

Private Subnet (IPv4: 10.0.1.0/24)

- (IPv6: 2001:db8:1234:1a02::/64)
- Public Subnet (IPv4: 10.0.0.0/24)
- (IPv6: 2001:db8:1234:1a00::/64)
- IPv6 Routing
- Region
- VPC
- (IPv4: 10.0.0.0/16)
- (IPv6: 2001:db8:1234:1a00::/56)
- Web server
- Private IPv4: 10.0.0.5
- EIP: 198.51.100.1
- IPv6: 2001:db8:1234:1a00::123
- Server

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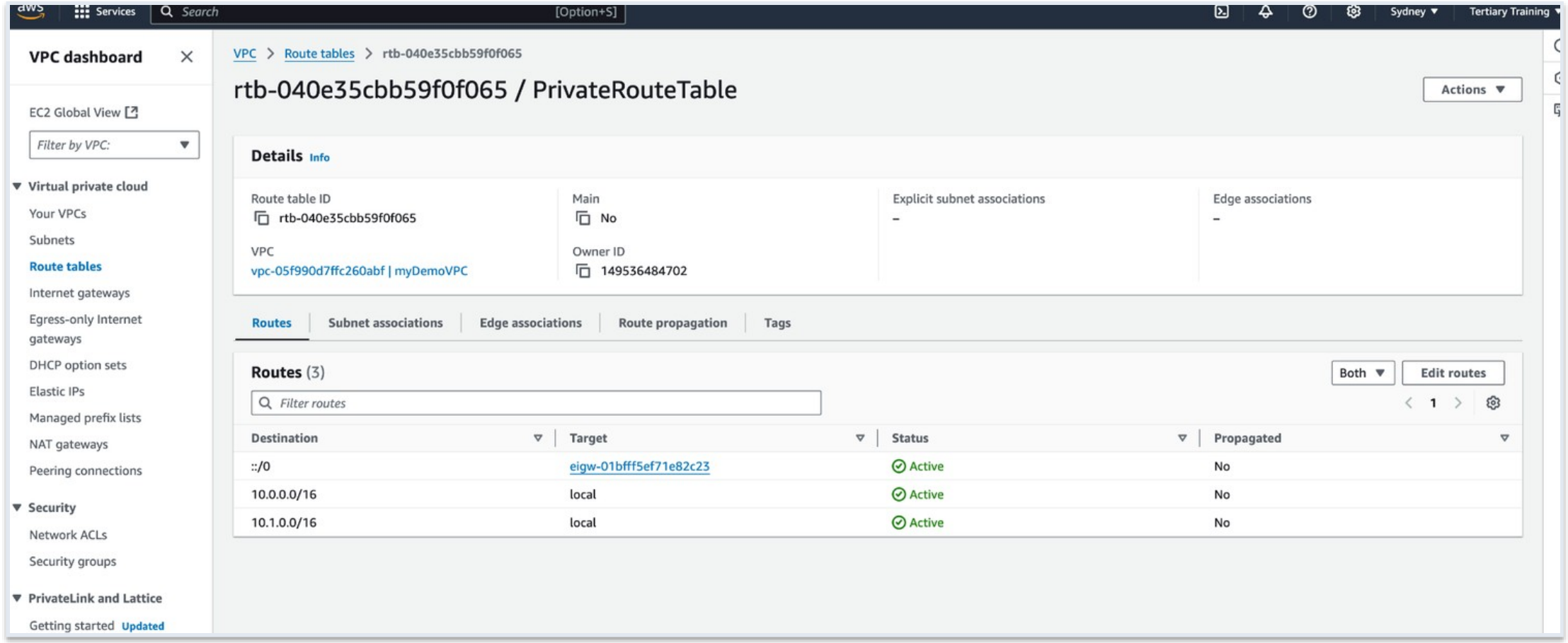
TOPIC 20

Activity: Egress-only Internet Gateway



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TOPIC 20

VPC Section Summary (1/3)

- CIDR - IP Range
- VPC - Virtual Private Cloud => we define a list of IPv4 & IPv6 CIDR
- Subnets - tied to an AZ, we define a CIDR
- Internet Gateway - at the VPC level, provide IPv4 & IPv6 Internet Access
- Route Tables - must be edited to add routes from subnets to the IGW,VPC Peering Connections,VPC Endpoints, …
- Bastion Host - public EC2 instance to SSH into, that has SSH connectivity to EC2 instances in private subnets
- NAT instances - gives Internet access to EC2 instances in private subnets. Old, must be setup in a public subnet, disable Source / Destination check flag
- NAT Gateway - managed by AWS, provides scalable Internet access to private EC2 instances, when the target is an IPv4 address
- Private DNS + Route 53 enable DNS Resolution + DNS Hostnames (VPC)

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TOPIC 20

VPC Section Summary (2/3)

- NACL - stateless, subnet rules for inbound and outbound, don’t forget Ephemeral Ports
- Security Groups - stateful, operate at the EC2 instance level
- Reachability Analyzer - perform network connectivity testing between AWS resources
- VPC Peering - connect two VPCs with non overlapping CIDR, non-transitive
- VPC Endpoints - provide private access to AWS Services (S3, DynamoDB, CloudFormation, SSM) within a VPC
- VPC Flow points - can be setup at the VPC / Subnet / ENI Level, for ACCEPT and
- REJECT traffic, helps identifying attacks, analyze using Athena or CloudWatch Logs Insights
- Site-to-Site VPN - setup a Customer Gateway on DC, a Virtual Private Gateway on VPC, and site-to-site VPN over public Internet
- AWS VPN Cloudhub - hub-and-spoke VPN model to connect your sites

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TOPIC 20

VPC Section Summary (3/3)

- Direct connect - setup a Virtual Private Gateway on VPC, and establish a direct private connection to an AWS Direct Connect Location
- Direct connect gateway - setup a Direct Connect to many VPCs in different AWS regions
- AWS PrivateLink / VPC Endpoint Services:
- Connect services privately from your service VPC to customers VPC
- Doesn’t need VPC Peering, public Internet, NAT Gateway, Route Tables
- Must be used with Network Load Balancer & ENI
- CLassic Link - connect EC2-Classic EC2 instances privately to your VPC
- Transit gateway - transitive peering connections for VPC,VPN & DX
- Traffic Mirroring - copy network traffic from ENIs for further analysis
- Egress-only Internet Gateway - like a NAT Gateway, but for IPv6

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TOPIC 20

Networking Costs in AWS per GB - Simplified

- Region
- Availability Zone
- Availability Zone
- Region
- Availability Zone
- $0.02
- Inter-region
- $0.02 if using Public IP / Elastic IP
- Free if using private IP
- $0.01 if
- Using private IP
- Use Private IP instead of Public IP for good savings and better network performance
- Use same AZ for maximum savings (at the cost of high availability)

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TOPIC 20

Minimizing egress traffic network cost

- Egress traffic: outbound traffic (from AWS to outside)
- Ingress traffic: inbound traffic - from outside to AWS (typically free)
- Try to keep as much internet traffic within AWS to minimize costs
- Direct Connect Location that are co-located in the same AWS Region result in lower cost for egress
- network
- Corporate data center
- AWS Cloud
- Database
- Application
- Egress cost is minimized
- Corporate data center
- AWS Cloud
- DB Query

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TOPIC 20

S3 Data Transfer Pricing - Analysis for USA

- S3 ingress: free
- S3 to internet: $0.09 per GB
- S3 Transfer Acceleration:
- Faster transfer times (50 to 500% better)
- Additional cost on top of Data Transfer Pricing: +$0.04 to $0.08 per GB
- S3 to Cloudfront: $0.00 per GB
- Cloudfront to Internet: $0.085 per GB (slightly cheaper than S3)
- Caching capability (lower latency)
- Reduce costs associated with S3 Requests Pricing (7x cheaper with CloudFront)
- S3 Cross Region Replication: $0.02 per GB
- Replication $0.02
- Edge location
- Transfer acceleration +$0.04

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TOPIC 20

Private subnet 2 (10.0.1.0/24)

- Public subnet
- Pricing: NAT Gateway vs Gateway VPC Endpoint
- Region
- (us-east-1)
- VPC (10.0.0.0/16)
- Private subnet 1 (10.0.0.0/24)
- EC2 Instance
- NAT Gateway
- Internet Gateway
- VPC Endpoint
- Internet
- S3 Bucket
- EC2 Instance

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TOPIC 20

Network Protection on AWS

- To protect network on AWS, we've seen
- Network Access Control Lists (NACLs)
- Amazon VPC security groups
- AWS WAF (protect against malicious requests)
- AWS Shield & AWS Shield Advanced
- AWS Firewall Manager (to manage them across accounts)
- But what if we want to protect in a sophisticated way our entire VPC?

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TOPIC 20

AWS Network Firewall

- Protect your entire Amazon VPC
- From Layer 3 to Layer 7 protection
- Any direction, you can inspect
- VPC to VPC traffic
- Outbound to internet
- Inbound from internet
- To / from Direct Connect & Site-to-Site VPN
- Internally, the AWS Network Firewall uses the AWS Gateway Load Balancer
- Rules can be centrally managed cross-
- account by AWS Firewall Manager to apply to many VPCs
- VPC
- Private subnet
- Peered VPC

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TOPIC 20

Network Firewall - Fine Grained Controls

- Supports 1000s of rules
- IP & port - example: 10,000s of IPs filtering
- Protocol – example: block the SMB protocol for outbound communications
- Stateful domain list rule groups: only allow outbound traffic to *.mycorp.com or third-party software repo
- General pattern matching using regex
- Traffic filtering: Allow, drop or alert for the traffic that matches the rules
- Active flow inspection to protect against network threats with intrusion- prevention capabilities (like Gateway Load Balancer, but all managed by AWS)
- Send logs of rule matches to Amazon S3, CloudWatch Logs, Kinesis Data Firehose

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OTHER SERVICES

Other Services

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AWS X-Ray

- Debugging in Production, the good old way:
- Test locally
- Add log statements everywhere
- Re-deploy in production
- Log formats differ across applications and log analysis is hard.
- Debugging: one big monolith “easy”, distributed services “hard”
- No common views of your entire architecture
- Enter… AWS X-Ray!

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AWS X-Ray

- Visual analysis of our applications

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AWS X-Ray advantages

- Troubleshooting performance (bottlenecks)
- Understand dependencies in a microservice architecture
- Pinpoint service issues
- Review request behavior
- Find errors and exceptions
- Are we meeting time SLA?
- Where I am throttled?
- Identify users that are impacted

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AWS Amplify - web and mobile applications

- A set of tools and services that helps you develop and deploy scalable full stack web and mobile applications
- Authentication, Storage, API (REST, GraphQL), CI/CD, PubSub, Analytics, AI/ML Predictions, Monitoring, …
- Connect your source code from GitHub, AWS CodeCommit, Bitbucket, GitLab, or upload directly
- Frontend
- configure backend using
- Amplify CLI
- Amazon S3 Amazon Cognito
- AWS
- AppSync
- API
- Gateway

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EXAM REVIEW & TIPS

Exam Review & Tips

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State of learning checkpoint

1. Let’s look how far we’ve gone on our learning journey

2. https://aws.amazon.com/certification/certified-sysops-admin-associate/

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AWS Certified SysOps Exam Labs

- Starting with SOA-C02, the exam will contain 3 Labs
- Each exam lab consists of several different tasks
- AWS recommends to allocate 20 minutes for each exam lab
- First, you'll need to answer the multiple-choice/multiple-responses questions (you can't go back once you have completed this section)
- Second, you'll need to answer the exam labs
- You must complete all the work on the exam lab, before you move to the next one (you won't be able to go back to a lab once you have completed it)

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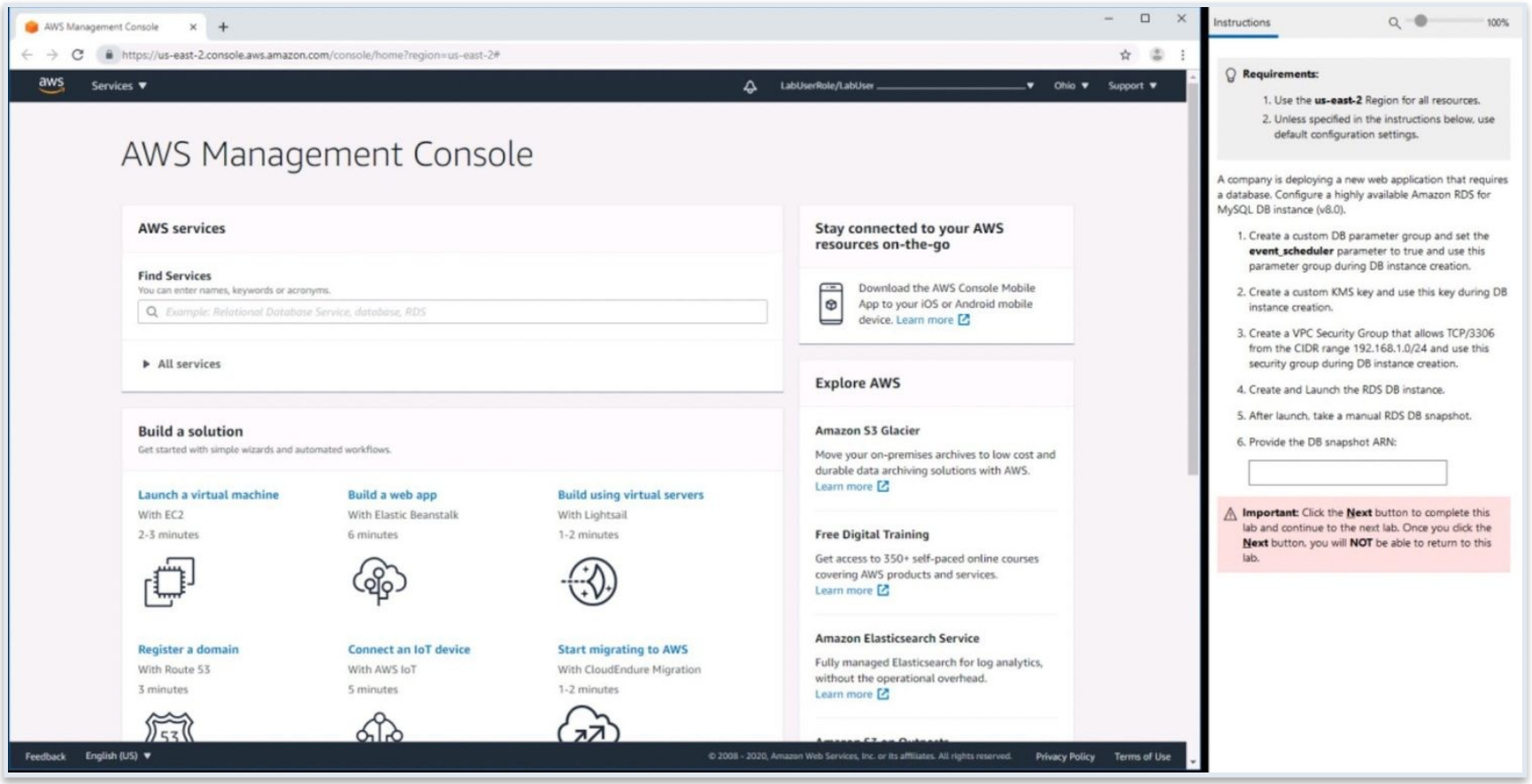
AWS Certified SysOps Exam Sample Lab



A company is deploying a new web application.

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Practice Exams

Access the Practice Exams from the Google Classroom

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SUMMARY

Q&A

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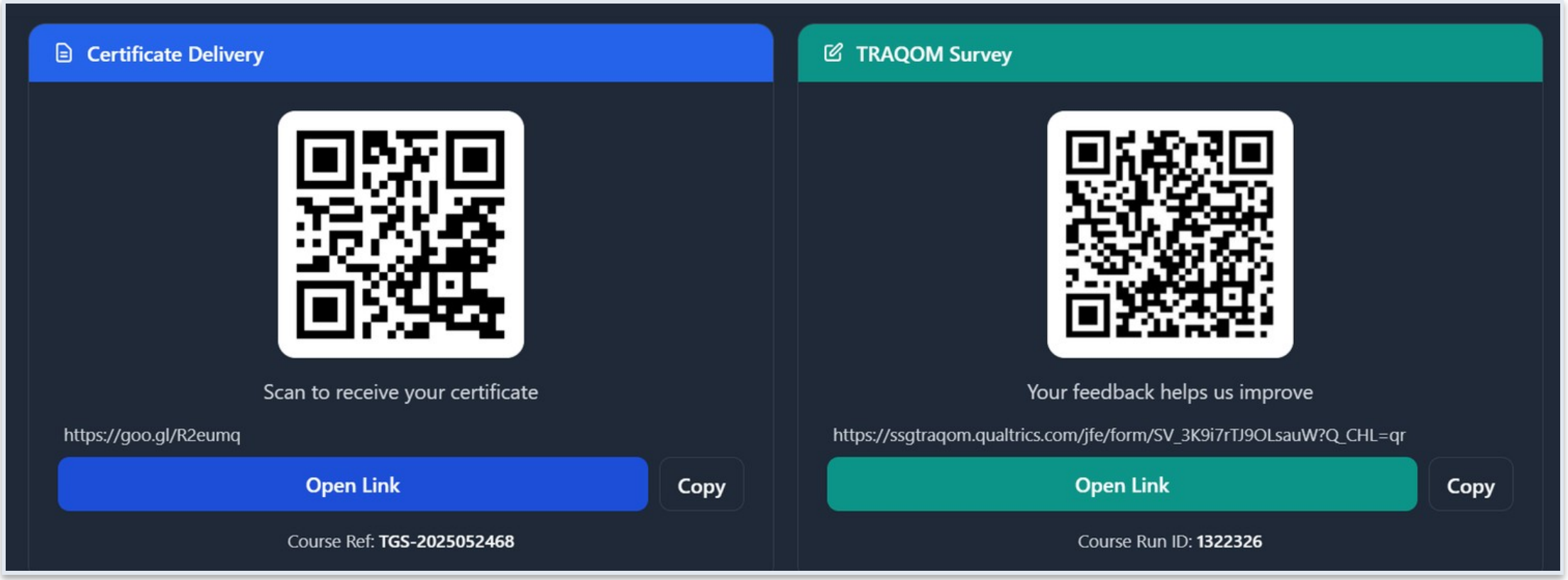
Cert & TRAQOM Survey (Mandatory)



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It is mandatory for you to take both AM, PM and Assessment digital attendance for WSQ funded courses.

- The trainer or administrator will show you the digital attendance QR code generated from SSG portal.
- Please scan the QR code from your mobile phone camera and submit your attendance.
- Digital Attendance (Mandatory)

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FINAL

Assessment

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Recommended Courses

1. WSQ - AWS Certified Solutions Architect Associate Training
2. WSQ - AWS Certified AI Practitioner Training
3. WSQ - AWS Certified Cloud Practitioner Training
4. WSQ - AWS Certified DevOps Engineer Training (Synchronous e-Learning)
5. WSQ - AWS Certified Machine Learning Engineer Associate Training

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Support

- If you have any enquiries during and after the class, you can contact us below
- Email: enquiry@tertiaryinfotech.com
- Tel: 61000613

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THANK YOU!

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